Astrophysical Fluid Dynamics

Lecture notes for ASTRO 442

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Contents

Preface . . . 9
How to use these notes . . . 9
Accessibility . . . 9
Use of AI . . . 9
License . . . 9

I Foundations

1 Fluids as Continua . . . 15
1.1 Fluid elements . . . 15
1.2 The continuity equation . . . 17
1.3 Radial flows . . . 20
1.4 In-class worksheet . . . 22
1.5 Exercises . . . 23
1.6 Further reading . . . 23

2 Thermodynamics Refresher . . . 25
2.1 Equation of state for ideal gases . . . 25
2.2 The first law of thermodynamics . . . 27
2.3 Radiation pressure . . . 28
2.4 Energy equation . . . 30
2.5 In-class worksheet . . . 30
2.6 Exercises . . . 31
2.7 Further reading . . . 31

3 Core-Collapse Supernovae . . . 33
3.1 Observational facts . . . 33
3.2 Shock breakout . . . 34
3.3 Radiation diffusion . . . 35
3.4 Ejecta expansion . . . 37
3.5 Radioactive heating . . . 39
3.6 Recombination and nebular phase . . . 41
3.7 In-class worksheet . . . 42
3.8 Exercises . . . 42
3.9 Further reading . . . 43

4 The Euler Equation . . . 45
4.1 Internal stresses . . . 45

4.2 Euler's equation . . . 46
4.3 In-class worksheet . . . 48
4.4 Further reading . . . 49

5 Eulerian and Lagrangian Descriptions . . . 51
5.1 Motivation . . . 51
5.2 Advective derivative . . . 52
5.3 Eulerian formulation of fluid mechanics . . . 53
5.4 Pressureless spherical accretion . . . 54
5.5 In-class worksheet . . . 55
5.6 Exercises . . . 55
5.7 Further reading . . . 55

6 Conservation Laws . . . 57
6.1 Momentum equation in conservation form . . . 57
6.2 Energy equation . . . 58
6.3 Heat conduction . . . 61
6.4 In-class worksheet . . . 62
6.5 Exercises . . . 62
6.6 Further reading . . . 63

7 The Gravitational Potential . . . 65
7.1 In-class worksheet . . . 65
7.2 Exercises . . . 65
7.3 Further reading . . . 66

8 Hydrostatic Equilibrium and the Virial Theorem . . . 67
8.1 The virial theorem . . . 67
8.2 Hydrostatic equilibria . . . 70
8.3 Application to star formation . . . 71
8.4 In-class worksheet . . . 72
8.5 Exercises . . . 72
8.6 Further reading . . . 73

II Waves, Shocks, and Outflows

9 Sound Waves and the Doppler Effect . . . 77
9.1 Oscillation Equation . . . 77
9.2 Wave Energy and Momentum . . . 81
9.3 In-class worksheets . . . 83
9.3.1 Sound waves . . . 83
9.3.2 The Doppler effect . . . 84
9.4 Exercises . . . 84
9.5 Further reading . . . 85

10 Waves in Stratified Atmospheres . . . 87
10.1 Waves in Isothermal Atmospheres . . . 87
10.2 Gravity Waves . . . 90
10.3 In-class worksheet . . . 91
10.4 Exercises . . . 92
10.5 Further reading . . . 93

11 Shock Waves . . . 95
11.1 In-class worksheet . . . 95
11.1.1 Shock waves I . . . 95
11.1.2 Shock waves II . . . 95
11.2 Exercises . . . 96
11.3 Further reading . . . 97

12 Radiative Shocks . . . 99
12.1 Exercises . . . 99
12.2 Further reading . . . 99

13 Blast Waves . . . 101
13.1 In-class worksheet . . . 101
13.2 Exercises . . . 102
13.3 Further reading . . . 102

14 Stellar Winds and Spherical Accretion . . . 103
14.1 In-class worksheet . . . 103
14.2 Exercises . . . 103
14.3 Further reading . . . 104

III Instabilities, Viscosity, Disks, and Turbulence

15 Fluid Instabilities I: Convection . . . 107
15.1 In-class worksheet . . . 107
15.2 Exercises . . . 107
15.3 Further reading . . . 107

16 Fluid Instabilities II: Rayleigh–Taylor and Kelvin–Helmholtz . . . 109
16.1 In-class worksheet . . . 109
16.2 Exercises . . . 110
16.3 Further reading . . . 110

17 Fluid Instabilities III: Thermal and Gravitational Instabilities . . . 111
17.1 Exercises . . . 111
17.2 Further reading . . . 111

18 Diffusion and Viscous Flows . . . 113
18.1 In-class worksheet . . . 113
18.2 Exercises . . . 113
18.3 Further reading . . . 114

19 Accretion Disks I: Structure and Viscous Evolution . . . 115
19.1 In-class worksheet . . . 115
19.2 Exercises . . . 115
19.3 Further reading . . . 117

20 Accretion Disks II: Steady Disks and Spectra . . . 119
20.1 In-class worksheet . . . 119
20.2 Exercises . . . 119
20.3 Further reading . . . 119

21 Turbulence . . . 121
21.1 What is Turbulence? . . . 121
21.2 Kolmogorov's Theory . . . 121
21.3 Shredding of Clouds in a Turbulent Flow . . . 126
21.4 Turbulence Modeling: Reynolds Stress . . . 127
21.5 In-class worksheet . . . 129
21.6 Exercises . . . 129
21.7 Further reading . . . 130

IV Plasma and Magnetohydrodynamics

22 Introduction to Plasma . . . 133
22.1 Electromagnetism Refresher . . . 133
22.1.1 Lorentz Force and Maxwell Equations . . . 133
22.1.2 Electrostatics . . . 134
22.1.3 Electromagnetic Waves . . . 136
22.2 Plasma . . . 137
22.2.1 What is a Plasma? . . . 137
22.2.2 When is a Plasma a Fluid? . . . 139
22.2.3 Cold Plasma Approximation . . . 140
22.3 In-class worksheet . . . 142
22.4 Exercises . . . 143
22.5 Further reading . . . 143

23 The MHD Equations . . . 145
23.1 Lorentz Force . . . 145
23.2 Ohm's Law . . . 145
23.3 Faraday's Law . . . 147
23.4 Conservation Laws . . . 148
23.5 In-class worksheet . . . 149

23.6 Exercises . . . 150
23.7 Further reading . . . 150

24 Flux Freezing . . . 151
24.1 The Ideal-MHD Limit . . . 151
24.1.1 Magnetic Field Amplification . . . 152
24.1.2 Resistive Evolution . . . 154
24.2 Star Formation . . . 155
24.3 Io and Jupiter . . . 157
24.4 In-class worksheet . . . 162
24.5 Exercises . . . 163
24.6 Further reading . . . 164

25 Magnetic Tension and Hydromagnetic Equilibria . . . 165
25.1 Magnetic Tension Force . . . 165
25.2 Hydromagnetic Equilibria . . . 168
25.3 In-class worksheet . . . 171
25.4 Exercises . . . 172
25.5 Further reading . . . 173

26 MHD Waves . . . 175
26.1 Waves in a Homogeneous Plasma . . . 175
26.2 Families of MHD Waves . . . 176
26.2.1 Alfvén Waves . . . 176
26.2.2 Magnetosonic Waves . . . 178
26.3 Effect of Propagation Direction . . . 179
26.3.1 Parallel Propagating Waves ($\theta = 0$) . . . 179
26.3.2 Perpendicularly Propagating Waves ($\theta = 90^\circ$) . . . 180
26.3.3 Friedrichs Diagrams . . . 180
26.4 Weak and Strong Field Limits . . . 180
26.5 Shock Waves . . . 181
26.6 In-class worksheet . . . 182
26.7 Exercises . . . 183
26.8 Further reading . . . 183

27 The Magnetorotational Instability . . . 185
27.1 Qualitative Picture . . . 185
27.2 Detailed Calculation . . . 186
27.3 In-class worksheet . . . 189
27.4 Exercises . . . 190
27.5 Further reading . . . 191

Appendices . . . 193

A Vector Calculus . . . 195
A.1 In-class worksheet . . . 195

Bibliography **197**

Preface

These are the lecture notes for ASTRO 442, an introduction to the fluid dynamics and magnetohydrodynamics of astrophysical systems. They are written to be self-contained: the textbook, Clarke and Carswell's *Principles of Astrophysical Fluid Dynamics* [1], is optional further reading. Each chapter ends with pointers to the relevant sections of the textbook.

> ! Work in progress
>
> These notes are a work in progress. I revise them as I teach the course, so you may still find typos, errors, and rough edges. Chapters marked "in preparation" will be completed before the corresponding lecture.
>
> Corrections and suggestions for improvement are welcome. You can report them by opening an issue on GitHub, or propose a change directly with a pull request.

How to use these notes

- **One chapter per lecture topic.** Canvas lists which chapter goes with each lecture.
- **Everything is numbered by chapter.** Equation (9.14) is the fourteenth numbered equation of Chapter 9, and Exercise 8.4 is the fourth exercise of Chapter 8. When you ask a question about the notes, quote these numbers.
- **In-class worksheets** appear near the end of each chapter, followed by the **exercises**. Homework assignments on Canvas list the exercises to solve, for example "Exercises 2.1, 7.3".
- **Units.** We use SI units, together with a few units that are customary in astronomy: number densities in cm^{-3}, mass densities in $\mathrm{g\,cm}^{-3}$, opacities in $\mathrm{cm}^2\,\mathrm{g}^{-1}$, large energies in erg, magnetic fields in gauss (given in parentheses after the value in tesla), and pc, AU, $M_\odot$, eV. Many astrophysics books and papers use Gaussian units for electromagnetism: Chapter 22 explains how to translate their formulas.
- **Optional material** is set apart in boxes titled "Going further"; you can skip it on a first reading.

Accessibility

These notes are designed to meet the Web Content Accessibility Guidelines (WCAG) 2.1, level AA. Mathematics is rendered with MathJax, which supports screen readers, speech, and braille, and lets you explore an equation term by term: click on an equation or focus it with the keyboard, then use the arrow keys. Every figure has a text description.

If something in these notes is not accessible to you, please let me know.

Use of AI

These notes started from my own lecture notes from previous years. They have been converted, corrected, and improved using AI tools (Anthropic's Claude models). I check and edit all AI-generated material.

License

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I Foundations

1 Fluids as Continua . . . 15
1.1 Fluid elements . . . 15
1.2 The continuity equation . . . 17
1.3 Radial flows . . . 20
1.4 In-class worksheet . . . 22
1.5 Exercises . . . 23
1.6 Further reading . . . 23

2 Thermodynamics Refresher . . . 25
2.1 Equation of state for ideal gases . . . 25
2.2 The first law of thermodynamics . . . 27
2.3 Radiation pressure . . . 28
2.4 Energy equation . . . 30
2.5 In-class worksheet . . . 30
2.6 Exercises . . . 31
2.7 Further reading . . . 31

3 Core-Collapse Supernovae . . . 33
3.1 Observational facts . . . 33
3.2 Shock breakout . . . 34
3.3 Radiation diffusion . . . 35
3.4 Ejecta expansion . . . 37
3.5 Radioactive heating . . . 39
3.6 Recombination and nebular phase . . . 41
3.7 In-class worksheet . . . 42
3.8 Exercises . . . 42
3.9 Further reading . . . 43

4 The Euler Equation . . . 45
4.1 Internal stresses . . . 45
4.2 Euler's equation . . . 46
4.3 In-class worksheet . . . 48
4.4 Further reading . . . 49

5 Eulerian and Lagrangian Descriptions . . . 51
5.1 Motivation . . . 51
5.2 Advective derivative . . . 52
5.3 Eulerian formulation of fluid mechanics . . . 53
5.4 Pressureless spherical accretion . . . 54
5.5 In-class worksheet . . . 55
5.6 Exercises . . . 55
5.7 Further reading . . . 55

6 Conservation Laws . . . 57
6.1 Momentum equation in conservation form . . . 57
6.2 Energy equation . . . 58
6.3 Heat conduction . . . 61
6.4 In-class worksheet . . . 62

6.5 Exercises . . . 62
6.6 Further reading . . . 63

7 The Gravitational Potential . . . 65
7.1 In-class worksheet . . . 65
7.2 Exercises . . . 65
7.3 Further reading . . . 66

8 Hydrostatic Equilibrium and the Virial Theorem . . . 67
8.1 The virial theorem . . . 67
8.2 Hydrostatic equilibria . . . 70
8.3 Application to star formation . . . 71
8.4 In-class worksheet . . . 72
8.5 Exercises . . . 72
8.6 Further reading . . . 73

1. Fluids as Continua

1.1 Fluid elements

Astrophysical systems contain very large numbers of elementary particles. However, for most applications, we are only interested in the dynamics of the gas or plasma on scales that are large compared to the typical length scale of interaction between its constituent particles. In this context, it is convenient to treat matter as a continuum. That is, instead of keeping track of the momentum and position of each particle in the system, we describe it using equations for the evolution of macroscopic, average quantities.

We imagine matter as being composed of **fluid elements**. These are parcels of gas, or plasma, that have a characteristic size that is much smaller than the scales in which we are interested, but that still contain a large enough number of fluid constituents, molecules, or ions, so that it makes sense to talk about their properties, such as temperature. In other words, we require a fluid element $\mathcal{V}$ to be a parcel of matter that can be considered to be in thermodynamic equilibrium. We will call **fluid** a system that is composed of fluid elements. As we will see later, the equations of motion for fluids do not explicitly depend on the size or shape of the fluid elements, so we will not attempt to define them more precisely. The only requirement is that fluid elements should exist. We will also discuss later what happens when the concept of fluid element breaks down. For the moment, we will assume that the system we are studying can be decomposed in fluid elements and we will derive equations of motion for these fluid elements.

How large should a fluid element be? The particles in a fluid element need to collide with each other frequently, so that they can exchange energy and momentum and reach thermodynamic equilibrium. The average distance traveled by a particle between two collisions is called the **mean free path**

$$\lambda = \frac{1}{n\sigma}, \tag{1.1}$$

where n is the number density (Equation (1.3)) and σ is the cross section for collisions. Fluid elements need to be much larger than λ, but also much smaller than the length scale L over which the properties of the system change. This is possible only if

$$\lambda \ll L. \tag{1.2}$$

The ratio λ/L is called the **Knudsen number**. In other words, the fluid approximation requires the Knudsen number to be small.

> Example
>
> Consider a cloud of atomic hydrogen in the interstellar medium with number density $n \sim 1\,\mathrm{cm}^{-3} = 10^6\,\mathrm{m}^{-3}$. The cross section for collisions between hydrogen atoms is $\sigma \sim 10^{-19}\,\mathrm{m}^2$, so the mean free path is $\lambda \sim 10^{13}\,\mathrm{m}$, or about 70 AU. This is much smaller than the typical size of interstellar clouds, $L \sim 1\,\mathrm{pc} \approx 3 \times 10^{16}\,\mathrm{m}$. So $\lambda/L \sim 3 \times 10^{-4}$ and the cloud can be treated as a fluid.

For plasmas, the mean free path does not tell the whole story. Magnetic fields force charged particles to gyrate around the field lines, so they cannot drift freely across the field lines even in the absence of collisions. For this reason, plasmas can sometimes be treated as fluids even when λ is large. We will come back to this point in Chapter 22.

We define the **number density** of a fluid element $\mathcal{V}$ to be

$$n = \frac{\text{“number of particles in } \mathcal{V}\text{ ”}}{\text{“volume of } \mathcal{V}\text{ ”}} \qquad [\mathrm{m}^{-3}]. \tag{1.3}$$

As the system evolves, the number of particles inside the volume element $\mathcal{V}$ remains constant, but the size of the volume changes, so n depends on the fluid element and on time.

An associated concept is that of the **mass density** (or simply density)

$$\rho = \frac{\text{“mass contained in } \mathcal{V} \text{ ”}}{\text{“volume of } \mathcal{V} \text{ ”}} \qquad [\mathrm{kg\,m^{-3}}]. \tag{1.4}$$

We use SI units in these notes, but we follow the astrophysical custom of quoting number densities in $\mathrm{cm^{-3}}$ and mass densities in $\mathrm{g\,cm^{-3}}$, with $1\,\mathrm{cm^{-3}} = 10^6\,\mathrm{m^{-3}}$ and $1\,\mathrm{g\,cm^{-3}} = 10^3\,\mathrm{kg\,m^{-3}}$. Note that we always convert them to SI before we use them in a formula. If the gas is composed of identical particles of mass m, then $\rho = mn$. For mixtures with many constituents

$$\rho = \sum_{i=1}^{N} m_i n_i = \sum_{i=1}^{N} m_i Y_i n = \mu m_u n, \tag{1.5}$$

where we have introduced the particle number fractions $Y_i = n_i/n$, the atomic mass unit $m_u \approx 1.66 \times 10^{-27}\,\mathrm{kg}$, and the **mean molecular weight**

$$\mu = \sum_{i=1}^{N} Y_i \frac{m_i}{m_u}, \tag{1.6}$$

which is the average mass of the particles in units of m_u. Note that μ is dimensionless. For example, $\mu \approx 1$ for atomic hydrogen, while $\mu \approx 1/2$ for fully ionized hydrogen, since there are one proton and one electron per atom. We can also decompose ρ in terms of the mass densities of the different species $\rho_i = m_i n_i$:

$$\rho = \sum_{i=1}^{N} \rho_i = \sum_{i=1}^{N} X_i \rho. \tag{1.7}$$

The $X_i = \rho_i/\rho$ are the mass fractions. Y_i measures the relative contribution of each species to the number density, while X_i measures the relative contribution to the mass density. In general, $X_i \neq Y_i$, unless all particle species have the same mass.

We also assume that each element has a well-defined velocity

$$\mathbf{v} = \text{“velocity of } \mathcal{V} \text{ ”} \qquad [\mathrm{m\,s^{-1}}] \tag{1.8}$$

and temperature

$$T = \text{“temperature of } \mathcal{V} \text{ ”} \qquad [\mathrm{K}]. \tag{1.9}$$

With these definitions in place, we are now ready to state explicitly the assumptions that we are making when we treat a gas or a plasma as a fluid. Consider region $\mathcal{R}$ filled with matter. We say that this system can be treated as a fluid if the following assumption holds.

Assumption 1.1 (Fluid assumption). *Every point $P \in \mathcal{R}$ belongs to one fluid element, with given values of density ρ, composition $\{Y_i\}$, velocity $\mathbf{v}$, and temperature T.*

This assumption tells us that it makes sense to say that there is a single value of density, temperature, and velocity at each point. This assumption is violated, for example, in systems in which different particle species move independently. Such systems can sometimes be described as **multi-fluids**, that is systems in which there are different kinds of fluid elements that interact with each other. Plasmas in weakly collisional regimes can sometimes be treated as multi-fluids. A gas of photons in vacuum, instead, cannot be treated as a (multi-)fluid, since at each point in space there are photons traveling in infinitely many different directions.

The assumption also implies that sufficiently small parcels of material are in thermodynamic equilibrium, so that fluid elements have a well defined temperature. This is also called **local thermodynamic equilibrium (LTE)** condition. This condition is violated in some impulsive phenomena, such as in shock waves. Violations of LTE can sometimes be included in the fluid model. We will see two examples later: the effect of viscosity and heat conduction and the treatment of shocks. In other cases, it is necessary to describe non-LTE effects using a combination of fluid and non-fluid models. For example, to simulate the effect of cosmic rays on the dynamics of clouds in the interstellar medium it is necessary to solve separate equations for the cosmic rays, which cannot be treated as a fluid, and for the rest of the matter, for which the fluid approximation holds very well.

It is important to note that whether or not the fluid assumption holds is not an absolute statement, but also depends on the time and length scale of interest. For example, the dynamics of dark matter in the Bullet cluster (Figure 1.1) cannot be described using the fluid approximation. This is because dark matter does not flow with a single velocity. On the other hand, on cosmological scales, dark matter dynamics can be reasonably well described as a pressure-less fluid, as long as streams of dark matter do not cross. The Zel'dovich approximation is a famous application of this model.

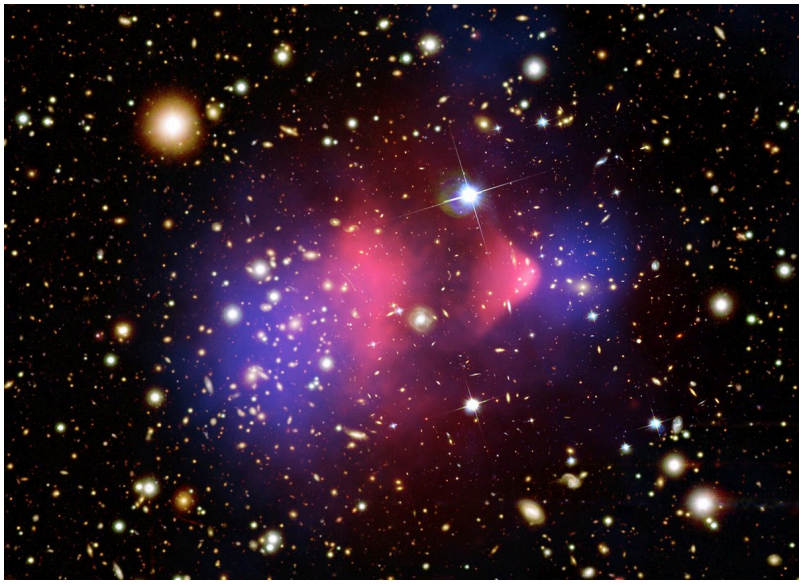


Figure 1.1: The Bullet cluster is the product of the head-on collision of two galaxy clusters (optical data). The gas in the two clusters has interacted producing a shock-wave (X-ray data in red) and trails the two groups of galaxies. Dark matter (blue halos) is nearly collisionless, so the two dark matter clouds associated with the clusters have passed through each other. Image credit: ESA, used under the ESA Standard Licence; not covered by the license of these notes.

1.2 The continuity equation

We now consider a system that can be described as a fluid. We wish to derive equations to predict how ρ, $\mathbf{v}$, and T evolve for all fluid elements. To this aim, we will use three fundamental laws in classical physics: 1) the law of conservation of mass; 2) Newton's second law; and 3) the first law of thermodynamics. The law of conservation of mass gives rise to the continuity equation, which we will now derive.

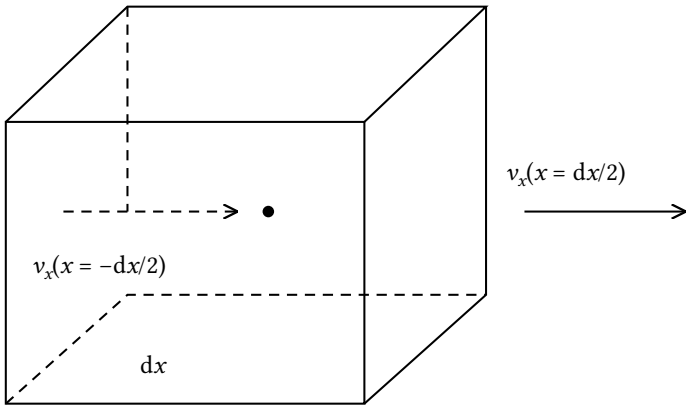


Figure 1.2: Fluid element stretching in the x-direction.

Consider the infinitesimal fluid element $\mathcal{V}$ shown in Figure 1.2. Let $\mathrm{d}x$ be the size of the fluid element along the x-direction at some time t. Let v_x be the x-component of the velocity. If we take the origin to

be the center of the fluid element, the velocity with which the left face is moving is $v_x\left(-\frac{\mathrm{d}x}{2}\right)$, while the velocity of the right face is $v_x\left(\frac{\mathrm{d}x}{2}\right)$. For the moment, we assume $v_y = v_z = 0$. The initial volume of $\mathcal{V}$ is $V = \mathrm{d}x\mathrm{d}y\mathrm{d}z$. At time $t + \mathrm{d}t$, $\mathcal{V}$ is deformed and has volume

$$V' = \left(\mathrm{d}x + \left[v_x\left(\frac{\mathrm{d}x}{2}\right) - v_x\left(-\frac{\mathrm{d}x}{2}\right)\right]\mathrm{d}t\right)\mathrm{d}y\mathrm{d}z = \left(\mathrm{d}x + \frac{\partial v_x}{\partial x}\mathrm{d}x\mathrm{d}t\right)\mathrm{d}y\mathrm{d}z, \tag{1.10}$$

so the relative rate in the change of volume of $\mathcal{V}$ is

$$\frac{1}{V}\frac{\mathrm{d}V}{\mathrm{d}t} = \frac{1}{V}\frac{V' - V}{\mathrm{d}t} = \frac{\partial v_x}{\partial x}. \tag{1.11}$$

In an analogous way, we can find the rate of change of an infinitesimal volume V if v_y and v_z are nonzero:

$$\frac{1}{V}\frac{\mathrm{d}V}{\mathrm{d}t} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} = \nabla \cdot \mathbf{v}. \tag{1.12}$$

We have derived this equation for a cubical fluid element. However, it turns out that this equation holds for infinitesimal fluid elements of any shape. The extension of this result to finite (as opposed to infinitesimal) regions is a special case of the **Reynolds transport theorem**, which states that

$$\frac{\mathrm{d}V_\Omega}{\mathrm{d}t} = \oint_{\partial\Omega} \mathbf{v} \cdot \mathbf{n}\mathrm{d}A = \int_\Omega \nabla \cdot \mathbf{v}\mathrm{d}V, \tag{1.13}$$

where Ω is some extended region comoving with the flow, V_Ω is its volume, $\partial\Omega$ is the surface enclosing Ω, $\mathbf{n}$ is the outwards pointing normal to it, and $\mathrm{d}A$ and $\mathrm{d}V$ are the area and volume elements.

The second equality follows from the Gauss divergence theorem. To see why the theorem holds, we can think of Ω as being composed of many infinitesimal fluid elements. Each of them changes its volume at the rate $(\nabla \cdot \mathbf{v})\mathrm{d}V$ (Equation (1.12)), so the rate of change of V_Ω is obtained by summing over all of them, which gives the volume integral of $\nabla \cdot \mathbf{v}$. The geometrical interpretation of the first equality is illustrated in Figure 1.3. The rate of change of the volume of Ω is equal to the surface integral of the component of the velocity normal to its boundary. This makes sense, because the tangential part of the velocity only slides the boundary along itself, while the normal component moves it outwards or inwards. Because $\nabla \cdot \mathbf{v}$ determines the rate of change of the fluid element volumes, it is also called the **expansion**.

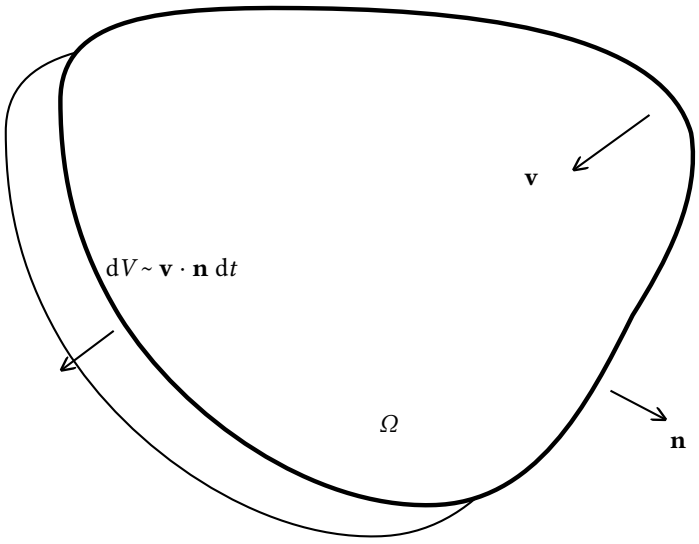


Figure 1.3: The instantaneous rate of change in the volume of a region is equal to the flux of the velocity field through its boundary.

If we denote by $\mathcal{M}$ the mass of a fluid element $\mathcal{V}$, and use the fact that $\mathcal{M}$ is conserved, then we find

$$\rho = \frac{\mathcal{M}}{V} \Rightarrow \frac{\mathrm{d}\rho}{\mathrm{d}t} = -\frac{\mathcal{M}}{V^2}\frac{\mathrm{d}V}{\mathrm{d}t} = -\frac{\mathcal{M}}{V}\frac{1}{V}\frac{\mathrm{d}V}{\mathrm{d}t} = -\rho\nabla\cdot\mathbf{v}. \tag{1.14}$$

We can rearrange the terms to obtain

$$\boxed{\frac{1}{\rho}\frac{\mathrm{d}\rho}{\mathrm{d}t} = -\nabla\cdot\mathbf{v}} \tag{1.15}$$

This is the **continuity equation**, the first of the fundamental equations of motion for fluids. Note that $\mathrm{d}/\mathrm{d}t$ denotes the rate of change of a quantity as we follow a fluid element. We will see in Chapter 5 how this is related to the rate of change at a fixed point in space.

The continuity equation can also be written in terms of n, as long as the number of particles in each fluid element is conserved, that is, in the absence of ionization, recombination, or reactions:

$$\frac{1}{n}\frac{\mathrm{d}n}{\mathrm{d}t} = -\nabla\cdot\mathbf{v}. \tag{1.16}$$

The continuity equation also holds separately for the n_i and ρ_i of each species in the fluid, as long as there are no reactions converting between the species, i.e., $\dot{X}_i = \dot{Y}_i = 0$, and all species move with the same velocity.

Example

A flow in which all fluid elements have the same velocity has $\nabla\cdot\mathbf{v} = 0$, since $\mathbf{v}$ is a constant, so the density of each fluid element does not change ($\mathrm{d}\rho/\mathrm{d}t = 0$).

Example

Consider a planar flow $\mathbf{v} = (-y, x, 0)$ [arbitrary units]. This is an example of a rotating flow, as shown in the visualization below:

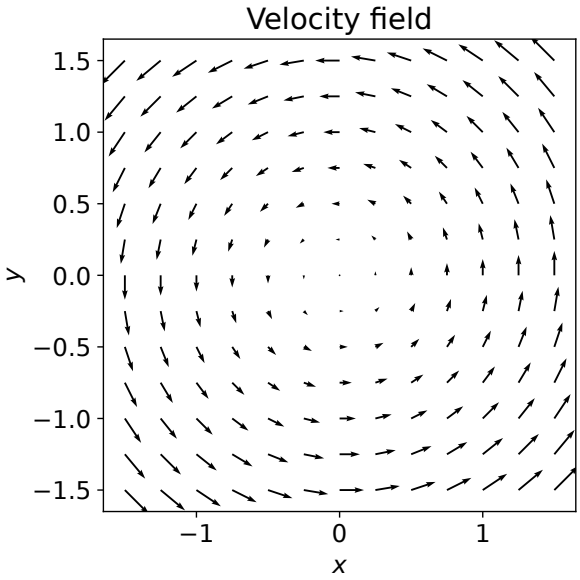


Figure 1.4: Velocity field $\mathbf{v} = (-y, x, 0)$.

This flow also preserves the density of each fluid element:

$$\nabla \cdot \mathbf{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} = \frac{\partial(-y)}{\partial x} + \frac{\partial(x)}{\partial y} = 0. \tag{1.17}$$

Example

In cylindrical coordinates (R, ϕ, z), a generic rotating flow with rotational axis along z takes the form

$$\mathbf{v} = v_\phi \hat{\boldsymbol{\phi}}. \tag{1.18}$$

Rotating flows preserve the volume of fluid elements if

$$\nabla \cdot \mathbf{v} = \frac{1}{R}\frac{\partial(R v_R)}{\partial R} + \frac{1}{R}\frac{\partial v_\phi}{\partial \phi} + \frac{\partial v_z}{\partial z} = \frac{1}{R}\frac{\partial v_\phi}{\partial \phi} = 0. \tag{1.19}$$

That is, as long as all fluid elements with fixed (R, z) rotate with the same angular frequency.

Flows for which $\nabla \cdot \mathbf{v} = 0$ everywhere are said to be **incompressible**. An important property of incompressible flows is that the net flux of the velocity field across any closed surface should vanish:

$$\oint_{\partial\Omega} \mathbf{v} \cdot \mathbf{n} \mathrm{d}A = \int_\Omega \nabla \cdot \mathbf{v} \mathrm{d}V = 0. \tag{1.20}$$

We can interpret this result graphically by introducing the velocity field lines, also called **streamlines**. These are curves $(x(s), y(s), z(s))$ satisfying

$$\frac{\mathrm{d}x}{\mathrm{d}s} = v_x, \quad \frac{\mathrm{d}y}{\mathrm{d}s} = v_y, \quad \frac{\mathrm{d}z}{\mathrm{d}s} = v_z. \tag{1.21}$$

In an incompressible flow there are no sources or sinks for the streamlines. This means that every streamline that enters a closed surface must also exit from it (Figure 1.5).

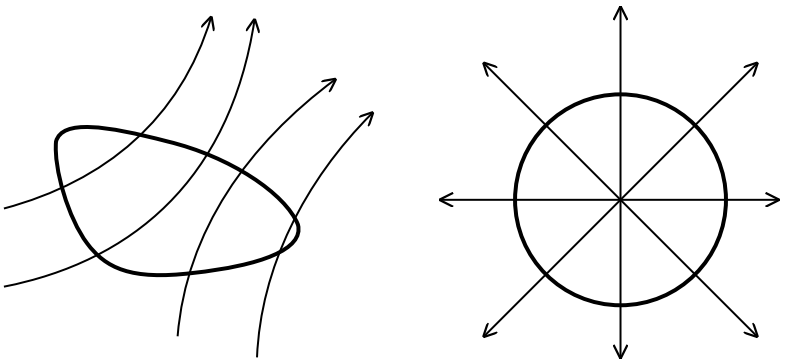


Figure 1.5: Streamlines in an incompressible flow (left) leave every closed surface they enter. Flows with expansion or compression (right) can have sources or sinks of streamlines. Note the analogy with the solenoidal geometry of magnetic field lines in magnetism.

1.3 Radial flows

Most astrophysical transients are the result of explosions that generate approximately radial outflows. As such, an important class of flows are those for which the gas is flowing in the radial direction:

$$\mathbf{v} = v_r \hat{\mathbf{r}}. \tag{1.22}$$

These are called **radial flows**. In the previous equation, v_r can depend on r and on the direction (θ, ϕ).

Using the divergence in spherical coordinates (Appendix A), we find that radial flows are generically compressible:

$$\begin{aligned}\nabla \cdot \mathbf{v} &= \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 v_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial \theta}(\sin\theta v_\theta) \\ &\quad + \frac{1}{r\sin\theta}\frac{\partial v_\phi}{\partial \phi} \\ &= \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 v_r) \\ &= \frac{\partial v_r}{\partial r} + \frac{2}{r} v_r.\end{aligned} \tag{1.23}$$

This expression vanishes only if $v_r \propto 1/r^2$, which is singular at the origin. So radial flows that are regular at the origin are always compressible.

Note that the radial direction, $\hat{\mathbf{r}}$, is not well defined at the origin, since all directions are radial there. This implies that $v_r(r=0) = 0$. This regularity condition also ensures that the fluid element at the origin is not "broken" (transported in different directions), but only stretched.

An important example of radial flow is that of a *homologously expanding* (or contracting) sphere of gas. Here, "homologous" means that the relative rate of change of the volume of each fluid element, $\frac{1}{V}\frac{\mathrm{d}V}{\mathrm{d}t}$, is equal for all fluid elements. To see how this works, we consider a sphere of gas that at time $t = 0$ has a constant density ρ_0. The assumption of constant density is not necessary, as all calculations will remain valid even for variable density, but it will make the results more transparent.

To see what this implies for the velocity, note that for a radial flow the condition that $\nabla \cdot \mathbf{v}$ is the same for all fluid elements reads $\partial_r v_r + 2v_r/r = 3H$, where H does not depend on r. The solutions of this equation are $v_r = Hr + B/r^2$. The regularity condition at the origin requires $B = 0$, so $v_r \propto r$.

Accordingly, in a homologous flow, the velocity of fluid elements that are at $r = q$ at $t = 0$ is given by

$$v_r(r = q, t = 0) = Hq, \tag{1.24}$$

where H is a constant. At later times, we assume that each fluid element continues to move with the velocity it had *initially*. In other words, we are assuming that fluid elements are not subject to any force and they coast at their original velocity. This is Newton's first law applied to the fluid elements.

To find the change in density in the system, we could be tempted to write

$$\nabla \cdot \mathbf{v} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 v_r) \overset{!!!}{=} \frac{1}{r^2}\frac{\partial}{\partial r}(r^3 H) = 3H, \tag{1.25}$$

but this is **incorrect**. The issue is that at any time $t \neq 0$, the velocity of a fluid element at the location r is not Hr. Instead, its velocity is Hq, where q is the position that that fluid element had at time $t = 0$.

To find it, we consider the equation of motion of the fluid element originally located at $r = q$:

$$\frac{\mathrm{d}r(t;q)}{\mathrm{d}t} = Hq, \qquad r(0;q) = q. \tag{1.26}$$

This can be integrated to find

$$r(t;q) = q + Hqt. \tag{1.27}$$

Solving for q gives

$$q = \frac{r}{1+Ht}. \tag{1.28}$$

This means that

$$v_r(r,t) = \frac{rH}{1+Ht} \neq rH. \tag{1.29}$$

The continuity equation gives

$$\frac{1}{\rho}\frac{\mathrm{d}\rho}{\mathrm{d}t} = -\nabla \cdot \mathbf{v} = -3\frac{H}{1+Ht}. \tag{1.30}$$

Note that the expansion (or compression if $H < 0$) rates depend only on H and t and not on the initial position of the fluid element q. In other words, all fluid elements are expanding/contracting at the same rate.

To make this result more transparent, we define $R(t)$ to be the radius of the expanding/contracting gas sphere at time t. Evidently, $\dot{R}(t) = HR_0$ and $R(t) = R_0(1 + Ht)$. So Equation (1.30) can be written as

$$\boxed{\frac{\dot{\rho}}{\rho} = -3\frac{\dot{R}}{R}} \tag{1.31}$$

This equation can be integrated to find $\rho \propto R^{-3}$. This should not be surprising, because the mass of the sphere, $\frac{4\pi}{3}\rho R^3$, is conserved. The expansion is illustrated in Figure 1.6.

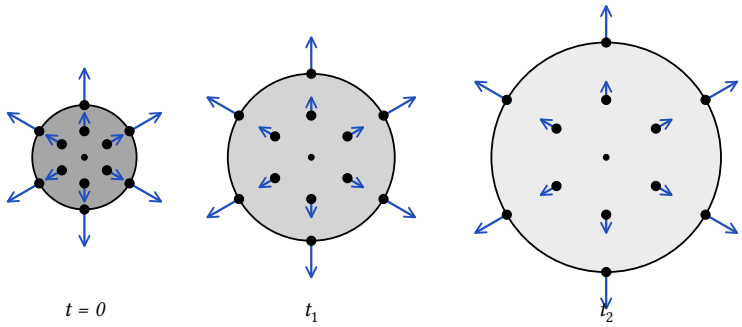


Figure 1.6: Homologous expansion of a sphere of gas, at $t = 0$ and at two later times. Each marked fluid element keeps its initial velocity (arrows), which is proportional to its initial distance from the centre. The pattern of fluid elements is the same at all times, only scaled up, and the density (shading) decreases as R^{-3}.

The flows produced by many types of astronomical explosions can be described well as homologously expanding flows at sufficiently late times, typically once the gas has expanded to several times its initial radius. This is because at early times pressure gradients accelerate the gas, converting internal energy into kinetic energy. As the gas expands, it cools adiabatically and the pressure forces become negligible. Afterwards, the fluid elements are not subject to any net force, so they coast at a fixed velocity. Fluid elements that started from a small region then end up at $r \approx vt$, which is a homologous flow. This is the basis for the models of supernova light curves that we will discuss in Chapter 3.

Example

Consider a supernova that ejects $M = 10\,M_\odot$ of gas, with the outer layers moving at $v = 5000\,\mathrm{km\,s^{-1}}$. After one year, the radius of the ejecta is $R \approx vt \approx 1.6 \times 10^{14}\,\mathrm{m}$, or about 1000 AU, so their mean density is

$$\bar{\rho} = \frac{3M}{4\pi R^3} \approx 1 \times 10^{-12}\,\mathrm{kg\,m^{-3}} = 1 \times 10^{-15}\,\mathrm{g\,cm^{-3}}. \tag{1.32}$$

This corresponds to about 10^9 particles per cm^3. Since $R \propto t$ at late times, the density decreases as t^{-3}, so after ten years it is a thousand times smaller.

1.4 In-class worksheet

1. Consider a gas with particle number density n and particle fractions $\{Y_i\}$. Let $\{m_i\}$ be the masses of the different atomic species in the gas. Write an explicit expression for
   a. The mean molecular weight μ;
   b. The mass fractions $\{X_i\}$.

2. Consider a gas composed of atomic hydrogen H and helium He. The particle fraction of hydrogen is 70%.
   a. What is the mean molecular weight of this gas?
   b. What is the mass fraction of He?
   c. The Sun has a helium mass fraction $X_{\mathrm{He}} \approx 0.27$, with the rest mostly hydrogen. What fraction of its atoms are helium?
3. Consider a radial flow $\mathbf{v} = v_r \hat{\mathbf{r}}$.
   a. Find $\nabla \cdot \mathbf{v}$ using the expression for the divergence of $\mathbf{v}$ in spherical coordinates:

$$\begin{aligned} \nabla \cdot \mathbf{v} = &\frac{1}{r^2}\frac{\partial}{\partial r}(r^2 v_r) \\ &+ \frac{1}{r\sin\theta}\frac{\partial}{\partial \theta}(\sin\theta v_\theta) \\ &+ \frac{1}{r\sin\theta}\frac{\partial v_\phi}{\partial \phi}, \end{aligned} \tag{1.33}$$

   b. Is the flow incompressible if $v_r = \text{const}$?

1.5 Exercises

Exercise 1.1 (Differential rotation of a star). A spherical star of radius R_* is differentially rotating with angular velocity

$$\Omega(R) = \frac{\Omega_0}{1 + \left(\frac{R}{R_*}\right)^2}, \tag{1.34}$$

where R is the cylindrical radius.

1. Is the flow incompressible? Why?
2. What is the velocity of the surface of the star as a function of the latitude angle θ?
3. The star has the radius of the Sun and an equatorial rotation period of 25 days. What is the rotation velocity of its surface at the equator and at a latitude of 45°?

1.6 Further reading

- C. Clarke and B. Carswell [1], Chapter 1 (introduction to concepts) and Chapter 2 (the fluid equations, including conservation of mass).

2. Thermodynamics Refresher

2.1 Equation of state for ideal gases

In our discussion so far, we have considered a gas to be a set of fluid elements, each with its own macroscopic velocity $\mathbf{v}$. However, gases are actually composed of particles, each with its own microscopic velocity. How do we reconcile these two pictures? For simplicity, let's consider a fluid element $\mathcal{V}$ containing a single species of atoms, each having mass m. Let $\mathbf{u}_i$ be the velocity of the ith particle. Then we can decompose $\mathbf{u}_i$ as

$$\mathbf{u}_i = \mathbf{v} + \mathbf{w}_i, \tag{2.1}$$

where

$$\mathbf{v} = \langle \mathbf{u} \rangle = \frac{1}{N} \sum_{i=1}^{N} \mathbf{u}_i, \tag{2.2}$$

is the average velocity and $\mathbf{w}_i$ is the peculiar velocity of each particle. By definition $\langle \mathbf{w} \rangle = 0$.

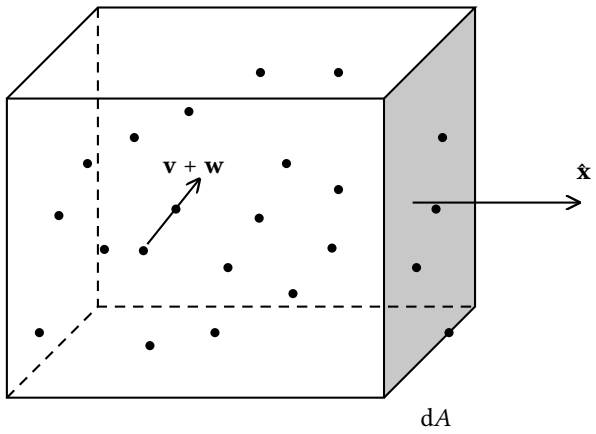


Figure 2.1: Decomposition of the velocity of particles in a fluid element.

Because the random component of the velocity averages to zero, it does not impact the continuity equation. To see this, consider the net mass crossing the grayed surface in Figure 2.1. Over any given $\mathrm{d}t$, only the particles in the region with volume $\langle u_x \rangle \mathrm{d}t\mathrm{d}A$ cross the surface, so the mass crossing the surface is

$$nm\langle u_x \rangle \mathrm{d}t\mathrm{d}A = \rho v_x \mathrm{d}t\mathrm{d}A. \tag{2.3}$$

However, the random motion of particles contributes to a net *momentum* flux. This is because each particle contributes a momentum mu_x, so the momentum flux is

$$F\mathrm{d}t = nm\langle u_x^2 \rangle \mathrm{d}t\mathrm{d}A = \rho v_x^2 \mathrm{d}t\mathrm{d}A + \rho \langle w_x^2 \rangle \mathrm{d}t\mathrm{d}A. \tag{2.4}$$

Note that the momentum flux per unit time has units of

$$[F] = [\rho v_x^2 \mathrm{d}A] = \mathrm{kg\,m\,s^{-2}} = \mathrm{N}, \tag{2.5}$$

where N (newton) is the SI unit of force, so the quantity $\rho\langle w_x^2\rangle$ is a force per unit area. Pressure is measured in pascal, $1\,\mathrm{Pa} = 1\,\mathrm{N\,m^{-2}}$. If the random motions are isotropic, that is, if they have no preferred direction, then $\langle w_x^2\rangle = \langle w_y^2\rangle = \langle w_z^2\rangle = \frac{1}{3}\langle|\mathbf{w}|^2\rangle$. Accordingly, we define as *gas pressure* the quantity

$$p = \frac{1}{3}\rho\langle|\mathbf{w}|^2\rangle. \tag{2.6}$$

The factor $1/3$ can be understood to come from the fact that pressure forces are the result of random motion in three directions. In analogy, the quantity ρv_x^2 is called the ram pressure. If you are swimming in a river the ram pressure is the force that tries to drag you along with the flow, while the pressure is responsible for the buoyancy you experience.

For a monoatomic gas with particles of mass m, the rms velocity can be computed using the equipartition theorem[1]:

$$m\langle|\mathbf{w}|^2\rangle = 3k_B T, \tag{2.7}$$

where T is the temperature in kelvin and k_B is the Boltzmann constant,

$$k_B = 1.380649 \times 10^{-23}\,\mathrm{J\,K^{-1}}. \tag{2.8}$$

Combining this with Equation (2.6) we find the equation of state for an ideal gas:

$$\boxed{p = nk_B T.} \tag{2.9}$$

Often, the equation of state for ideal gases is written in terms of the ideal gas constant $\mathcal{R}$ as

$$p = \frac{\mathcal{R}}{N_A m}\rho T, \tag{2.10}$$

where N_A is Avogadro's number. Since $n = \frac{\rho}{m}$, we see that $\mathcal{R}$ is just an alias for $N_A k_B$. For mixtures of monoatomic gases, Equation (2.9) remains unchanged, with n now being the number density of particles of any species, while Equation (2.10) becomes

$$p = \frac{\mathcal{R}}{N_A \mu m_u}\rho T = \frac{\rho k_B T}{\mu m_u}, \tag{2.11}$$

where m_u is the atomic mass unit and μ is the mean molecular weight (Equation (1.6)):

$$\mu = \frac{\rho}{n m_u}. \tag{2.12}$$

Thermal motion is not only a source of pressure, but also a reservoir of energy for the gas. In other words, because of their motion, the particles in the gas have average kinetic energy per unit mass

$$\varepsilon = \frac{1}{2}\langle|\mathbf{w}|^2\rangle = \frac{3}{2m}k_B T \Rightarrow k_B T = \frac{2}{3}m\varepsilon. \tag{2.13}$$

Using this definition we can eliminate the temperature from the ideal-gas equation of state:

$$p = \frac{2}{3}\rho\varepsilon, \tag{2.14}$$

where we have used the definition $\rho = nm$.

The last equation is a particular case of a Γ-law equation of state:

$$\boxed{p = (\Gamma - 1)\rho\varepsilon,} \tag{2.15}$$

where $\Gamma = \frac{5}{3}$ for a monoatomic gas. The Γ-law equation of state is a good approximation also for non monoatomic gases or mixture of gases, for which Equation (2.14) does not hold. For example, air can be

[1] That is, we assign an energy $\frac{1}{2}k_B T$ to each of the 3 translational degrees of freedom of the gas particles.

described with a Γ-law equation of state with $\Gamma = 1.4$. Another example is given by radiation-dominated gases, which we discuss below. Note that it is possible to rewrite Equation (2.15) as

$$p = (\Gamma - 1)u, \tag{2.16}$$

where $u = \rho\varepsilon$ is the internal energy density, measured in $\mathrm{J/m^3}$.

2.2 The first law of thermodynamics

The first law of thermodynamics applied to a fluid element of volume V and mass $\mathcal{M}$ reads

$$\mathrm{d}E = \delta Q - \delta W. \tag{2.17}$$

It states that the amount of energy in the fluid element increases by the amount of heat absorbed by the fluid element, δQ, and decreases by the amount of mechanical work done, δW.

For a perfect fluid, the mechanical work is given by

$$\delta W = p\mathrm{d}V, \tag{2.18}$$

where V is the volume of the fluid element. In the case of nonperfect fluids, there are corrections that arise from the anisotropic components of the stress tensor.

For a reversible process, the amount of absorbed heat is given by

$$\delta Q = T\mathrm{d}S. \tag{2.19}$$

In many cases, it is convenient to work with the first law of thermodynamics using specific (per unit mass) quantities:

$$\varepsilon = \frac{E}{\mathcal{M}}, \qquad w = \frac{W}{\mathcal{M}}, \qquad q = \frac{Q}{\mathcal{M}}, \qquad s = \frac{S}{\mathcal{M}}, \qquad v = \frac{V}{\mathcal{M}} = \frac{1}{\rho}. \tag{2.20}$$

Note that here v denotes the specific volume, that is, the volume per unit mass, and not the velocity. In terms of the specific quantities, the first law reads

$$\mathrm{d}\varepsilon = \delta q - \delta w = T\mathrm{d}s - p\mathrm{d}v = T\mathrm{d}s + \frac{p}{\rho^2}\mathrm{d}\rho. \tag{2.21}$$

In the case of adiabatic flows $\delta q = 0$, the first law simplifies to

$$\mathrm{d}\varepsilon = \frac{p}{\rho^2}\mathrm{d}\rho. \tag{2.22}$$

For a Γ-law equation of state $\varepsilon = \frac{p}{(\Gamma-1)\rho}$, so we find

$$\frac{1}{\rho}\mathrm{d}p - \frac{p}{\rho^2}\mathrm{d}\rho = (\Gamma - 1)\frac{p}{\rho^2}\mathrm{d}\rho \Rightarrow \frac{\mathrm{d}p}{p} = \Gamma\frac{\mathrm{d}\rho}{\rho}. \tag{2.23}$$

If Γ is constant, this can be integrated to yield

$$\boxed{p = K\rho^\Gamma,} \tag{2.24}$$

where K is an integration constant. Equation (2.24) represents a so-called **polytropic equation of state**. Note that for an adiabatic flow, K is constant with time in each fluid element, but K does not have to be the same in each fluid element. A comparison between Equation (2.24) and Equation (2.15) shows that

$$K = (\Gamma - 1)\rho^{1-\Gamma}\varepsilon. \tag{2.25}$$

For an ideal gas, K is a function of the specific entropy only, so we can think of K as a measure of the entropy of a fluid element.

Example

Consider a fluid element of monoatomic ideal gas ($\Gamma = 5/3$) that is compressed adiabatically, so that its density doubles. Since K is constant, the pressure increases by a factor $2^{5/3} \approx 3.2$. From the ideal-gas law, $T \propto p/\rho \propto \rho^{\Gamma-1}$, so the temperature increases by a factor $2^{2/3} \approx 1.6$.

Note that, if K is the same for all fluid elements, Equation (2.24) is barotropic, that is, the pressure depends only on the density. In this case, this equation of state, together with the continuity equation and the equation for the velocity (Euler's equation, Chapter 4), form a closed system of equations, using which it is, in principle, possible to compute all quantities in the flow. If K differs between fluid elements, the system is still closed, since K is constant along each fluid element ($\mathrm{d}K/\mathrm{d}t = 0$). However, the polytropic equation of state cannot be applied if $\delta q \neq 0$, due to the presence of heating or cooling of the gas. Even if the flow is adiabatic ($\delta q = 0$), there are irreversible thermodynamic processes that operate in shocks. When this happens, the steps we took to get Equation (2.24) from Equation (2.21) are no longer valid, because in irreversible processes the entropy of a fluid element increases even if no heat is exchanged, $T\mathrm{d}s > \delta q$. In all these situations, it is necessary to derive a new equation, in addition to the equation of state and the velocity and continuity equations, to close the system. In the presence of heating and cooling, this is the energy equation discussed below. Shocks require a different treatment, which we will discuss in Chapter 6 and Chapter 11.

2.3 Radiation pressure

Gas-pressure is not the only relevant source of pressure in astrophysical flows. Radiation pressure is another important component. While gas pressure arises due to the random (or thermal) component of the velocity of particles, radiation pressure arises as a result of momentum deposition by photons in optically thick conditions, in which radiation can be considered to be isotropic and in thermal equilibrium with the gas. We can derive the radiation pressure using a similar method as for the gas pressure, if we assume that photons carry momentum. However, to relate the radiation pressure to the temperature we need to use thermodynamics, and the value of the constant of proportionality can only be derived using quantum mechanics.

Consider a region filled with radiation and let u be the energy density in radiation. We assume that the radiation field is isotropic, meaning that there is an equal probability to find a photon traveling in each direction. In other words, the energy density of photons per unit solid angle of direction is $J = \frac{u}{4\pi}$. Let's compute the flux of momentum crossing an area $\mathrm{d}A$ with normal $\hat{\mathbf{n}}$ (shown in Figure 2.2).

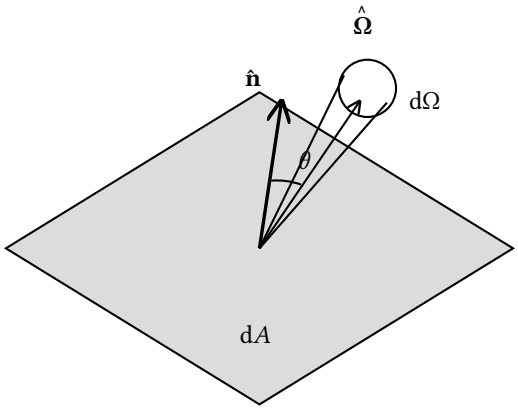


Figure 2.2: Area element for the calculation of the radiation pressure.

Each photon with energy $\varepsilon_{\mathrm{ph}}$ carries a momentum $\varepsilon_{\mathrm{ph}}/c$. Over the time interval $\mathrm{d}t$ photons in the volume $\mathrm{d}V = c\mathrm{d}t\mathrm{d}A\cos\theta$ will cross $\mathrm{d}A$, each carrying a normal component of the momentum equal to $(\varepsilon_{\mathrm{ph}}/c)\cos\theta$, so the overall flux of momentum is

$$\int J(\cos\theta)^2 \mathrm{d}t\mathrm{d}A\mathrm{d}\Omega = \frac{u\mathrm{d}t\mathrm{d}A}{4\pi}\int_0^{2\pi}\mathrm{d}\phi\int_0^{\pi}\mathrm{d}\theta\sin\theta(\cos\theta)^2 = \frac{1}{3}u\mathrm{d}t\mathrm{d}A \Rightarrow \boxed{p = \frac{1}{3}u.} \tag{2.26}$$

So we have found that the radiation pressure can be described with a Γ-law equation of state with $\Gamma = 4/3$. However, to be able to actually compute the pressure, we need to find a relation between u, p, and the temperature T. Consider the following thought experiment (Figure 2.3):

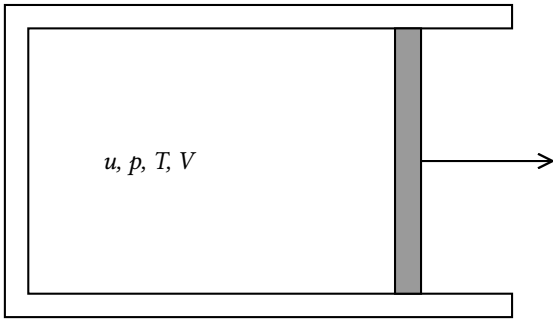


Figure 2.3: Blackbody enclosure.

The first and second law of thermodynamics read

$$\delta Q = \mathrm{d}E + p\mathrm{d}V, \quad T\mathrm{d}S = \delta Q. \tag{2.27}$$

The energy can be written in terms of the energy density as $E = uV$, so

$$\mathrm{d}E = V\mathrm{d}u + u\mathrm{d}V = V\frac{\mathrm{d}u}{\mathrm{d}T}\mathrm{d}T + u\mathrm{d}V \tag{2.28}$$

Solving Equation (2.27) for $\mathrm{d}S$ and using Equation (2.26) to eliminate p, we find

$$\mathrm{d}S = \frac{V}{T}\frac{\mathrm{d}u}{\mathrm{d}T}\mathrm{d}T + \frac{4}{3}\frac{u}{T}\mathrm{d}V. \tag{2.29}$$

We can rewrite the previous equation as:

$$\left(\frac{\partial S}{\partial T}\right)_V = \frac{V}{T}\frac{\mathrm{d}u}{\mathrm{d}T}, \qquad \left(\frac{\partial S}{\partial V}\right)_T = \frac{4}{3}\frac{u}{T}. \tag{2.30}$$

Since $\mathrm{d}S$ is an exact differential:

$$\frac{\partial^2 S}{\partial V\partial T} = \frac{1}{T}\frac{\mathrm{d}u}{\mathrm{d}T} = \frac{4}{3}\frac{1}{T}\frac{\mathrm{d}u}{\mathrm{d}T} - \frac{4}{3}\frac{u}{T^2} = \frac{\partial^2 S}{\partial T\partial V}. \tag{2.31}$$

Combining the terms we find

$$\frac{\mathrm{d}u}{\mathrm{d}T} = 4\frac{u}{T} \Rightarrow \boxed{u = aT^4.} \tag{2.32}$$

The constant of integration is the *radiation constant*. It can be derived using quantum mechanics:

$$a \cong 7.5657 \times 10^{-16}\,\mathrm{J\,m^{-3}\,K^{-4}}. \tag{2.33}$$

The radiation constant is related to the Stefan–Boltzmann constant σ by

$$a = \frac{4\sigma}{c}, \tag{2.34}$$

where c is the speed of light.

In general, both radiation and gas pressure can be present. For example, for ionized stellar matter the pressure and energy densities can be computed as

$$p = \frac{\rho k_B T}{\mu m_u} + \frac{1}{3} a T^4, \qquad u = \rho\varepsilon = \frac{3}{2}\frac{\rho k_B T}{\mu m_u} + a T^4. \tag{2.35}$$

Example

Consider the center of the Sun, where $\rho \approx 150\,\mathrm{g\,cm^{-3}} = 1.5 \times 10^5\,\mathrm{kg\,m^{-3}}$ and $T \approx 1.5 \times 10^7\,\mathrm{K}$. For fully ionized solar matter $\mu \approx 0.6$, so the gas pressure is

$$p_{\mathrm{gas}} = \frac{\rho k_B T}{\mu m_u} \approx 3 \times 10^{16}\,\mathrm{Pa}, \tag{2.36}$$

while the radiation pressure is

$$p_{\mathrm{rad}} = \frac{1}{3} a T^4 \approx 1.3 \times 10^{13}\,\mathrm{Pa}. \tag{2.37}$$

So radiation contributes only about 0.04% of the pressure at the center of the Sun. As we will see in Chapter 3, the situation is very different in the ejecta of supernovae, which are radiation dominated.

2.4 Energy equation

The first law of thermodynamics can be used to derive the energy equation for a perfect fluid. We start from the first law written for the energy per unit mass

$$\mathrm{d}\varepsilon = \delta q + \frac{p}{\rho}\frac{\mathrm{d}\rho}{\rho}. \tag{2.38}$$

Dividing both terms by $\mathrm{d}t$, multiplying by ρ, and using the continuity equation (Equation (1.15)) gives

$$\boxed{\rho \frac{\mathrm{d}\varepsilon}{\mathrm{d}t} = \rho\dot{q} - p\nabla\cdot\mathbf{v}.} \tag{2.39}$$

The first term on the RHS contains the net heating rate per unit mass, $\dot{q} = \dot{q}_{\mathrm{heat}} - \dot{q}_{\mathrm{cool}}$. For example, $\dot{q}_{\mathrm{heat}}$ might represent the energy generation due to nuclear burning in the gas, while $\dot{q}_{\mathrm{cool}}$ might be the energy lost by emitting radiation. The second term is the so-called pdV term. It represents the energy spent by doing work on the environment, or on other fluid elements. A list of common heating and cooling processes in astrophysics is given in C. Clarke and B. Carswell [1], Section 4.5.

2.5 In-class worksheet

For a reversible process in a gas with fixed composition, the first law of thermodynamics reads

$$\mathrm{d}E = T\mathrm{d}S - p\mathrm{d}V. \tag{2.40}$$

Define the specific (per unit mass) quantities:

$$\varepsilon = \frac{E}{\mathcal{M}}, \qquad w = \frac{W}{\mathcal{M}}, \qquad q = \frac{Q}{\mathcal{M}}, \qquad s = \frac{S}{\mathcal{M}}, \qquad v = \frac{V}{\mathcal{M}} = \frac{1}{\rho}. \tag{2.41}$$

1. Find an expression for $\mathrm{d}\varepsilon$ in terms of T, s, p, ρ.
2. Consider an ideal-gas with $p = (\Gamma - 1)\rho\varepsilon$ undergoing an adiabatic process ($\delta q = 0$).
   a. Show that
   $$p = K\rho^{\Gamma}. \tag{2.42}$$
   b. What is the relation between K and ε?

3. Consider an adiabatically expanding radiation-pressure dominated gas sphere. Let T and V be the temperature and volume of the gas at a given time t.
   a. Find an expression for the entropy of the gas S.
   b. Show that $pV^{4/3} = \text{const}$ as the gas expands.

2.6 Exercises

Exercise 2.1 (Equation of state of an H/He plasma). Consider a mixture of ionized hydrogen and helium with $X_{\mathrm{H}} = 70\%$ and $X_{\mathrm{He}} = 30\%$. The pressure is given by

$$p = p_{\mathrm{ions}} + p_e + p_{\mathrm{rad}}, \tag{2.43}$$

where p_{ions} is the partial pressure of the ions, p_e is the partial pressure of the electrons, and p_{rad} is the radiation pressure. Treat ions and electrons as ideal monoatomic gasses (with the appropriate mean molecular weights), and assume that the radiation is in thermal equilibrium with the gas.

1. Write a code to compute p_{ions}, p_e, and p_{rad} for any given density ρ and temperature T.
2. Plot p_{ions}, p_e, and p_{rad} as a function of temperature for $\rho = 150\,\mathrm{g\,cm^{-3}}$ (central density of the Sun) and for $\rho = 10^{-24}\,\mathrm{g\,cm^{-3}}$ (ISM density). At what temperature does radiation become dominant in each case?

2.7 Further reading

- C. Clarke and B. Carswell [1], Chapter 4 (the energy equation), in particular Sections 4.4–4.5 on energy transport and on heating and cooling processes.

3. Core-Collapse Supernovae

3.1 Observational facts

Supernovae are among the most energetic and luminous transients in the sky. They are the result of the explosion of massive stars, or of accreting white dwarfs at the end of their lives. Here, we use the tools of fluid mechanics we have developed so far to construct a simple model for these and similar explosive transients, such as novae and kilonovae. Our goal is to understand the basic physics, not to develop precise models, or to discuss all the nuances of the problem, for which we refer to the specialized literature.

Observationally, supernovae are characterized by a fast rise in the luminosity, on a timescale of a few days. They peak on a timescale of days to weeks and then decline over a timescale of months. A famous example of a core-collapse supernova light curve is given by the SN 1987A light curve in Figure 3.1. At peak magnitude, supernovae have luminosity $L \sim 10^{10} L_\odot$, where $L_\odot$ is the luminosity of the Sun. They remain bright for weeks to months, before beginning to fade. At peak, their typical effective temperature is $T_{\rm eff} \sim 2T_\odot \approx 12,000\,\mathrm{K}$, $T_\odot$ being the effective temperature of the Sun. For a black body, the Stefan–Boltzmann law gives a relation between temperature and luminosity:

$$L = 4\pi R^2 \sigma T^4, \tag{3.1}$$

where σ is the Stefan–Boltzmann constant

$$\sigma = \frac{ac}{4} = 5.6704 \times 10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}}. \tag{3.2}$$

Assuming in first approximation supernovae to be black body emitters, this implies that the characteristic radius of a supernova at peak magnitude is

$$R_p \sim 2.5 \times 10^4 R_\odot \approx 10^{13}\,\mathrm{m}. \tag{3.3}$$

This is about twenty times larger than the radius of the largest stars. The duration of the peak is of a few weeks, say $\sim 2 \times 10^6\,\mathrm{s}$, so the total energy release in photons is

$$\mathcal{E}_r \sim (2 \times 10^6\,\mathrm{s}) \times 10^{10} L_\odot \approx 8 \times 10^{49}\,\mathrm{erg}. \tag{3.4}$$

Here and in the rest of the chapter, we follow the custom of quoting large energies in erg and luminosities in $\mathrm{erg\,s^{-1}}$, with $1\,\mathrm{erg} = 10^{-7}\,\mathrm{J}$. From the observation of Doppler shifted emission and absorption lines, we know that the typical velocities are in the range of $2,000\,\mathrm{km\,s^{-1}}$ to $10,000\,\mathrm{km\,s^{-1}}$. These velocities are consistent with the estimated length and temporal scales of the supernova phenomenon:

$$v \sim \frac{10^{13}\,\mathrm{m}}{10^6\,\mathrm{s}} = 10^7\,\mathrm{m\,s^{-1}} = 10^4\,\mathrm{km\,s^{-1}}. \tag{3.5}$$

The associated kinetic energy is

$$K \sim \frac{1}{2} M v^2 \approx 10^{51} \left(\frac{M}{M_\odot}\right) \mathrm{erg} \gg \mathcal{E}_r. \tag{3.6}$$

From these basic observational facts, we know that supernovae must be powered by objects with large surface areas and require large energies. All these numbers already suggest that supernovae could be powered by the explosion of stars. This is most dramatically confirmed by the disappearance of stars previously observed at the location of supernovae.

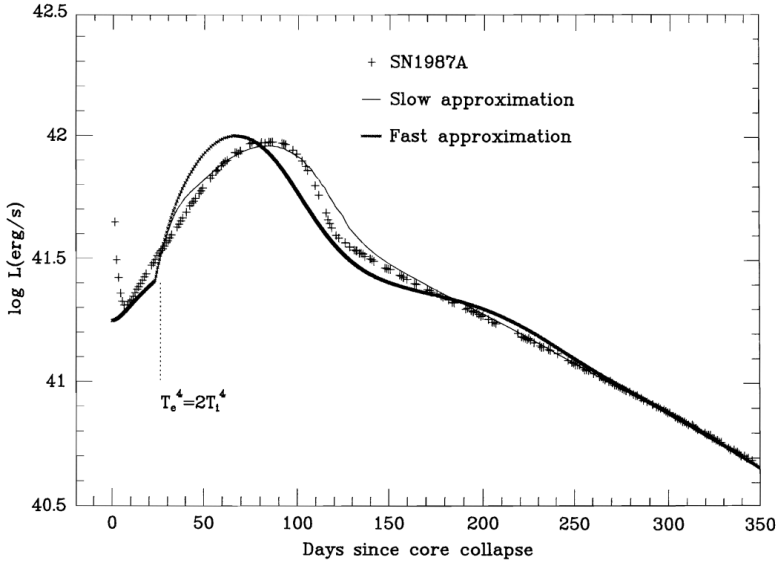


Figure 3.1: Bolometric light curve for SN 1987A (dots) and two semi-analytic models (lines). From D. Arnett [2]; © Princeton University Press, reproduced for non-commercial educational use and not covered by the license of these notes.

3.2 Shock breakout

Our supernova model begins with a star of radius R_0 in which a large amount of energy $\mathcal{E}$, much larger than the gravitational binding energy of the star, is injected. The source of this energy might be gravitational, in the case of core-collapse supernovae, or thermonuclear, but the outcome is similar: the progenitor is disrupted and matter starts expanding with large velocities. To fix our ideas, we will consider the explosion of a red supergiant with $R_0 \sim 10^{12}$ m, however all our results can be rescaled for other radii.

Immediately after the energy has been injected into the star, kinetic and internal energy are expected to be in equipartition $K \sim E \sim \mathcal{E} \sim 10^{51}$ erg $= 10^{44}$ J, so we can estimate the temperature, assuming the gas to be radiation dominated:

$$E = \frac{4}{3}\pi R_0^3 a T^4. \tag{3.7}$$

This estimate gives a temperature of

$$T \approx 4.2 \times 10^5 \,\mathrm{K} \left(\frac{R_0}{10^{12}\,\mathrm{m}}\right)^{-3/4} \left(\frac{E}{10^{51}\,\mathrm{erg}}\right)^{1/4}. \tag{3.8}$$

The energy is injected close to the center of the star and it is carried outwards by a shock wave (Chapter 11), which reaches the surface of the star after a time $\sim R_0/v \sim 10^5$ s, about a day (see Equation (3.32) below). When this happens, the hot radiation behind the shock is able to escape. This is the so-called *shock breakout*, which is the first light emanating from the supernova. Because of the high temperature, the radiation spectrum at breakout peaks in the UV and X-ray bands.

We can check if the assumption of radiation dominated gas is consistent by computing the ratio of radiation pressure to gas pressure:

$$\beta = \frac{\frac{1}{3}aT^4}{nk_BT} = \frac{a}{3k_B}\frac{T^3}{n} \approx \frac{a}{3k_B}\frac{4\pi R_0^3}{3}\frac{\mu m_u}{M}T^3, \tag{3.9}$$

where we have approximated n with the average number density, $n = \rho/(\mu m_u)$ (Equation (1.5)), and M is the mass of the progenitor. We find β:

$$\beta \approx 4.8 \times 10^3 \mu \left(\frac{1 M_\odot}{M}\right) \left(\frac{R_0}{10^{12}\,\mathrm{m}}\right)^{3/4} \left(\frac{E}{10^{51}\,\mathrm{erg}}\right)^{3/4}, \tag{3.10}$$

confirming that radiation dominates at early times. As we will see later, for homologous expansion, $E \sim R^{-1}$, so β is independent of the radius to which the ejecta has expanded, and matter remains radiation dominated.

3.3 Radiation diffusion

After the breakout, the star is full of radiation, but the photons need to make their way out of the explosion debris, before they can reach our telescopes. The dominant source of opacity is Thomson scattering on free electrons, but photons can also be absorbed and then re-emitted through various free-free, bound-free, and bound-bound processes. Their path out of the ejecta can be described as a random walk, such as the one illustrated in Figure 3.2.

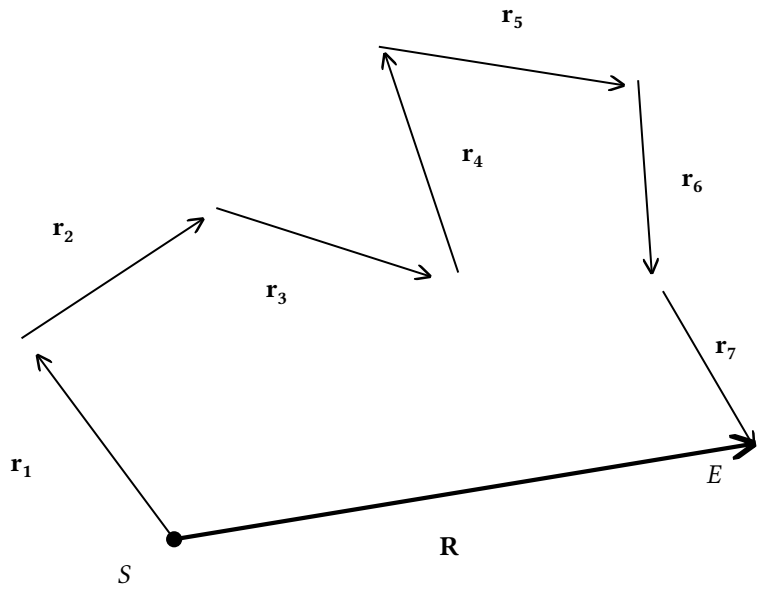


Figure 3.2: Random walk of a photon in the supernova ejecta.

Consider a photon emitted at some location S deep inside the supernova ejecta. The photon is not able to escape immediately. Instead, it propagates over a distance λ_1 before interacting with matter, either because it is scattered, or because it is absorbed and subsequently re-emitted in a different direction. The photon then moves in a new straight line over another distance λ_2 and interacts again. This process is repeated N times (7 in the figure). We want to estimate how many steps are necessary for the photon to escape.

Let $\mathbf{r}_1$, $\mathbf{r}_2$, etc., be the displacement vectors between emission and absorption locations. We want to compute the total displacement

$$\mathbf{R} = \mathbf{r}_1 + \mathbf{r}_2 + \mathbf{r}_3 + \cdots, \tag{3.11}$$

and compare $|\mathbf{R}|$ with the size of the expanding ejecta cloud R. Since the direction of each step is random

$$\langle \mathbf{R} \rangle = \langle \mathbf{r}_1 + \mathbf{r}_2 + \mathbf{r}_3 + \cdots \rangle = \langle \mathbf{r}_1 \rangle + \langle \mathbf{r}_2 \rangle + \langle \mathbf{r}_3 \rangle + \cdots = 0. \tag{3.12}$$

However, the mean square of the displacement does not vanish:

$$\begin{aligned}\langle \mathbf{R}^2 \rangle &= \langle (\mathbf{r}_1 + \mathbf{r}_2 + \mathbf{r}_3 + \cdots) \\ &\qquad \cdot (\mathbf{r}_1 + \mathbf{r}_2 + \mathbf{r}_3 + \cdots) \rangle \\ &= \langle \mathbf{r}_1^2 \rangle + \langle \mathbf{r}_2^2 \rangle + \langle \mathbf{r}_3^2 \rangle + \cdots \\ &\quad + 2\langle \mathbf{r}_1 \cdot \mathbf{r}_2 \rangle + 2\langle \mathbf{r}_1 \cdot \mathbf{r}_3 \rangle + \cdots\end{aligned} \tag{3.13}$$

The mixed products disappear, since the orientation of the displacement vectors is random, but

$$\langle \mathbf{r}_i^2 \rangle = \lambda^2 > 0, \tag{3.14}$$

up to a factor of order unity, where λ is the photon mean free path. We conclude that

$$\langle \mathbf{R}^2 \rangle = N\lambda^2. \tag{3.15}$$

Equating this to the size R of the supernova, we find that the number of interactions experienced by the photon before escaping is

$$N = \left(\frac{R}{\lambda}\right)^2. \tag{3.16}$$

Neglecting the time between absorption and re-emission, the time required for the photon to escape, the so-called radiative diffusion time, is given by

$$\tau_d = N\frac{\lambda}{c} = \frac{R^2}{\lambda c}. \tag{3.17}$$

The mean free path can be expressed as $\lambda = 1/(\kappa\rho)$, where κ is the opacity (compare with Equation (1.1), $\kappa\rho$ plays the role of $n\sigma$), and $\rho = \frac{3M}{4\pi R^3}$, so

$$\tau_d = \frac{\kappa\rho}{c}R^2 = \frac{3}{4\pi}\frac{\kappa M}{Rc}. \tag{3.18}$$

Our calculation did not account for geometrical effects, but captured the essence of the problem. A more careful numerical calculation reveals that the characteristic radiation diffusion time scale is

$$\boxed{\tau_d = \frac{\kappa M}{\beta R c}}, \tag{3.19}$$

where β is a geometrical factor, not to be confused with the ratio of radiation to gas pressure. For spherical geometry, $\beta \approx 13.8$.

Note that, since $R = R_0 + vt \approx vt$, the diffusion time scale decays with time. On the other hand, the hydrodynamic time scale $\tau_h = R/\dot{R} \approx t$ increases with time. Most of the radiation will escape at the time at which $t_p = \tau_d(t_p)$, which is also the time at which diffusion and hydrodynamic time scales become equal:

$$\boxed{t_p = \left(\frac{\kappa M}{\beta c v}\right)^{1/2}}. \tag{3.20}$$

This is the peak time of the *bolometric luminosity*. For Thomson scattering in a hydrogen-rich, ionized gas $\kappa = 0.4\,\mathrm{cm^2\,g^{-1}}$, that is, $0.04\,\mathrm{m^2\,kg^{-1}}$ (as is customary, we quote opacities in $\mathrm{cm^2\,g^{-1}}$), so

$$t_p = 1.4 \times 10^6\,\mathrm{s}\left(\frac{M}{M_\odot}\right)^{1/2}\left(\frac{\kappa}{0.4\,\mathrm{cm^2\,g^{-1}}}\right)^{1/2}\left(\frac{v}{10^4\,\mathrm{km\,s^{-1}}}\right)^{-1/2}. \tag{3.21}$$

The characteristic radius of the supernova at this time is

$$R_p = R_0 + t_p v \approx t_p v \sim 10^{13}\,\mathrm{m}. \tag{3.22}$$

This time scale and characteristic radius are in good agreement with observations. These formulae also show that the typical duration of the rise time of the supernova is proportional to the square root of the mass and of the opacity, and inversely proportional to the square root of the velocity. This implies that the explosion of less massive stars produces faster transients, while more massive stars produce light curves that evolve over longer timescales. These relations are key to interpret not only supernovae, but also other explosive transients, such as kilonovae. For SN 1987A the inferred velocity from spectroscopy is $v \sim 3000\,\mathrm{km\,s^{-1}}$ and the peak time is $t_p \approx 100$ days (Figure 3.1), so we can estimate $M \approx 10 M_\odot$. This gives an explosion energy of $\mathcal{E} = \frac{1}{2} M v^2 \sim 10^{51}$ erg. These estimates are only correct to within an order of magnitude, since they neglect important effects, such as recombination, which we discuss below.

3.4 Ejecta expansion

We describe the supernova outflow as a single fluid element undergoing homologous expansion (Section 1.3), the so-called "one-zone approximation". Accordingly, the equation for energy balance in the ejecta is (Equation (2.39))

$$\rho \frac{\mathrm{d}\varepsilon}{\mathrm{d}t} + p \nabla \cdot \mathbf{v} = \rho \dot{q}. \tag{3.23}$$

For the moment, we neglect radioactive heating, so $\dot{q}$ only accounts for the loss of energy due to the escape of photons. Using the fact that $\rho = M/V$ and M is constant, we can rewrite this as

$$\frac{\mathrm{d}(\varepsilon M)}{\mathrm{d}t} + p V \nabla \cdot \mathbf{v} = M \dot{q}. \tag{3.24}$$

The first term is just the time derivative of the total internal energy in the ejecta $E = \varepsilon M$, while the term on the RHS is minus the total luminosity, $L = -M\dot{q}$, since $\dot{q}$ only accounts for the energy lost by the escape of photons. Using the fact that $\nabla \cdot \mathbf{v} = \frac{1}{V}\frac{\mathrm{d}V}{\mathrm{d}t}$ (Equation (1.12)), the second term in the LHS can be rewritten as

$$p V \nabla \cdot \mathbf{v} = p \frac{\mathrm{d}V}{\mathrm{d}t}, \tag{3.25}$$

so the energy equation becomes

$$\dot{E} + p\dot{V} = -L. \tag{3.26}$$

This is, of course, just the first law of thermodynamics applied to the entire supernova.

Using dimensional arguments, we can estimate the luminosity as

$$\boxed{L \approx \frac{E}{\tau_d}}, \tag{3.27}$$

where E is the total thermal energy in the ejecta and τ_d is the radiative diffusion time. This formula is valid when $t \gtrsim \tau_d$, that is after the peak of the light curve, because photons are trapped for $t \ll \tau_d$. We should be careful with the interpretation of the results we will find using Equation (3.27). For example, we will find that the light curve decays for all times, but, as previously discussed, we actually expect the supernova light curve to rise and then decline. However, Equation (3.27) is only valid after the peak.

It is useful to rewrite the pdV term as

$$p\dot{V} = pV\frac{\dot{V}}{V} = 3pV\left(\frac{\dot{R}}{R}\right) = 3pV\frac{1}{\tau_h}, \tag{3.28}$$

where we have used the hydrodynamical timescale $\tau_h = R/\dot{R} = R/v$, and the fact that

$$\frac{\dot{V}}{V} = 3\frac{\dot{R}}{R}. \tag{3.29}$$

For a radiation dominated gas, $p = u/3$ (Equation (2.26)), so

$$p = \frac{1}{3}\rho\varepsilon \Rightarrow 3pV = M\varepsilon = E. \tag{3.30}$$

Combining the last two equations with Equation (3.27), we obtain

$$\dot{E} + \frac{E}{\tau_h} = -\frac{E}{\tau_d} \Rightarrow \boxed{\frac{\dot{E}}{E} = -\left(\frac{1}{\tau_h} + \frac{1}{\tau_d}\right).} \tag{3.31}$$

Remember that this expression is only valid for $t \gtrsim \tau_d$. Also note that both τ_h and τ_d depend on time.

At the time of break out, the hydrodynamical time scale is

$$\tau_{h,0} = \frac{R}{v} = 10^5\,\mathrm{s}\left(\frac{R_0}{10^{12}\,\mathrm{m}}\right)\left(\frac{v}{10^4\,\mathrm{km\,s^{-1}}}\right)^{-1}. \tag{3.32}$$

The radiation diffusion time scale can be estimated using $\kappa \approx 0.4\,\mathrm{cm^2\,g^{-1}}$ to be

$$\tau_{d,0} = 2\times 10^7\,\mathrm{s}\left(\frac{M}{M_\odot}\right)\left(\frac{R_0}{10^{12}\,\mathrm{m}}\right)^{-1} \gg \tau_{h,0}. \tag{3.33}$$

So at early times radiation is trapped in the expanding ejecta, however most of the thermal energy is "lost" to pdV work (it is converted to kinetic energy) on a time scale $t \sim \tau_h$, before it has a chance to escape as radiation. Indeed, at early times we can neglect the radiative losses to find

$$\frac{\dot{E}}{E} + \frac{\dot{R}}{R} = 0 \Rightarrow ER \sim \text{const} \Rightarrow E \sim \frac{1}{R}. \tag{3.34}$$

So after matter has expanded by a factor 10, the thermal energy is reduced to a tenth of the initial value. This explains why $\mathcal{E}_r \ll \mathcal{E}$ in supernovae. Indeed, since $R_p \sim 10R_0$ (Equation (3.22)), we expect $\mathcal{E}_r \sim \mathcal{E}/10 \sim 10^{50}$ erg, in agreement with Equation (3.4).

To calculate the bolometric light curve, we write the energy as

$$\frac{\dot{E}}{E} + \frac{\dot{R}}{R} = -\frac{1}{\tau_d(t)} \Rightarrow \frac{\mathrm{d}\ln(ER)}{\mathrm{d}t} = -\frac{1}{\tau_d(t)}. \tag{3.35}$$

and expand $\tau_d(t)$ as:

$$\tau_d(t) = \tau_{d,p}\frac{R_p}{R_p + vt} = \tau_{d,p}\frac{R_p/v}{R_p/v + t} = \frac{\tau_{d,p}\tau_{h,p}}{\tau_{h,p} + t}, \tag{3.36}$$

where $\tau_{d,p}$ and $\tau_{h,p}$ are the diffusion and hydrodynamic time scale at the peak of the light curve, respectively, and where we now measure t from the time of the peak.

In this way we find

$$\frac{\mathrm{d}\ln(ER)}{\mathrm{d}t} = -\frac{t + \tau_{h,p}}{\tau_{d,p}\tau_{h,p}} \Rightarrow ER = (E_pR_p)\exp\left(-\frac{\tau_{h,p}t + t^2/2}{\tau_{d,p}\tau_{h,p}}\right). \tag{3.37}$$

The last expression gives the expression for the luminosity, because

$$L = \frac{E}{\tau_d} = \left(\frac{\beta c}{\kappa M}\right)ER, \tag{3.38}$$

so

$$\boxed{L = \left(\frac{\beta c E_p R_p}{\kappa M}\right)\exp\left(-\frac{\tau_{h,p}t + t^2/2}{\tau_{d,p}\tau_{h,p}}\right) \quad \text{for } t \geq 0.} \tag{3.39}$$

The luminosity at peak can be estimated as

$$
\begin{aligned}
L_0 &= \frac{\beta c E_p R_p}{\kappa M} = \frac{E_p}{\tau_{d,p}} \\
&\approx 5.2 \times 10^{43}\,\mathrm{erg\,s^{-1}} \left(\frac{E_p}{10^{50}\,\mathrm{erg}}\right) \\
&\quad \times \left(\frac{R_p}{10^{13}\,\mathrm{m}}\right)\left(\frac{M_\odot}{M}\right)\left(\frac{0.4\mathrm{cm^2\,g^{-1}}}{\kappa}\right)
\end{aligned}
\tag{3.40}
$$

For the explosion of a red supergiant with $M \sim 10 M_\odot$, we have $R_p \sim 10 R_0 \sim 10^{13}$ m and $E_p \sim \mathcal{E}/10 \sim 10^{50}$ erg, so this formula predicts $L_0 \sim 5 \times 10^{42}\,\mathrm{erg\,s^{-1}}$, comparable to the luminosity of Type II-P supernovae (Figure 3.5). For SN 1987A, whose progenitor was a compact blue supergiant with $R_0 \approx 3 \times 10^{10}$ m, this estimate is about ten times smaller than the observed peak. As we will see below, the light curve of SN 1987A is powered by radioactive decays.

This shows that, in absence of radioactive heating, or when radioactive heating is subdominant, the luminosity declines rapidly after having peaked at $t = t_p$. At late times, $t \gtrsim \tau_{h,p}$, the luminosity decreases as $\exp(-t^2/\tau_{\rm lc}^2)$, where the characteristic decay time for the light curve is given by

$$
\begin{aligned}
\tau_{\rm lc} &= \sqrt{2\tau_{d,p}\tau_{h,p}} \\
&\approx 2 \times 10^6\,\mathrm{s} \left(\frac{\kappa}{0.4\,\mathrm{cm^2\,g^{-1}}}\right)^{1/2} \left(\frac{M}{M_\odot}\right)^{1/2} \\
&\quad \times \left(\frac{v}{10^4\,\mathrm{km\,s^{-1}}}\right)^{-1/2}.
\end{aligned}
\tag{3.41}
$$

Note that $\tau_{\rm lc}$ does not depend on the expansion radius, but remains constant. Since $\tau_{d,p} = t_p$ and $\tau_{h,p} \approx t_p$, we also have $\tau_{\rm lc} \approx \sqrt{2} t_p$, which explains why Equation (3.41) and Equation (3.21) have the same scaling.

This simple model with no radioactive heating already predicts several key properties of supernovae and other explosive transients. It shows that the luminosity is proportional to the explosion energy per unit mass and to the radius of the progenitor, and inversely proportional to the opacity. Indeed, using $E_p \approx \mathcal{E} R_0/R_p$ in Equation (3.40), we find $L_0 \approx \beta c \mathcal{E} R_0/(\kappa M)$. It also shows that the typical duration of the transient is proportional to the square root of the mass and of the opacity, and inversely proportional to the square root of the velocity. For example, we have found that the explosion of less massive stars produces faster transients, while more massive stars produce light curves that evolve over longer timescales.

3.5 Radioactive heating

The radioactive decay of ^{56}Ni to ^{56}Co and of ^{56}Co to ^{56}Fe powers the light curve of supernovae. These decays produce γ-rays and positrons. Some of these particles escape from the ejecta, so their energy does not contribute to the optical light curve. However, especially in the first weeks, the majority of the decay products are reprocessed into thermal radiation. After having included this new heating source, the first law of thermodynamics is:

$$
\dot{E} + p\dot{V} = \dot{Q}_{\rm heat} - L, \tag{3.42}
$$

where $\dot{Q}_{\rm heat}$ is the energy deposited by the radioactive decays in $\mathrm{erg\,s^{-1}}$.

Proceeding as before, we can introduce a heating time scale

$$
\tau_{\rm heat} = \frac{E}{\dot{Q}_{\rm heat}}, \tag{3.43}
$$

and obtain the energy equation

$$
\frac{\dot{E}}{E} + \frac{\dot{R}}{R} = \frac{1}{\tau_{\rm heat}} - \frac{1}{\tau_d} \Rightarrow \boxed{\frac{\mathrm{d}\ln(ER)}{\mathrm{d}t} = \frac{1}{\tau_{\rm heat}} - \frac{1}{\tau_d}.} \tag{3.44}
$$

With the inclusion of radioactive heating, the light curve changes qualitatively. In particular, the luminosity can continue to grow even after $t \gtrsim \tau_d$ and reach a higher maximum. To see this, we use the fact

that $ER = (\kappa M/\beta c)L \propto L$ (Equation (3.38)), which implies that the peak is reached when the RHS of Equation (3.44) vanishes:

$$\frac{1}{\tau_{\rm heat}} = \frac{1}{\tau_d} \Rightarrow \frac{\dot{Q}_{\rm heat}}{E} = \frac{1}{\tau_d} \Rightarrow \boxed{\dot{Q}_{\rm heat} = \frac{E}{\tau_d} = L.} \tag{3.45}$$

This shows that the luminosity at peak equals the heating rate. This is a key result in supernova theory called "Arnett's rule"[2]. Since the energy released by each nuclear decay is known from laboratory experiments, Arnett's rule provides a way to estimate the amount of radioactive material produced by the supernova. Type Ia supernovae (Figure 3.4), which are the result of the thermonuclear explosion of white dwarfs, are an example of supernovae in which radioactivity plays the dominant role in the light curve over all times and for which Arnett's rule can be applied.

The effect of radioactive heating is illustrated in Figure 3.3, which shows the solution of the one-zone model, Equation (3.44) with $L = E/\tau_d$, for the explosion of a red and of a blue supergiant. The explosion of the compact blue supergiant would be very faint without radioactive heating, because most of its thermal energy is lost to $p\mathrm{d}V$ work. With radioactive heating, the light curve has a broad peak after about four months, which is qualitatively similar to the one of SN 1987A (Figure 3.1). Note that the peak is reached when the luminosity equals the heating rate, as predicted by Arnett's rule.

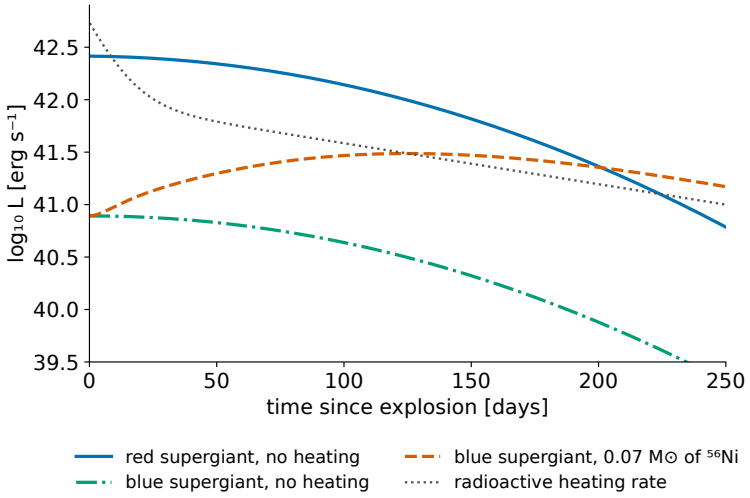


Figure 3.3: Light curves from the one-zone model for the explosion of a red supergiant ($R_0 = 10^{12}$ m) and of a compact blue supergiant ($R_0 = 3 \times 10^{10}$ m), with $M = 10\,M_\odot$, $v = 3000\,\mathrm{km\,s^{-1}}$, and an initial thermal energy of 5×10^{50} erg. The dashed line includes the heating by the decay of $0.07\,M_\odot$ of ^{56}Ni, whose rate is shown by the dotted line. The model is not accurate at early times, before the peak (see text).

[2]

1.

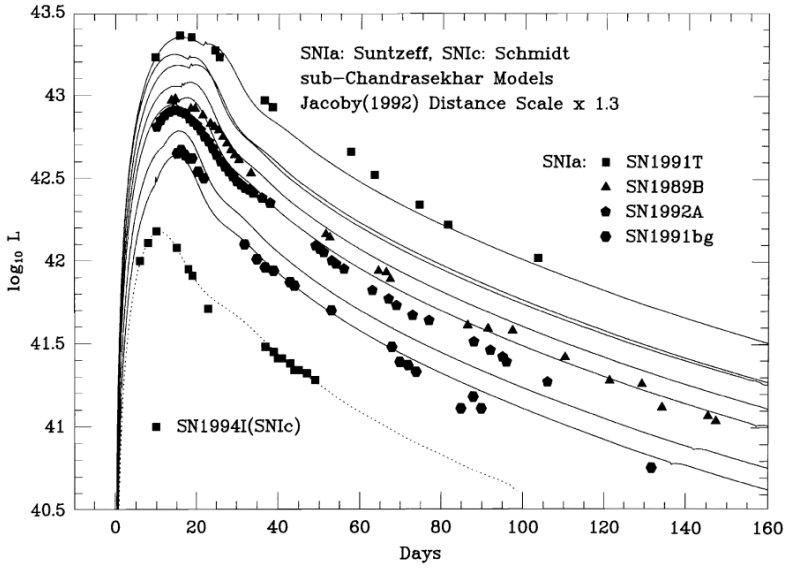


Figure 3.4: Sample of Type Ia thermonuclear supernova light curves: data and theoretical models. From D. Arnett [2]; © Princeton University Press, reproduced for non-commercial educational use and not covered by the license of these notes.

3.6 Recombination and nebular phase

After the peak of the light curve, as the explosion debris expand and cool, other physical effects start to become important. First, a larger fraction of gamma rays and positrons escape and/or are not thermalized. Secondly, ions begin to recombine. This removes the free electrons that were responsible for the Thomson opacity from the gas which then becomes transparent in the infrared/optical bands. Thus, as matter recombines, radiation quickly escapes, rapidly cooling down the material. This process launches a recombination wave into the supernova debris. The photons we see at this stage are no longer coming from the outer edge of the ejecta, but instead are emitted at a photosphere that is very close to the recombination front.

Recombination of hydrogen and helium is important for many classes of supernovae, including for SN 1987A, but their impact is particularly spectacular in hydrogen rich, Type II-P supernovae. Because recombination of hydrogen happens at a fixed temperature[3], the luminosity can be estimated as

$$L = 4\pi R_r^2(t)\sigma T_r^4, \tag{3.46}$$

where $T_r \approx 6000\,\mathrm{K}$ is the recombination temperature (a constant), and $R_r(t)$ is the position of the recombination front (which nearly coincides with the photosphere). In Type II-P supernovae, $R_r(t) \sim$ const for ~ 100 days and the bolometric luminosity plateaus to a nearly constant value (P stands for plateau). A prototypical example is SN 1999em, shown in Figure 3.5.

Example

Consider the plateau of SN 1999em. If we take the radius of the recombination front to be $R_r \sim 10^{13}$ m, as estimated in Equation (3.22), and $T_r \approx 6000\,\mathrm{K}$, Equation (3.46) gives

$$L \approx 4\pi(10^{13}\,\mathrm{m})^2\sigma(6000\,\mathrm{K})^4 \approx 9\times10^{34}\,\mathrm{W} = 9\times10^{41}\,\mathrm{erg\,s^{-1}}, \tag{3.47}$$

or $\log L \approx 42.0$, in good agreement with the plateau in Figure 3.5.

After the recombination wave has swept through the ejecta, the supernova enters the so called nebular phase. In this stage there is no longer a photosphere, because the material is transparent. The black body approximation fails, because radiation is produced by various decays and matter is no longer in

[3] But note that the onset of recombination depends on the composition.

thermodynamic equilibrium. Often, the luminosity decays exponentially, with characteristic time scale equal to the mean lifetime of ^{56}Co, $\tau \approx 111$ days (its half life is 77 days), so it is possible to fit the decay tail to estimate the amount of radioactive material. This has to be done carefully, accounting for the thermalization efficiency of positrons and gamma-rays.

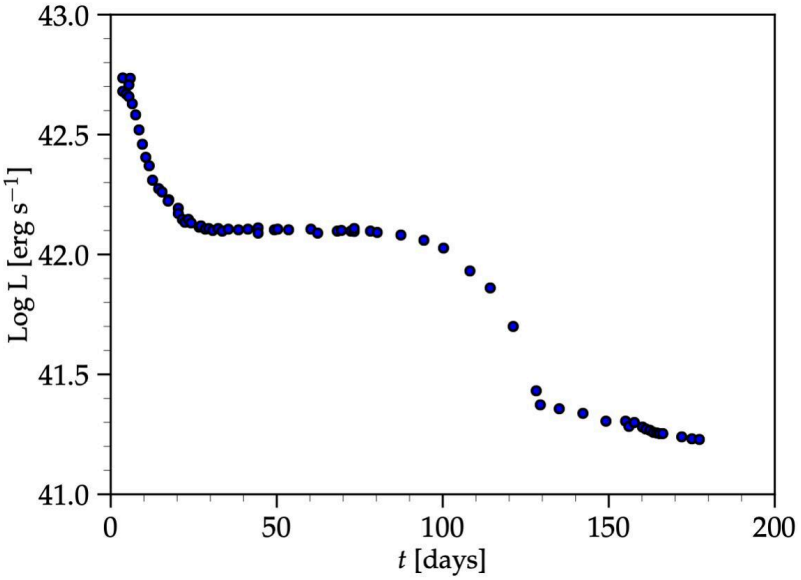


Figure 3.5: Bolometric light curve for SN 1999em, a classical example of a Type II-P supernova. Data from A. Elmhamdi *et al.* [3]; figure courtesy of Viktoriya Morozova, used with permission and not covered by the license of these notes.

3.7 In-class worksheet

1. Consider the explosion of a red supergiant with radius $R_0 = 10^{12}$ m and mass $M \sim 10\,M_\odot$. At time $t = 0$, we inject a large amount of energy into the star, $\sim 10^{51}$ erg $= 10^{44}$ J.

   Constants: $a = 7.6 \times 10^{-16}$ J m^{-3} K^{-4}, $k_B = 1.4 \times 10^{-23}$ J/K, $h = 6.6 \times 10^{-34}$ J s.

   a. Find the temperature of the gas, and determine the waveband in which the black body emission peaks.

   b. Show that the gas is radiation dominated. Compute $\beta = P_{\rm rad}/P_{\rm gas}$.

2. Given the mean free path $\lambda = 1/(\kappa\rho)$, and the definition of the photon diffusion time through a system of size R, $\tau_d = R^2/\lambda c$, find the time needed for a photon to escape the Sun, after being created in the core. Take $\kappa \sim 25$ cm^2 g^{-1}.

3. Show that:

$$\tau_d(t) = \frac{\tau_d(0)\tau_h(0)}{\tau_h(0) + t}. \tag{3.48}$$

3.8 Exercises

Exercise 3.1 (The Katz integral). Consider a homologously expanding radioactive cloud of debris, with radius $R \approx vt$, launched by an explosion of some kind, e.g., a supernova.

1. Assume that the gas is described by the gamma-law equation of state

$$p = (\Gamma - 1)\rho\varepsilon. \tag{3.49}$$

   Show that the first law of thermodynamics becomes

$$\frac{\mathrm{d}E}{\mathrm{d}t} = -L + \dot{Q} - 3(\Gamma - 1)\frac{E}{t}, \tag{3.50}$$

where L is the luminosity (energy radiated away in erg/s), and $\dot{Q}$ is the radioactive heating rate in erg/s.

2. Show that the following exact relation holds *under suitable assumptions*:

$$\int_0^t L(t')t'\,\mathrm{d}t' = \int_0^t \dot{Q}(t')t'\,\mathrm{d}t'. \tag{3.51}$$

The quantity on the LHS is called the Katz integral and can be determined observationally. Discuss the validity of the approximations made to derive this result.

3. The radioactive heating can be written as

$$\dot{Q} = M(t)\frac{q}{\tau}, \tag{3.52}$$

where q is the radioactive energy release in J/kg, τ is the radioactive decay timescale, and $M(t) = M_0 \mathrm{e}^{-t/\tau}$ is the mass of the radioactive material at time t. Show how M_0 can be *estimated* from the Katz integral, assuming q and τ to be known from nuclear physics experiments.

3.9 Further reading

- 1 derives the light-curve model used in this chapter.
- D. Arnett [2] covers supernova light curves and nucleosynthesis in depth.

4. The Euler Equation

4.1 Internal stresses

Up to this point we have described the motion of fluids in which the velocity field is prescribed, or known. Now we derive the equation of motion for fluids. The acceleration of a fluid element $\mathcal{V}$ with mass $\mathcal{M}$ is given by Newton's law:

$$\mathcal{M}\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = \mathbf{F}, \tag{4.1}$$

where $\mathbf{F}$ is the sum of all forces (the net force) acting on $\mathcal{V}$. To derive an equation that does not explicitly make reference to the fluid elements, we can divide Equation (4.1) by the volume of $\mathcal{V}$ to find

$$\rho\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = \mathbf{f}, \tag{4.2}$$

where $\mathbf{f}$ is the force per unit volume acting on the fluid:

$$[\mathbf{f}] = \mathrm{N\,m^{-3}} = \mathrm{kg\,m^{-2}\,s^{-2}}, \tag{4.3}$$

where N is the newton, the SI unit of force (Equation (2.5)). There are two types of forces that can act on a fluid element: external and internal forces. In astrophysical applications, gravity is the most common external force. The force per unit volume (**body force**) due to gravity is

$$\mathbf{f}_g = \rho\mathbf{g} = -\rho\nabla\Psi, \tag{4.4}$$

where $\mathbf{g}$ is the gravitational acceleration and Ψ is the gravitational potential (Chapter 7). The internal forces are due to the interaction of $\mathcal{V}$ with other fluid elements.

Consider a container with water at rest on the surface of the Earth. The container has been left undisturbed for sufficiently long time that the velocity of all fluid elements is zero. Let $\mathcal{V}$ be a fluid element of water somewhere away from the edge of the container. Clearly, $\mathcal{V}$ is subject to the force of gravity. Since $\mathcal{V}$ is not in contact with the container, the gravitational force cannot be balanced by a contact force with the walls of the container. Instead, the force on $\mathcal{V}$ must be balanced by a force exchanged by $\mathcal{V}$ with the neighboring fluid elements. This shows that the equation of motion of a fluid must include the acceleration due to its internal forces. What is their nature?

We consider an infinitesimal fluid element $\mathcal{V}$ in a fluid at rest, as illustrated in Figure 4.1. We can take the origin of the coordinate system to be the center of $\mathcal{V}$ and its extent in the three directions to be $\mathrm{d}x$, $\mathrm{d}y$, and $\mathrm{d}z$. We also take the z axis to point upwards, so that the gravitational acceleration is $\mathbf{g} = -g\hat{\mathbf{z}}$, with $g > 0$.

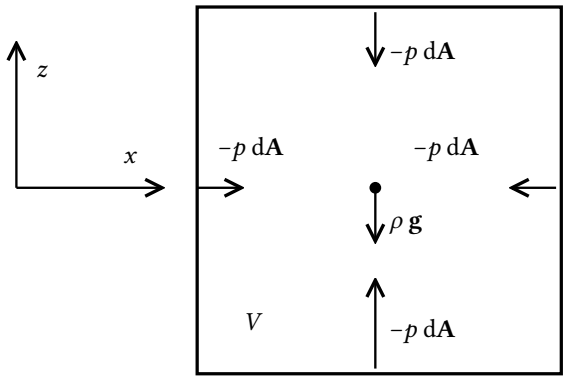


Figure 4.1: Force diagram for a fluid element in a static water container.

The fluid element is subject to forces applied by the rest of the fluid to its six square faces and to the force of gravity, which we can take to be applied to its center of mass, as shown in Figure 4.1. Our goal is to determine these forces. What distinguishes a fluid from a solid is that a fluid at rest cannot sustain shear stresses: if we try to shear it, it flows. So, for a fluid at rest, the surface forces must be perpendicular to each face. In a moving fluid, instead, these surface forces can in principle have many possible magnitudes and directions, and we cannot make much progress without knowing more about the nature of the fluid.

However, there is a class of fluids, known as **perfect fluids**, for which the internal forces are particularly simple. In a perfect fluid the surface forces are assumed to be always directed along the perpendicular to each face, also when the fluid is in motion (as is the case in Figure 4.1). The problem of finding the internal forces thus reduces to that of finding a scalar quantity, called the **pressure**, that gives the magnitude of the force per unit area across each face of the fluid element. One can show, by considering the forces acting on a small wedge of fluid, that this magnitude does not depend on the orientation of the face[4]:

$$p = \text{“force per unit area”} \qquad [\mathrm{Pa} = \mathrm{N\,m^{-2}}]. \tag{4.5}$$

In particular, for a perfect fluid the force exerted by the fluid on an infinitesimal surface with outward-directed area element $\mathrm{d}\mathbf{A}$ is

$$\mathrm{d}\mathbf{F} = -p\mathrm{d}\mathbf{A}. \tag{4.6}$$

In most cases the pressure is positive $p > 0$, so a fluid element, or equivalently an object immersed in a fluid, is subject to compression.

The concept of perfect fluid is an idealization of reality, but, in practice, liquids and gases can be considered as perfect fluids to very good approximation in most situations. More generally, because the force needs to transform under basis change in the same way as a vector, it can be shown that internal forces can be written as

$$\mathrm{d}\mathbf{F} = \mathbf{T}\mathrm{d}\mathbf{A}, \tag{4.7}$$

where $\mathbf{T}$ is a matrix that depends on the properties of the fluid, the so-called **stress tensor** of the fluid. The perfect fluid is the special case of this general theory in which

$$\mathbf{T} = -p\mathbf{I}, \tag{4.8}$$

where $\mathbf{I}$ is the identity matrix. We will encounter examples of non-perfect fluids later in the course (Chapter 21 and Chapter 18).

4.2 Euler's equation

We can now consider the net acceleration of the fluid element in the vertical direction. There are three forces to consider: the force of gravity, and the forces at the bottom and at the top of $\mathcal{V}$. The forces from the side do not contribute, since they are orthogonal to the z-direction. Putting everything together we find:

$$\mathcal{M}\frac{\mathrm{d}v_z}{\mathrm{d}t} = -\left[p\left(z = \frac{\mathrm{d}z}{2}\right) - p\left(z = -\frac{\mathrm{d}z}{2}\right)\right]\mathrm{d}A + \mathcal{M}g_z, \tag{4.9}$$

where $\mathrm{d}A = \mathrm{d}x\mathrm{d}y$, and $\mathcal{M} = \rho V = \rho \mathrm{d}x\mathrm{d}y\mathrm{d}z$. Dividing both terms by V gives

$$\rho\frac{\mathrm{d}v_z}{\mathrm{d}t} = -\frac{1}{\mathrm{d}z}\left[p\left(z = \frac{\mathrm{d}z}{2}\right) - p\left(z = -\frac{\mathrm{d}z}{2}\right)\right] + \rho g_z = -\partial_z p + \rho g_z. \tag{4.10}$$

A similar derivation can be done for the momentum balance in the x- and y-directions to find

$$\rho\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = -\nabla p + \rho\mathbf{g}. \tag{4.11}$$

[4]Consider a small wedge-shaped fluid element of size ℓ. The surface forces acting on its faces are proportional to their areas, so they scale as ℓ^2, while body forces, such as gravity, and the inertia of the element are proportional to its volume, so they scale as ℓ^3. As $\ell \to 0$, the surface forces must balance each other on their own. Because the areas of the faces of the wedge are related by geometry, this is only possible if the force per unit area is the same on all faces.

In the more general case, in which we consider an arbitrary volume force applied onto the fluid, we obtain the so-called Euler, or momentum equation for a perfect fluid:

$$\boxed{\rho \frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = -\nabla p + \mathbf{f}.} \tag{4.12}$$

Note that it is the *gradient* of the pressure, and not the pressure directly that determines the mechanical response of the fluid. We will see that this has important implications in a number of situations. Also note that $\mathrm{d}\mathbf{v}/\mathrm{d}t$ is the acceleration of the fluid element. In Chapter 5 we will rewrite Euler's equation in terms of derivatives at fixed points in space.

In the case of non-perfect fluids, Equation (4.12) becomes

$$\rho \frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = \nabla \cdot \mathbf{T} + \mathbf{f}, \tag{4.13}$$

where we have introduced the divergence of a tensor, defined to be a vector with components

$$\nabla \cdot \mathbf{T} = \begin{pmatrix} \partial_x T_{xx} + \partial_y T_{xy} + \partial_z T_{xz} \\ \partial_x T_{yx} + \partial_y T_{yy} + \partial_z T_{yz} \\ \partial_x T_{zx} + \partial_y T_{zy} + \partial_z T_{zz} \end{pmatrix}. \tag{4.14}$$

The momentum equation, Equation (4.13), holds even for solids, which can rarely be described as perfect fluids, but for which the stress tensor can also be defined.

Example

Consider a liquid at rest ($\mathbf{v} = 0$) in a tank and subject to gravity. The tank is not pressurized, so the surface of the liquid is at the atmospheric pressure p_0, in order to be in mechanical equilibrium with the air. Liquids are nearly incompressible, so, if the tank is not too deep, we can neglect the change of the density with depth and take $\rho = \text{const}$. What is the pressure at the bottom of the tank?

We can take z to be the vertical coordinate, pointing upwards, with $z = 0$ being the bottom of the tank and $z = d > 0$ being the surface of the liquid. The force balance in the z-direction is

$$\partial_z p = -\rho g, \tag{4.15}$$

where $g > 0$ is the gravitational acceleration. Integrating this equation from the surface, we find

$$p(z) = p_0 + \int_z^d \rho g \mathrm{d}z' = p_0 + \rho g (d - z). \tag{4.16}$$

The pressure at the bottom of the tank is $p_0 + \rho g d$. For water, $\rho \approx 1\,\mathrm{g\,cm^{-3}} = 10^3\,\mathrm{kg\,m^{-3}}$, so for $d = 10\,\mathrm{m}$ we find $\rho g d \approx 10^5\,\mathrm{Pa}$, which is about one atmosphere. In other words, every 10 m of water adds about one atmosphere of pressure.

Example

Consider a body of volume V fully immersed in a fluid at rest. What is the net force exerted by the fluid on the body?

The force is obtained by integrating Equation (4.6) over the surface ∂V of the body. Using a variant of the divergence theorem, $\oint_{\partial V} p \mathrm{d}\mathbf{A} = \int_V \nabla p \mathrm{d}V$, we find

$$\mathbf{F}_b = -\oint_{\partial V} p \mathrm{d}\mathbf{A} = -\int_V \nabla p \mathrm{d}V. \tag{4.17}$$

The pressure in the fluid does not know about the body: it is the same pressure that would be there if the body were replaced by fluid. So, from Equation (4.11) with $\mathbf{v} = 0$, $\nabla p = \rho_f \mathbf{g}$, where ρ_f is the density of the fluid. If ρ_f is constant, we find

$$\mathbf{F}_b = -\rho_f V \mathbf{g} = \rho_f V g \hat{\mathbf{z}}. \tag{4.18}$$

The body is pushed upwards with a force equal to the weight of the fluid it displaces. This is Archimedes' principle. Note that only the gradient of p enters in this result. For example, increasing the atmospheric pressure p_0 at the surface of the fluid does not change the force on the body.

Let's recap our situation. We have started with the continuity equation (Equation (1.15)), which describes the evolution of the density ρ given the velocity $\mathbf{v}$. To find the velocity we have now derived an equation of motion for it, Euler's equation, which however involves the pressure. As we have seen in Chapter 2, the pressure is given by the **equation of state** of the fluid, which depends on its density, temperature or entropy, and composition

$$p = p(\rho, T, \{Y_i\}). \tag{4.19}$$

This should not be surprising, because water and interstellar gas behave very differently, even though they satisfy the same equations. Their different dynamics is the result of the fact that they have different equations of state. To close the system, we also need an equation for the temperature, or the internal energy: this is the energy equation (Equation (2.39)). There are some special cases, however, in which the pressure can be taken to be a function of only the density:

$$p = p(\rho). \tag{4.20}$$

In this case we say that the equation of state is **barotropic**, and the continuity and Euler equations form a closed system, without the need for an energy equation. Some important cases that we will encounter in our course are

- pressureless fluids, in which $p = 0$;
- isothermal gases, in which $p = c_s^2 \rho$, where $c_s > 0$ is a constant, which we will see is equal to the speed of sound in the gas;
- polytropic gases, in which $p = K\rho^\Gamma$ (Equation (2.24)), where K and Γ are some known positive constants.

The pressureless approximation can sometimes be employed for highly supersonic flows, or for collisionless matter, such as dark matter or dust, when its velocity dispersion is small (that is, when it is cold). Note that collisionless matter can only be described as a fluid as long as it flows with a single velocity at each point. When different streams cross, the fluid approximation itself breaks down (Chapter 1). The isothermal and polytropic equations of state hold when temperature or entropy are constant in the gas, so the equation of state becomes barotropic. We will discuss this in more detail later.

4.3 In-class worksheet

1. A tank of water, open at the top, is accelerated horizontally with constant acceleration a. Once the water has come to rest with respect to the tank, what is the shape of its free surface? Treat the water as incompressible.
2. Core-collapse supernovae are triggered when the iron core of a massive star exceeds the Chandrasekhar mass. Initially, the core is in hydrostatic equilibrium:

   $$\partial_r p = -\frac{GM}{r^2}\rho, \tag{4.21}$$

   where $M = M(r)$ is the enclosed mass. After the core becomes unstable, it starts to contract. Show that, if $\Gamma \approx 4/3$, and if the collapse is homologous and adiabatic, then the iron core remains close to hydrostatic equilibrium as it contracts, meaning that Equation (4.21) remains approximately true. [Hint: compare how the two sides of Equation (4.21) scale with the radius of the core.] What happens if $\Gamma > 4/3$? And if $\Gamma < 4/3$? In reality, the flow is no longer adiabatic, once it becomes supersonic, so the homologous collapse only describes the region with $v < c_s$.

4.4 Further reading

- C. Clarke and B. Carswell [1], Chapter 2 (the fluid equations, including the equation of motion).

5. Eulerian and Lagrangian Descriptions

5.1 Motivation

Up to this point, we have considered the equation of motion for individual fluid elements. This is the so-called Lagrangian description of fluid mechanics. However, there are many problems in which it is more convenient to work with equations that describe the density, velocity, and temperature of the fluid at a specific point in space, and not for individual fluid elements. In other words, we want to solve for $\rho(t, \mathbf{x})$ and $\mathbf{v}(t, \mathbf{x})$, as opposed to $\rho(t; \mathcal{V})$ and $\mathbf{v}(t; \mathcal{V})$. This is the so-called Eulerian approach.

As motivation, we consider a celestial body of mass M, for example a planet or a star, that is accreting matter at a constant rate $\dot{M}$. M and $\dot{M}$ are unknown, but we know density and velocity of the fluid at a given radius r_0 from spatially resolved observations of an emission line. How can we estimate M and $\dot{M}$? We assume the in-falling matter to have no angular momentum, so that the flow is radial. We also neglect radiation pressure and fluid pressure ($p = 0$). We will consider the more general case in which pressure forces are included later in the course (Chapter 14). However, we remark that the pressureless approximation provides an accurate description of the flow when it is highly supersonic (close to the surface of the accretor).

The equations of motion for the gas (assuming spherical symmetry) are

$$\frac{1}{\rho}\frac{\mathrm{d}\rho}{\mathrm{d}t} = -\frac{1}{r^2}\partial_r(r^2 v), \qquad \frac{\mathrm{d}v}{\mathrm{d}t} = -\frac{GM}{r^2}. \tag{5.1}$$

In the new language we have just introduced, these are the Lagrangian equations of motion. That is, they describe the density and velocity of each fluid element. Note that, even though the problem describes a steady state system, we cannot drop the time derivatives, because each individual fluid element is not stationary, but it is in free fall.

Consider a fluid element that starts at rest at large distance from the accretor. Since the fluid elements are not subject to any force besides gravity, they are in free fall

$$v = -\sqrt{\frac{2GM}{r}}. \tag{5.2}$$

Inserting this into the continuity equation yields:

$$\frac{1}{\rho}\frac{\mathrm{d}\rho}{\mathrm{d}t} = \frac{3\sqrt{2GM}}{2r^{3/2}}. \tag{5.3}$$

Here, we need to be careful, because in the Lagrangian framework $r = r(t)$, since we are traveling with each fluid element. So, before we can solve the continuity equation, we need to find r as solution of

$$\frac{\mathrm{d}r}{\mathrm{d}t} = -\sqrt{\frac{2GM}{r}}. \tag{5.4}$$

This equation can be solved analytically assuming the initial radius to be $r(0) = r_0$, but its solution is rather complex:

$$r(t) = \left(-\frac{3\sqrt{2GM}t}{2} + r_0^{3/2}\right)^{2/3}. \tag{5.5}$$

Plugging this back into the continuity equation, we find

$$\frac{1}{\rho}\frac{\mathrm{d}\rho}{\mathrm{d}t} = -\frac{3\sqrt{2GM}}{3\sqrt{2GM}t - 2r_0^{3/2}}. \tag{5.6}$$

This can be solved with initial conditions $\rho(0) = \rho_0$ to yield

$$\rho(t) = \frac{2r_0^{3/2}\rho_0}{2r_0^{3/2} - 3\sqrt{2GM}t}. \tag{5.7}$$

After all of this work, we still do not have ρ as a function of radius: for that we need to solve t from Equation (5.5) and plug the result back into Equation (5.7). Note, however, that the denominator in Equation (5.7) is just $2r^{3/2}$, so $\rho = \rho_0(r_0/r)^{3/2}$ along each fluid element. Since the flow is steady, every fluid element has density ρ_0 when it crosses r_0, so this is also $\rho(r)$. We got there, but only after solving for the trajectory of each fluid element. We also still need to relate ρ_0 to $\dot{M}$... Wouldn't our lives have been much simpler if we could have solved directly for $\rho(t, r)$? As we will see shortly, the Eulerian formulation of fluid mechanics gives us $\rho(r)$ directly, without ever computing a trajectory.

5.2 Advective derivative

We leave for a moment the problem of the spherical accretion, to develop the general theory of the Eulerian formulation of fluid mechanics. Let $A(t; \mathcal{V})$ be some quantity that is specific to a fluid element $\mathcal{V}$. For example, A could be the density or the velocity of the fluid element $\mathcal{V}$. The same quantity can also be thought of as being a function of position:

$$A(t, \mathbf{x}) \coloneqq A(t; \text{“fluid element at location } \mathbf{x} \text{ at time } t \text{ ”}). \tag{5.8}$$

If the law of motion of each fluid element is known, then it is possible to compute $A(t, \mathbf{x})$ from $A(t; \mathcal{V})$. We have done this in the case of the homologously expanding flow (Section 1.3), when we needed to compute $v(t, r)$ given $v(t, q)$, where q was the “label” we used to identify the fluid elements.

Consider a fluid element $\mathcal{V}$ with trajectory $\mathbf{x}(t)$. Then, at time t

$$A(t; \mathcal{V}) = A(t, \mathbf{x}(t)). \tag{5.9}$$

We can differentiate both sides with respect to t to find

$$\begin{aligned}
\frac{\mathrm{d}A(t; \mathcal{V})}{\mathrm{d}t} &= \frac{\mathrm{d}A(t, \mathbf{x}(t))}{\mathrm{d}t} \\
&= \frac{\partial A(t, \mathbf{x}(t))}{\partial t} + \Big(\frac{\mathrm{d}x}{\mathrm{d}t}\partial_x + \frac{\mathrm{d}y}{\mathrm{d}t}\partial_y \\
&\qquad + \frac{\mathrm{d}z}{\mathrm{d}t}\partial_z \Big) A(t, \mathbf{x}(t)) \\
&= \frac{\partial A(t, \mathbf{x}(t))}{\partial t} + \left(\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} \cdot \nabla \right) A(t, \mathbf{x}(t)) \\
&= \frac{\partial A(t, \mathbf{x}(t))}{\partial t} + (\mathbf{v} \cdot \nabla) A(t, \mathbf{x}(t)),
\end{aligned} \tag{5.10}$$

where we used the chain rule for differentiation and the definition of velocity. Equation (5.10) gives us a way to relate the time derivative of a quantity defined with respect to fluid elements to the same quantity taken to be a function of spatial position and time. The reason why

$$\frac{\mathrm{d}A(t; \mathcal{V})}{\mathrm{d}t} \neq \frac{\partial A(t, \mathbf{x}(t))}{\partial t} \tag{5.11}$$

is that the partial derivative with respect to time considers the rate of change of A at a fixed location, but there are *different* fluid elements crossing that location as a function of time. This is illustrated in Figure 5.1. The operator $\frac{\mathrm{d}}{\mathrm{d}t}$ that we have used so far is called the **advective derivative** (also material, or Lagrangian, derivative; many books write it as D/Dt), because it represents the rate of change of a quantity measured when riding along a fluid element.

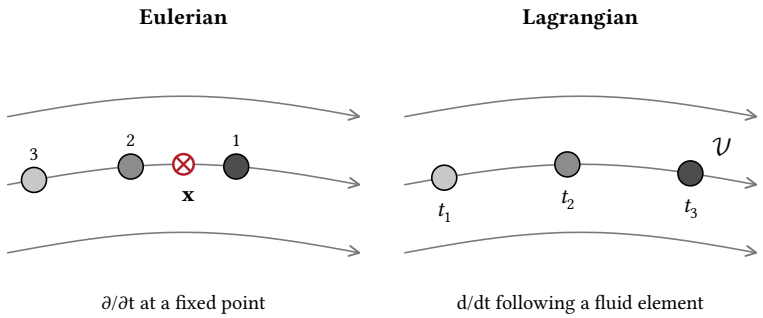


Figure 5.1: Eulerian (left) and Lagrangian (right) descriptions of a flow. In the Eulerian description, we measure the rate of change of a quantity at a fixed point $\mathbf{x}$, across which different fluid elements (1, 2, 3) pass over time. In the Lagrangian description, we follow a single fluid element $\mathcal{V}$ along its trajectory.

Equation (5.10) can be written more succinctly as

$$\boxed{\frac{\mathrm{d}A}{\mathrm{d}t} = \frac{\partial A}{\partial t} + \mathbf{v}\cdot\nabla A.} \tag{5.12}$$

To be mathematically precise, we should keep in mind that this relation represents the equivalence between two *different* mathematical operations performed on *different* objects. The A on the LHS is a function of only time for a fixed fluid element $\mathcal{V}$, while the A on the RHS is a function of position and time and is not specific to any single fluid element.

Example

Consider a steady temperature field $T(\mathbf{x}) = T_0 + \alpha x$, where T_0 and α are constants, in a fluid moving with uniform velocity $\mathbf{v} = v_0\hat{\mathbf{x}}$. At any fixed point, the temperature does not change: $\partial_t T = 0$. However, each fluid element moves towards regions with higher temperature (if $\alpha v_0 > 0$), so its temperature increases at the rate

$$\frac{\mathrm{d}T}{\mathrm{d}t} = \partial_t T + \mathbf{v}\cdot\nabla T = \alpha v_0. \tag{5.13}$$

5.3 Eulerian formulation of fluid mechanics

The Eulerian equations can be obtained from the Lagrangian equations by replacing all advective derivatives:

$$\begin{aligned}
\frac{\mathrm{d}\rho}{\mathrm{d}t} &\to \frac{\partial\rho}{\partial t} + (\mathbf{v}\cdot\nabla)\rho,\\
\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} &\to \frac{\partial\mathbf{v}}{\partial t} + (\mathbf{v}\cdot\nabla)\mathbf{v},\\
\frac{\mathrm{d}\varepsilon}{\mathrm{d}t} &\to \frac{\partial\varepsilon}{\partial t} + (\mathbf{v}\cdot\nabla)\varepsilon.
\end{aligned} \tag{5.14}$$

The continuity equation (Equation (1.15)) becomes

$$\frac{\partial\rho}{\partial t} + (\mathbf{v}\cdot\nabla)\rho + \rho\nabla\cdot\mathbf{v} = 0. \tag{5.15}$$

In Cartesian coordinates, we can expand all derivatives as

$$\begin{aligned}
&(\mathbf{v}\cdot\nabla)\rho + \rho\nabla\cdot\mathbf{v} \\
&\quad = v_x\partial_x\rho + v_y\partial_y\rho + v_z\partial_z\rho \\
&\qquad\quad +\rho\partial_x v_x + \rho\partial_y v_y + \rho\partial_z v_z \\
&\quad = \partial_x(\rho v_x) + \partial_y(\rho v_y) + \partial_z(\rho v_z) \\
&\quad = \nabla\cdot(\rho\mathbf{v}).
\end{aligned} \tag{5.16}$$

So the continuity equation can be written as

$$\boxed{\frac{\partial\rho}{\partial t} + \nabla\cdot(\rho\mathbf{v}) = 0.} \tag{5.17}$$

The velocity equation (Equation (4.12)) can be written as

$$\boxed{\rho\frac{\partial\mathbf{v}}{\partial t} + \rho(\mathbf{v}\cdot\nabla)\mathbf{v} = -\nabla p + \mathbf{f}.} \tag{5.18}$$

The term $(\mathbf{v}\cdot\nabla)\mathbf{v}$ is the acceleration of a fluid element due to the fact that it moves to places where the velocity is different. For this reason, fluid elements can accelerate even in a steady flow, in which $\partial_t\mathbf{v} = 0$. The spherical accretion flow of the next section is an example.

In the same way, the energy equation (Equation (2.39)) becomes

$$\rho\frac{\partial\varepsilon}{\partial t} + \rho(\mathbf{v}\cdot\nabla)\varepsilon = \rho\dot{q} - p\nabla\cdot\mathbf{v}. \tag{5.19}$$

We will write these equations in conservative form in Chapter 6.

5.4 Pressureless spherical accretion

We are now ready to solve the problem of the spherical accretion of a pressureless fluid that we have attempted to solve at the beginning of this chapter. Because the Eulerian equations deal with quantities as a function of position and time, all time derivatives vanish for a steady flow and we find:

$$\partial_r(r^2\rho v) = 0, \qquad v\partial_r v = -\frac{GM}{r^2}. \tag{5.20}$$

The second equation can be written as

$$\frac{1}{2}\partial_r(v^2) = -\frac{GM}{r^2}, \tag{5.21}$$

which can be integrated to give, for an infalling flow that is at rest at large distances,

$$v(r) = -\sqrt{\frac{2GM}{r}}. \tag{5.22}$$

The continuity equation shows that

$$r^2\rho v = \text{const}. \tag{5.23}$$

Apart from a 4π factor, this is the accretion rate, because the inflow of mass across a surface of radius r is (assuming spherical symmetry):

$$\dot{M} = -\oint r^2\rho v\,\mathrm{d}\Omega = -4\pi r^2\rho v. \tag{5.24}$$

So

$$\rho(r) = -\frac{\dot{M}}{4\pi r^2 v} = \frac{\dot{M}}{4\pi\sqrt{2GM}r^{3/2}}. \tag{5.25}$$

We can now solve for M using Equation (5.22) and for $\dot{M}$ using Equation (5.25). Evaluating them at r_0, where $\rho = \rho_0$ and $v = v_0$, we find

$$M = \frac{r_0 v_0^2}{2G}, \qquad \dot{M} = 4\pi r_0^2 \rho_0 |v_0|, \tag{5.26}$$

which answers the question we asked at the beginning of the chapter.

5.5 In-class worksheet

1. Consider the planar rotating flow $\mathbf{v} = (-y, x, 0)$ of Figure 1.4 and a steady temperature field $T(\mathbf{x}) = T_0 + \alpha x$, where T_0 and α are constants.
   a. What is the rate of change of the temperature, $\partial_t T$, at a fixed point?
   b. What is the rate of change of the temperature, $\mathrm{d}T/\mathrm{d}t$, of the fluid element located at (x, y)?
   c. Explain the difference between the two results.
2. An emission line from gas falling radially onto a star is observed at the radius $r_0 = 10\,R_\odot$, where the gas has velocity $v_0 = -200\,\mathrm{km\,s^{-1}}$ and density $\rho_0 = 10^{-15}\,\mathrm{g\,cm^{-3}} = 10^{-12}\,\mathrm{kg\,m^{-3}}$. Assuming pressureless free fall, estimate the mass of the star in units of $M_\odot$ and the accretion rate in units of $M_\odot\,\mathrm{yr}^{-1}$.

5.6 Exercises

Exercise 5.1 (Eulerian and Lagrangian frames). A fluid moves so that, at time t, the position $\mathbf{x} = (x_1, x_2, x_3)$ of a fluid element initially located at $\mathbf{q} = (q_1, q_2, q_3)$ is

$$\begin{aligned} x_1 &= \frac{1}{2}(q_1 + q_2)\mathrm{e}^{t} + \frac{1}{2}(q_1 - q_2)\mathrm{e}^{-t}, \\ x_2 &= \frac{1}{2}(q_1 + q_2)\mathrm{e}^{t} - \frac{1}{2}(q_1 - q_2)\mathrm{e}^{-t}, \\ x_3 &= q_3. \end{aligned} \tag{5.27}$$

1. Find the initial position (q_1, q_2, q_3) of a fluid element currently at (x_1, x_2, x_3). [You need to invert Equation (5.27) to find $\mathbf{q} = \mathbf{q}(t, \mathbf{x})$.]
2. Compute the velocity $\dot{\mathbf{x}}(t; \mathbf{q})$ and acceleration $\ddot{\mathbf{x}}(t; \mathbf{q})$ of the fluid elements.
3. Find the Eulerian velocity $\mathbf{v}(t, \mathbf{x}) = \dot{\mathbf{x}}[t; \mathbf{q}(t, \mathbf{x})]$ and acceleration $\mathbf{a}(t, \mathbf{x}) = \ddot{\mathbf{x}}[t; \mathbf{q}(t, \mathbf{x})]$.
4. Verify that $\mathbf{a} = \partial_t \mathbf{v} + (\mathbf{v} \cdot \nabla)\mathbf{v}$.
5. Is the flow incompressible? Sketch the streamlines.

5.7 Further reading

- C. Clarke and B. Carswell [1], Chapter 2 (the fluid equations).

6. Conservation Laws

6.1 Momentum equation in conservation form

In this chapter, we wish to reformulate the hydrodynamics equations in a way that makes conservation of momentum and energy manifest. These equations will also be useful later on, when we discuss shock waves (Chapter 11). We start with the equation for the momentum.

The starting point is the velocity equation (Equation (4.13)), which reads

$$\rho\left[\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v}\cdot\nabla)\mathbf{v}\right] = \nabla\cdot\mathbf{T} + \rho\mathbf{g}. \tag{6.1}$$

We can bring ρ inside the time derivative at the price of including new terms to the equation:

$$\rho\frac{\partial \mathbf{v}}{\partial t} + \rho(\mathbf{v}\cdot\nabla)\mathbf{v} = \frac{\partial(\rho\mathbf{v})}{\partial t} - \mathbf{v}\frac{\partial \rho}{\partial t} + \rho(\mathbf{v}\cdot\nabla)\mathbf{v}. \tag{6.2}$$

Using the continuity equation (Equation (5.17)) we find

$$\begin{aligned}\rho\frac{\partial \mathbf{v}}{\partial t} + \rho(\mathbf{v}\cdot\nabla)\mathbf{v} = \frac{\partial(\rho\mathbf{v})}{\partial t} + \mathbf{v}\nabla\cdot(\rho\mathbf{v}) \\ +\rho(\mathbf{v}\cdot\nabla)\mathbf{v}.\end{aligned} \tag{6.3}$$

Let's write down explicitly the last two terms in Cartesian coordinates:

$$\begin{aligned}&\mathbf{v}\nabla\cdot(\rho\mathbf{v}) + \rho(\mathbf{v}\cdot\nabla)\mathbf{v} \\ &= \mathbf{v}[\partial_x(\rho v_x) + \partial_y(\rho v_y) + \partial_z(\rho v_z)] \\ &\qquad +\rho v_x\partial_x\mathbf{v} + \rho v_y\partial_y\mathbf{v} + \rho v_z\partial_z\mathbf{v} \\ &= \partial_x(\rho v_x\mathbf{v}) + \partial_y(\rho v_y\mathbf{v}) + \partial_z(\rho v_z\mathbf{v}) \\ &= \nabla\cdot[\rho\mathbf{v}\otimes\mathbf{v}],\end{aligned} \tag{6.4}$$

where we have recognized the second to last line as being the divergence of the tensor

$$\mathbf{v}\otimes\mathbf{v} := \begin{pmatrix} v_xv_x & v_xv_y & v_xv_z \\ v_yv_x & v_yv_y & v_yv_z \\ v_zv_x & v_zv_y & v_zv_z \end{pmatrix}. \tag{6.5}$$

Putting everything together we can write the momentum equation

$$\frac{\partial(\rho\mathbf{v})}{\partial t} + \nabla\cdot[\rho\mathbf{v}\otimes\mathbf{v} - \mathbf{T}] = \rho\mathbf{g}. \tag{6.6}$$

In the special case in which $\mathbf{T} = -p\mathbf{I}$ we obtain

$$\boxed{\frac{\partial(\rho\mathbf{v})}{\partial t} + \nabla\cdot[\rho\mathbf{v}\otimes\mathbf{v} + p\mathbf{I}] = \rho\mathbf{g}.} \tag{6.7}$$

The meaning of this equation is more clear if we integrate over some fixed region Ω (not moving with the flow):

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_\Omega \rho\mathbf{v}\mathrm{d}V + \int_\Omega \nabla\cdot[\rho\mathbf{v}\otimes\mathbf{v} + p\mathbf{I}]\mathrm{d}V = \int_\Omega \rho\mathbf{g}\mathrm{d}V. \tag{6.8}$$

Moving the second term to the RHS and using the divergence theorem, we find

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_\Omega \rho \mathbf{v}\mathrm{d}V = -\int_{\partial\Omega}[\rho(\mathbf{v}\cdot\mathbf{n})\mathbf{v} + p\mathbf{n}]\mathrm{d}A + \int_\Omega \rho\mathbf{g}\mathrm{d}V, \tag{6.9}$$

where $\partial\Omega$ is the surface bounding Ω and $\mathbf{n}$ its outward directed normal.

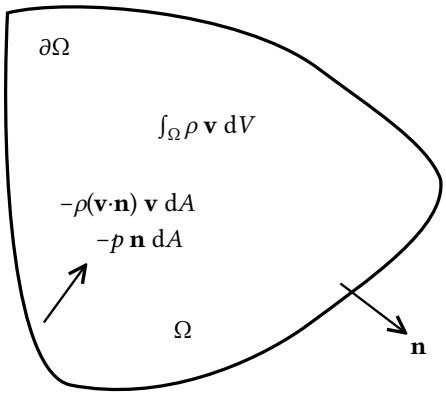


Figure 6.1: Diagram illustrating the balance of momentum in Ω.

In words, Equation (6.9) states that the total momentum in some region changes due to the action of volumetric forces and because of the flux of momentum across the boundary. For example, if we take Ω to be the region with $x > 0$ and neglect gravity, then $\partial\Omega$ is the yz-plane. If we also assume that this region is isolated, so that no momentum flows in or out through the surface at infinity, then Equation (6.9) becomes:

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\int_{x>0}\rho\begin{pmatrix}v_x\\v_y\\v_z\end{pmatrix}\mathrm{d}x\mathrm{d}y\mathrm{d}z \\ = \int_{x=0}\left[\rho v_x\begin{pmatrix}v_x\\v_y\\v_z\end{pmatrix} + p\begin{pmatrix}1\\0\\0\end{pmatrix}\right]\mathrm{d}y\mathrm{d}z.\end{aligned} \tag{6.10}$$

Note that the pressure across the $x = 0$ surface only influences the x-component of the momentum.

The last equation also gives an alternative interpretation of pressure. We see that in the RHS the pressure has the same effect as the term $\rho v_x v_x$. This term describes the flow of momentum through the yz-plane and is also called the **ram pressure**. This suggests that we can think of pressure not only as force per unit area, but also as a flux of momentum per unit area. This is the same interpretation that we found in Chapter 2, when we derived the pressure as the flux of momentum due to the random motion of the particles (Equation (2.4)).

6.2 Energy equation

The energy equation (Equation (2.39)) reads

$$\rho\frac{\mathrm{d}\varepsilon}{\mathrm{d}t} = \rho\dot{q} - p\nabla\cdot\mathbf{v}. \tag{6.11}$$

We can expand the advective derivative in Equation (6.11) to obtain the energy equation in the Eulerian frame (Equation (5.19)):

$$\rho\frac{\partial\varepsilon}{\partial t} + \rho(\mathbf{v}\cdot\nabla)\varepsilon = \rho\dot{q} - p\nabla\cdot\mathbf{v}. \tag{6.12}$$

Using a similar approach to what we have done before with the momentum equation, we bring the density into the time derivative to find

$$\frac{\partial(\rho\varepsilon)}{\partial t} + \nabla \cdot (\rho\varepsilon\mathbf{v}) = \rho\dot{q} - p\nabla \cdot \mathbf{v}. \tag{6.13}$$

The form of this equation is similar to that of the Eulerian continuity equation and to the momentum equation above. They are all examples of **balance laws**: equations describing the conservation of some quantity. A balance law reads:

$$\frac{\partial U}{\partial t} + \nabla \cdot \mathbf{F} = S, \tag{6.14}$$

where U, $\mathbf{F}$, and S are called the conserved quantities, the fluxes, and the sources of the balance law, respectively. If $S = 0$, then Equation (6.14) is said to be a **conservation law**. The name balance law is more clear if we convert the equation to integral form by integrating it over a fixed volume Ω:

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_\Omega U\mathrm{d}V + \int_\Omega \nabla \cdot \mathbf{F}\mathrm{d}V = \int_\Omega S\mathrm{d}V. \tag{6.15}$$

Using the divergence theorem, we obtain

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_\Omega U\mathrm{d}V = \int_\Omega S\mathrm{d}V - \int_{\partial\Omega} \mathbf{F} \cdot \mathbf{n}\mathrm{d}A. \tag{6.16}$$

This equation signifies that the total amount of a quantity U in the volume Ω changes by the amount of U that flows in/out through the boundary $\partial\Omega$, plus the amount of U created/destroyed by the source/sink term S (see Figure 6.2). Equations in the form of Equation (6.14) are also said to be in **conservation form**.

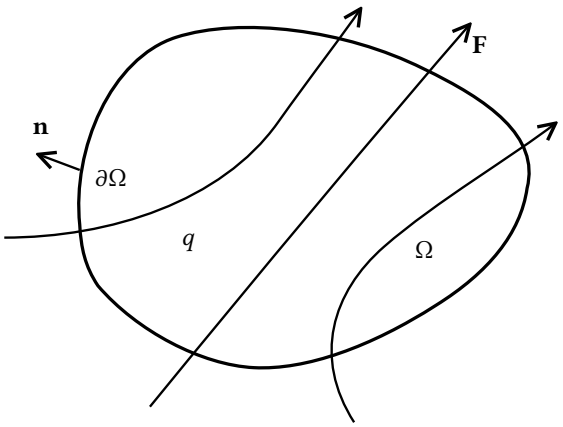


Figure 6.2: Graphical illustration of a balance law.

At this point, one might ask in what sense Equation (6.13) describes the conservation of the energy in the fluid. First, this equation does not contain any terms describing the work done on the system by external forces. Second, ε is only part of the energy in the flow: we are not including kinetic and potential energy. The issue is that Equation (6.13) only describes the evolution of the internal energy. Mechanical work done by gravity, or other forces, can only directly affect the mechanical energy of the fluid (its kinetic energy). The conversion between internal and kinetic energy is operated by the pdV term. We postpone for the moment the issue of the gravitational energy of the system (Chapter 7), which will turn out to be more complex than expected, to focus on the interplay between kinetic and internal energy.

To derive an equation for the kinetic energy, we start from the Euler equation

$$\rho\frac{\partial \mathbf{v}}{\partial t} + \rho(\mathbf{v} \cdot \nabla)\mathbf{v} = -\nabla p + \mathbf{f}. \tag{6.17}$$

We multiply this equation by $\mathbf{v}$ to find

$$\frac{1}{2}\rho\left[\frac{\partial v^2}{\partial t} + (\mathbf{v}\cdot\nabla)v^2\right] = -\nabla p\cdot\mathbf{v} + \mathbf{f}\cdot\mathbf{v}. \tag{6.18}$$

Once again, we can use the continuity equation to bring ρ inside the time derivative to find

$$\frac{\partial\left(\frac{1}{2}\rho v^2\right)}{\partial t} + \nabla\cdot\left(\frac{1}{2}\rho v^2\mathbf{v}\right) = -\mathbf{v}\cdot\nabla p + \mathbf{f}\cdot\mathbf{v}. \tag{6.19}$$

Using the identity

$$\nabla\cdot(p\mathbf{v}) = \mathbf{v}\cdot\nabla p + p\nabla\cdot\mathbf{v}, \tag{6.20}$$

we can rewrite this equation as

$$\frac{\partial\left(\frac{1}{2}\rho v^2\right)}{\partial t} + \nabla\cdot\left[\left(\frac{1}{2}\rho v^2 + p\right)\mathbf{v}\right] = p\nabla\cdot\mathbf{v} + \mathbf{f}\cdot\mathbf{v}. \tag{6.21}$$

This is the wanted equation for the kinetic energy. We notice two things. First, this equation shows that pressure can be interpreted also as an energy density. Second, this equation has the same pdV term as the internal energy equation, but with opposite sign! This is the term describing the conversion of internal energy into kinetic energy and vice versa. Also note that the work done by external forces appears in the kinetic energy equation, but heating and cooling terms do not.

Adding Equation (6.19) to Equation (6.13) gives the equation describing conservation of the total energy (internal + kinetic):

$$\boxed{\begin{aligned}&\frac{\partial\left[\rho\left(\frac{1}{2}v^2 + \varepsilon\right)\right]}{\partial t} + \nabla\cdot\left[\left(\frac{1}{2}\rho v^2 + \rho\varepsilon + p\right)\mathbf{v}\right]\\ &\quad = \rho\dot{q} + \mathbf{f}\cdot\mathbf{v}.\end{aligned}} \tag{6.22}$$

This shows that the total energy in the fluid is conserved in absence of external forces and heating/cooling. If $\mathbf{f}$ is a conservative force, it is possible to rewrite the energy equation in a way that is manifestly conservative by including the potential inside the time derivative. For example, consider gravity, $\mathbf{f} = -\rho\nabla\Psi$, and assume that the potential Ψ does not depend on time, as is the case for an external, static potential. Then, using the continuity equation,

$$\mathbf{f}\cdot\mathbf{v} = -\rho\mathbf{v}\cdot\nabla\Psi = -\nabla\cdot(\rho\Psi\mathbf{v}) + \Psi\nabla\cdot(\rho\mathbf{v}) = -\nabla\cdot(\rho\Psi\mathbf{v}) - \frac{\partial(\rho\Psi)}{\partial t}, \tag{6.23}$$

so that Equation (6.22) becomes

$$\begin{aligned}&\frac{\partial\left[\rho\left(\frac{1}{2}v^2 + \varepsilon + \Psi\right)\right]}{\partial t}\\ &\quad +\nabla\cdot\left[\left(\frac{1}{2}\rho v^2 + \rho\varepsilon + p + \rho\Psi\right)\mathbf{v}\right] = \rho\dot{q}.\end{aligned} \tag{6.24}$$

If the gas is self-gravitating, instead, Ψ depends on time through the density, and this simple manipulation does not work. We will come back to this in Chapter 7.

We can now summarize the equations of hydrodynamics in conservation form (Equation (6.14)). In the absence of heating and cooling, and with a body force $\mathbf{f}$, we have:

Table 6.1: The equations of hydrodynamics in conservation form.

Conservation of	U	$\mathbf{F}$	S
mass	ρ	$\rho\mathbf{v}$	0
momentum	$\rho\mathbf{v}$	$\rho\mathbf{v}\otimes\mathbf{v} + p\mathbf{I}$	$\mathbf{f}$
energy	$\rho\left(\frac{1}{2}v^2 + \varepsilon\right)$	$\left(\frac{1}{2}\rho v^2 + \rho\varepsilon + p\right)\mathbf{v}$	$\mathbf{f}\cdot\mathbf{v}$

We will use these equations in Chapter 11 to derive the jump conditions across shock waves.

6.3 Heat conduction

Heat conduction can be included in Equation (6.22) by adding a heat flux $\mathbf{F}_{\text{heat}}$:

$$\begin{aligned}&\frac{\partial\left[\rho\left(\frac{1}{2}v^2+\varepsilon\right)\right]}{\partial t}\\&+\nabla\cdot\left[\left(\frac{1}{2}\rho v^2+\rho\varepsilon+p\right)\mathbf{v}+\mathbf{F}_{\text{heat}}\right]\\&=\rho\dot{q}+\mathbf{f}\cdot\mathbf{v}.\end{aligned}\tag{6.25}$$

In many astrophysical applications, energy is transported mostly by photons, and the heat flux can be approximated as

$$\mathbf{F}_{\text{heat}}=-\frac{c}{3\rho\kappa_R}\nabla(aT^4),\tag{6.26}$$

where c is the speed of light, a the radiation constant (Equation (2.33)), and κ_R is the so-called *Rosseland mean opacity*. The term in parentheses is an energy density. Indeed, it is the energy density of radiation for a black body (Equation (2.32)):

$$u_{\text{rad}}=aT^4.\tag{6.27}$$

The opacity has units of $\mathrm{m^2\,kg^{-1}}$ (we quote it in $\mathrm{cm^2\,g^{-1}}$, as is customary), so $\lambda=1/(\rho\kappa_R)$ has units of a length. Indeed, λ is the photon mean free path. Combining these two facts, we can write Equation (6.26) as

$$\mathbf{F}_{\text{heat}}=-\frac{1}{3}\lambda c\nabla u_{\text{rad}}=-\mathcal{D}\nabla u_{\text{rad}},\qquad\mathcal{D}=\frac{\lambda c}{3}.\tag{6.28}$$

This is a special case of Fick's law of diffusion:

$$\textbf{diffusive flux}=-\mathcal{D}\nabla(\text{density of quantity being diffused}).\tag{6.29}$$

The diffusion coefficient has units of

$$[\mathcal{D}]=\mathrm{m^2\cdot s^{-1}}.\tag{6.30}$$

The time needed to diffuse over a distance R is $\sim R^2/\mathcal{D}$, which is the diffusion time we found in Equation (3.17) with the random-walk argument (up to a factor of 3).

If energy due to radiation was advected with velocity $\mathbf{v}_{\text{eff}}$, then

$$\mathbf{F}_{\text{heat}}=u_{\text{rad}}\mathbf{v}_{\text{eff}}.\tag{6.31}$$

Comparing this to Equation (6.28) suggests that

$$v_{\text{eff}}\sim\left(\frac{\lambda}{3H}\right)c,\tag{6.32}$$

where H is the radiation energy density scale-height:

$$H=\frac{u_{\text{rad}}}{|\nabla u_{\text{rad}}|}.\tag{6.33}$$

In the interior of stars $H\gg\lambda$, so radiation diffuses very slowly: the time it needs to cross the star is much longer than the light-crossing time.

Example

Consider the Sun. With an opacity $\kappa\approx 25\,\mathrm{cm^2\,g^{-1}}=2.5\,\mathrm{m^2\,kg^{-1}}$ and the mean density of the Sun, $\bar{\rho}\approx 1.4\,\mathrm{g\,cm^{-3}}=1.4\times10^3\,\mathrm{kg\,m^{-3}}$, the photon mean free path is $\lambda\approx 3\times10^{-4}\,\mathrm{m}$. Taking $H\sim R_\odot$, Equation (6.32) gives

$$v_{\text{eff}} \sim \frac{3 \times 10^{-4}\,\text{m}}{3 \times 7 \times 10^{8}\,\text{m}}\, c \approx 4 \times 10^{-5}\,\text{m}\,\text{s}^{-1}, \tag{6.34}$$

or about $40\,\mu\text{m}\,\text{s}^{-1}$. At this speed, energy takes $R_\odot / v_{\text{eff}} \sim 5 \times 10^5$ years to reach the surface from the center, while light would cross the Sun in about 2 s.

Close to the surface of stars $H \lesssim \lambda/3$ and Equation (6.28) predicts that radiant energy propagates faster than the speed of light. This is a symptom that the diffusion approximation breaks down. More generally, the transport of energy by radiation can be described using Equation (6.26) only if photons and matter are in local thermodynamic equilibrium. If the diffusion approximation cannot be used, matter and radiation need to be treated separately using the equations of radiation-hydrodynamics[5].

6.4 In-class worksheet

1. Show that the equation for energy conservation in the Eulerian frame can be written as

$$\frac{\partial(\rho\varepsilon)}{\partial t} + \nabla \cdot (\rho\varepsilon \mathbf{v}) = \rho\dot{q} - p\nabla \cdot \mathbf{v}. \tag{6.35}$$

   Discuss the physical meaning of each term.

2. *Ram pressure.* Consider the steady flow around a rigid body. Let $\partial\Omega$ be a closed surface in the fluid that encloses the body, with outward normal $\mathbf{n}$, as shown in Figure 6.3.

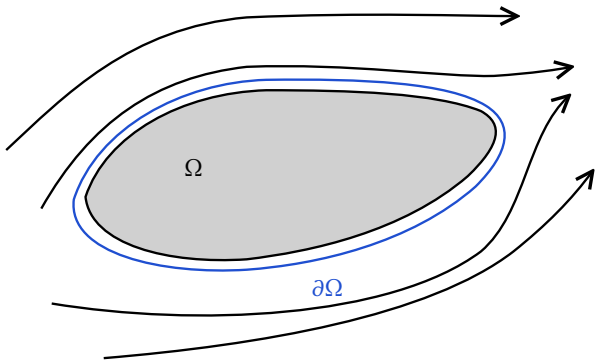


Figure 6.3: Flow past a rigid body. The outer outline is the closed surface $\partial\Omega$ that encloses the body.

   a. Use the momentum equation and the divergence theorem to show that the total force acting on the body is

$$\mathbf{F} = -\oint_{\partial\Omega} [\rho(\mathbf{v}\cdot\mathbf{n})\mathbf{v} + p\mathbf{n}]\,\mathrm{d}A. \tag{6.36}$$

   b. Assume that $p \approx$ const and that the flow in the wake of the body is slower than on the front. What kind of force does the body experience?

6.5 Exercises

Exercise 6.1 (Angular momentum). The angular momentum of a fluid element of mass $\mathcal{M}$ is $\mathcal{M}\mathbf{r} \times \mathbf{v}$, so the angular momentum density for a fluid is (upon division by $\mathcal{V}$):

$$\mathbf{j} = \rho\mathbf{r} \times \mathbf{v}. \tag{6.37}$$

1. Starting from the momentum equation, derive a Lagrangian equation for the evolution of $\mathbf{r} \times \mathbf{v}$.
2. Show that, in absence of external forces, $\mathbf{j}$ satisfies the conservation law

[5]

2.

$$\frac{\partial \mathbf{j}}{\partial t} + \nabla \cdot [\mathbf{j} \otimes \mathbf{v} + p\mathbf{R}] = 0, \tag{6.38}$$

where

$$\mathbf{R} = \begin{pmatrix} 0 & -z & y \\ z & 0 & -x \\ -y & x & 0 \end{pmatrix}. \tag{6.39}$$

[Hint. Write out explicitly the expression $\mathbf{r} \times \nabla p$.]

3. Consider a fluid that is subject to an external force per unit volume $\mathbf{f}$. How does

$$\mathbf{J} = \int \mathbf{j} \mathrm{d}V \tag{6.40}$$

evolve with time?

4. Show that, for an isolated fluid in an axisymmetric potential $\Psi(R, z)$, the z-component of $\mathbf{J}$ is conserved.

6.6 Further reading

- C. Clarke and B. Carswell [1], Chapter 2 (the fluid equations) and Chapter 4 (the energy equation).
- 2 for the equations of radiation hydrodynamics.

7. The Gravitational Potential

> **Note**
>
> This chapter is in preparation.

7.1 In-class worksheet

1. **Gravitational binding energy.**
   a. Consider a spherically-symmetric body of mass M and radius R. How much work $\mathrm{d}W$ is required to carry the outer shell of mass $m = 4\pi\rho R^2 \mathrm{d}\ell$ to infinity?
   b. Show that the gravitational binding energy of a spherical body with constant density, mass M and radius R is
   $$W = -\frac{3GM^2}{5R}. \tag{7.1}$$
   c. The iron core of a massive star, with mass $M = 1.4\,M_\odot$ and radius $R \approx 5000\,\mathrm{km}$, collapses to form a protoneutron star with a radius $R \approx 10\,\mathrm{km}$. How much gravitational energy is liberated in the process?

7.2 Exercises

Exercise 7.1 (Homologous collapse and the Friedmann equation). A homogeneous gas sphere is collapsing under its self-gravity. Assume the velocity to be given by

$$\mathbf{v}(t, \mathbf{r}) = H(t)\mathbf{r}, \tag{7.2}$$

where $\mathbf{r}$ is the displacement of the fluid element from the center of the cloud.

1. Argue that the density remains constant throughout the sphere (although it is changing with time) and show that the momentum equation implies
   $$\dot{H}(t) + H(t)^2 = -\frac{4\pi G\rho(t)}{3}. \tag{7.3}$$
   Assume the gas to be described by a barotropic EOS $p = p(\rho)$.
2. Introduce the scale factor $a(t)$ defined so that
   $$\frac{\dot{a}(t)}{a(t)} = H(t). \tag{7.4}$$
   Show that
   $$\frac{\mathrm{d}}{\mathrm{d}t}\left(H^2a^2 - \frac{8\pi}{3}G\rho a^2\right) = 2Ha^2\left(\dot{H} + H^2 + \frac{4\pi G}{3}\rho\right) = 0 \tag{7.5}$$
   and show that this implies that the dynamics is described by Friedmann equation
   $$\dot{a}(t)^2 - \frac{8\pi G\rho(t)}{3}a(t)^2 = -k, \tag{7.6}$$
   where k is a constant of motion.

3. Show that $\rho(t) \propto a(t)^{-3}$ using the continuity equation. Use this to cast the Friedmann equation into the equation of a particle rolling down an effective potential

$$\frac{1}{2}\dot{a}(t)^2 + U_{\text{eff}}(a) = \mathcal{E}. \tag{7.7}$$

Sketch a graph for $\rho(t)$ and $a(t)$ as a function of time.

Exercise 7.2 (Free fall onto a point mass and a uniform sphere). A particle is released from rest at a distance R_0 from the centre of a body of mass M. In each of the two cases listed below, find

1. the initial acceleration of the particle, and
2. the time the particle takes to fall to the centre.

The two cases are:

1. the body is a point mass;
2. the body is a uniform sphere of radius R_0.

Adapted from C. Clarke and B. Carswell [1], Problem 9.

Exercise 7.3 (Collapse of a stellar cluster). A cluster consists initially of stars at rest, distributed in a uniform sphere. Find how long it takes a star to reach the center as a function of its initial radius in the cluster and comment on your results.

7.3 Further reading

- C. Clarke and B. Carswell [1], Chapter 3 (gravitation).

8. Hydrostatic Equilibrium and the Virial Theorem

8.1 The virial theorem

We consider a self-gravitating cloud of gas in equilibrium, described by the momentum equation in conservation form (Equation (6.7))

$$\frac{\partial(\rho\mathbf{v})}{\partial t} + \nabla\cdot[\rho\mathbf{v}\otimes\mathbf{v} + p\mathbf{I}] = -\rho\nabla\Psi, \tag{8.1}$$

where Ψ is the gravitational potential. We consider the scalar moment of inertia

$$I = \frac{1}{2}\int_V \rho(\mathbf{r})r^2\mathrm{d}V, \tag{8.2}$$

where V is a volume that encloses the whole gas cloud. The time derivative of I is (using the continuity equation)

$$\begin{aligned}\frac{\mathrm{d}I}{\mathrm{d}t} &= \frac{1}{2}\int_V \frac{\partial\rho}{\partial t}r^2\mathrm{d}V \\ &= -\frac{1}{2}\int_V \nabla\cdot(\rho\mathbf{v})r^2\mathrm{d}V \\ &= -\frac{1}{2}\int_V \nabla\cdot(\rho\mathbf{v}r^2)\mathrm{d}V \\ &\quad +\frac{1}{2}\int_V \rho\mathbf{v}\cdot\nabla r^2\mathrm{d}V.\end{aligned} \tag{8.3}$$

The total divergence term is equal to a boundary integral

$$\int_V \nabla\cdot(\rho\mathbf{v}r^2)\mathrm{d}V = \int_{\partial V}\rho\mathbf{v}r^2\cdot\mathbf{n}\mathrm{d}A, \tag{8.4}$$

but the latter vanishes since, by assumption, $\rho\to 0$ at the boundary of V. So we find

$$\frac{\mathrm{d}I}{\mathrm{d}t} = \frac{1}{2}\int_V \rho\mathbf{v}\cdot\nabla r^2\mathrm{d}V = \int_V \rho\mathbf{v}\cdot\mathbf{r}\mathrm{d}V. \tag{8.5}$$

Here, we have used the fact that $\nabla r = \mathbf{r}/r$.

The second time derivative of I is

$$\begin{aligned}\frac{\mathrm{d}^2 I}{\mathrm{d}t^2} &= \int_V \frac{\partial(\rho\mathbf{v})}{\partial t}\cdot\mathbf{r}\mathrm{d}V \\ &= -\int_V \rho\nabla\Psi\cdot\mathbf{r}\mathrm{d}V \\ &\quad -\int_V \nabla\cdot[\rho\mathbf{v}\otimes\mathbf{v}+p\mathbf{I}]\cdot\mathbf{r}\mathrm{d}V \\ &= -\int_V \rho\nabla\Psi\cdot\mathbf{r}\mathrm{d}V \\ &\quad -\int_V \nabla\cdot[\rho(\mathbf{v}\cdot\mathbf{r})\mathbf{v}+p\mathbf{r}]\mathrm{d}V \\ &\quad +\int_V [\rho\mathbf{v}\otimes\mathbf{v}+p\mathbf{I}]:\nabla\mathbf{r}\mathrm{d}V,\end{aligned} \tag{8.6}$$

where the notation $\mathbf{A}:\mathbf{B}$ denotes the inner product between two tensors:

$$\mathbf{A}:\mathbf{B}=\sum_{i=1}^{3}\sum_{k=1}^{3}A_{ik}B_{ki}=\mathrm{Trace}\left[\begin{pmatrix}A_{xx} & A_{xy} & A_{xz}\\ A_{yx} & A_{yy} & A_{yz}\\ A_{zx} & A_{zy} & A_{zz}\end{pmatrix}\begin{pmatrix}B_{xx} & B_{xy} & B_{xz}\\ B_{yx} & B_{yy} & B_{yz}\\ B_{zx} & B_{zy} & B_{zz}\end{pmatrix}\right]. \tag{8.7}$$

Once again, we can drop the total divergence terms, since, by the divergence theorem, they are equal to boundary integrals that vanish because ρ and p vanish at the boundary of V. We find

$$\frac{\mathrm{d}^2 I}{\mathrm{d}t^2}=-\int_V \rho\nabla\Psi\cdot\mathbf{r}\mathrm{d}V+\int_V[\rho\mathbf{v}\otimes\mathbf{v}+p\mathbf{I}]:\nabla\mathbf{r}\mathrm{d}V. \tag{8.8}$$

The second term in the RHS is simpler than it looks, because

$$\nabla\mathbf{r}=\begin{pmatrix}\partial_x x & \partial_x y & \partial_x z\\ \partial_y x & \partial_y y & \partial_y z\\ \partial_z x & \partial_z y & \partial_z z\end{pmatrix}=\begin{pmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix}=\mathbf{I}. \tag{8.9}$$

So everything simplifies to

$$\int_V[\rho\mathbf{v}\otimes\mathbf{v}+p\mathbf{I}]:\nabla\mathbf{r}\mathrm{d}V=\int_V\rho v^2\mathrm{d}V+3\int_V p\mathrm{d}V=:2K+3\Pi, \tag{8.10}$$

where K is the kinetic energy of the gas and Π is the integrated pressure.

The first term in the RHS of Equation (8.8) requires a little more work. First, we need to compute an expression for the gravitational potential in terms of the mass density ρ. To do this, we remember that the gravitational potential generated by a distribution of point masses with mass m_i at locations $\mathbf{r}_i$ is given by

$$\Psi(\mathbf{r})=-\sum_{i=1}^{N}\frac{Gm_i}{|\mathbf{r}-\mathbf{r}_i|}. \tag{8.11}$$

This can be generalized to the case of a mass density distribution by converting the sum into an integral

$$\Psi(\mathbf{r})=-\int_V\frac{G\rho(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}\mathrm{d}^3 r'. \tag{8.12}$$

Inserting this into the first term in the RHS of Equation (8.8) yields

$$\int_V\rho\nabla\Psi\cdot\mathbf{r}\mathrm{d}^3 r=-\int_V\mathrm{d}^3 r\rho(\mathbf{r})\nabla_{\mathbf{r}}\int_V\mathrm{d}^3 r'\frac{G\rho(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}\cdot\mathbf{r}, \tag{8.13}$$

where we have used the notation $\nabla_{\mathbf{r}}$ to remind ourselves that the derivatives are computed with respect to the unprimed position $\mathbf{r}$. In particular, note that $\rho(\mathbf{r}')$ in the integral is not affected by the derivative, since it is a function of $\mathbf{r}'$ and not of $\mathbf{r}$. The only term that we need to differentiate is the denominator:

$$\begin{aligned}
&\nabla_{\mathbf{r}} \frac{1}{|\mathbf{r}-\mathbf{r}'|} \\
&\quad = \nabla_{\mathbf{r}} \frac{1}{\sqrt{(x-x')^2+(y-y')^2+(z-z')^2}} \\
&\quad = \frac{-1}{[(x-x')^2+(y-y')^2+(z-z')^2]^{3/2}} \\
&\qquad \times \begin{pmatrix} x-x' \\ y-y' \\ z-z' \end{pmatrix} \\
&\quad = -\frac{\mathbf{r}-\mathbf{r}'}{|\mathbf{r}-\mathbf{r}'|^3}.
\end{aligned} \tag{8.14}$$

Inserting the last expression into Equation (8.13), we obtain

$$\begin{aligned}
&\int_V \rho \nabla\Psi \cdot \mathbf{r} \mathrm{d}^3 r \\
&\quad = \int_V \mathrm{d}^3 r \int_V \mathrm{d}^3 r' \frac{G\rho(\mathbf{r}')\rho(\mathbf{r})}{|\mathbf{r}-\mathbf{r}'|^3} (\mathbf{r}-\mathbf{r}') \cdot \mathbf{r} \\
&\quad = \int_V \mathrm{d}^3 r \int_V \mathrm{d}^3 r' \frac{G\rho(\mathbf{r}')\rho(\mathbf{r})}{|\mathbf{r}-\mathbf{r}'|^3} |\mathbf{r}-\mathbf{r}'|^2 \\
&\qquad + \int_V \mathrm{d}^3 r \int_V \mathrm{d}^3 r' \frac{G\rho(\mathbf{r}')\rho(\mathbf{r})}{|\mathbf{r}-\mathbf{r}'|^3} (\mathbf{r}-\mathbf{r}') \cdot \mathbf{r}' \\
&\quad = \int_V \mathrm{d}^3 r \int_V \mathrm{d}^3 r' \frac{G\rho(\mathbf{r}')\rho(\mathbf{r})}{|\mathbf{r}-\mathbf{r}'|} \\
&\qquad + \int_V \mathrm{d}^3 r' \rho(\mathbf{r}') \mathbf{r}' \cdot \int_V \mathrm{d}^3 r \frac{G\rho(\mathbf{r})(\mathbf{r}-\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|^3} \\
&\quad = -\int_V \mathrm{d}^3 r \Psi(\mathbf{r})\rho(\mathbf{r}) \\
&\qquad - \int_V \rho(\mathbf{r}') \nabla_{\mathbf{r}'} \Psi(\mathbf{r}') \cdot \mathbf{r}' \mathrm{d}^3 r'.
\end{aligned} \tag{8.15}$$

We have used the fact that (note the difference in sign compared to Equation (8.14))

$$\nabla_{\mathbf{r}'} \frac{1}{|\mathbf{r}-\mathbf{r}'|} = \frac{\mathbf{r}-\mathbf{r}'}{|\mathbf{r}-\mathbf{r}'|^3}. \tag{8.16}$$

The last term on the RHS of Equation (8.15) is identical to the LHS, so we find

$$-\int_V \rho \nabla\Psi \cdot \mathbf{r} \mathrm{d}^3 r = \frac{1}{2} \int_V \mathrm{d}^3 r \Psi(\mathbf{r})\rho(\mathbf{r}) = W, \tag{8.17}$$

W being the gravitational potential energy.

For a **stationary configuration** $\mathrm{d}^2 I/\mathrm{d}t^2 = 0$, so Equation (8.8) yields the *virial theorem*:

$$\boxed{W + 2K + 3\Pi = 0.} \tag{8.18}$$

For an ideal gas $p = (\Gamma - 1)\rho\varepsilon$ (Equation (2.15)), so

$$\Pi = \int_V p \mathrm{d}V = (\Gamma - 1) \int_V \rho\varepsilon \mathrm{d}V = (\Gamma - 1)U, \tag{8.19}$$

where U is the total internal energy. In this case, the virial theorem reads

$$\boxed{W + 2K + 3(\Gamma - 1)U = 0.} \tag{8.20}$$

8.2 Hydrostatic equilibria

A fluid is in hydrostatic equilibrium when $\mathbf{v} = 0$, so that Euler's equation reduces to $\nabla p = -\rho \nabla \Psi$. For a spherical star, this reads $\mathrm{d}p/\mathrm{d}r = -GM(r)\rho/r^2$, where $M(r)$ is the mass enclosed within the radius r (see also Equation (4.21)).

The virial theorem has an important consequence for the stability of hydrostatic equilibria. Consider a cloud of gas or star in hydrostatic equilibrium, that is with no large scale motion (no rotation). Then the virial theorem states that

$$W + 3(\Gamma - 1)U = 0. \tag{8.21}$$

On the other hand, the energy of the cloud is

$$E = W + U. \tag{8.22}$$

The energy must be negative for the system to be bound, $E \leq 0$. This implies

$$U \leq -W = 3(\Gamma - 1)U \Rightarrow [-1 + 3(\Gamma - 1)]U \geq 0, \tag{8.23}$$

since $U \geq 0$, this requires

$$\Gamma \geq 4/3. \tag{8.24}$$

Monoatomic gas has $\Gamma = 5/3$, so regular stars are stable. Matter dominated by relativistic particles, such as radiation or the relativistic degenerate electrons in white dwarfs and in the iron cores of massive stars, has $\Gamma \approx 4/3$. Stars made of such matter are only marginally stable, and small effects that reduce Γ below $4/3$ can make them collapse. This happens in the iron cores of massive stars, where electron captures reduce the pressure, triggering core-collapse supernovae (Chapter 3 and Section 4.3), and in very massive stars, where the production of electron–positron pairs triggers so-called pair-instability supernovae.

A surprising consequence of the virial theorem is that self-gravitating systems have *negative specific heat*. Consider a self-gravitating cloud (a protostar) emitting infrared radiation. As its energy E becomes more negative its gravitational binding energy $-W$ must increase, because $U \geq 0$. We can use Equation (8.21) and Equation (8.22) to solve for U in terms of E:

$$U = -\frac{E}{3\Gamma - 4}. \tag{8.25}$$

If $\Gamma > 4/3$, Equation (8.25) predicts that U will increase as E becomes more negative. Meaning that the system heats up as it loses energy! Eventually the cloud contracts to the point that nuclear fusion starts occurring and a star is born. Nuclear burning *prevents* the star from heating up further.

Example

Could the Sun be powered by gravitational contraction alone? In this case, the Sun would radiate its gravitational binding energy, $|W| \sim GM_\odot^2/R_\odot$, over a time

$$t_{\mathrm{KH}} \sim \frac{GM_\odot^2}{R_\odot L_\odot} \approx 3 \times 10^7 \,\mathrm{yr}, \tag{8.26}$$

where $L_\odot \approx 3.8 \times 10^{26}\,\mathrm{W}$ ($3.8 \times 10^{33}\,\mathrm{erg\,s^{-1}}$) is the luminosity of the Sun. This is the so-called *Kelvin–Helmholtz time*. It is much shorter than the age of the Earth, about 4.5×10^9 years, so the Sun must have another source of energy: nuclear fusion.

Example

The hot gas in a galaxy cluster is in approximate virial equilibrium in the gravitational potential of the cluster. Consider a cluster with mass $M \sim 10^{15}\, M_\odot$ and radius $R \sim 1\,\mathrm{Mpc}$, and assume, for simplicity, that the gas has the same distribution as the total mass. For a uniform sphere $W = -\frac{3}{5}GM^2/R$, while $3\Pi = 3Mk_BT/(\mu m_u)$, so the virial theorem gives

$$k_BT \sim \frac{GM\mu m_u}{5R} \approx 5\,\mathrm{keV}, \tag{8.27}$$

or $T \sim 6 \times 10^7\,\mathrm{K}$, where we have taken $\mu \approx 0.6$ for fully ionized gas. This is why galaxy clusters are bright X-ray sources, as in the case of the Bullet cluster (Figure 1.1).

8.3 Application to star formation

Consider a spherical cloud of radius R. For simplicity, we take its density $\bar{\rho}$ to be uniform. Then the gravitational potential energy is

$$-W = G\int_0^R 4\pi r\rho(r)M(r)\mathrm{d}r = \frac{16\pi^2}{15}G\bar{\rho}^2R^5 = \frac{3}{5}\frac{GM^2}{R}, \tag{8.28}$$

where $M = \frac{4\pi}{3}\bar{\rho}R^3$ is the mass of the cloud and

$$M(r) = \int_0^r 4\pi r'^2\rho(r')\mathrm{d}r' \tag{8.29}$$

is the mass enclosed within the radius r. For an ideal gas, the pressure is given by

$$p = \frac{k_BT}{\mu m_u}\rho, \tag{8.30}$$

where μ is the mean molecular weight (Equation (1.6)), m_u is the atomic mass unit, k_B is the Boltzmann constant and T is the temperature. We can assume T to be constant. This is a good approximation for optically thin clouds which can efficiently cool by emitting radiation. With this approximation

$$\Pi = \int_0^R 4\pi r^2 p(r)\mathrm{d}r = \frac{4}{3}\pi c_s^2\bar{\rho}R^3, \tag{8.31}$$

where

$$c_s^2 = \frac{k_BT}{\mu m_u} \tag{8.32}$$

is a constant. As we will see later, c_s turns out to be the sound speed of the gas (Chapter 9). The condition for collapse of the cloud is that $3\Pi < |W|$, which implies that collapse will occur for a cloud with

$$R > R_J, \qquad R_J^2 = \frac{15c_s^2}{4\pi G\bar{\rho}}. \tag{8.33}$$

R_J is called the *Jeans length*. Typical molecular clouds have $R \gg R_J$, so instead of collapsing to an object with mass equal to the mass of the cloud, they fragment into smaller pieces. The mass of these fragments can be estimated as the mass contained within one Jeans length, the so-called *Jeans mass*[6]:

[6]Note that this definition is slightly different from the one given in C. Clarke and B. Carswell [1]; however, this difference is inconsequential given the level of approximation of these arguments.

$$M_J = \frac{4\pi}{3}\bar{\rho}R_J^3 = \frac{15^{3/2}c_s^3}{3(4\pi\bar{\rho})^{1/2}G^{3/2}} = \frac{15^{3/2}k_B^{3/2}}{3(4\pi)^{1/2}m_u^2 G^{3/2}}\frac{T^{3/2}}{\mu^2 n^{1/2}}. \tag{8.34}$$

Inserting the values of the constants, we find

$$M_J \approx 94\,M_\odot \left(\frac{T}{\mathrm{K}}\right)^{3/2}\left(\frac{\mathrm{cm}^{-3}}{n}\right)^{1/2}\frac{1}{\mu^2}. \tag{8.35}$$

For a typical molecular cloud with $T = 10\,\mathrm{K}$, $n = 100\,\mathrm{cm}^{-3}$, and $\mu = 2$ (for molecular hydrogen) the Jeans length is $R_J \approx 1.5\,\mathrm{pc}$ and the Jeans mass is $M_J \approx 70\,M_\odot$. This is much larger than the mass of a typical star. Note however that the Jeans mass decreases with increasing n. So, as long as the gas can efficiently cool and remain at low temperature, the fragments collapse and undergo further fragmentation. In the dense cores of molecular clouds, with $n \sim 10^4\,\mathrm{cm}^{-3}$, the Jeans mass is $M_J \approx 7\,M_\odot$, and at even higher densities it approaches the mass of typical stars.

While the arguments we have given here give an order of magnitude estimate for the mass of protostars, star formation is strongly influenced by a number of factors that we have neglected. These include turbulence and magnetic fields. Moreover, star formation strongly depends on the ability of the gas to cool to low temperature, as shown by Equation (8.35).

8.4 In-class worksheet

For a magnetic star the virial theorem becomes:

$$W + 2K + 3\Pi + \mathfrak{M} = 0, \tag{8.36}$$

where the new term is the magnetic energy of the star, which is

$$\mathfrak{M} = \frac{1}{2\mu_0}\int B^2 \mathrm{d}V. \tag{8.37}$$

1. Find the condition for stability of a nonrotating star in terms of Γ and $\mathfrak{M}$.
2. What is the maximum magnetic field strength for a Sun-like star ($R = 7\times10^5\,\mathrm{km}$, $M = 2\times10^{30}\,\mathrm{kg}$)? Neglect factors of order unity, and give the answer in tesla and in gauss ($1\,\mathrm{T} = 10^4\,\mathrm{G}$).
3. What is the maximum magnetic field strength for a neutron star ($R = 12\,\mathrm{km}$, $M = 3\times10^{30}\,\mathrm{kg}$)?

8.5 Exercises

Exercise 8.1 (Self-gravitating slab). An infinite, static slab of incompressible fluid with uniform density ρ_0 fills the region $|z| < a$ and is held together by its own gravity.

1. Compute the gravitational acceleration inside and outside the slab, and the pressure profile inside it. Use Poisson's equation, $\nabla^2\Psi = 4\pi G\rho$.
2. Model the disk of a galaxy as such a slab, with $\rho_0 = 10^{-23}\,\mathrm{g\,cm}^{-3} = 10^{-20}\,\mathrm{kg\,m}^{-3}$ and $a = 10^{19}\,\mathrm{m}$. A star starts at rest at $z = a$. How fast does it cross the mid-plane, and what is the period of its vertical oscillation?

Adapted from C. Clarke and B. Carswell [1], Problem 8.

Exercise 8.2 (Mass of a molecular cloud). A self-gravitating cloud of molecular hydrogen $\left(p = nk_BT = \frac{2}{5}\rho\varepsilon\right)$ is measured to have radius $R = 0.1\,\mathrm{pc}$ and temperature $T = 10\,\mathrm{K}$. *Estimate* the mass of the cloud.

Exercise 8.3 (Temperature of the solar core). Estimate the temperature at the center of the Sun, assuming that density and pressure change over a length scale comparable to the solar radius. Make the

estimate twice: first assuming that the Sun is supported by gas pressure, then assuming that it is supported by radiation pressure. Which assumption is more plausible, and why?

Adapted from C. Clarke and B. Carswell [1], Problem 11.

Exercise 8.4 (Numerical solution of the Lane–Emden equation). The Lane–Emden equation describes the structure of a star with a polytropic equation of state, $p = K\rho^{1+1/n}$. It reads

$$\frac{1}{\xi^2}\frac{\mathrm{d}}{\mathrm{d}\xi}\left(\xi^2\frac{\mathrm{d}\theta}{\mathrm{d}\xi}\right) = -\theta^n, \qquad \theta(0) = 1, \qquad \theta'(0) = 0, \tag{8.38}$$

where θ^n is the density in units of the central density, and ξ is the radius in suitable units. Write a code to compute numerically the solution of the Lane–Emden equation for a given n.

1. Validate your code by comparing numerical and analytic solution for $n = 0, 1, 5$.
2. Plot the solution for $n = 1.5$ and find the radius of the polytrope, i.e., the value ξ_r at which $\theta(\xi_r) = 0$.

Hint: use the `solve_ivp` function from `scipy.integrate`.

8.6 Further reading

- C. Clarke and B. Carswell [1], Section 3.6 (the virial theorem) and Chapter 5 (hydrostatic equilibrium, polytropes, and the Lane–Emden equation).

II

Waves, Shocks, and Outflows

9 Sound Waves and the Doppler Effect . . . 77
9.1 Oscillation Equation . . . 77
9.2 Wave Energy and Momentum . . . 81
9.3 In-class worksheets . . . 83
9.4 Exercises . . . 84
9.5 Further reading . . . 85

10 Waves in Stratified Atmospheres . . . 87
10.1 Waves in Isothermal Atmospheres . . . 87
10.2 Gravity Waves . . . 90
10.3 In-class worksheet . . . 91
10.4 Exercises . . . 92
10.5 Further reading . . . 93

11 Shock Waves . . . 95
11.1 In-class worksheet . . . 95
11.2 Exercises . . . 96
11.3 Further reading . . . 97

12 Radiative Shocks . . . 99
12.1 Exercises . . . 99
12.2 Further reading . . . 99

13 Blast Waves . . . 101
13.1 In-class worksheet . . . 101
13.2 Exercises . . . 102
13.3 Further reading . . . 102

14 Stellar Winds and Spherical Accretion . . . 103
14.1 In-class worksheet . . . 103
14.2 Exercises . . . 103
14.3 Further reading . . . 104

9. Sound Waves and the Doppler Effect

9.1 Oscillation Equation

The equations of fluid mechanics are nonlinear and only few analytic solutions are known. However, significant insight into astrophysical systems can be obtained by considering small perturbations around a background flow. Typically the background is chosen to be a hydrostatic solution, e.g., the density profile of an atmosphere. However, this method can be applied to more general situations. This is the key working principle of *asteroseismology* (*helioseismology* in the case of the Sun). Sound waves are also one of the key tools to learn about the early Universe, as they generate the baryon acoustic oscillations seen in the CMB and in the distribution of galaxies.

To set the stage, consider a gas flow with density, pressure, and velocity

$$\rho_0(t;\mathcal{V}), \quad p_0(t;\mathcal{V}), \quad \mathbf{v}_0(t;\mathcal{V}). \tag{9.1}$$

We describe its perturbation in terms of a small displacement vector $\boldsymbol{\xi}(t;\mathcal{V})$ describing the offset of each fluid element from its background state.

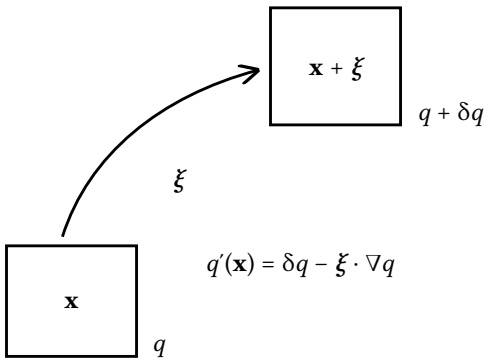


Figure 9.1: Lagrangian displacement of a fluid element from position $\mathbf{x}$ to position $\mathbf{x}+\boldsymbol{\xi}$.

The density, pressure, and velocity *of each fluid element* change according to

$$\rho(t;\mathcal{V}) = \rho_0(t;\mathcal{V}) + \delta\rho(t;\mathcal{V}), \quad p(t;\mathcal{V}) = p_0(t;\mathcal{V}) + \delta p(t;\mathcal{V}), \quad \mathbf{v}(t;\mathcal{V}) = \mathbf{v}_0(t;\mathcal{V}) + \frac{\mathrm{d}\boldsymbol{\xi}(t;\mathcal{V})}{\mathrm{d}t} \tag{9.2}$$

The quantities with the δ in front are the Lagrangian perturbations[7]. The Eulerian perturbation of a quantity is given by

$$q'(t,\mathbf{x}) = q_0(t,\mathbf{x}-\boldsymbol{\xi}) + \delta q(t,\mathbf{x}) - q_0(t,\mathbf{x}) = \delta q(t,\mathbf{x}) - \boldsymbol{\xi}\cdot\nabla q_0. \tag{9.3}$$

The difference between Lagrangian and Eulerian perturbations is that the Lagrangian perturbations are specific to individual fluid elements, while the Eulerian perturbations record the difference in the flow at a fixed position. This distinction turns out to be crucial when perturbing non-homogeneous backgrounds, for which $\nabla q_0 \neq 0$, as we will see in Chapter 10.

For the moment, we consider a homogeneous medium in which ρ_0, p_0, and $\mathbf{v}_0 = 0$ are constant. We perturb it by displacing the fluid elements by a small amount $\boldsymbol{\xi}$ and let it evolve freely. How do the

[7] I use the common convention of denoting with δq the Lagrangian perturbations of a quantity q and with q' the Eulerian perturbations. The same convention is used in R. M. Kulsrud [4], but C. Clarke and B. Carswell [1] use Δq for Lagrangian perturbation and δq for Eulerian perturbation.

perturbations evolve? To find out, we need to insert the *Lagrangian* perturbations into the *Lagrangian* equations (Chapter 5).

The continuity equation (Equation (1.15)) reads

$$\frac{\mathrm{d}\rho}{\mathrm{d}t} = -\rho\nabla\cdot\mathbf{v} \to \frac{\mathrm{d}(\rho_0+\delta\rho)}{\mathrm{d}t} = -(\rho_0+\delta\rho)\nabla\cdot\left(\frac{\mathrm{d}\boldsymbol{\xi}}{\mathrm{d}t}\right). \tag{9.4}$$

Dropping quantities that are quadratic or higher order in the perturbation, and integrating in time with the condition that $\delta\rho = 0$ when $\boldsymbol{\xi} = 0$, we obtain

$$\frac{\mathrm{d}\delta\rho}{\mathrm{d}t} = -\rho_0\nabla\cdot\frac{\mathrm{d}\boldsymbol{\xi}}{\mathrm{d}t} \to \boxed{\delta\rho = -\rho_0\nabla\cdot\boldsymbol{\xi}}. \tag{9.5}$$

To find the perturbed momentum equation, first note that

$$\frac{\mathrm{d}\boldsymbol{\xi}}{\mathrm{d}t} = \frac{\partial\boldsymbol{\xi}}{\partial t} + (\mathbf{v}\cdot\nabla)\boldsymbol{\xi} = \frac{\partial\boldsymbol{\xi}}{\partial t} + (\mathbf{v}_0\cdot\nabla)\boldsymbol{\xi} + \left(\frac{\mathrm{d}\boldsymbol{\xi}}{\mathrm{d}t}\cdot\nabla\right)\boldsymbol{\xi} = \frac{\partial\boldsymbol{\xi}}{\partial t} + O(|\boldsymbol{\xi}|^2), \tag{9.6}$$

so

$$\mathbf{v} = \frac{\partial\boldsymbol{\xi}}{\partial t}, \qquad \frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = \frac{\partial\mathbf{v}}{\partial t} + (\mathbf{v}_0\cdot\nabla)\mathbf{v} + \left(\frac{\mathrm{d}\boldsymbol{\xi}}{\mathrm{d}t}\cdot\nabla\right)\mathbf{v} = \frac{\partial^2\boldsymbol{\xi}}{\partial t^2} + O(|\boldsymbol{\xi}|^2). \tag{9.7}$$

The momentum equation (Equation (4.12)) becomes

$$\rho\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = -\nabla p \to \frac{\partial^2\boldsymbol{\xi}}{\partial t^2} = -\frac{1}{\rho_0}\nabla p = -\frac{1}{\rho_0}\nabla\delta p. \tag{9.8}$$

Here we have used the fact that, for a homogeneous background, the Eulerian and Lagrangian perturbations are the same, $p' = \delta p$ (Equation (9.3)). This is not true in a stratified atmosphere, as we will see in Chapter 10.

We will consider only the case of adiabatic perturbations, for which

$$p = p(\rho). \tag{9.9}$$

In other words, we are taking the pressure at constant entropy. For example, for an ideal gas, we have already seen that for adiabatic processes (Equation (2.24))

$$p = (\Gamma-1)\rho\varepsilon \to p = K\rho^\Gamma, \tag{9.10}$$

with $K = (\Gamma-1)\rho^{1-\Gamma}\varepsilon$.

Why should sound waves be adiabatic? A sound wave compresses some regions of the fluid and expands others, so the compressed regions become hotter. Heat flows from the hot to the cold regions, but this takes time. Heat needs to diffuse over a distance comparable to the wavelength λ_w of the wave, which takes a time $t_{\mathrm{diff}} \sim \lambda_w^2/D$, where $D \sim \lambda v_{\mathrm{th}}$ and λ is the mean free path (Chapter 6). The period of the wave is $\sim \lambda_w/c_s$, where c_s is the sound speed defined below, which is comparable to the thermal velocity of the particles v_{th}. So we have

$$\frac{t_{\mathrm{diff}}}{\lambda_w/c_s} \sim \frac{\lambda_w}{\lambda} \gg 1, \tag{9.11}$$

whenever the fluid description is valid. In other words, heat does not have time to leave the compressed regions during one period, and the perturbations are adiabatic. The opposite limit, isothermal perturbations, happens when the gas can cool very quickly, for example by radiation. This is the case in cold molecular clouds (Chapter 8, Chapter 12).

In this case, we can use the chain rule to rewrite $\nabla\delta p$ in terms of $\nabla\delta\rho$:

$$\frac{\partial^2\boldsymbol{\xi}}{\partial t^2} = -\frac{1}{\rho_0}\frac{\mathrm{d}p}{\mathrm{d}\rho}\nabla\delta\rho = \frac{\mathrm{d}p}{\mathrm{d}\rho}\nabla\nabla\cdot\boldsymbol{\xi}. \tag{9.12}$$

Defining the *sound speed* $c_s = \sqrt{\mathrm{d}p/\mathrm{d}\rho}$, we obtain the **oscillation equation**:

$$\boxed{\frac{\partial^2 \boldsymbol{\xi}}{\partial t^2} - c_s^2 \nabla \nabla \cdot \boldsymbol{\xi} = 0.} \tag{9.13}$$

In general, the sound speed needs to be evaluated at constant entropy, that is

$$\boxed{c_s = \sqrt{\left(\frac{\partial p}{\partial \rho}\right)_s}.} \tag{9.14}$$

Example

For an ideal gas, with $p = (\Gamma - 1)\rho\varepsilon = \frac{\rho}{\mu m_u} k_B T$, the sound speed is

$$c_s^2 = \left(\frac{\partial p}{\partial \rho}\right)_s = \Gamma K \rho^{\Gamma - 1} = (\Gamma - 1)\Gamma\varepsilon = \frac{\Gamma k_B T}{\mu m_u}. \tag{9.15}$$

If, instead, the perturbations are isothermal, we need to take the derivative of the pressure at constant temperature. For an ideal gas, this gives the **isothermal sound speed**

$$c_{\text{iso}} = \sqrt{\left(\frac{\partial p}{\partial \rho}\right)_T} = \sqrt{\frac{k_B T}{\mu m_u}} = \frac{c_s}{\sqrt{\Gamma}}. \tag{9.16}$$

This is the sound speed that appears in the equation of state of an isothermal gas, $p = c_{\text{iso}}^2 \rho$ (Chapter 4, Chapter 8).

Example

Consider air at room temperature, $T = 293\,\text{K}$. Air is mostly made of N_2 and O_2, so $\mu \approx 29$ and $\Gamma = 7/5$. From Equation (9.15) we find $c_s \approx 343\,\text{m}\,\text{s}^{-1}$, which is the measured value. Newton was the first to compute the speed of sound, but he assumed that the perturbations were isothermal. He found $c_{\text{iso}} \approx 290\,\text{m}\,\text{s}^{-1}$, about 15% too small. The discrepancy was only explained a century later by Laplace, who realized that sound waves are adiabatic.

In astrophysics, sound speeds span a large range. In a cold molecular cloud with $T = 10\,\text{K}$ and $\mu \approx 2.3$, we have $c_s \approx 0.2\,\text{km}\,\text{s}^{-1}$. At the center of the Sun, $T \approx 1.6 \times 10^7\,\text{K}$ and $\mu \approx 0.85$, so $c_s \approx 500\,\text{km}\,\text{s}^{-1}$. This means that sound takes about one hour to cross the Sun (Exercise 9.1).

The oscillation Equation (9.13) is the fundamental equation to describe sound waves. Taking the divergence on both sides and using Equation (9.5), we find that the density perturbation $\delta\rho$ satisfies the wave equation:

$$\frac{\partial^2 \delta\rho}{\partial t^2} - c_s^2 \nabla^2 \delta\rho = 0. \tag{9.17}$$

To solve the oscillation equation, we consider the ansatz

$$\boldsymbol{\xi} = \boldsymbol{\xi}_1 \exp[\mathrm{i}(\mathbf{k} \cdot \mathbf{x} - \omega t)], \tag{9.18}$$

which describes a plane sound wave propagating in the direction $\mathbf{k}$ and having frequency ω. $\boldsymbol{\xi}_1$ is the *polarization vector*, describing the way in which fluid elements oscillate as the wave propagates. As usual, the physical displacement is the real part of Equation (9.18). Inserting this ansatz into Equation (9.13), we find

$$\omega^2 \boldsymbol{\xi}_1 = c_s^2 \mathbf{k}(\mathbf{k} \cdot \boldsymbol{\xi}_1). \tag{9.19}$$

This means that plane wave solutions to Equation (9.13) exist only for specific choices of $\boldsymbol{\xi}_1, \omega$ and $\mathbf{k}$. We see that the RHS is parallel to $\mathbf{k}$, so solutions with $\omega \neq 0$ can exist only if $\boldsymbol{\xi}_1 \parallel \mathbf{k}$. This means that sound waves are longitudinal. Displacements with $\boldsymbol{\xi}_1 \perp \mathbf{k}$ are also solutions, but with $\omega = 0$. They do not compress the fluid, so there is no restoring force (recall that a fluid at rest cannot sustain shear, Chapter 4).

This is opposite from electromagnetic waves in vacuum, in which the electric and magnetic fields oscillate in the plane perpendicular to the direction of propagation. Setting $\boldsymbol{\xi}_1 = \xi_1 \hat{\mathbf{k}}$, we find

$$\omega^2 = c_s^2 k^2. \tag{9.20}$$

This is the **dispersion relation** for sound waves. It means that both phase and group velocity for sound waves are given by c_s

$$v_{\text{phase}} = \frac{\omega}{k} = c_s, \qquad v_{\text{group}} = \frac{\mathrm{d}\omega}{\mathrm{d}k} = c_s. \tag{9.21}$$

This justifies the name *sound speed* for c_s. Since v_{phase} does not depend on k, sound waves are *non-dispersive*: a pulse made of many waves with different wavelengths propagates without changing its shape.

What happens to fluid elements when a sound wave passes through them? They are displaced along the direction of the wave. This causes a compression of the fluid. From Equation (9.5), we find:

$$\delta\rho = -\mathrm{i}\rho_0(\mathbf{k}\cdot\boldsymbol{\xi}_1)\mathrm{e}^{\mathrm{i}(\mathbf{k}\cdot\mathbf{x}-\omega t)}. \tag{9.22}$$

Example

Consider a sound wave propagating along z with wave number k. Then $\mathbf{k} = k\hat{\mathbf{z}}$ and $\omega = c_s k$. The only non-zero component of $\boldsymbol{\xi}$ is the z-component

$$\xi_z = \mathrm{Re}\{\xi_1 \mathrm{e}^{\mathrm{i}k(z-c_s t)}\} = A\cos[k(z-c_s t)] + B\sin[k(z-c_s t)] \Leftrightarrow \xi_1 = A - \mathrm{i}B \tag{9.23}$$

The density perturbation is given by

$$\frac{\delta\rho}{\rho_0} = -\frac{\partial \xi_z}{\partial z} = k\{A\sin[k(z-c_s t)] - B\cos[k(z-c_s t)]\}. \tag{9.24}$$

Note that the density perturbation is shifted by a quarter of a wavelength with respect to the displacement. The gas is most compressed where ξ_z changes sign, not where it is largest. This is shown in Figure 9.2.

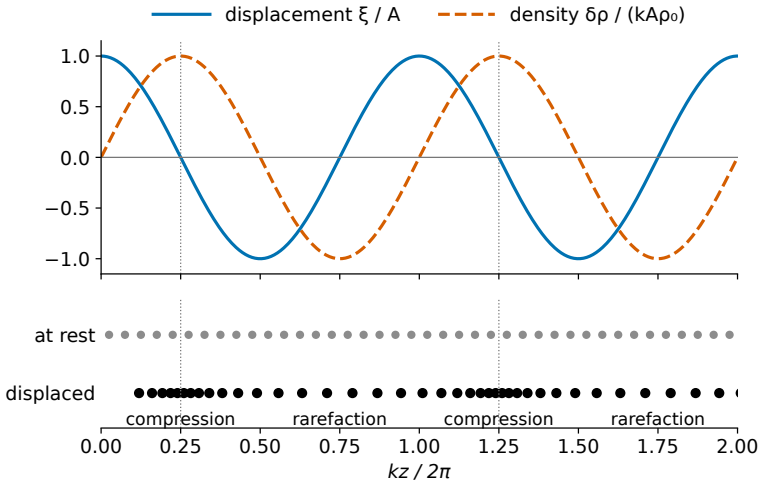


Figure 9.2: A plane sound wave at a fixed time. *Top*: displacement ξ_z and density perturbation $\delta\rho/\rho_0$. *Bottom*: positions of fluid elements, initially equally spaced, after being displaced by ξ_z.

The same method can be applied to a homogeneous fluid moving with constant velocity $\mathbf{v}_0$. In this case the sound waves are carried by the flow, and the dispersion relation becomes

$$\omega = \mathbf{v}_0 \cdot \mathbf{k} \pm c_s k. \tag{9.25}$$

This means that the frequency of the sound measured by an observer at rest is shifted with respect to the frequency $c_s k$ measured by an observer moving with the fluid. This is the **Doppler effect**. We derive Equation (9.25) in the in-class worksheet (Section 9.3).

9.2 Wave Energy and Momentum

Sound waves also transport energy and momentum. The deposition of energy by breaking waves (Chapter 11) is thought to play an important role in phenomena such as core-collapse supernovae, stellar eruptions, and the evolution of giant molecular clouds.

i Going further: Normal Modes

Before we can compute the energy carried by waves, we need to develop some general perturbation theory. In general, the oscillation equation takes the form

$$\rho_0 \frac{\partial^2 \boldsymbol{\xi}}{\partial t^2} = \hat{\mathbf{F}}\boldsymbol{\xi}, \tag{9.26}$$

where $\hat{\mathbf{F}}$ is a differential operator. For example, for sound waves in a homogeneous medium $\hat{\mathbf{F}}\boldsymbol{\xi} = \rho_0 c_s^2 \nabla\nabla\cdot\boldsymbol{\xi}$. We can find a dispersion relation by looking for solutions such that

$$\boldsymbol{\xi}(t,\mathbf{x}) = \boldsymbol{\eta}(\mathbf{x})\mathrm{e}^{-\mathrm{i}\omega t}. \tag{9.27}$$

Inserting this ansatz in Equation (9.26), we find an eigenvalue problem

$$-\omega^2\rho_0\boldsymbol{\eta}(\mathbf{x}) = \hat{\mathbf{F}}\boldsymbol{\eta}(\mathbf{x}). \tag{9.28}$$

In most applications, $\hat{\mathbf{F}}$ is self-adjoint:

$$\int \boldsymbol{\zeta}(\mathbf{x})\cdot\hat{\mathbf{F}}\boldsymbol{\xi}(\mathbf{x})\mathrm{d}V = \int \boldsymbol{\xi}(\mathbf{x})\cdot\hat{\mathbf{F}}\boldsymbol{\zeta}(\mathbf{x})\mathrm{d}V. \tag{9.29}$$

For sound waves, this can be shown by integrating by parts: both sides are equal to $-\int \rho_0 c_s^2 (\nabla\cdot\boldsymbol{\zeta})(\nabla\cdot\boldsymbol{\xi})\mathrm{d}V$. An important consequence of Equation (9.29) is that ω^2 is always real. So each mode either oscillates, if $\omega^2 > 0$, or grows exponentially, if $\omega^2 < 0$. We will use this to study instabilities later in the course (Chapter 15, Chapter 16, Chapter 17).

The general theory of linear operators then implies that Equation (9.28) has an infinite set of eigenvalues $\{\omega_n\}$ and eigenfunctions $\{\boldsymbol{\eta}_n\}$ and that any perturbation (not just plane waves) can be written as an infinite superposition of elementary solutions:

$$\boldsymbol{\xi}(t,\mathbf{x}) = \sum_n \boldsymbol{\eta}_n(\mathbf{x})\mathrm{e}^{-\mathrm{i}\omega_n t}. \tag{9.30}$$

The quantities $\boldsymbol{\eta}_n(\mathbf{x})$ are also called the *normal modes of oscillations*. Note that the normal modes need not be countable. For example, for sound waves in a homogeneous infinite medium all positive sound frequencies ω are allowed. In this case, Equation (9.30) takes the form of an integral. For plane waves in a homogeneous medium, Equation (9.30) is just the Fourier transform:

$$\boldsymbol{\xi}(t,\mathbf{x}) = \int \mathrm{d}^3\mathbf{k}\boldsymbol{\eta}(\mathbf{k})\mathrm{e}^{\mathrm{i}\mathbf{k}\cdot\mathbf{x}}\mathrm{e}^{-\mathrm{i}c_s k t}, \tag{9.31}$$

where the real part is understood. However, when we consider sound waves in a finite medium, for example a star of a given radius, standing waves must fit an integer number of half wavelengths inside the star. This makes the frequencies $\{\omega_n\}$ quantized.

Example

Consider a rigid, cylindrical pipe of length L filled with gas at constant density and temperature. The ends of the pipe are sealed. The displacement equation for sound waves propagating in the pipe is

$$\begin{cases} \partial_t^2 \xi(t,z) = c_s^2 \partial_z^2 \xi(t,z), 0 < z < L, \\ \xi(t,0) = \xi(t,L) = 0. \end{cases} \tag{9.32}$$

If we look for solutions in the form $\xi(t,z) = \eta(z)\mathrm{e}^{-\mathrm{i}\omega t}$, we find the eigenvalue problem

$$\begin{cases} \partial_z^2 \eta(z) + \left(\frac{\omega}{c_s}\right)^2 \eta(z) = 0, 0 < z < L \\ \eta(0) = \eta(L) = 0. \end{cases} \tag{9.33}$$

This equation is just the equation for a harmonic oscillator. The general solution is

$$\eta(z) = A\cos(kz) + B\sin(kz), \qquad k = \frac{\omega}{c_s}. \tag{9.34}$$

However, not all solutions satisfy the boundary conditions. To satisfy the boundary conditions, we require

$$A = 0, \qquad k = k_n = \frac{n\pi}{L} \Rightarrow \omega = c_s\left(\frac{n\pi}{L}\right), \quad n = 1, 2, \ldots \tag{9.35}$$

For example, for a pipe of length $L = 1\,\mathrm{m}$ filled with air, the frequency of the fundamental mode is $\omega_1/2\pi = c_s/(2L) \approx 170\,\mathrm{Hz}$. This is how organ pipes work. The first few normal modes are plotted in Figure 9.3.

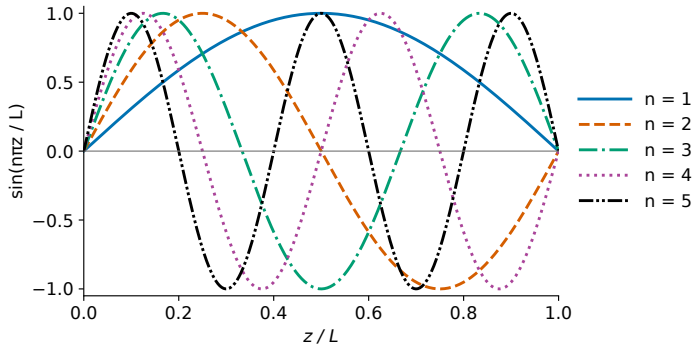


Figure 9.3: Normal modes for gas in a cylindrical pipe.

This means that all sound waves propagating along the pipe are in the form

$$\xi(t,z) = \sum_{n=1}^{\infty}\left[A_n \sin\left(\frac{n\pi c_s t}{L}\right) + B_n \cos\left(\frac{n\pi c_s t}{L}\right)\right]\sin\left(\frac{n\pi z}{L}\right), \tag{9.36}$$

where A_n and B_n are determined by the initial velocity $\partial_t \xi(0,z)$ and displacement $\xi(0,z)$, respectively.

i Going further: Work Integral

We might be tempted to define the energy density of a wave in the general form $\boldsymbol{\xi}(t,\mathbf{x}) = \boldsymbol{\eta}(\mathbf{x})\mathrm{e}^{-\mathrm{i}\omega t}$ as

$$K_{\text{wave}} = \frac{1}{2}\rho_0 \left\langle \left|\frac{\partial \boldsymbol{\xi}}{\partial t}\right|^2 \right\rangle = \frac{1}{2}\rho_0 |\boldsymbol{\eta}|^2 \omega^2 \langle \sin^2(\omega t)\rangle = \frac{\omega^2}{4}\rho_0 |\boldsymbol{\eta}|^2, \tag{9.37}$$

where $\langle\cdot\rangle$ denotes averaging over a period $T = 2\pi/\omega$ of the wave. Equation (9.37) is the kinetic energy density associated with the wave. However, it turns out that the kinetic energy is just one portion of the wave energy.

The energy of the wave is related to ξ^2, so it cannot be computed directly from linear theory. However, we can use a trick:

1. adiabatically perturb the system to create a wave with an external force;
2. compute the work done by the external force on the system.

To achieve this we add a source term to the oscillation equation:

$$\rho_0 \frac{\partial^2 \boldsymbol{\xi}}{\partial t^2} = \hat{\mathbf{F}}\boldsymbol{\xi} + \mathbf{F}_{\text{ext}}(t). \tag{9.38}$$

Suppose we want to excite the mode $\boldsymbol{\xi}(t, \mathbf{x}) = \boldsymbol{\eta}(\mathbf{x})\mathrm{e}^{-\mathrm{i}\omega t}$, where $\boldsymbol{\eta}$ is such that $\hat{\mathbf{F}}\boldsymbol{\eta} = -\rho_0\omega^2\boldsymbol{\eta}$. Then we can take $\mathbf{F}_{\text{ext}}(t) = F_0\mathrm{e}^{\gamma t}\boldsymbol{\eta}(\mathbf{x})$, where F_0 is chosen so that for $t \lesssim 0$, $\boldsymbol{\xi}(t, \mathbf{x}) = \mathrm{e}^{\gamma t}\boldsymbol{\eta}(\mathbf{x})$. Inserting this into the oscillation equation yields

$$\rho_0 \frac{\partial^2 \boldsymbol{\xi}}{\partial t^2} = \rho_0\gamma^2\mathrm{e}^{\gamma t}\boldsymbol{\eta} = \mathrm{e}^{\gamma t}\hat{\mathbf{F}}\boldsymbol{\eta} + F_0\mathrm{e}^{\gamma t}\boldsymbol{\eta} = (-\rho_0\omega^2 + F_0)\mathrm{e}^{\gamma t}\boldsymbol{\eta}. \tag{9.39}$$

We can solve for F_0 to find that

$$F_0 = \rho_0(\omega^2 + \gamma^2). \tag{9.40}$$

The work done by the external force per unit volume is given by the *work integral*

$$W_{\text{ext}} = \int_{-\infty}^{0} \mathrm{d}t \frac{\partial \boldsymbol{\xi}}{\partial t} \cdot \mathbf{F}_{\text{ext}} = \int_{-\infty}^{0} \mathrm{d}t \rho_0\gamma(\gamma^2 + \omega^2)\mathrm{e}^{2\gamma t}\boldsymbol{\eta} \cdot \boldsymbol{\eta} = \rho_0|\boldsymbol{\eta}|^2 \int_{-\infty}^{0} \mathrm{d}t \gamma(\gamma^2 + \omega^2)\mathrm{e}^{2\gamma t} \tag{9.41}$$

where we have used the fact that $\boldsymbol{\eta}$ is time independent. The last integral can be computed analytically, so the previous expression yields

$$W_{\text{ext}} = \rho_0|\boldsymbol{\eta}|^2\left(\frac{\gamma^2}{2} + \frac{\omega^2}{2}\right). \tag{9.42}$$

The energy density of the wave is obtained in the limit in which $\gamma \downarrow 0$:

$$\boxed{E_{\text{wave}} = \frac{\omega^2}{2}\rho_0|\boldsymbol{\eta}|^2.} \tag{9.43}$$

Note that $E_{\text{wave}} = 2K_{\text{wave}}$, meaning that half of the wave energy is potential energy. For example, for a sound wave with $\boldsymbol{\eta} = \mathbf{A}\cos(\mathbf{k} \cdot \mathbf{x})$, the mean energy density (averaging over one wave-length) is

$$E_{\text{wave}} = \frac{\omega^2}{4}\rho_0 A^2. \tag{9.44}$$

It is possible to compute the linear momentum density $\mathbf{P}_{\text{wave}}$ carried by waves in a similar way. One finds

$$\boxed{\mathbf{P}_{\text{wave}} = \frac{\mathbf{k}}{\omega}E_{\text{wave}}.} \tag{9.45}$$

9.3 In-class worksheets

9.3.1 Sound waves

Consider an ideal gas with

$$p = (\Gamma - 1)\rho\varepsilon = \frac{\rho}{\mu m_u} k_B T. \tag{9.46}$$

What is the sound speed c_s as a function of ρ and T? Does it depend on ρ?

Show that

$$\boldsymbol{\xi} = f(z - c_s t)\hat{\mathbf{z}} \tag{9.47}$$

is a solution of

$$\frac{\partial^2 \boldsymbol{\xi}}{\partial t^2} - c_s^2 \nabla\nabla \cdot \boldsymbol{\xi} = 0. \tag{9.48}$$

Sketch the solution for different times if $f(x) = \mathrm{e}^{-x^2/2}$. What is the corresponding density perturbation? Is $f(z + c_s t)\hat{\mathbf{z}}$ also a solution? What does it describe?

9.3.2 The Doppler effect

Consider an initially homogeneous flow with

$$\rho = \rho_0 = \text{const}, \qquad p = p_0 = \text{const}, \qquad \mathbf{v} = \mathbf{v}_0 = \text{const}. \tag{9.49}$$

Introduce a small perturbation in the form of a Lagrangian displacement $\boldsymbol{\xi}$.

1. Show that the Lagrangian density perturbation is given by
$$\delta\rho = -\rho_0 \nabla \cdot \boldsymbol{\xi}. \tag{9.50}$$
2. Use the results from part 1) to show that the oscillation equation reads
$$\left[\frac{\partial}{\partial t} + (\mathbf{v}_0 \cdot \nabla)\right]^2 \boldsymbol{\xi} = c_s^2 \nabla\nabla \cdot \boldsymbol{\xi}. \tag{9.51}$$
3. Introduce the ansatz $\boldsymbol{\xi} = \boldsymbol{\eta} \mathrm{e}^{\mathrm{i}(\mathbf{k}\cdot\mathbf{x} - \omega t)}$ and show that the dispersion relation is
$$\omega_\pm = \mathbf{v}_0 \cdot \mathbf{k} \pm c_s k. \tag{9.52}$$
What is the physical meaning of the $\pm$ solutions?
4. Consider a sound wave propagating at an angle θ with respect to $\mathbf{v}_0$. Let $\omega_0 = c_s k$ be the frequency of sound in the fluid frame. Show that the frequency of sound in the lab frame is
$$\omega = \omega_0 \left(1 + \frac{v_0}{c_s} \cos\theta\right), \tag{9.53}$$
where $v_0 = |\mathbf{v}_0|$.

9.4 Exercises

Exercise 9.1 (Helioseismology). The surface of the Sun oscillates with periods of about 5 minutes. These oscillations are due to sound waves trapped inside the Sun. In this exercise, we estimate their frequencies with a very simple model. Assume $\Gamma = 5/3$.

1. Estimate the sound speed at the center of the Sun, where $T \approx 1.6 \times 10^7$ K and $\mu \approx 0.85$, and at the base of the convection zone, at $r \approx 0.7\, R_\odot$, where $T \approx 2 \times 10^6$ K and $\mu \approx 0.6$.
2. Estimate the time t_s it takes for sound to travel from the center to the surface of the Sun. For $r < 0.7\, R_\odot$, use the average of the two sound speeds you found in part 1. For $r > 0.7\, R_\odot$, assume that the temperature decreases linearly to zero at the surface, and that μ is constant.
3. Model the Sun as a pipe of length $R_\odot$, closed at both ends. What is the frequency of the fundamental mode? What is the order n of a mode with a period of 5 minutes?
4. The observed frequencies of consecutive modes of this type differ by about 135 μHz. Compare with your estimate. What do these observations measure?

9.5 Further reading

- C. Clarke and B. Carswell [1], Sections 6.1 and 6.3.

10. Waves in Stratified Atmospheres

10.1 Waves in Isothermal Atmospheres

Consider a stratified, static, isothermal atmosphere. We assume constant acceleration g in the negative z direction. The hydrostatic equilibrium equations are (Chapter 4)

$$\partial_z p_0 = -\rho_0 g, \qquad p_0 = c_{\text{iso}}^2 \rho_0, \tag{10.1}$$

where $c_{\text{iso}} = \sqrt{k_B T/(\mu m_u)}$ is the isothermal sound speed (Equation (9.16)).

Combining these two we find

$$\partial_z \rho_0 = -\frac{g}{c_{\text{iso}}^2}\rho_0 \Rightarrow \rho_0 = \tilde{\rho} \mathrm{e}^{-z/H}. \tag{10.2}$$

$H = c_{\text{iso}}^2/g$ is called the scale height of the atmosphere and has units of a length.

We want to study the propagation of waves in this atmosphere. The continuity equation still gives (Equation (9.5))

$$\delta\rho = -\rho_0 \nabla \cdot \boldsymbol{\xi}, \tag{10.3}$$

and the velocity perturbation is

$$\mathbf{v}' = \delta\mathbf{v} - (\boldsymbol{\xi}\cdot\nabla)\mathbf{v}_0 = \delta\mathbf{v} = \frac{\mathrm{d}\boldsymbol{\xi}}{\mathrm{d}t} = \frac{\partial\boldsymbol{\xi}}{\partial t} + (\mathbf{v}_0\cdot\nabla)\boldsymbol{\xi} = \frac{\partial\boldsymbol{\xi}}{\partial t}. \tag{10.4}$$

The sound speed is given by

$$c_s^2 = \left(\frac{\partial p}{\partial \rho}\right)_s = \Gamma c_{\text{iso}}^2, \tag{10.5}$$

which implies that $\delta p = c_s^2 \delta\rho$ for adiabatic perturbations. However, $p' \neq c_s^2 \rho'$, because the Eulerian and Lagrangian perturbations differ when $\nabla p_0 \neq 0$ (Equation (9.3)). Instead we have

$$\delta p = c_s^2 \delta\rho \Rightarrow (p' + \boldsymbol{\xi}\cdot\nabla p_0) = c_s^2\delta\rho \Rightarrow p' = c_s^2\delta\rho + g\rho_0\xi_z. \tag{10.6}$$

The momentum equation reads

$$\rho\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = -\nabla p - g\rho\hat{\mathbf{z}}. \tag{10.7}$$

Setting $\mathbf{v} = \mathbf{v}'$, $p = p_0 + p'$, and $\rho = \rho_0 + \rho'$ yields

$$\rho_0\frac{\partial^2\boldsymbol{\xi}}{\partial t^2} = -\nabla p_0 - g\rho_0\hat{\mathbf{z}} - \nabla p' - g\rho'\hat{\mathbf{z}} + \ldots, \tag{10.8}$$

where the dots denote terms of second order in the perturbation. The first two terms on the RHS cancel out. Dropping the higher order terms we find

$$\rho_0\frac{\partial^2\boldsymbol{\xi}}{\partial t^2} = -c_s^2\nabla\delta\rho + g\frac{\rho_0}{H}\xi_z\hat{\mathbf{z}} - g\rho_0\nabla\xi_z - g\left(\delta\rho + \frac{\rho_0}{H}\xi_z\right)\hat{\mathbf{z}}, \tag{10.9}$$

where we have used Equation (10.6) and

$$\rho' = \delta\rho - (\boldsymbol{\xi}\cdot\nabla)\rho_0 = \delta\rho + \frac{\rho_0}{H}\xi_z. \tag{10.10}$$

The second and last term on the RHS of Equation (10.9) cancel out leaving

$$\rho_0 \frac{\partial^2 \boldsymbol{\xi}}{\partial t^2} = -c_s^2 \nabla \delta\rho - g\rho_0 \nabla \xi_z - g\delta\rho \hat{\mathbf{z}}. \tag{10.11}$$

Using the fact that

$$\begin{aligned} \nabla \delta\rho &= \nabla(-\rho_0 \nabla \cdot \boldsymbol{\xi}) \\ &= -\rho_0 \nabla\nabla \cdot \boldsymbol{\xi} - \nabla\rho_0 \nabla \cdot \boldsymbol{\xi} \\ &= -\rho_0 \nabla\nabla \cdot \boldsymbol{\xi} + \frac{\rho_0}{H} \nabla \cdot \boldsymbol{\xi} \hat{\mathbf{z}}, \end{aligned} \tag{10.12}$$

we obtain

$$\begin{aligned} \rho_0 \frac{\partial^2 \boldsymbol{\xi}}{\partial t^2} = c_s^2 \rho_0 \nabla\nabla \cdot \boldsymbol{\xi} - \rho_0 \Gamma g \nabla \cdot \boldsymbol{\xi} \hat{\mathbf{z}} \\ -g\rho_0 \nabla \xi_z + g\rho_0 \nabla \cdot \boldsymbol{\xi} \hat{\mathbf{z}}. \end{aligned} \tag{10.13}$$

This can be simplified to

$$\boxed{\begin{aligned} \frac{\partial^2 \boldsymbol{\xi}}{\partial t^2} = c_s^2 \left(\nabla\nabla \cdot \boldsymbol{\xi} - \frac{c_{\rm iso}^2}{c_s^2} \frac{1}{H} \nabla \xi_z \right) \\ - \frac{(c_s^2 - c_{\rm iso}^2)}{H} \nabla \cdot \boldsymbol{\xi} \hat{\mathbf{z}}. \end{aligned}} \tag{10.14}$$

We are going to analyze the consequences of this equation shortly. Before we do that, we show that this equation reduces to Eq. (6.31) of C. Clarke and B. Carswell [1] in the limit of vertical oscillations in the z direction. To do that, we take the divergence of Equation (10.14) and use $\nabla \cdot \boldsymbol{\xi} = -\delta\rho/\rho_0$ to obtain

$$\begin{aligned} \frac{\partial^2 \delta\rho}{\partial t^2} &= c_s^2 \left(\nabla^2 \delta\rho + \frac{2}{H} \partial_z \delta\rho + \frac{\delta\rho}{H^2} \right) \\ &\quad + g\rho_0 \nabla^2 \xi_z \\ &\quad - \frac{(c_s^2 - c_{\rm iso}^2)}{H} \left(\partial_z \delta\rho + \frac{\delta\rho}{H} \right) \\ &= c_s^2 \nabla^2 \delta\rho + \frac{c_s^2 + c_{\rm iso}^2}{H} \partial_z \delta\rho \\ &\quad + \frac{c_{\rm iso}^2}{H} \rho_0 \nabla^2 \xi_z + c_{\rm iso}^2 \frac{\delta\rho}{H^2}. \end{aligned} \tag{10.15}$$

Assuming that $\delta\rho = \delta\rho(t, z)$ and $\boldsymbol{\xi} = \xi_z \hat{\mathbf{z}}$, we obtain,

$$\frac{\partial^2 \delta\rho}{\partial t^2} - c_s^2 \frac{\partial^2 \delta\rho}{\partial z^2} - \frac{c_s^2}{H} \frac{\partial \delta\rho}{\partial z} = 0, \tag{10.16}$$

where we have used the fact that

$$\nabla^2 \xi_z = \partial_z^2 \xi_z = \partial_z \left(-\frac{\delta\rho}{\rho_0} \right) = -\frac{\partial_z \delta\rho}{\rho_0} - \frac{\delta\rho}{\rho_0 H}. \tag{10.17}$$

Equation (10.16) is identical to Eq. (6.31) of C. Clarke and B. Carswell [1]. It describes the propagation of density modes in a stratified atmosphere, but neglects an entirely different class of modes, the gravity waves, in which the fluid elements move up and down, while the wave propagates mostly in the horizontal direction. We discuss them in Section 10.2.

As before, we look for traveling wave solutions

$$\boldsymbol{\xi} = \boldsymbol{\eta} e^{\mathrm{i}(\mathbf{k}\cdot\mathbf{x} - \omega t)}, \tag{10.18}$$

where now we allow the components of $\mathbf{k}$ to be complex. Inserting this into Equation (10.14) we obtain

$$\boxed{\omega^2\boldsymbol{\eta} = c_s^2\left(\mathbf{k}\cdot\boldsymbol{\eta} + \mathrm{i}\frac{c_{\mathrm{iso}}^2}{c_s^2}\frac{1}{H}\eta_z\right)\mathbf{k} + \mathrm{i}\frac{(c_s^2 - c_{\mathrm{iso}}^2)}{H}\mathbf{k}\cdot\boldsymbol{\eta}\hat{\mathbf{z}}.} \tag{10.19}$$

- In the limit $kH \gg 1$, that is, when the wavelength is much smaller than the scale height of the atmosphere, the terms proportional to $1/H$ can be neglected and we recover the dispersion relation for waves in a homogeneous medium, namely $\omega^2 = c_s^2 k^2$, as expected.
- Purely horizontal waves, with $\boldsymbol{\eta}, \mathbf{k} \perp \hat{\mathbf{z}}$, cannot satisfy the dispersion relation, unless $c_s = c_{\mathrm{iso}}$, that is for isothermal as opposed to adiabatic perturbations. Physically this means that density fluctuations become buoyant, so they acquire a $k_z \neq 0$ component in their wave vector.
- For waves with $\boldsymbol{\eta} \parallel \mathbf{k} \parallel \hat{\mathbf{z}}$, the dispersion relation simplifies to

$$\omega^2 = c_s^2\left(k^2 + \mathrm{i}\frac{1}{H}k\right) \Rightarrow k_{1,2} = -\frac{\mathrm{i}}{2H} \pm \sqrt{\left(\frac{\omega}{c_s}\right)^2 - \left(\frac{1}{2H}\right)^2}. \tag{10.20}$$

  These are the so-called pressure or p-modes of the atmosphere. They are the generalization of sound waves. Note that k has a real part only if ω is larger than the **acoustic cutoff frequency**

$$\omega_{\mathrm{ac}} = \frac{c_s}{2H}. \tag{10.21}$$

  – If $\omega \gg \omega_{\mathrm{ac}}$, then the dispersion relation for p-modes can be approximated as:

$$k_{1,2} \approx -\frac{\mathrm{i}}{2H} \pm \frac{\omega}{c_s}. \tag{10.22}$$

  In this case the displacement grows exponentially with height:

$$\xi_z = \eta_z \mathrm{e}^{\mathrm{i}(\mathbf{k}\cdot\mathbf{x} - \omega t)} = \eta_z \exp\left[\mathrm{i}\omega\left(\pm\frac{z}{c_s} - t\right)\right]\exp\left(\frac{z}{2H}\right). \tag{10.23}$$

  This makes sense, because the energy flux carried by the wave, which is the energy density (Equation (9.43)) times c_s, $\sim \rho_0\omega^2|\boldsymbol{\xi}|^2 c_s$, must be the same at all heights. Since ρ_0 decreases with height, $|\boldsymbol{\xi}|$ must increase as $\rho_0^{-1/2} \propto \mathrm{e}^{z/2H}$. The density perturbation of the wave propagating upward is

$$\delta\rho = -\rho_0\nabla\cdot\boldsymbol{\xi} = -\tilde{\rho}\eta_z\left(\frac{\mathrm{i}\omega}{c_s} + \frac{1}{2H}\right)\exp\left[\mathrm{i}\omega\left(\frac{z}{c_s} - t\right)\right]\exp\left(-\frac{z}{2H}\right). \tag{10.24}$$

  Note that the density perturbation itself decreases with height. However, the background density decreases faster, so the relative perturbation $\delta\rho/\rho_0 \propto \mathrm{e}^{z/2H}$ grows. This means that waves propagating upward through an isothermal atmosphere eventually become nonlinear, $\delta\rho \sim \rho_0$, and break, forming shocks (Chapter 11). Waves breaking in this way are thought to contribute to the heating of the solar chromosphere.

  – If $\omega < \omega_{\mathrm{ac}}$, then k is purely imaginary and the waves do not propagate. In the limit $\omega \ll \omega_{\mathrm{ac}}$, we find

$$k_{1,2} \approx 0, -\frac{\mathrm{i}}{H} \qquad \Rightarrow \qquad [\xi_z]_{1,2} = \eta_z\mathrm{e}^{-\mathrm{i}\omega t}, \eta_z\mathrm{e}^{z/H}\mathrm{e}^{-\mathrm{i}\omega t}. \tag{10.25}$$

  In other words, the atmosphere responds to the perturbation as a whole, rather than with a wave propagating upward.

- Let us now consider the case of perturbations with horizontal displacement. We can align the x-axis with $\boldsymbol{\eta}$, meaning that $\boldsymbol{\eta} = \eta\hat{\mathbf{x}}$. In this case, the dispersion relation becomes

$$\omega^2\eta\hat{\mathbf{x}} = c_s^2(k_x\eta)\mathbf{k} + \mathrm{i}\frac{(c_s^2 - c_{\mathrm{iso}}^2)}{H}k_x\eta\hat{\mathbf{z}}. \tag{10.26}$$

This means that there are two components of $\mathbf{k} = (k_x, 0, k_z)$. The x-component of Equation (10.26) is

$$\omega^2 = c_s^2 k_x^2 \Rightarrow k_x = \pm\frac{\omega}{c_s}. \tag{10.27}$$

The z-component of Equation (10.26) is

$$k_z = -\mathrm{i}\left(1 - \frac{c_{\mathrm{iso}}^2}{c_s^2}\right)\frac{1}{H}. \tag{10.28}$$

The corresponding displacement is

$$\boldsymbol{\xi}(t, \mathbf{x}) = \eta \exp\left[\mathrm{i}\omega\left(\pm\frac{x}{c_s} - t\right)\right] \exp\left[\left(1 - \frac{c_{\mathrm{iso}}^2}{c_s^2}\right)\frac{z}{H}\right]\hat{\mathbf{x}}. \tag{10.29}$$

These are called **Lamb waves**. They are sound waves that propagate horizontally and are trapped near the bottom of the atmosphere. Indeed, their energy density $\rho_0|\boldsymbol{\xi}|^2 \propto \mathrm{e}^{(1-2/\Gamma)z/H}$ decreases with height for $\Gamma < 2$. Lamb waves generated by the eruption of the Hunga Tonga volcano in 2022 were observed to travel around the Earth several times. In the limit in which $c_s \to c_{\mathrm{iso}}$, that is $\Gamma \to 1$ (isothermal perturbations), these waves reduce to horizontal sound waves.

Example

Consider the atmosphere of the Earth. It is not isothermal, but we can estimate its properties taking $T \approx 288\,\mathrm{K}$, $\mu \approx 29$, and $\Gamma = 7/5$. We find $H = k_B T/(\mu m_u g) \approx 8\,\mathrm{km}$, $c_s \approx 340\,\mathrm{m\,s^{-1}}$, and an acoustic cutoff period $2\pi/\omega_{\mathrm{ac}} \approx 5\,\mathrm{min}$.

For the solar photosphere, $T \approx 5800\,\mathrm{K}$, $\mu \approx 1.3$, $g \approx 2.7 \times 10^2\,\mathrm{m\,s^{-2}}$, and $\Gamma = 5/3$. We find $H \approx 140\,\mathrm{km}$, $c_s \approx 8\,\mathrm{km\,s^{-1}}$, and $\omega_{\mathrm{ac}}/2\pi \approx 4.6\,\mathrm{mHz}$, corresponding to a period of about 3.6 minutes. The measured value is about 5.3 mHz. The solar oscillations with periods of about 5 minutes have lower frequencies, about 3.3 mHz, so they cannot propagate into the solar atmosphere. They are reflected at the surface and remain trapped inside the Sun. This is why the Sun rings like a pipe (Exercise 9.1).

10.2 Gravity Waves

To understand the second class of waves, consider a fluid element that is displaced upward by a small distance ξ_z. We assume that the displacement is slow enough for the fluid element to stay in pressure equilibrium with its surroundings, and that it is adiabatic. The pressure of the fluid element changes by $\delta p = \xi_z \partial_z p_0 = -\rho_0 g \xi_z$, so its density changes by

$$\delta\rho = \frac{\delta p}{c_s^2} = -\frac{g}{c_s^2}\rho_0\xi_z. \tag{10.30}$$

The density of the surrounding gas at the new position is $\rho_0 + \xi_z\partial_z\rho_0 = \rho_0 - (g/c_{\mathrm{iso}}^2)\rho_0\xi_z$. Since $c_s > c_{\mathrm{iso}}$, the fluid element is denser than its surroundings, by

$$\Delta\rho = \delta\rho - \xi_z\partial_z\rho_0 = g\left(\frac{1}{c_{\mathrm{iso}}^2} - \frac{1}{c_s^2}\right)\rho_0\xi_z. \tag{10.31}$$

So gravity pulls it back down. The equation of motion of the fluid element is

$$\rho_0\frac{\mathrm{d}^2\xi_z}{\mathrm{d}t^2} = -g\Delta\rho = -\rho_0 N^2\xi_z, \qquad N^2 = g^2\left(\frac{1}{c_{\mathrm{iso}}^2} - \frac{1}{c_s^2}\right) = \frac{(\Gamma-1)g^2}{c_s^2}. \tag{10.32}$$

In other words, the fluid element oscillates up and down with frequency N, which is called the **buoyancy frequency** or **Brunt–Väisälä frequency**. Note that $N = 0$ if $\Gamma = 1$. This makes sense, because, if the perturbations are isothermal, a displaced fluid element always has the same density as its surroundings. If the atmosphere were stratified in such a way that $N^2 < 0$, the displaced fluid element would be lighter than its surroundings and would keep rising. This is the onset of convection, which we will discuss in Chapter 15.

To find the waves associated with this oscillation, we go back to Equation (10.19) and consider a general wave vector. We can always align the x-axis with the horizontal component of $\mathbf{k}$, so that $\mathbf{k} = (k_x, 0, k_z)$. Equation (10.19) is a linear system of equations for $\boldsymbol{\eta}$, which has non-trivial solutions only if its determinant vanishes. To simplify the result, it is useful to write $k_z = K - \mathrm{i}/(2H)$, with K real. This takes into account the growth of the amplitude with height, $\mathrm{e}^{z/2H}$, that we found for the p-modes. After some algebra, we find the dispersion relation

$$\boxed{\omega^4 - \omega^2\left[c_s^2(k_x^2 + K^2) + \omega_{\mathrm{ac}}^2\right] + N^2 c_s^2 k_x^2 = 0.} \tag{10.33}$$

This is a quadratic equation for ω^2, so for each wave vector there are two solutions:

- The larger solution describes the acoustic waves, or p-modes. One can show that these have $\omega > \omega_{\mathrm{ac}}$. For $c_s k \gg \omega_{\mathrm{ac}}$ they reduce to sound waves, $\omega^2 \approx c_s^2(k_x^2 + K^2)$.
- The smaller solution describes the **gravity waves**, or g-modes, in which the restoring force is buoyancy. These have $\omega < N$. For $c_s k \gg \omega_{\mathrm{ac}}$ we find

$$\omega^2 \approx N^2 \frac{k_x^2}{k_x^2 + K^2}. \tag{10.34}$$

  Interestingly, the frequency of gravity waves depends only on the direction of $\mathbf{k}$, not on the wavelength. Gravity waves with $K = 0$, propagating horizontally, have the largest frequency, $\omega \approx N$. In these waves, the fluid elements move up and down, like the fluid element we considered above.

For an isothermal atmosphere, $\omega_{\mathrm{ac}}^2 - N^2 = (g/c_s)^2(\Gamma/2 - 1)^2 \geq 0$, so $N \leq \omega_{\mathrm{ac}}$. Waves with frequencies between N and ω_{ac} cannot propagate vertically. The two branches are shown in Figure 10.1. The Lamb waves we found before lie between the two branches, because they do not propagate vertically.

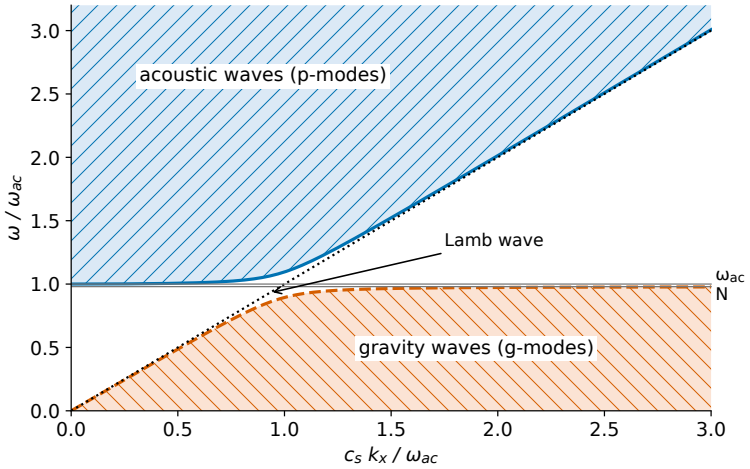


Figure 10.1: Dispersion relation for waves in an isothermal atmosphere with $\Gamma = 5/3$, for vertical wave number $K = 0$. Acoustic waves (p-modes) can propagate in the region above the upper curve and gravity waves (g-modes) in the region below the lower curve. Waves with $K \neq 0$ move further into these regions. The dotted line is the Lamb wave, $\omega = c_s k_x$.

10.3 In-class worksheet

We consider small disturbances in a plane parallel, isothermal atmosphere. We will assume z to be the vertical direction and g to be the gravitational acceleration, assumed to be constant and towards the negative z direction. Let $c_{\mathrm{iso}} = \sqrt{k_B T/(\mu m_u)}$ be the isothermal sound speed.

1. Show that the static background density ρ_0 satisfies

$$\rho_0(z) = \tilde{\rho} \mathrm{e}^{-z/H}, \tag{10.35}$$

where $H = c_{\mathrm{iso}}^2/g$ is the scale height of the atmosphere.

2. Let $\delta\rho$ and ρ' be the Lagrangian and Eulerian perturbations of the density. Show that

$$\rho' = \delta\rho + \frac{\rho_0}{H}\xi_z. \tag{10.36}$$

3. Let p' be the Eulerian pressure perturbation. Show that

$$p' = c_s^2 \delta\rho + g\rho_0 \xi_z, \tag{10.37}$$

where $c_s^2 = \Gamma c_{\mathrm{iso}}^2$ is the square of the sound speed and ξ_z is the z-component of the displacement.

4. Use the results from the previous two points to show that

$$\rho_0 \frac{\partial^2 \boldsymbol{\xi}}{\partial t^2} = -c_s^2 \nabla\delta\rho - g\rho_0 \nabla\xi_z - g\delta\rho\hat{\mathbf{z}}. \tag{10.38}$$

5. If time allows: consider vertical waves, $\boldsymbol{\xi} = \xi_z(t,z)\hat{\mathbf{z}}$, and look for solutions $\xi_z \propto \mathrm{e}^{\mathrm{i}(kz-\omega t)}$. Show that

$$\omega^2 = c_s^2\left(k^2 + \frac{\mathrm{i}k}{H}\right), \tag{10.39}$$

and that the waves can propagate, that is, k has a real part, only if $\omega > c_s/(2H)$.

10.4 Exercises

Exercise 10.1 (Waves in a polytropic atmosphere). An isentropic atmosphere sits in a uniform gravitational field g, with pressure $p_0 = K\rho_0^{1+1/n}$ and K constant. Let x be the depth below the top of the atmosphere, and suppose that the unperturbed density is $\rho_0 = (bx)^n$ with b constant.

1. Show that hydrostatic equilibrium requires $g = Kb(1+n)$.
2. Perturb the atmosphere with a small vertical velocity v, taken positive upward, together with adiabatic perturbations $p' \ll p_0$ of the pressure and $\rho' \ll \rho_0$ of the density. Show that $p' = (gx/n)\rho'$.
3. Show that the linearized continuity and momentum equations are

$$\frac{\partial \rho'}{\partial t} - v\frac{\partial \rho_0}{\partial x} - \rho_0 \frac{\partial v}{\partial x} = 0 \tag{10.40}$$

and

$$\frac{\partial v}{\partial t} + \left(\frac{\rho'}{\rho_0}\right)g - \frac{g}{n\rho_0}\frac{\partial(x\rho')}{\partial x} = 0. \tag{10.41}$$

4. For $n = 1/2$, combine these equations to show that

$$\frac{\partial^2 v}{\partial t^2} = 2gx\frac{\partial^2 v}{\partial x^2} + 3g\frac{\partial v}{\partial x}. \tag{10.42}$$

5. Look for solutions $v(t,x) = v_1(x)\mathrm{e}^{-\mathrm{i}\omega t}$ and change variable to $\tau = (2x/g)^{1/2}$. Show that Equation (10.42) becomes

$$\frac{1}{\tau}\frac{\mathrm{d}^2}{\mathrm{d}\tau^2}(\tau v_1) + \omega^2 v_1 = 0. \tag{10.43}$$

Find the solution that stays finite at $x = 0$ and interpret it physically.

Adapted from C. Clarke and B. Carswell [1], Problem 22.

10.5 Further reading

- C. Clarke and B. Carswell [1], Section 6.2.

11. Shock Waves

> **Note**
>
> This chapter is in preparation.

11.1 In-class worksheet

11.1.1 Shock waves I

1. Consider a shock wave in an ideal gas with adiabatic index Γ. Let ρ_1, p_1, and T_1 be the upstream (unshocked) gas density, pressure, and temperature. Let ρ_2, p_2, and T_2 be the corresponding downstream (post-shock) quantities.

   a. Show that the ratio of downstream to upstream temperature across the shock is

   $$\frac{T_2}{T_1} = \frac{p_2}{p_1}\frac{(\Gamma+1)p_1 + (\Gamma-1)p_2}{(\Gamma-1)p_1 + (\Gamma+1)p_2}. \tag{11.1}$$

   b. If $T_2 = 4T_1$, what are p_2 and ρ_2 as a function of ρ_1, p_1, and T_1?

11.1.2 Shock waves II

1. The Bullet Cluster (1E 0657-56) consists of two colliding clusters of galaxies. The name comes from the characteristic shape of a shock formed by the interaction of the gas from the two clusters, which is visible in X-ray. The image in Figure 11.1 has been obtained with the Chandra X-ray space telescope.

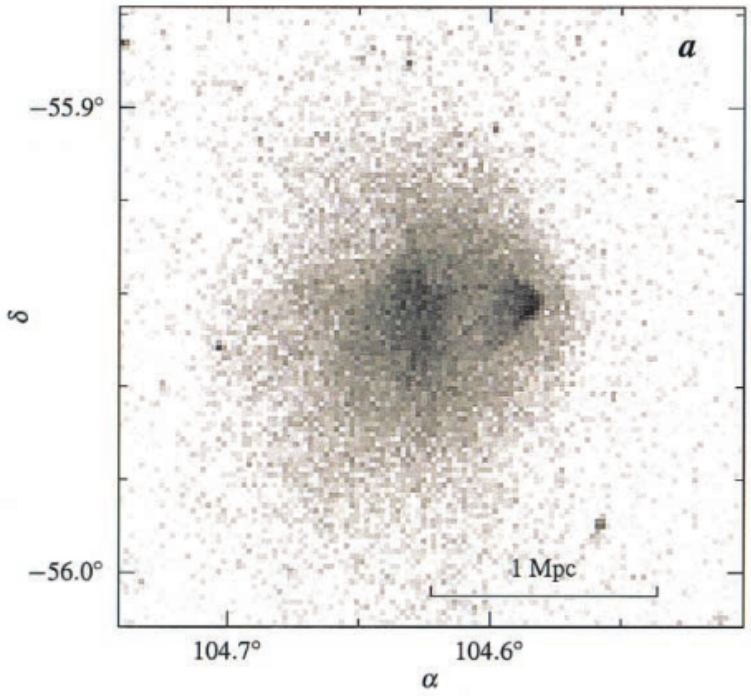


Figure 11.1: Chandra/ACIS 0.5–5 keV image of the Bullet Cluster. From M. Markevitch *et al.* [5]; © The American Astronomical Society, reproduced for non-commercial educational use and not covered by the license of these notes.

This data can be fitted to infer the following parameters for the upstream and downstream electron density and temperature:

$$\begin{aligned} n_1 &= 3.7 \times 10^{-3}\,\mathrm{cm}^{-3}, T_1 = 9.3 \times 10^7\,\mathrm{K}, \\ n_2 &= 9.3 \times 10^{-3}\,\mathrm{cm}^{-3}, T_2 = 2.2 \times 10^8\,\mathrm{K}. \end{aligned} \tag{11.2}$$

Estimate

a. the Mach number of the shock (defined as the Mach number of the upstream flow);

b. the relative velocity of approach of the two clusters of galaxies.

11.2 Exercises

Exercise 11.1 (Piston-driven shock). A piston drives a strong shock wave through a cold gas that is initially at rest; similar considerations are used to describe young supernova remnants. The piston moves at speed v_p and has a mass per unit surface area Σ_p. The gas follows a gamma-law equation of state

$$p = (\Gamma - 1)\rho\varepsilon. \tag{11.3}$$

1. Express the shock speed s, and the downstream density ρ_2 and pressure p_2, in terms of the upstream density ρ_1. Keep Γ general (do not set $\Gamma = 5/3$) and do not use the expression for s given by Kulsrud, which contains a typo.
2. Calculate the work that the piston does on the gas and estimate when the piston begins to decelerate. Compare the mass of the piston with the mass of the gas that the shock has swept up by that time. For this estimate, take the piston velocity to stay constant until the piston has lost a significant fraction of its kinetic energy.

Adapted from R. M. Kulsrud [4], Problem 4.6.

11.3 Further reading

- C. Clarke and B. Carswell [1], Chapter 7 (supersonic flows).
- L. D. Landau and E. M. Lifshitz [6], Sections 84–89 (shock waves).

12. Radiative Shocks

> **Note**
>
> This chapter is in preparation.

12.1 Exercises

Exercise 12.1 (Colliding clouds). Two identical clouds, each of radius 3×10^{16} m (about 1 pc) and temperature 10 K, collide with each other at a relative velocity of $4\,\mathrm{km\,s^{-1}}$.

1. *Estimate* the time t_{coll} that each cloud needs to run entirely into the shock.
2. The shocked gas cools at a rate $\dot{Q} = 10^{-7}\,\mathrm{W\,kg^{-1}}$. Decide whether the shock is approximately adiabatic or isothermal.
3. Suppose the collision produces an isothermal shock. How thick is the shocked layer, x, at the moment when the whole of each cloud has been shocked?
4. At later times the layer settles into a structure that can be approximated by a hydrostatic isothermal slab with column density $10^{-2}\,\mathrm{g\,cm^{-2}} = 0.1\,\mathrm{kg\,m^{-2}}$. In this hydrostatic structure, what fraction of the clouds' mass lies within the thickness x? Neglect edge effects and variations of the column density across the plane of the slab. [1, Section 5.2.]

Adapted from C. Clarke and B. Carswell [1], Problem 24.

12.2 Further reading

- C. Clarke and B. Carswell [1], Chapter 7 (supersonic flows).

13. Blast Waves

> **Note**
>
> This chapter is in preparation.

13.1 In-class worksheet

1. **Atmospheric explosion.** These are experimental measurements of the expansion of a blast wave due to an atmospheric explosion.

t (ms)	0.09	0.24	0.38	0.52	0.66	0.80	0.94
R(m)	24	42	51	60	66	70	77

This data is plotted for your convenience in Figure 13.1 (log-log scale). The data file `fireball.dat` and the plotting script `fireball.py` are available in the repository.

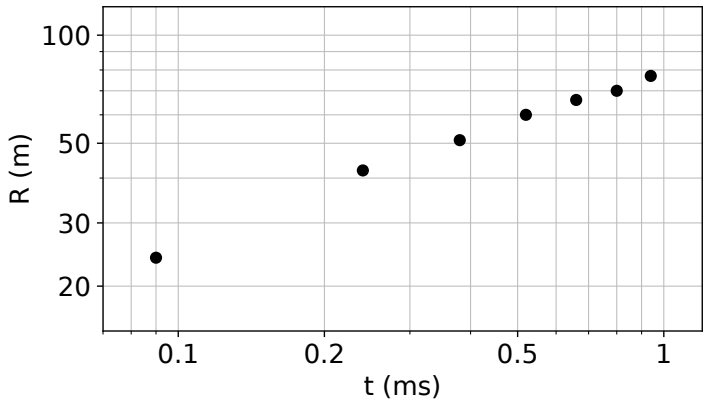


Figure 13.1: Radius of the blast wave as a function of time for the atmospheric explosion (log-log scale).

a. Argue that the radius of the shock should scale as $R = \xi_0 \rho_1^{\alpha} E^{\beta} t^{\gamma}$, where ρ_1 is the atmospheric density, E is the explosion energy, t is the time, and ξ_0 is a dimensionless parameter of order one. Is this ansatz supported by the data? Compute α, β, γ.

b. Show that the shock expansion speed is

$$U_{\rm sh} = \frac{2}{5}\frac{r_{\rm sh}}{t}. \tag{13.1}$$

c. *Estimate* the explosion energy E. You can assume $\rho_1 = 1.25 \times 10^{-3}\,\mathrm{g\,cm^{-3}} = 1.25\,\mathrm{kg\,m^{-3}}$ for the density of the atmosphere.

d. Estimate the temperature of the fireball 1 ms after the explosion. Use $\Gamma = \frac{7}{5}$ and $\mu \approx 29$ for air. Is this estimate an upper limit or a lower limit?

13.2 Exercises

Exercise 13.1 (Ionized cavities around star clusters). A cluster of ionizing stars with total luminosity L clears out a cavity in the interstellar medium, which has an undisturbed density ρ_0.

1. Use the similarity-solution method to find how the cavity radius r grows with time t. [Hint. Take $r \propto L^\alpha \rho_0^\beta t^\gamma$ and determine α, β, γ.]
2. Such bubbles stall once their expansion speed becomes comparable to the sound speed of the interstellar medium. Show that the area covered by a stalled bubble is proportional to L. Use this to discuss how the porosity of the interstellar medium (the fraction of the galaxy's area, seen from above the disk, that is covered by stalled bubbles) depends on the way in which ionizing stars of a given total luminosity are grouped into clusters.
3. Suppose the galactic disk can be approximated by a uniform-density gas slab with a sharp edge at height z. Discuss how clusters with small L and clusters with large L behave differently.

Adapted from C. Clarke and B. Carswell [1], Problem 28.

13.3 Further reading

- C. Clarke and B. Carswell [1], Chapter 8 (blast waves).

14. Stellar Winds and Spherical Accretion

> **Note**
>
> This chapter is in preparation.

14.1 In-class worksheet

1. The radial velocity v for an isothermal wind with isothermal sound speed $c_{\rm iso}$ satisfies the equation

$$(v^2 - c_{\rm iso}^2)\frac{1}{v}\frac{\mathrm{d}v}{\mathrm{d}r} = \frac{2c_{\rm iso}^2}{r} - \frac{GM}{r^2}. \tag{14.1}$$

   Find the acceleration of the wind at the sonic point

$$r_c = \frac{GM}{2c_{\rm iso}^2}. \tag{14.2}$$

14.2 Exercises

Exercise 14.1 (Radiation-driven winds). Consider an isothermal steady stellar wind. Radiation pressure in optically thin conditions produces an additional force per unit mass that can be parameterized as

$$f = \frac{A}{r^2}, \tag{14.3}$$

where $A > 0$.

1. Show that the momentum equation becomes

$$(v^2 - c_{\rm iso}^2)\frac{1}{v}\frac{\mathrm{d}v}{\mathrm{d}r} = \frac{2c_{\rm iso}^2}{r} - \frac{GM}{r^2} + \frac{A}{r^2}, \tag{14.4}$$

   where $c_{\rm iso}$ is the isothermal sound speed.
2. Assuming $A < GM$, how does the structure of the wind change?
3. How does the radial profile of the velocity change if $A > GM$? Give a physical interpretation of the results.

Exercise 14.2 (Isothermal Bondi accretion). Far from a star of mass M located at the origin, isothermal gas has pressure $p_\infty = c_{\rm iso}^2 \rho_\infty$ and density ρ_∞. This gas is steadily accreted onto the star.

1. Find the accretion rate, assuming that the gas remains isothermal. At what radius does the infalling gas reach the sound speed?
2. Let $c_{\rm iso} = 1\,\mathrm{km/s}$ and $n_\infty = 10^3$ hydrogen atoms per cm^3, and take the mass of the hydrogen atom to be 1.66×10^{-27} kg. Express this radius in units of the solar radius $R_\odot = 7 \times 10^8$ m.
3. Calculate the time that a star of mass M needs to double its mass.

Adapted from C. Clarke and B. Carswell [1], Problem 30.

14.3 Further reading

- C. Clarke and B. Carswell [1], Chapter 9 (Bernoulli's equation).

III Instabilities, Viscosity, Disks, and Turbulence

15 Fluid Instabilities I: Convection . . . 107
15.1 In-class worksheet . . . 107
15.2 Exercises . . . 107
15.3 Further reading . . . 107

16 Fluid Instabilities II: Rayleigh–Taylor and Kelvin–Helmholtz . 109
16.1 In-class worksheet . . . 109
16.2 Exercises . . . 110
16.3 Further reading . . . 110

17 Fluid Instabilities III: Thermal and Gravitational Instabilities . 111
17.1 Exercises . . . 111
17.2 Further reading . . . 111

18 Diffusion and Viscous Flows . . . 113
18.1 In-class worksheet . . . 113
18.2 Exercises . . . 113
18.3 Further reading . . . 114

19 Accretion Disks I: Structure and Viscous Evolution . . . 115
19.1 In-class worksheet . . . 115
19.2 Exercises . . . 115
19.3 Further reading . . . 117

20 Accretion Disks II: Steady Disks and Spectra . . . 119
20.1 In-class worksheet . . . 119
20.2 Exercises . . . 119
20.3 Further reading . . . 119

21 Turbulence . . . 121
21.1 What is Turbulence? . . . 121
21.2 Kolmogorov's Theory . . . 121
21.3 Shredding of Clouds in a Turbulent Flow . . . 126
21.4 Turbulence Modeling: Reynolds Stress . . . 127
21.5 In-class worksheet . . . 129
21.6 Exercises . . . 129
21.7 Further reading . . . 130

15. Fluid Instabilities I: Convection

> **Note**
>
> This chapter is in preparation.

15.1 In-class worksheet

1. Is an isothermal atmosphere stable against convection? Why?

15.2 Exercises

Exercise 15.1 (Vertical structure and convective stability of a thin disk). A thin disk of gas orbits an object of mass M. In the vertical (z) direction, the disk is held up by a balance between the pressure gradient and the z-component of the gravitational force exerted by the central object.

1. Write down the equation of hydrostatic equilibrium in the z-direction.
2. Suppose that the density profile is $\rho(z) \propto (z_m^2 - z^2)^2$. Determine the corresponding pressure profile and the polytropic index n of the disk.
3. Is the disk stable against convection if it consists of ionized hydrogen?

Adapted from C. Clarke and B. Carswell [1], Problem 40.

15.3 Further reading

- C. Clarke and B. Carswell [1], Chapter 10 (fluid instabilities).

16. Fluid Instabilities II: Rayleigh–Taylor and Kelvin–Helmholtz

> **Note**
>
> This chapter is in preparation.

16.1 In-class worksheet

1. A heavy fluid with density ρ_2 is layered on top of a lighter fluid with density ρ_1 in a constant vertical gravitational field g ($\rho_2 > \rho_1$). The system is perturbed by rotating a cylinder of fluid of radius r, as shown in Figure 16.1.

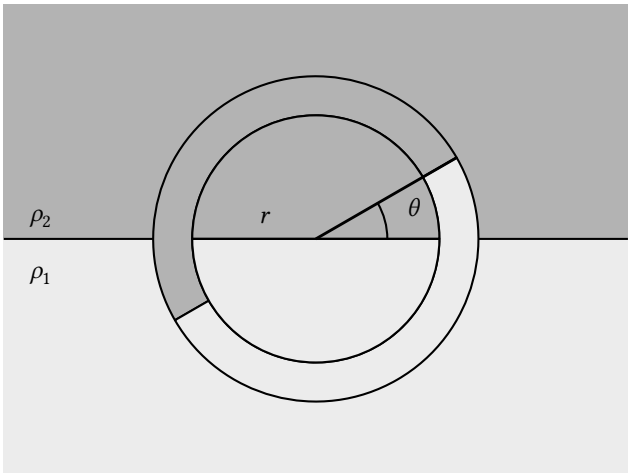


Figure 16.1: A cylinder of fluid of radius r straddling the interface between the heavy fluid (density ρ_2, above) and the light fluid (density ρ_1, below). The ring around the cylinder has been rotated rigidly by an angle θ.

Assume that the fluid is incompressible and that the fluid ring rotates rigidly.

a. Derive an equation for $\ddot{\theta}$.

b. Show that the growth rate of the instability is

$$\lambda \sim \sqrt{\frac{Ag}{r}}, \tag{16.1}$$

where A is the Atwood number

$$A = \frac{\rho_2 - \rho_1}{\rho_2 + \rho_1}. \tag{16.2}$$

16.2 Exercises

Exercise 16.1 (Instability in a pipe). A smooth pipe of square cross-section, with side a, lies on a horizontal surface in a uniform vertical gravitational field g. The pipe is filled with incompressible fluid: the upper half of the pipe contains fluid of density ρ_1 and the lower half contains fluid of density ρ_0, with $\rho_1 > \rho_0$.

1. Name the instability that develops, and identify the energy source that powers it.
2. Compute the energy E_1 per unit length of the pipe that is available to the instability.

Adapted from C. Clarke and B. Carswell [1], Problem 41.

16.3 Further reading

- C. Clarke and B. Carswell [1], Chapter 10 (fluid instabilities).

17. Fluid Instabilities III: Thermal and Gravitational Instabilities

> **Note**
>
> This chapter is in preparation.

17.1 Exercises

Exercise 17.1 (Thermal balance of a slab). A slab of gas held together by its own gravity is heated by cosmic rays, at a constant rate per unit mass. It cools by optically thin thermal bremsstrahlung, with a cooling rate per unit mass proportional to $\rho T^{1/2}$.

1. Using the discussion of thermal instability in C. Clarke and B. Carswell [1] (Section 10.4), analyse the thermal stability of the slab.
2. Is the slab stable against the Rayleigh–Taylor instability (see Chapter 16)?

Adapted from C. Clarke and B. Carswell [1], Problem 36.

17.2 Further reading

- C. Clarke and B. Carswell [1], Chapter 10 (fluid instabilities).

18. Diffusion and Viscous Flows

> **Note**
>
> This chapter is in preparation.

18.1 In-class worksheet

1. Core-collapse supernovae inject metals, such as oxygen, into the interstellar medium (ISM). After the supernova remnant has stalled its expansion, the diffusion of these elements throughout the galaxy can be modeled through an advection-diffusion equation:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = \nabla \cdot (D \nabla \rho), \tag{18.1}$$

   where ρ is the mass density of the newly synthesized elements, $\mathbf{v}$ is the local velocity of the ISM, and D is a diffusion coefficient.

   a. Show that, under suitable assumptions,

$$\frac{\mathrm{d}}{\mathrm{d}t} \int_V \rho \mathrm{d}^3\mathbf{r} = 0. \tag{18.2}$$

   b. Assume that $\mathbf{v} = \text{const}$, $D = \text{const} > 0$, and look for solutions in the form

$$\rho(t, \mathbf{x}) = \rho_0 \mathrm{e}^{\mathrm{i}(\mathbf{k}\cdot\mathbf{x} - \omega t)}. \tag{18.3}$$

   Sketch the solution for different times and give a physical interpretation of the results.

   c. How do the results change if $D < 0$?

   d. Show that Equation (18.1) can be written as

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \left[\rho \left(\mathbf{v} - D \frac{\nabla \rho}{\rho} \right) \right] = 0. \tag{18.4}$$

   Interpret the term in round brackets as an effective propagation velocity $\mathbf{v}_{\text{eff}}$. How does $\mathbf{v}_{\text{eff}}$ depend on the wave number $\mathbf{k}$ if ρ is as in Equation (18.3)?

18.2 Exercises

Exercise 18.1 (Steady viscous vortex). Consider a steady-state, axisymmetric solution of the incompressible, viscous, Navier–Stokes equations in cylindrical coordinates (R, ϕ, z) in the form

$$v_R = -\alpha R, \qquad v_\phi = \frac{\mathcal{C}}{2\pi R} g(R), \qquad v_z = 2\alpha z, \tag{18.5}$$

where $\alpha > 0$ and $\mathcal{C} > 0$ are given constants, while $g(R)$ is a function satisfying

$$\lim_{R\to 0} g(R) = 0, \qquad \lim_{R\to\infty} g(R) = 1. \tag{18.6}$$

Let $\nu > 0$ be the kinematic viscosity.

1. Show that the flow is incompressible.
2. Derive an equation for $g(R)$ assuming the flow to be steady state.

3. Compute the vorticity ω of the flow.

18.3 Further reading

- C. Clarke and B. Carswell [1], Chapter 11 (viscous flows).

19. Accretion Disks I: Structure and Viscous Evolution

> **Note**
>
> This chapter is in preparation.

19.1 In-class worksheet

1. Cylindrical coordinates (R, ϕ, z) are defined as follows

$$x = R\cos\phi, \qquad y = R\sin\phi, \qquad z = z. \tag{19.1}$$

The unit vectors in the R, ϕ, and z directions are

$$\begin{aligned} \hat{\mathbf{R}} &= \frac{1}{\sqrt{x^2+y^2}}\begin{pmatrix} x \\ y \\ 0 \end{pmatrix}, \\ \hat{\boldsymbol{\phi}} &= \frac{1}{\sqrt{x^2+y^2}}\begin{pmatrix} -y \\ x \\ 0 \end{pmatrix}, \\ \hat{\mathbf{z}} &= \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}. \end{aligned} \tag{19.2}$$

   a. Given the fluid velocity written as $\mathbf{v} = v_R\hat{\mathbf{R}} + v_\phi\hat{\boldsymbol{\phi}} + v_z\hat{\mathbf{z}}$, find its Cartesian components v_x, v_y, and v_z.

   b. Use the chain rule for derivatives to show that

$$\frac{\partial}{\partial R} = \frac{x}{R}\frac{\partial}{\partial x} + \frac{y}{R}\frac{\partial}{\partial y}, \qquad \frac{\partial}{\partial \phi} = -y\frac{\partial}{\partial x} + x\frac{\partial}{\partial y}. \tag{19.3}$$

   c. Using the results from the previous points show that

$$\nabla\cdot\mathbf{v} = \frac{1}{R}\frac{\partial(Rv_R)}{\partial R} + \frac{1}{R}\frac{\partial v_\phi}{\partial\phi} + \frac{\partial v_z}{\partial z}. \tag{19.4}$$

   d. Conclude that the continuity equation in cylindrical coordinates is

$$\frac{\partial\rho}{\partial t} + \frac{1}{R}\left[\frac{\partial(\rho R v_R)}{\partial R} + \frac{\partial(\rho v_\phi)}{\partial\phi}\right] + \frac{\partial(\rho v_z)}{\partial z} = 0. \tag{19.5}$$

19.2 Exercises

Exercise 19.1 (α-disks: scale height, viscosity, Reynolds number).

1. Consider gas in an accretion disk that is in hydrostatic equilibrium in the vertical direction, with the vertical pressure gradient balancing the vertical component of the gravitational acceleration due to the central object. Show that the angular frequency of rotation Ω, the sound speed c_s, and the vertical scale height H of the disk satisfy $c_s \sim H\Omega$.

2. Suppose now that the $R\phi$ component of the viscous stress tensor equals a factor α times the thermal pressure. Show that (i) $\nu \sim \alpha c_s H$ and (ii) the Reynolds number is $\mathrm{Re} \sim \frac{1}{\alpha}\left(\frac{R}{H}\right)^2$.

Adapted from C. Clarke and B. Carswell [1], Problem 47.

Exercise 19.2 (Similarity form of the disk equation).

1. Show that the viscous evolution equation of a Keplerian accretion disk,

$$\frac{\partial \Sigma}{\partial t} = \frac{3}{R}\frac{\partial}{\partial R}\left[R^{1/2}\frac{\partial}{\partial R}(\nu \Sigma R^{1/2})\right], \tag{19.6}$$

   can be recast as

$$\frac{\partial S}{\partial t} = \frac{\partial^2 Y}{\partial X^2}, \tag{19.7}$$

   with $S = \Sigma R^{3/2}$, $Y = 3\nu\Sigma R^{1/2}$, and $X = 2R^{1/2}$.
2. Obtain also a partial differential equation for $S(t, X)$ when $Y = Y_0 + \beta(S - S_0)$, where Y_0, S_0, and β are constants.
3. Describe how S evolves in the two cases (a) $\beta > 0$ and (b) $\beta < 0$.

Adapted from C. Clarke and B. Carswell [1], Problem 49.

Exercise 19.3 (Numerical evolution of an accretion disk).

1. Show that Equation (19.6) can be written as

$$\frac{\partial S}{\partial T} = \frac{\partial^2 (S X^{-2})}{\partial X^2}, \tag{19.8}$$

   where S and X are defined as in Exercise 19.2 and $T = 12\nu t$.
2. To solve Equation (19.8) numerically, introduce a grid

$$\{(T_n, X_k) = (n\Delta T, X_{\min} + k\Delta X) : n = 0, 1, ..., N_T, k = 0, 1, ..., N_X\}, \tag{19.9}$$

   where ΔT, $X_{\min}$, and ΔX are fixed parameters. Let S_k^n be the approximation of S at (T_n, X_k)[8]. Then S_k^n for $k = 1, ..., N_X - 1$ can be written as

$$\frac{S_k^{n+1} - S_k^n}{\Delta T} = \frac{S_{k+1}^{n+1} X_{k+1}^{-2} - 2S_k^{n+1} X_k^{-2} + S_{k-1}^{n+1} X_{k-1}^{-2}}{\Delta X^2}. \tag{19.10}$$

   The boundary conditions for this problem are

$$S_0 = S_{N_X} = 0, \tag{19.11}$$

   corresponding to Dirichlet boundary condition for S.

   Show that Equation (19.10) can be written as

$$(I - \Delta T A)\mathbf{S}^{n+1} = \mathbf{S}^n, \tag{19.12}$$

   where $\mathbf{S}^n = (S_1^n, S_2^n, ..., S_{N_X-1}^n)$ is a vector of values, I is the identity matrix, and A is a tridiagonal matrix. Find the explicit form of A.
3. Write code to evolve in time the density in the disk by repeatedly solving Equation (19.12) starting from some initial condition $\mathbf{S}^0$. You can use `scipy.linalg.solve_banded` to solve the linear equations.
4. Consider as initial conditions

[8] Note that the n in S_k^n is an index and not a power.

$$S(t=0, X) = \mathrm{e}^{-(X-10)^2}, \tag{19.13}$$

with $X_{\min} = 1$. Plot the numerical solutions at successive times. Interpret the results in a few sentences.

19.3 Further reading

- C. Clarke and B. Carswell [1], Chapter 12 (accretion discs).

20. Accretion Disks II: Steady Disks and Spectra

> **Note**
>
> This chapter is in preparation.

20.1 In-class worksheet

1. **Viscous instability in accretion disks.** The equation describing the evolution of the surface density Σ in an accretion disk is given by

$$\partial_t \Sigma = \frac{3}{R}\partial_R[R^{1/2}\partial_R(\nu\Sigma R^{1/2})]. \tag{20.1}$$

   Assume that the viscosity ν is in the form $\nu = \nu(\Sigma, R)$.

   a. Introduce the quantity $\psi = \nu\Sigma$ and consider a small perturbation $\delta\Sigma$ in the surface density of the disk. Find the equation describing the evolution of the corresponding perturbation $\delta\psi$ of ψ.

   b. Under what condition is the disk stable?

20.2 Exercises

Exercise 20.1 (Accretion-disk spectrum). Model an accretion disk as a set of concentric rings $i = 1, 2, \ldots, N$, each with median radius R_i and radial extent ΔR. Let the disk extend from an inner radius $R_{\rm in}$ to an outer radius $R_{\rm out}$ and the temperature be

$$T(R) = T_{\rm in}\left(\frac{R_{\rm in}}{R}\right)^{3/4}. \tag{20.2}$$

Assume that the ith ring emits as a blackbody with a fixed temperature $T_i = T(R_i)$

$$B_\nu(T) = \frac{2h\nu^3}{c^2}\frac{1}{e^{h\nu/k_B T} - 1}, \tag{20.3}$$

where h is the Planck constant, c the speed of light, and k_B the Boltzmann constant.

1. Write a code to compute and plot the emergent spectrum from a single ring at radius R_i. Plot the emergent spectrum for several radii. You can work in units in which $k_B = h = c = 1$. You can take $T_{\rm in} = 1$ in these units.
2. Write a code to compute the emergent spectrum from the full disk by summing up the contribution from each ring. Plot the resulting spectrum and discuss its physical interpretation. Verify that the spectrum approximately scales as $\nu^{1/3}$ for $T_{\rm out} \lesssim \nu \lesssim T_{\rm in}$. You can work in the same units as in the previous point. Additionally assume that $R_{\rm out}/R_{\rm in} = 10^4$, so that $T_{\rm out} = 10^{-3}\, T_{\rm in}$.

20.3 Further reading

- C. Clarke and B. Carswell [1], Chapter 12 (accretion discs).

21. Turbulence

21.1 What is Turbulence?

Turbulence is the chaotic, complex motion observed in fluids across all scales, from the chaotic motion of the quark–gluon plasma produced in heavy-ion collisions (fm scale) to the motion of gas in the intracluster medium (Mpc scale). Turbulence plays an important role in many terrestrial and astrophysical flows. For example, turbulence in the interstellar medium is thought to generate overdensities that eventually grow up to become stars. Turbulence also shapes the evolution and death of stars. Turbulent flows efficiently mix fluid: think of the way in which milk and coffee mix once turbulence is generated with a spoon. Turbulent flows also efficiently dissipate kinetic energy and develop large *effective* shear viscosity, e.g., triggering angular momentum transport.



Figure 21.1: Study of turbulence by Leonardo da Vinci (ca. 1510), Royal Collection Trust (public domain).

Despite its importance for a number of applications, including many of practical, industrial impact, and despite the fact that turbulence has been studied for more than 500 years, we still lack a rigorous theory of turbulence. The closest thing we have to a theory is Kolmogorov's 1941 theory (K41). These notes contain an introductory-level description of K41 and some example applications. For a more in depth description, I would recommend Landau–Lifshitz Vol. 6 [6] and Pope, *Turbulent Flows*, 2000 [7].

21.2 Kolmogorov's Theory

Fluid motion is driven by external forces, such as gravity, which have a characteristic scale ℓ_0 comparable to the size of the system of interest. For example, for a star $\ell_0 \sim R_*$, the radius of the star. The starting point of Kolmogorov's theory is to decompose the flow velocity in some patch of the fluid $\mathbf{v}$ into a mean flow $\langle \mathbf{v} \rangle$ and a fluctuating component $\mathbf{v}'$, not necessarily assumed to be small. The theory then assumes that, if the size ℓ of the region under consideration is sufficiently small, $\ell \ll \ell_0$, $\mathbf{v}'$ is *universal*, at least in a statistical sense, meaning that it does not depend on the particular system, but that is set by the equations of motion.

To keep our notation simple, we set $\langle \mathbf{v} \rangle = 0$, meaning that $\mathbf{v}' = \mathbf{v}$. Note that we can always set $\langle \mathbf{v} \rangle = 0$ by working in the appropriate reference frame, so this is not a restriction on the theory we are developing. We are also going to assume that the flow is incompressible, which is a good approximation if the velocity fluctuations are small compared to the sound speed (Chapter 9). As we will see, the velocity fluctuations become smaller at smaller scales, so this is always true at sufficiently small scales, except in shocks. The flow is then described by the Navier–Stokes equations (Chapter 18)

$$\partial_t \mathbf{v} + (\mathbf{v} \cdot \nabla)\mathbf{v} = \mathbf{g} - \frac{\nabla p}{\rho} + \nu \nabla^2 \mathbf{v}, \qquad \nabla \cdot \mathbf{v} = 0, \tag{21.1}$$

where $\mathbf{g}$ is some external acceleration field, typically gravity in astrophysical applications.

The equation for the evolution of the kinetic energy is obtained by multiplying Equation (21.1) by $\mathbf{v}$:

$$\frac{1}{2}[\partial_t v^2 + (\mathbf{v} \cdot \nabla)v^2] = \mathbf{g} \cdot \mathbf{v} - \frac{1}{\rho}\nabla p \cdot \mathbf{v} + \nu \nabla^2 \mathbf{v} \cdot \mathbf{v}. \tag{21.2}$$

Consider the first term on the RHS, representing the work done by the external force on the flow. If $\ell \ll \ell_0$, then we can take $\mathbf{g} \sim$ const over this region and

$$\langle \mathbf{g} \cdot \mathbf{v} \rangle \sim \mathbf{g} \cdot \langle \mathbf{v} \rangle = 0. \tag{21.3}$$

This suggests that the evolution of v for $\ell \ll \ell_0$ is independent of the details of the external force $\mathbf{g}$, so v can be universal. Of course, if $\ell \sim \ell_0$, then $\langle \mathbf{g} \cdot \mathbf{v} \rangle > 0$ in general, meaning that kinetic energy is injected at the scale ℓ_0 by gravity and then evolves according to Equation (21.2).

Let's now consider the role of pressure (note that $\rho = $ const):

$$\langle \nabla p \cdot \mathbf{v} \rangle = \frac{1}{V}\int_V \nabla p \cdot \mathbf{v} \mathrm{d}V = \frac{1}{V}\int_V [\nabla \cdot (p\mathbf{v}) - p\nabla \cdot \mathbf{v}]\mathrm{d}V = \frac{1}{V}\oint_S p\mathbf{v} \cdot \mathbf{n} \mathrm{d}S = 0. \tag{21.4}$$

In the last equation we have assumed that $\mathbf{v}$ is isotropic (no preferred direction), so $\mathbf{v} \cdot \mathbf{n}$ is zero on average.

In a similar way, we can rewrite the average of the last term in the RHS of Equation (21.2) as

$$\int_V \nabla^2 \mathbf{v} \cdot \mathbf{v} \mathrm{d}V = \oint_S \mathbf{n} \cdot (\nabla \mathbf{v}\mathbf{v})\mathrm{d}S - \int_V \nabla \mathbf{v} : \nabla \mathbf{v} \mathrm{d}V = -\int_V |\nabla \mathbf{v}|^2 \mathrm{d}V. \tag{21.5}$$

This means that, on average, we can simplify the kinetic energy equation to read

$$\partial_t v^2 = 2\mathbf{g} \cdot \mathbf{v} - (\mathbf{v} \cdot \nabla)v^2 - 2\nu|\nabla \mathbf{v}|^2. \tag{21.6}$$

As already mentioned, the first term on the RHS is responsible for energy injection. The last term describes kinetic energy *dissipation*. We can estimate this term as

$$\nu|\nabla \mathbf{v}|^2 \sim \nu\left(\frac{v(\ell)}{\ell}\right)^2, \tag{21.7}$$

where $v(\ell)$ is the characteristic velocity fluctuation on a length scale ℓ:

$$v(\ell) = \langle |\mathbf{v}(\mathbf{x} + \mathbf{l}) - \mathbf{v}(\mathbf{x})| \rangle = \frac{1}{V}\int_V |\mathbf{v}(\mathbf{x} + \mathbf{l}) - \mathbf{v}(\mathbf{x})| \mathrm{d}V. \tag{21.8}$$

We use the notation $A \sim B$ to signify that two quantities, A and B, are equal up to factors of order unity.

The quantity on the RHS of Equation (21.7) has units of power per unit mass, as expected, since it is the rate at which the kinetic energy per unit mass is dissipated. Equation (21.7) suggests that the energy is dissipated at a scale ℓ_d (the dissipation or Kolmogorov scale) at which the dissipation rate is equal to the mean power of the global flow:

$$\nu\left(\frac{v(\ell_d)}{\ell_d}\right)^2 \sim \frac{v^3(\ell_0)}{\ell_0} \Rightarrow \ell_d \sim \ell_0 \left[\frac{\nu}{\ell_0 v(\ell_0)}\right]^{1/2}\left[\frac{v(\ell_d)}{v(\ell_0)}\right] = \frac{\ell_0}{\mathrm{Re}^{1/2}}\left[\frac{v(\ell_d)}{v(\ell_0)}\right]. \tag{21.9}$$

In the previous equation, we have introduced the Reynolds number

$$\mathrm{Re} = \frac{\ell_0 v(\ell_0)}{\nu}. \tag{21.10}$$

Experimental observations of fluids in pipes suggest that turbulence sets in when $\mathrm{Re} \gtrsim 10^3$. In astrophysical situations, Re can be much larger. For example, in the interstellar medium $\mathrm{Re} \sim 10^9$. In neutron star mergers we contend with flows having $\mathrm{Re} \sim 10^{15}$. Consequently, Equation (21.9) implies that $\ell_d \ll \ell_0$ for turbulent flows.

Summarizing, in our analysis of Equation (21.6) we have found that kinetic energy is injected at large scale ℓ_0 (first term on the RHS) and dissipated at the dissipation scale $\ell_d \ll \ell_0$ (last term on the RHS). This means that there is a range of scales $\ell_d \ll \ell \ll \ell_0$, the so-called *inertial range*, in which the dynamics is dominated by the nonlinear term $(\mathbf{v}\cdot\nabla)v^2$. We can estimate it as

$$(\mathbf{v}\cdot\nabla)v^2 \sim \frac{v^3(\ell)}{\ell}. \tag{21.11}$$

Note that, for an incompressible flow, the nonlinear term can also be written as a divergence, $(\mathbf{v}\cdot\nabla)v^2 = \nabla\cdot(\mathbf{v}v^2)$. So, by the same argument we used for the pressure, its average over a region vanishes. In other words, the nonlinear term does not create or destroy kinetic energy. Instead, it moves kinetic energy between different scales. In a steady state, the energy per unit mass and per unit time injected at the scale ℓ_0,

$$\mathcal{P} = \frac{v^3(\ell_0)}{\ell_0}, \tag{21.12}$$

must flow through all the scales in the inertial range, without losses, until it reaches the dissipation scale, where it is dissipated. At the scale ℓ, the kinetic energy per unit mass $v^2(\ell)$ is passed on to smaller scales in a time $\ell/v(\ell)$, so the flux of energy is $v^3(\ell)/\ell$. Since this must be the same at all scales, we have

$$\frac{v^3(\ell)}{\ell} \sim \mathcal{P} \Rightarrow \boxed{v(\ell) \sim (\mathcal{P}\ell)^{1/3} = v(\ell_0)\left(\frac{\ell}{\ell_0}\right)^{1/3}}. \tag{21.13}$$

This result is key to the theory of turbulence and is called Kolmogorov's second similarity hypothesis[9]. It has been verified many times through direct numerical simulations and experiments. Note that $v(\ell)$ decreases at smaller scales, so the flow is subsonic at sufficiently small scales, as we anticipated.

Often this hypothesis is formulated in terms of the *energy spectrum* $E(k)$. This is defined so that $E(k)\mathrm{d}k$ is the kinetic energy per unit mass in eddies with wave numbers between k and $k + \mathrm{d}k$:

$$\frac{1}{2}\langle|\mathbf{v}|^2\rangle = \int_0^\infty E(k)\mathrm{d}k. \tag{21.14}$$

Eddies of size ℓ have wave numbers $k \sim 1/\ell$. The kinetic energy per unit mass in eddies of size $\sim \ell$ is the energy in a range of wave numbers of width $\sim k$ around k, so

$$v^2(\ell) \sim kE(k) \Rightarrow E(k) \sim \frac{v^2(1/k)}{k} \sim \mathcal{P}^{2/3}k^{-5/3}. \tag{21.15}$$

This is the famous Kolmogorov $k^{-5/3}$ spectrum. A comparison between data and theory over about 16 orders of magnitude in scale is shown in Figure 21.2. The figure collects measurements of density fluctuations in the interstellar medium from several types of observations, with spacecraft data, including Voyager 1, at the smallest scales. It shows the three-dimensional spectrum of density fluctuations, which in Kolmogorov's theory scales as $k^{-11/3}$. The extra factor k^{-2} with respect to $E(k)$ comes from the area of a shell of radius k in Fourier space. Deviations from the Kolmogorov scaling only appear at very small scales, when plasma kinetic effects become important.

[9] The first Kolmogorov similarity hypothesis concerns the universality of the flow at the dissipation scale. This is not relevant to astrophysical applications, because astrophysical plasmas become collisionless at small scale, i.e., they do not behave like a fluid anymore. However, Kolmogorov's theory still applies for $\ell \gg \ell_d$.

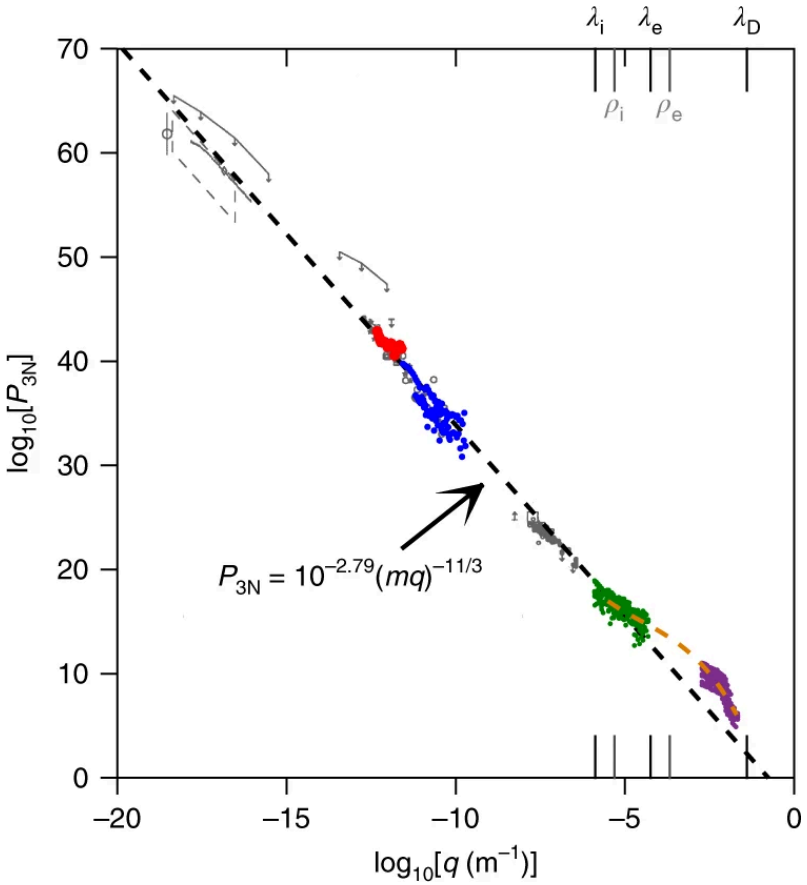


Figure 21.2: Power spectrum of electron density fluctuations in the interstellar medium from several types of observations, including Voyager 1 (points), and Kolmogorov spectrum (dashed). From K. H. Lee and L. C. Lee [8]; © Springer Nature, reproduced for non-commercial educational use and not covered by the license of these notes.

A physical interpretation of the K41 model is given by the Richardson cascade (which actually predates K41). In this model, energy is injected at large scale by $\mathbf{g}$ and dissipated at small scales by the viscosity. The nonlinear term $(\mathbf{v} \cdot \nabla)v^2$ couples different scales and is responsible for the transport of energy from large scale to small scale, as illustrated in Figure 21.3.

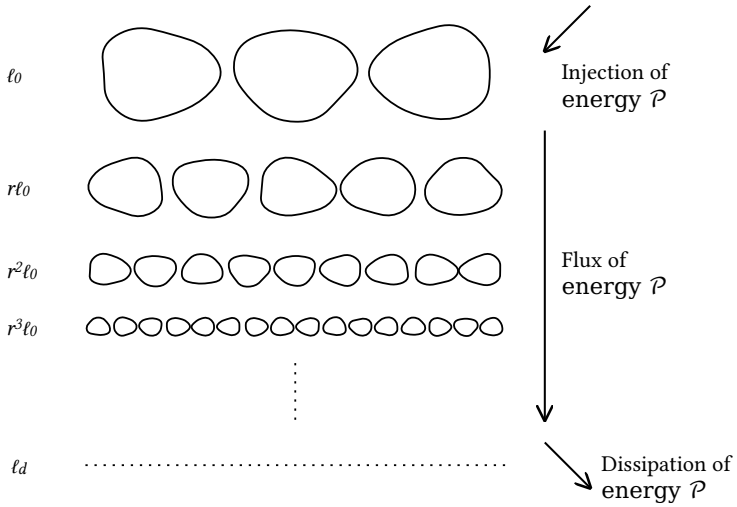


Figure 21.3: The Richardson cascade. Redrawn after U. Frisch [9].

The main mechanisms for the transport of energy are vortex stretching and shear (Kelvin–Helmholtz) instabilities (Chapter 16).

We can interpret $v^3(\ell)/\ell$ as a term that controls the kinetic energy dissipation:

$$\frac{v^3(\ell)}{\ell} = \frac{v^2(\ell)}{\tau}, \qquad \tau = \frac{\ell}{v(\ell)}, \tag{21.16}$$

where τ is the eddy turnover time, the time over which the specific kinetic energy $v^2(\ell)$ is transferred to smaller scales. To see this, notice that for $\ell_d \ll \ell \ll \ell_0$, Equation (21.6) reduces to

$$\boxed{\frac{\mathrm{d}v^2(\ell)}{\mathrm{d}t} \sim -\frac{v^2(\ell)}{\tau}.} \tag{21.17}$$

The constancy of $v^3(\ell)/\ell$ implies that **the rate at which turbulence dissipates kinetic energy is independent of ν** or of the detailed form of the physical processes responsible for dissipation! Instead, it is set by the dynamics at large scale. In other words, when kinetic energy is injected by mechanical work at large scale in a turbulent flow, turbulence efficiently transports this energy to small scales (generating large velocity gradients) at which point it is dissipated. This is the reason why turbulence theory applies equally well to the flow of water around the propeller of a ship and to the motion of rarefied gas in the intergalactic medium. This finding also justifies turbulence modeling approaches used in simulations.

With Kolmogorov's second similarity law, we can provide a better estimate for ℓ_d. Using Equation (21.13) into Equation (21.9), we can eliminate $v(\ell_d)$ to find

$$\boxed{\ell_d \sim \left(\frac{\nu}{\ell_0 v(\ell_0)}\right)^{3/4} \ell_0 = \mathrm{Re}^{-3/4}\ell_0}. \tag{21.18}$$

This shows that the scale separation in developed turbulence can be enormous. A direct numerical simulation needs to resolve all scales between ℓ_d and ℓ_0, so it needs $(\ell_0/\ell_d)^3$ grid cells. It also needs a number of time steps $\sim \ell_0/\ell_d$, because the time step must be smaller than the time it takes for the flow to cross one cell. So the cost of a direct numerical simulation scales as

$$\text{cost} \sim \left(\frac{\ell_0}{\ell_d}\right)^4 \sim \mathrm{Re}^3, \tag{21.19}$$

which makes it impossible to directly simulate very turbulent flows. The largest simulations today have about 10^4 grid cells in each direction, so they can reach $\mathrm{Re} \sim 10^5$. To study flows with higher Reynolds numbers, it is necessary to give up the idea of simulating all scales in the flow and instead resort to modeling the dynamics below the numerical grid scale using turbulence theory.

Example

Consider the wind in the atmosphere of the Earth, with $\ell_0 \sim 1\,\mathrm{km}$ and $v(\ell_0) \sim 10\,\mathrm{m\,s^{-1}}$. The kinematic viscosity of air is $\nu \approx 1.5 \times 10^{-5}\,\mathrm{m^2\,s^{-1}}$, so $\mathrm{Re} \sim 7 \times 10^8$. The dissipation scale is $\ell_d \sim \mathrm{Re}^{-3/4}\ell_0 \sim 0.2\,\mathrm{mm}$, and a direct numerical simulation would cost $\sim \mathrm{Re}^3 \sim 10^{26}$, many orders of magnitude beyond what is possible. Even a cup of coffee stirred with a spoon, with $\ell_0 \sim 3\,\mathrm{cm}$, $v(\ell_0) \sim 0.1\,\mathrm{m\,s^{-1}}$, and $\nu \approx 10^{-6}\,\mathrm{m^2\,s^{-1}}$, has $\mathrm{Re} \sim 3000$ and $\ell_d \sim 70\,\mu\mathrm{m}$.

Before we discuss some applications of Kolmogorov's theory, it is worth mentioning its limitations. Astrophysical flows are often supersonic and highly magnetized. Compressibility and magnetic fields can have a profound role on the turbulent cascade. Shocks can dissipate kinetic energy directly at large scales (Chapter 11). Magnetic fields can give rise to *inverse cascade* whereby small magnetic field structures self-organize into large scale ordered fields (*turbulent dynamo process*, Chapter 24). The study of compressible, magnetized turbulence is still an active area of research and beyond the scope of this course. However, even for compressible, magnetized flows, Kolmogorov's theory provides a good starting point and can often be applied directly.

21.3 Shredding of Clouds in a Turbulent Flow

Consider a cloud of gas of radius R, for example a patch of material enriched by a supernova, that is carried by the turbulent flow of the interstellar medium. We want to compute the evolution of R with time. We will consider three cases, illustrated in Figure 21.4.

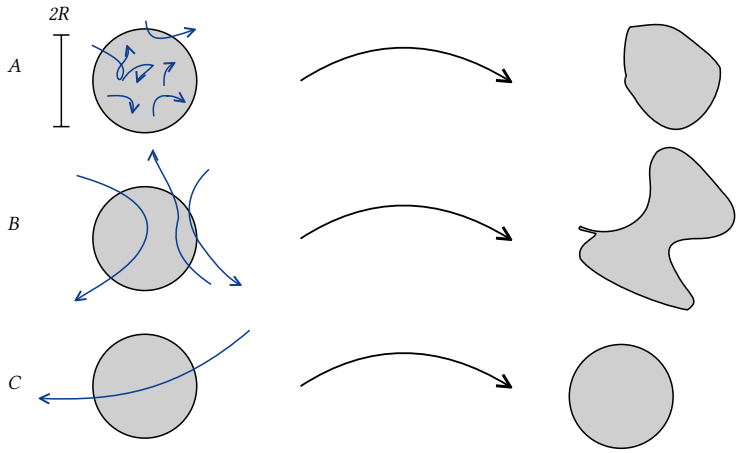


Figure 21.4: Distortion of a cloud in a turbulent medium. The three different cases are $R \sim \ell_d$ (case A), $\ell_d \ll R \ll \ell_0$ (case B), and $R \sim \ell_0$ (case C).

If $R \sim \ell_d$ the particles in the cloud will undergo Brownian random motion, so, as in the case of the random diffusion of photons (Chapter 6), we have $R \sim t^{1/2}$. If $R \sim \ell_0$, then the cloud is primarily advected by the mean flow. The edges of the cloud will still experience random diffusion, so $R \sim t^{1/2}$, but with an effective diffusion coefficient $\sim \ell_0 v(\ell_0)$, which is much larger than the molecular one. Indeed, eddies of size ℓ_0 act like particles with mean free path ℓ_0 (compare with Equation (1.1)) moving with velocity $v(\ell_0)$. We will see that this is the same estimate used for the turbulent viscosity.

The more interesting case is when $\ell_d \ll R \ll \ell_0$. In this case, the cloud undergoes rapid distortion:

$$\frac{\mathrm{d}R}{\mathrm{d}t} \sim v \sim R^{1/3} \Rightarrow \frac{\mathrm{d}R^2}{\mathrm{d}t} \sim Rv \sim R^{4/3}. \tag{21.20}$$

This is *Richardson's 4/3-law*. We can use the second Kolmogorov similarity hypothesis to find the time dependence for R:

$$\frac{\mathrm{d}R}{\mathrm{d}t} \sim v(\ell_0)\left(\frac{R}{\ell_0}\right)^{1/3} \Rightarrow R^{2/3} \sim \frac{v(\ell_0)}{\ell_0^{1/3}} t \Rightarrow R^2 \sim \mathcal{P} t^3. \tag{21.21}$$

In other words, $R \propto t^{3/2}$: the cloud grows faster and faster, because, as it grows, it is torn apart by larger and faster eddies.

Turbulence significantly accelerates the mixing of new elements produced in supernovae with the interstellar medium, where they can subsequently form new stars, planets, and people.

Example

Consider the metals ejected by a supernova into the warm interstellar medium, with $T \sim 10^4\,\mathrm{K}$ and $n \sim 1\,\mathrm{cm}^{-3}$. The turbulence in the interstellar medium is driven on scales $\ell_0 \sim 100\,\mathrm{pc}$, with $v(\ell_0) \sim 10\,\mathrm{km\,s}^{-1}$. So turbulence mixes the metals over $\sim 100\,\mathrm{pc}$ in a time $\sim \ell_0/v(\ell_0) \sim 10\,\mathrm{Myr}$. How long would it take without turbulence? The mean free path of the particles in this gas is $\lambda \sim 10^{10}\,\mathrm{m}$, and their thermal velocity is $v_{\mathrm{th}} \sim 10\,\mathrm{km\,s}^{-1}$, so the diffusion coefficient is $D \sim \lambda v_{\mathrm{th}}/3 \sim 3 \times 10^{13}\,\mathrm{m}^2\,\mathrm{s}^{-1}$ (Chapter 6). Molecular diffusion would take $\sim \ell_0^2/D \sim 10^{16}\,\mathrm{yr}$, much longer than

the age of the Universe. Note also that the ratio of the two timescales is $\ell_0 v(\ell_0)/D \sim 10^9$, which is the Reynolds number of the interstellar medium.

21.4 Turbulence Modeling: Reynolds Stress

Directly simulating turbulent flows is impossible, so one of the goals of turbulence theory is to develop effective strategies to deal with unresolved turbulence in simulations. Commonly used models are the Reynolds averaged Navier–Stokes (RANS) and Large-Eddy Simulations (LES) frameworks. Without going into the details, both approaches are based on the general idea of coarse graining the fluid equations. That is, we replace $\mathbf{v}$ with a suitable average $\bar{\mathbf{v}}$ and try to come up with equations for it. If we coarse grain the Navier–Stokes equations, we find:

$$\partial_t \mathbf{v} + \nabla \cdot \left(\mathbf{v} \otimes \mathbf{v} + \frac{p}{\rho} \mathbf{I} \right) = \mathbf{g} + \nu \nabla^2 \mathbf{v} \rightarrow \partial_t \bar{\mathbf{v}} + \nabla \cdot \overline{\left(\mathbf{v} \otimes \mathbf{v} + \frac{p}{\rho} \mathbf{I} \right)} = \bar{\mathbf{g}} + \nu \nabla^2 \bar{\mathbf{v}}. \quad (21.22)$$

These equations are exact, but unfortunately not closed, because we do not have a way to compute $\overline{\mathbf{v} \otimes \mathbf{v}}$ from $\bar{\mathbf{v}}$. This is the so-called turbulence closure problem. To work around this, we define the *Reynolds stress tensor*

$$\mathbf{R} = \overline{\mathbf{v} \otimes \mathbf{v}} - \bar{\mathbf{v}} \otimes \bar{\mathbf{v}} \quad (21.23)$$

and the momentum equation becomes:

$$\partial_t \bar{\mathbf{v}} + \nabla \cdot \left(\bar{\mathbf{v}} \otimes \bar{\mathbf{v}} + \mathbf{R} + \frac{\bar{p}}{\rho} \mathbf{I} \right) = \bar{\mathbf{g}} + \nu \nabla^2 \bar{\mathbf{v}}. \quad (21.24)$$

Equation (21.24) is the momentum equation for a fluid that has velocity equal to $\bar{\mathbf{v}}$, pressure $\bar{p}$, and a new component of the stress tensor $\mathbf{R}$. Because $\bar{\mathbf{v}}$ is some average of the true velocity, Equation (21.24) can be solved on a coarse computational mesh. The problem of modeling turbulence is now reduced to that of finding a good model for $\mathbf{R}$. There are many such models in the literature.

The simplest model, which is also the most commonly used for astrophysical applications, is the *turbulent* (or *eddy*) *viscosity* model. Accordingly, one takes

$$\mathbf{R} = \frac{1}{3}(\mathrm{tr}\mathbf{R})\mathbf{I} - \nu_T(\nabla \bar{\mathbf{v}} + \nabla^T \bar{\mathbf{v}}). \quad (21.25)$$

The first term is needed because $\mathrm{tr}\mathbf{R} = \overline{|\mathbf{v} - \bar{\mathbf{v}}|^2}$ is positive, while the trace of the second term vanishes for an incompressible flow. The first term is isotropic, so it acts like an additional pressure, the *turbulent pressure* $\rho \mathrm{tr}\mathbf{R}/3$, which we include in $\bar{p}$. We will meet the turbulent pressure again in the in-class worksheet. With this choice, and using $\nabla \cdot \bar{\mathbf{v}} = 0$, Equation (21.24) becomes

$$\partial_t \bar{\mathbf{v}} + \nabla \cdot \left(\bar{\mathbf{v}} \otimes \bar{\mathbf{v}} + \frac{\bar{p}}{\rho} \mathbf{I} \right) = \bar{\mathbf{g}} + (\nu + \nu_T) \nabla^2 \bar{\mathbf{v}}. \quad (21.26)$$

That is, turbulence is described simply as a dissipative process and ν_T is an effective turbulent viscosity. The turbulent viscosity, in turn, is often obtained using dimensional analysis arguments:

$$\nu_T = \ell_{\mathrm{mix}} V, \quad (21.27)$$

where ℓ_{mix} is a characteristic length, also called the *mixing length*, and V is a characteristic velocity. For example, in accretion disk theory (Chapter 19), $V = c_s$ and ℓ_{mix} is taken to be proportional to the scale-height of the disk H:

$$\ell_{\mathrm{mix}} = \alpha H = \alpha \frac{c_s}{\Omega} \Rightarrow \nu_T = \alpha \frac{c_s^2}{\Omega}, \quad (21.28)$$

where Ω is the angular velocity of the flow. This is the so-called α-viscosity model.

For shear flows along x with shear in the y-direction, one can use the velocity gradient instead of V:

$$\nu_T = \ell_{\rm mix}^2 \left| \frac{\partial \bar{v}_x}{\partial y} \right|. \tag{21.29}$$

This is Prandtl's mixing-length model. The Smagorinsky model, which is often used in large-eddy simulations, is a generalization of this model, in which $\ell_{\rm mix}$ is proportional to the size of the grid cells and $|\partial \bar{v}_x / \partial y|$ is replaced by the magnitude of the rate of strain of $\bar{\mathbf{v}}$.

For industrial applications, a popular strategy is to solve additional equations for $\partial_t \mathbf{R}$, with coefficients calibrated to experimental (wind tunnel) results. More sophisticated "subgrid" models have also been proposed for astrophysical applications, but their detailed description is beyond the scope of this course. See A. Brandenburg and Å. Nordlund [10] for more information.

Another popular approach is the Implicit Large-Eddy Simulations (ILES) method, in which $\mathbf{R}$ is not included explicitly in the equations. Instead, a numerical scheme is chosen that has a numerical viscosity $\nu_{\rm num} \approx \nu_T$. This approach is often effective, but ILES simulations need to be accompanied by careful resolution studies to assess their reliability, because accurate results can only be obtained if a sufficient portion of the inertial range is resolved by the simulations.



Figure 21.5: Density in a high-resolution simulation of compressible turbulence in the interstellar medium [11], as shown on the cover of *Nature Astronomy*, April 2021. © Springer Nature, reproduced for non-commercial educational use and not covered by the license of these notes.

21.5 In-class worksheet

Turbulence in Core-Collapse Supernovae. We construct here a simplified model showing the role of turbulence in aiding shock revival in core-collapse supernovae.

1. Consider a steady accretion shock near the protoneutron star. Assume that the in-falling material is cold and that the velocity downstream of the shock is small. Show that the Rankine–Hugoniot conditions (Chapter 11) simplify to

$$p_2 \approx \rho_1 v_1^2, \qquad \frac{\rho_2}{\rho_1} = \frac{\Gamma+1}{\Gamma-1}, \tag{21.30}$$

   where all quantities have their usual meaning.

2. Find the equilibrium pressure p_2 needed to support the accretion shock at a fixed radius r_s, given the mass of the protoneutron star M, and the accretion rate $\dot{M}$. Assume that the gas upstream of the shock falls freely from rest at infinity.

3. Consider angular averages of the flow, e.g.,

$$\langle \mathbf{v} \rangle = \frac{1}{4\pi} \oint \mathbf{v} \mathrm{d}\Omega. \tag{21.31}$$

   Decompose the velocity downstream of the shock as the sum of an average (laminar) component $\langle \mathbf{v} \rangle$ and a turbulent fluctuating component $\delta\mathbf{v}$, $\mathbf{v} = \langle \mathbf{v} \rangle + \delta\mathbf{v}$. Assume that ρ depends only on radius, so that $\langle \rho\mathbf{v} \rangle = \rho \langle \mathbf{v} \rangle$. Define the Reynolds stress as

$$\mathbf{R} = \langle \delta\mathbf{v} \otimes \delta\mathbf{v} \rangle. \tag{21.32}$$

   Show that the conservation of momentum across the shock can be written as

$$\rho_2 \langle v_2 \rangle^2 + \rho_2 R_{rr} + p_2 = \rho_1 v_1^2. \tag{21.33}$$

4. Define the specific turbulent kinetic energy as

$$\varepsilon_{\mathrm{turb}} = \frac{1}{2}|\delta\mathbf{v}|^2 = \frac{1}{2}(R_{rr} + R_{\theta\theta} + R_{\phi\phi}). \tag{21.34}$$

   Assume that $R_{rr} = R_{\theta\theta} + R_{\phi\phi}$ and show that with this definition Equation (21.33) can be written as

$$\begin{aligned} &\rho_2 \langle v_2 \rangle^2 \\ &+\rho_2[(\Gamma_{\mathrm{turb}} - 1)\varepsilon_{\mathrm{turb}} + (\Gamma - 1)\varepsilon_2] \\ &= \rho_1 v_1^2. \end{aligned} \tag{21.35}$$

   What is the effective adiabatic index of turbulence Γ_{turb}?

5. At early times, before the turbulence develops, the accretion shock is stalled at a radius r_s. What happens to the shock, once turbulence develops?

21.6 Exercises

Exercise 21.1 (Brownian and Lévy random walks). Consider the random walk of a particle. At each step i a particle at position $\mathbf{x}_i$ jumps to position

$$\mathbf{x}_{i+1} = \mathbf{x}_i + R_i \mathbf{d}_i, \tag{21.36}$$

where $\mathbf{d}_i$ is a random direction on the sphere (uniform in ϕ and $\cos\theta$) and R_i is either

1. a fixed constant (Brownian random walk); or
2. it has a Pareto distribution $p(R) \sim R^{-1-\beta}$, for a given β (Lévy random walk).

The Brownian random walk can be used to model ordinary diffusion process, while Lévy random walks can be used to model effective diffusion and mixing in a turbulent flow (superdiffusion).

Write a code to simulate Brownian and Lévy random walks with $\beta = 2/3$, then compute the median of the traveled distance $D_N = |\mathbf{x}_N - \mathbf{x}_0|$ as a function of N for the two. For $\beta < 1$ the mean length of a step is infinite, so the median is a better measure than the mean.

Show that $D_N \sim N^\gamma$ and estimate γ in the two cases. Compare the results with the predictions from Kolmogorov theory for diffusion in a turbulent flow ($R \sim t^{3/2}$). Explain why the Lévy random walk gives $D_N \sim N^{1/\beta}$.

[You can use `numpy.random.uniform` and `numpy.random.pareto` to generate R_i and $\mathbf{d}_i$.]

21.7 Further reading

- C. Clarke and B. Carswell [1], Chapter 14 (introduction to turbulence).
- L. D. Landau and E. M. Lifshitz [6] and S. B. Pope [7].

IV Plasma and Magneto-hydrodynamics

22 Introduction to Plasma . . . 133
22.1 Electromagnetism Refresher . . . 133
22.2 Plasma . . . 137
22.3 In-class worksheet . . . 142
22.4 Exercises . . . 143
22.5 Further reading . . . 143

23 The MHD Equations . . . 145
23.1 Lorentz Force . . . 145
23.2 Ohm's Law . . . 145
23.3 Faraday's Law . . . 147
23.4 Conservation Laws . . . 148
23.5 In-class worksheet . . . 149
23.6 Exercises . . . 150
23.7 Further reading . . . 150

24 Flux Freezing . . . 151
24.1 The Ideal-MHD Limit . . . 151
24.2 Star Formation . . . 155
24.3 Io and Jupiter . . . 157
24.4 In-class worksheet . . . 162
24.5 Exercises . . . 163
24.6 Further reading . . . 164

25 Magnetic Tension and Hydromagnetic Equilibria . . . 165
25.1 Magnetic Tension Force . . . 165
25.2 Hydromagnetic Equilibria . . . 168
25.3 In-class worksheet . . . 171
25.4 Exercises . . . 172
25.5 Further reading . . . 173

26 MHD Waves . . . 175
26.1 Waves in a Homogeneous Plasma . . . 175
26.2 Families of MHD Waves . . . 176
26.3 Effect of Propagation Direction . . . 179
26.4 Weak and Strong Field Limits . . . 180
26.5 Shock Waves . . . 181
26.6 In-class worksheet . . . 182
26.7 Exercises . . . 183
26.8 Further reading . . . 183

27 The Magnetorotational Instability . . . 185
27.1 Qualitative Picture . . . 185
27.2 Detailed Calculation . . . 186
27.3 In-class worksheet . . . 189
27.4 Exercises . . . 190
27.5 Further reading . . . 191

22. Introduction to Plasma

22.1 Electromagnetism Refresher

22.1.1 Lorentz Force and Maxwell Equations

Most of the baryonic matter in the Universe is in the form of plasma, a state of matter characterized by large electrical conductivity and charge mobility. Not surprisingly, plasma physics plays an important role in many astrophysical systems. Before we are ready to discuss plasmas, or even before we can precisely state what plasma is, it is a good idea to refresh some key concepts from electromagnetism. We will write the equations of electromagnetism in SI units, as in the rest of these notes.

The starting point for electromagnetism is the observation that some particles, electrons and protons, are endowed with electric charge, which is an intrinsic property, like mass. When two lumps of matter with net charge Q_1 and Q_2 are placed at a distance r, the two experience a force

$$F = \frac{Q_1 Q_2}{4\pi\epsilon_0 r^2}. \tag{22.1}$$

The force is repulsive if $Q_1 Q_2 > 0$ and attractive if $Q_1 Q_2 < 0$. In SI units, charge is measured in coulomb ($1\,\mathrm{C} = 1\,\mathrm{A\,s}$), and the constant ϵ_0, called the vacuum permittivity, is

$$\epsilon_0 = 8.854 \times 10^{-12}\,\mathrm{C^2\,N^{-1}\,m^{-2}}. \tag{22.2}$$

The charge of the electron is

$$e = 1.602176634 \times 10^{-19}\,\mathrm{C}. \tag{22.3}$$

The Coulomb force on particle 2 can also be thought of as the effect on particle 2 of an electrostatic field produced by particle 1. In a system of coordinates centered on particle 1, the electrostatic field produced by particle 1 is

$$\mathbf{E} = \frac{Q_1}{4\pi\epsilon_0 r^2}\hat{\mathbf{r}} \tag{22.4}$$

and the force on particle 2 is given by

$$\mathbf{F} = \frac{Q_1 Q_2}{4\pi\epsilon_0 r^2}\hat{\mathbf{r}} = Q_2\mathbf{E}. \tag{22.5}$$

In general, a charged particle with charge Q moving in an electromagnetic field experiences the so-called Lorentz force:

$$\mathbf{F} = Q(\mathbf{E} + \mathbf{v}\times\mathbf{B}). \tag{22.6}$$

The electric $\mathbf{E}$ and magnetic $\mathbf{B}$ fields are *defined* by Equation (22.6), as the force per unit charge on a particle at rest, and the additional force on a moving particle. Their units are

$$[\mathbf{E}] = \mathrm{N\,C^{-1}} = \mathrm{V\,m^{-1}}, \qquad [\mathbf{B}] = \mathrm{N\,s\,C^{-1}\,m^{-1}} = \mathrm{T\ (tesla)}. \tag{22.7}$$

In astronomy, magnetic fields are customarily measured in gauss, with $1\,\mathrm{G} = 10^{-4}\,\mathrm{T}$. We will give field strengths in tesla, followed by their value in gauss. Note that $\mathbf{E}$ and $c\mathbf{B}$, where c is the speed of light, have the same units. This makes sense, because electric and magnetic fields are different manifestations of a single field: the electromagnetic field, as is revealed by special relativity. The electromagnetic field can be found as solution of the Maxwell's field equations

$$\begin{aligned} \nabla \cdot \mathbf{E} = \frac{q}{\epsilon_0}, \nabla \cdot \mathbf{B} = 0, \\ \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \nabla \times \mathbf{B} = \mu_0 \mathbf{j} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}. \end{aligned} \tag{22.8}$$

In the previous equation, q is the charge density, $\mathbf{j}$ the current density, and $\mu_0 \approx 4\pi \times 10^{-7}\,\mathrm{N\,A^{-2}}$ is the vacuum permeability. The two constants are related to the speed of light by $\mu_0 \epsilon_0 = 1/c^2$. The equations in the first row are called the Gauss law for electric and magnetic fields. The second two equations are the Faraday, or induction, equation (left column), and the Ampère–Maxwell law (right column).

i Gaussian units

Most astrophysics texts and papers, including R. M. Kulsrud [4], write the equations of electromagnetism in Gaussian (CGS) units. In these units there is no ϵ_0 or μ_0, charge is measured in statcoulomb, and $\mathbf{E}$ and $\mathbf{B}$ have the same units. To translate a formula from Gaussian to SI units, replace $B/\sqrt{4\pi}$ with $B/\sqrt{\mu_0}$, e^2 with $e^2/(4\pi\epsilon_0)$, and $\mathbf{E}$ with $\sqrt{4\pi\epsilon_0}\,\mathbf{E}$, and write every $\mathbf{v} \times \mathbf{B}/c$ as $\mathbf{v} \times \mathbf{B}$. For example, the Gaussian plasma frequency $\sqrt{4\pi n e^2/m_e}$ becomes $\sqrt{ne^2/(\epsilon_0 m_e)}$ (Equation (22.50)), and the Gaussian magnetic pressure $B^2/8\pi$ becomes $B^2/2\mu_0$.

22.1.2 Electrostatics

If the fields do not depend on time, the Maxwell equations for $\mathbf{E}$ simplify to

$$\nabla \cdot \mathbf{E} = \frac{q}{\epsilon_0}, \qquad \nabla \times \mathbf{E} = 0. \tag{22.9}$$

The second equation tells us that $\mathbf{E} = -\nabla\Phi$, where Φ is the electrostatic potential. The latter satisfies the Poisson equation

$$\nabla^2 \Phi = -\frac{q}{\epsilon_0}. \tag{22.10}$$

Apart from a minus sign and the constant, $1/\epsilon_0$ instead of $4\pi G$, this is the same equation satisfied by the gravitational potential (Chapter 7).

The force experienced by a charge Q in an electrostatic field is

$$\mathbf{F} = Q\mathbf{E} = -Q\nabla\Phi. \tag{22.11}$$

Example

Consider the electrostatic potential generated by a spherical distribution of charge density q and the force experienced by an electron outside the sphere, as illustrated in Figure 22.1.

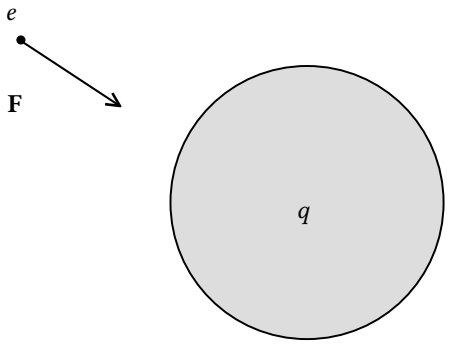


Figure 22.1: Spherical distribution of charge.

The radial component of the force experienced by an electron, which has charge $-e$, at a distance r from a charged sphere, with charge density q, is given by:

$$F_r = e\partial_r \Phi. \tag{22.12}$$

From the Poisson equation

$$\nabla \cdot \nabla \Phi = -\frac{q}{\epsilon_0}. \tag{22.13}$$

Integrating both sides over a sphere of radius r, we find

$$-\frac{1}{\epsilon_0}\int_V q \mathrm{d}V = -\frac{Q}{\epsilon_0} = \int_V \nabla \cdot \nabla \Phi \mathrm{d}V. \tag{22.14}$$

The RHS gives (using the divergence theorem)

$$\int_V \nabla \cdot \nabla \Phi \mathrm{d}V = \int_S \nabla \Phi \cdot \hat{\mathbf{r}} \mathrm{d}S = 4\pi r^2 \partial_r \Phi \Rightarrow \partial_r \Phi = -\frac{Q}{4\pi\epsilon_0 r^2} \Rightarrow F_r = -\frac{Qe}{4\pi\epsilon_0 r^2}. \tag{22.15}$$

So the electron is attracted by the sphere if $Q > 0$, and repelled if $Q < 0$.

Even though electrostatic and gravitational forces have a similar mathematical structure, these forces are very different for two reasons. First, gravity is always attractive, because no forms of matter are known to possess negative mass, while electric charge can be positive or negative and objects of similar charge repel each other. Second, gravity is a very weak force, as we explore with the next example.

Example

Consider a proton on the surface of the Sun. It is subject to an attractive gravitational force:

$$F_g = -\frac{GM_\odot m_p}{R_\odot^2}. \tag{22.16}$$

The proton is also subject to an electrostatic force:

$$F_e = \frac{Q_\odot e}{4\pi\epsilon_0 R_\odot^2}, \tag{22.17}$$

where $Q_\odot \approx \frac{M_\odot}{m_p}\delta e$ is the net charge of the Sun, and $\delta = (N_p - N_e)/N_p$ is the fractional charge imbalance. Gravity and electrostatic forces balance each other when

$$\delta = \frac{4\pi\epsilon_0 G m_p^2}{e^2} \sim 10^{-36}. \tag{22.18}$$

The Sun must be very close to neutral, because even a small charge imbalance would cause particles of the same charge as the Sun to be evaporated from the surface.

The argument we have just made for the Sun applies to most astrophysical systems. This implies that the charge imbalance in astronomical objects must be very small. In other words, they are *quasi-neutral*. As we will see shortly, quasi-neutrality is one of the key characteristics an ionized gas must have to be considered to be plasma. At this point, it is important to remark that, while we will be concerned only with systems that have zero net charge, these systems can still contain electric currents, which give rise to magnetic fields and internal stresses via the Lorentz force.

ℹ Going further: Conservation Laws in Electrodynamics

If we take the divergence of Ampère's law we find

$$\epsilon_0 \frac{\partial \nabla \cdot \mathbf{E}}{\partial t} + \nabla \cdot \mathbf{j} = 0. \tag{22.19}$$

Using Gauss' law

$$\frac{\partial q}{\partial t} + \nabla \cdot \mathbf{j} = 0. \tag{22.20}$$

So the charge density satisfies a continuity equation, just like the mass density for a fluid (Equation (1.15)). In other words, the Maxwell equations imply charge conservation.

It is also possible to show that the Maxwell equations imply conservation of the energy in the electromagnetic field. To see this, consider the work done by the electromagnetic field on a fluid with charge density q and velocity $\mathbf{v}$. The Lorentz force per unit volume is

$$\mathbf{f} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}). \tag{22.21}$$

So the work per unit volume is (using $\mathbf{j} = q\mathbf{v}$ and Ampère's law)

$$w = \mathbf{f} \cdot \mathbf{v} = q\mathbf{E} \cdot \mathbf{v} = \mathbf{E} \cdot \mathbf{j} = \frac{1}{\mu_0}(\nabla \times \mathbf{B}) \cdot \mathbf{E} - \epsilon_0 \mathbf{E} \cdot \frac{\partial \mathbf{E}}{\partial t}. \tag{22.22}$$

Using the vector identity

$$\mathbf{E} \cdot (\nabla \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{E}) - \nabla \cdot (\mathbf{E} \times \mathbf{B}) \tag{22.23}$$

and

$$\mathbf{E} \cdot \frac{\partial \mathbf{E}}{\partial t} = \frac{1}{2}\frac{\partial E^2}{\partial t}, \tag{22.24}$$

we can write

$$\mathbf{E} \cdot \mathbf{j} = \frac{1}{\mu_0}[\mathbf{B} \cdot (\nabla \times \mathbf{E}) - \nabla \cdot (\mathbf{E} \times \mathbf{B})] - \frac{\epsilon_0}{2}\frac{\partial E^2}{\partial t}. \tag{22.25}$$

We can use Faraday's law to get rid of $\nabla \times \mathbf{E}$ to rewrite the last equation as

$$\frac{\partial}{\partial t}\left(\frac{\epsilon_0 E^2}{2} + \frac{B^2}{2\mu_0}\right) + \nabla \cdot \left(\frac{1}{\mu_0}\mathbf{E} \times \mathbf{B}\right) = -\mathbf{E} \cdot \mathbf{j}. \tag{22.26}$$

We can recognize that this is a conservation law in the form

$$\frac{\partial u_{\mathrm{EM}}}{\partial t} + \nabla \cdot \mathbf{S} = -\mathbf{E} \cdot \mathbf{j}. \tag{22.27}$$

The last term is the work done by the electric field, so we can interpret $u_{\mathrm{EM}} = \frac{1}{2}\epsilon_0 E^2 + \frac{B^2}{2\mu_0}$ as the energy in the electromagnetic field and the vector $\mathbf{S} = \frac{1}{\mu_0}\mathbf{E} \times \mathbf{B}$, called the *Poynting flux*, as the flux of electromagnetic energy. Equation (22.26) is also called the Poynting theorem. As we will see in Chapter 23, the equation for energy conservation for a plasma is just the sum of the energy conservation equations for the matter, which we already know as the energy equation (Chapter 6), and the energy conservation equation for the electromagnetic field.

22.1.3 Electromagnetic Waves

The propagation of electromagnetic waves **in vacuum** can be studied with a similar formalism as that used to study sound waves (Chapter 9). Actually, since Maxwell equations are linear, the derivation of the dispersion relation for electromagnetic waves is simpler than that for sound waves. The starting point are the Maxwell equations in vacuum:

$$\nabla \cdot \mathbf{E} = 0, \qquad \nabla \cdot \mathbf{B} = 0,$$
$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \nabla \times \mathbf{B} = \frac{1}{c^2}\frac{\partial \mathbf{E}}{\partial t}. \tag{22.28}$$

We look for solutions in the form

$$\mathbf{E}(t, \mathbf{x}) = \hat{\mathbf{E}}_0 e^{i(\mathbf{k}\cdot\mathbf{x}-\omega t)}, \qquad \mathbf{B}(t, \mathbf{x}) = \hat{\mathbf{B}}_0 e^{i(\mathbf{k}\cdot\mathbf{x}-\omega t)}. \tag{22.29}$$

Inserting this ansatz into the first row of Equation (22.28), we obtain

$$\nabla \cdot \mathbf{E} = 0 \Rightarrow i\mathbf{k} \cdot \hat{\mathbf{E}}_0 e^{i(\mathbf{k}\cdot\mathbf{x}-\omega t)} = 0 \Rightarrow \mathbf{k} \perp \hat{\mathbf{E}}_0 \tag{22.30}$$

and, similarly, $\mathbf{k} \perp \hat{\mathbf{B}}_0$. This means that, differently from sound waves, electromagnetic waves in vacuum are transverse, meaning that $\mathbf{E}$ and $\mathbf{B}$ are perpendicular to the direction of propagation of the wave.

From Faraday's law, we obtain that

$$i\omega\hat{\mathbf{B}}_0 e^{i(\mathbf{k}\cdot\mathbf{x}-\omega t)} = i\mathbf{k} \times \hat{\mathbf{E}}_0 e^{i(\mathbf{k}\cdot\mathbf{x}-\omega t)} \Rightarrow \omega\hat{\mathbf{B}}_0 = \mathbf{k} \times \hat{\mathbf{E}}_0. \tag{22.31}$$

This implies that $\hat{\mathbf{E}}_0 \perp \hat{\mathbf{B}}_0$. From Ampère's law, we obtain

$$\mathbf{k} \times \hat{\mathbf{B}}_0 = -\frac{\omega}{c^2}\hat{\mathbf{E}}_0. \tag{22.32}$$

Substituting Equation (22.32) into Equation (22.31), we obtain (using the properties of the double vector product)

$$\omega\hat{\mathbf{B}}_0 = -\frac{c^2}{\omega}\mathbf{k} \times (\mathbf{k} \times \hat{\mathbf{B}}_0) = \cancel{-\frac{c^2}{\omega}\mathbf{k}(\mathbf{k} \cdot \hat{\mathbf{B}}_0)} + \frac{c^2}{\omega}k^2\hat{\mathbf{B}}_0 = \frac{c^2}{\omega}k^2\hat{\mathbf{B}}_0 \Rightarrow \boxed{\omega^2 = c^2k^2}. \tag{22.33}$$

So electromagnetic waves satisfy the same dispersion relation as sound waves (Equation (9.18)), but with c, the speed of light, instead of the sound speed c_s.

22.2 Plasma

22.2.1 What is a Plasma?

Consider a quasi-neutral gas made of electrons and ions. Assume that, due to a thermal fluctuation, a small overdensity of electrons δn is created in a region with characteristic size L. How large can this fluctuation be?

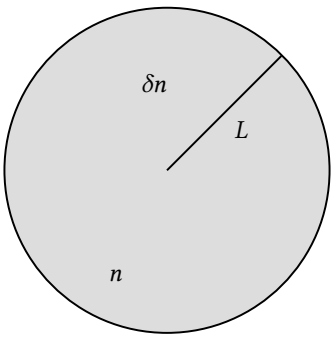


Figure 22.2: Schematic representation of an overdensity of electrons.

The electrostatic potential created by this overdensity is of order

$$\delta\Phi \sim \frac{L^2 e\delta n}{\epsilon_0}. \tag{22.34}$$

The potential energy of each electron is

$$e\delta\Phi \sim \frac{L^2 e^2 \delta n}{\epsilon_0}. \tag{22.35}$$

Their thermal energy is $\sim k_B T$ (in thermodynamic equilibrium). Thermal fluctuations should have potential energy comparable to the thermal energy, so

$$\frac{L^2 e^2 \delta n}{\epsilon_0} \sim k_B T \Rightarrow \frac{\delta n}{n} \sim \frac{1}{L^2}\left(\frac{\epsilon_0 k_B T}{n e^2}\right) =: \left(\frac{\lambda_D}{L}\right)^2, \tag{22.36}$$

where n is the density of free electrons (we will also write it as n_e) and we have introduced the *Debye length*

$$\boxed{\lambda_D = \sqrt{\frac{\epsilon_0 k_B T}{n e^2}}.} \tag{22.37}$$

The Debye length is a characteristic length scale for charge fluctuations in the gas. Over scales $L \gg \lambda_D$, one can assume that the medium is neutral, *if the density of the gas is sufficiently high.*

To see this, consider again a uniform gas of electrons and ions. The ion masses are of order $Am_p \approx 1836 A m_e$, so if ions and electrons have on average the same kinetic energy, as would be expected in equilibrium, $v_{\text{ions}} \ll v_e$. To a first approximation, we can consider the ions to be static and immersed in a sea of electrons. If electrons are abundant, each ion will then be surrounded by a sea of electrons, which effectively shield its charge hiding it from the other ions, as shown in Figure 22.3.

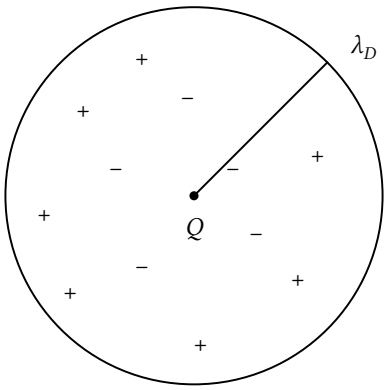


Figure 22.3: Electron cloud around an ion of charge Q.

It turns out that, in this situation, the electrostatic potential around the ion is

$$\Phi(r) = \frac{Q}{4\pi\epsilon_0 r} \mathrm{e}^{-r/\lambda_D}, \tag{22.38}$$

where λ_D is the Debye length (see the box below). In other words, the Debye length is also the characteristic screening length. However, screening can only take place if electrons are sufficiently mobile to screen each ion. To quantify this, we define the *plasma parameter*:

$$\Lambda = \frac{4}{3}\pi n \lambda_D^3 = \frac{4\pi}{3e^3}(\epsilon_0 k_B T)^{3/2} n^{-1/2}. \tag{22.39}$$

The plasma parameter is just the number of electrons in each Debye sphere. We say that an ionized gas for which $\lambda_D \ll L$ and $\Lambda \gg 1$ is a *plasma*. Typical plasma parameters for astrophysical systems are given in Table 22.1. As we can see, astrophysical plasmas span a tremendous range in scales.

Table 22.1: Temperature, free electron density, Debye length, and plasma parameters for various systems. The electrons in protoneutron stars and white dwarfs are degenerate (and, in protoneutron stars, relativistic), so for these systems the values computed with the classical formulas are only indicative.

	T [K]	n_e^{free} [cm^{-3}]	λ_D [m]	Λ	plasma?
The Sun (core)	10^7	10^{26}	2×10^{-11}	4.4	marginal
Warm ISM	10^4	1	7	1.4×10^9	yes
Molecular clouds	10	10^{-4}	20	4×10^6	yes
Intracluster medium	10^7	10^{-3}	7×10^3	1.4×10^{15}	yes
Protoneutron star	10^{11}	2×10^{38}	1.5×10^{-15}	3.1	marginal
White dwarf	10^5	6×10^{29}	3×10^{-14}	5.6×10^{-5}	no

i Going further: Debye Screening

We can derive Equation (22.38) as follows. Consider an ion of charge Q at the origin, surrounded by electrons with temperature T and average density n_e, and by other ions, which we assume to have uniform charge density en_e, so that the plasma is neutral on average. In thermal equilibrium, the electrons follow the Boltzmann distribution. An electron has potential energy $-e\Phi$, so

$$n(r) = n_e \mathrm{e}^{e\Phi/k_B T} \approx n_e \left(1 + \frac{e\Phi}{k_B T}\right), \tag{22.40}$$

where we have assumed $|e\Phi| \ll k_B T$, which is true sufficiently far from the ion. The charge density is then

$$q = en_e - en(r) = -\frac{n_e e^2}{k_B T}\Phi, \tag{22.41}$$

and the Poisson equation (Equation (22.10)) becomes

$$\nabla^2 \Phi = \frac{n_e e^2}{\epsilon_0 k_B T}\Phi = \frac{\Phi}{\lambda_D^2}. \tag{22.42}$$

For a spherically symmetric potential, $\nabla^2\Phi = \frac{1}{r}\frac{\mathrm{d}^2(r\Phi)}{\mathrm{d}r^2}$, so $r\Phi$ satisfies $\frac{\mathrm{d}^2(r\Phi)}{\mathrm{d}r^2} = \frac{r\Phi}{\lambda_D^2}$. The solution that does not grow at large r is $r\Phi = A\mathrm{e}^{-r/\lambda_D}$. Close to the ion, the potential must be that of the ion alone, $\Phi \approx Q/(4\pi\epsilon_0 r)$, so $A = Q/(4\pi\epsilon_0)$, and we recover Equation (22.38).

22.2.2 When is a Plasma a Fluid?

In Chapter 1 we have seen that a gas can be treated as a fluid if the mean free path λ of its particles is small compared to the scales of interest. In a plasma, the particles interact via the Coulomb force. Two particles are strongly deflected if they come so close that their electrostatic energy is comparable to their thermal energy, $e^2/(4\pi\epsilon_0 r) \sim k_B T$. This gives a cross section $\sigma \sim \pi[e^2/(4\pi\epsilon_0 k_B T)]^2$, and a mean free path $\lambda \sim 1/(n\sigma)$. Many weak collisions at larger distances make the particles lose their direction somewhat faster, and a more careful calculation gives

$$\lambda \approx 100\,\mathrm{m}\left(\frac{T}{1\,\mathrm{K}}\right)^2 \left(\frac{n}{1\,\mathrm{cm}^{-3}}\right)^{-1}. \tag{22.43}$$

Note that λ increases very quickly with temperature: hot plasmas are almost collisionless.

Magnetic fields change this picture. A particle with charge Q and mass m moving with velocity v perpendicular to a magnetic field B experiences a Lorentz force $|Q|vB$ perpendicular to its velocity. So it moves on a circle around the field lines, with radius r_L given by

$$\frac{mv^2}{r_L} = |Q|vB \Rightarrow r_L = \frac{mv}{|Q|B}. \tag{22.44}$$

r_L is called the *Larmor radius*, or *gyroradius*. In other words, particles cannot move freely across the magnetic field lines, even if they never collide. On scales much larger than r_L, a magnetized plasma behaves like a fluid in the directions perpendicular to the magnetic field. Along the field lines, instead, particles can travel a distance $\sim \lambda$. This is why magnetohydrodynamics (Chapter 23) works well even in plasmas in which collisions are rare. However, processes along the field lines, such as heat conduction, can be very different from those in an ordinary gas.

Example

Consider the warm interstellar medium, with $T \sim 10^4\,\mathrm{K}$, $n \sim 1\,\mathrm{cm}^{-3}$, and $B \sim 0.3\,\mathrm{nT}$ ($3\,\mu\mathrm{G}$). From Equation (22.43), $\lambda \sim 10^{10}\,\mathrm{m}$, which is much smaller than the scales of interest, $\sim 1\,\mathrm{pc} \approx 3 \times 10^{16}\,\mathrm{m}$. A thermal proton has velocity $v \sim \sqrt{k_B T/m_p} \approx 9\,\mathrm{km\,s}^{-1}$, so its Larmor radius is $r_L \approx 300\,\mathrm{km}$, even smaller.

The solar wind close to the Earth is different. It has $T \sim 10^5\,\mathrm{K}$ and $n \sim 5\,\mathrm{cm}^{-3}$, so $\lambda \sim 2 \times 10^{11}\,\mathrm{m}$, about the distance between the Earth and the Sun. The solar wind is essentially collisionless! However, the magnetic field is $B \sim 5\,\mathrm{nT}$ ($5 \times 10^{-5}\,\mathrm{G}$), so a thermal proton has $r_L \approx 60\,\mathrm{km}$. On large scales, the solar wind can still be described as a magnetized fluid.

22.2.3 Cold Plasma Approximation

The cold plasma approximation is used to study the propagation of electromagnetic waves, particularly radio waves, in the interstellar medium. As the name suggests, in the cold plasma approximation we ignore the thermal motion of the electrons. We also take the ions to be stationary. Let's consider the propagation of an electromagnetic wave through such a medium, as illustrated in Figure 22.4.

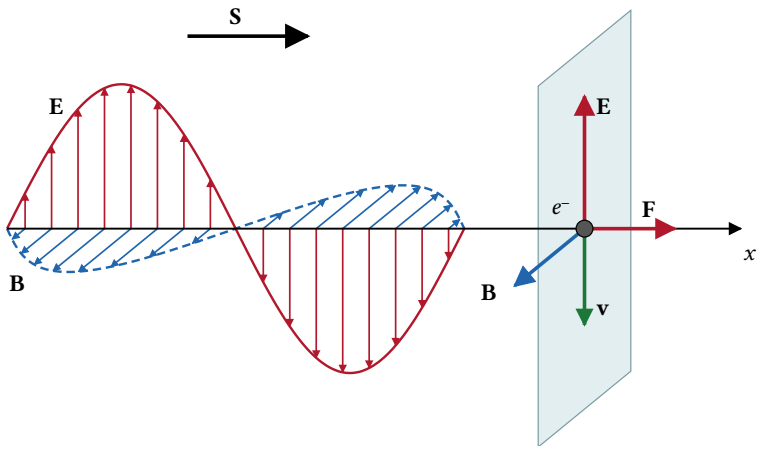


Figure 22.4: Lorentz force experienced by an electron due to a passing electromagnetic wave.

The Lorentz force experienced by an electron is (assuming $v \ll c$):

$$\mathbf{F} = -e(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \approx -e\mathbf{E}. \tag{22.45}$$

So, if we take $\mathbf{E} = \hat{\mathbf{E}}_0 \mathrm{e}^{\mathrm{i}(\mathbf{k}\cdot\mathbf{x} - \omega t)}$ and $\mathbf{v} = \hat{\mathbf{v}}_0 \mathrm{e}^{\mathrm{i}(\mathbf{k}\cdot\mathbf{x} - \omega t)}$, we have

$$\hat{\mathbf{v}}_0 = \frac{e\hat{\mathbf{E}}_0}{\mathrm{i}\omega m_e}. \tag{22.46}$$

The induced current density in the plasma is

$$\hat{\mathbf{j}} = -ne\hat{\mathbf{v}}_0 = \left(\frac{\mathrm{i}ne^2}{\omega m_e}\right)\hat{\mathbf{E}}_0 =: \sigma\hat{\mathbf{E}}_0, \tag{22.47}$$

where we have introduced the conductivity σ. Charge conservation is

$$\frac{\partial q}{\partial t} + \nabla\cdot\mathbf{j} = 0 \Rightarrow -\mathrm{i}\omega\hat{q}_0 + \mathrm{i}\mathbf{k}\cdot\hat{\mathbf{j}}_0 = 0 \Rightarrow \hat{q}_0 = \frac{\mathbf{k}\cdot\hat{\mathbf{j}}_0}{\omega} = \frac{\sigma}{\omega}\mathbf{k}\cdot\hat{\mathbf{E}}_0. \tag{22.48}$$

Gauss' law reads

$$\mathrm{i}\mathbf{k}\cdot\hat{\mathbf{E}}_0 = \frac{\hat{q}_0}{\epsilon_0} \Rightarrow \mathrm{i}\mathbf{k}\cdot\left(1 - \frac{\sigma}{\mathrm{i}\omega\epsilon_0}\right)\hat{\mathbf{E}}_0 = 0. \tag{22.49}$$

Introducing the dielectric constant

$$\begin{aligned}\epsilon &= 1 - \frac{\sigma}{\mathrm{i}\omega\epsilon_0} = 1 - \left(\frac{\omega_p}{\omega}\right)^2,\\ \omega_p &= \sqrt{\frac{ne^2}{\epsilon_0 m_e}}\\ &\approx 5.6\times10^4\,\mathrm{s}^{-1}\left(\frac{n}{1\,\mathrm{cm}^{-3}}\right)^{1/2}.\end{aligned} \tag{22.50}$$

To evaluate ω_p, we have used $1\,\mathrm{cm}^{-3} = 10^6\,\mathrm{m}^{-3}$. The corresponding frequency is $f_p = \omega_p/2\pi \approx 9\,\mathrm{kHz}\,(n/1\,\mathrm{cm}^{-3})^{1/2}$. ω_p is called the *plasma frequency*, and ϵ (not to be confused with the internal energy ε, or with the vacuum permittivity ϵ_0) is the dielectric constant. The plasma frequency has a simple physical meaning: if the electrons are displaced with respect to the ions, the electric field pulls them back, and they oscillate around their equilibrium positions with angular frequency ω_p (Exercise 22.1). Note also that $\lambda_D = v_{\mathrm{th}}/\omega_p$, where $v_{\mathrm{th}} = \sqrt{k_B T/m_e}$ is the thermal velocity of the electrons. In other words, the Debye length is the distance traveled by a thermal electron during one plasma oscillation.

With these definitions, we find that the Maxwell equations reduce to

$$\begin{aligned}\mathrm{i}\mathbf{k}\cdot\epsilon\hat{\mathbf{E}}_0 = 0, &\qquad \mathrm{i}\mathbf{k}\cdot\hat{\mathbf{B}}_0 = 0,\\ \mathrm{i}\mathbf{k}\times\hat{\mathbf{E}}_0 = \mathrm{i}\omega\hat{\mathbf{B}}_0, &\quad \mathrm{i}\mathbf{k}\times\hat{\mathbf{B}}_0 = -\mathrm{i}\frac{\omega}{c^2}\epsilon\hat{\mathbf{E}}_0.\end{aligned} \tag{22.51}$$

These can be solved in the same way as the vacuum Maxwell equations to find the dispersion relation for electromagnetic waves in a cold plasma

$$c^2k^2 = \epsilon\omega^2 \Rightarrow k = \frac{\omega}{c}\sqrt{1 - \left(\frac{\omega_p}{\omega}\right)^2} \Rightarrow \boxed{k = \frac{1}{c}\sqrt{\omega^2 - \omega_p^2}.} \tag{22.52}$$

For example, if the wave is propagating in the z-direction, $\mathbf{k} = k\hat{\mathbf{z}}$. Then

- If $\omega < \omega_p$, then k is imaginary and the wave cannot propagate. Its amplitude decays exponentially over a distance $\sim c/\omega_p$:

$$\mathbf{E} = \hat{\mathbf{E}}_0 \mathrm{e}^{-\mathrm{i}\omega t}\mathrm{e}^{-\frac{z}{c}\sqrt{\omega_p^2-\omega^2}}. \tag{22.53}$$

  Note that this is not due to dissipation. The electrons respond to the wave fast enough to cancel its electric field, so the wave is *reflected* by the plasma. This is the same mathematics we found for sound waves below the acoustic cutoff frequency (Equation (10.21)).

- If $\omega > \omega_p$ the wave can propagate, but different frequencies propagate at different speeds:

$$v_g = \frac{\partial\omega}{\partial k} = c\sqrt{1 - \frac{\omega_p^2}{\omega^2}}. \tag{22.54}$$

This leads to the phenomenon of dispersion, which can be used to measure the column density of free electrons between us and sources of radio pulses, such as pulsars and fast radio bursts, and hence to estimate their distance, as illustrated in Figure 22.5.

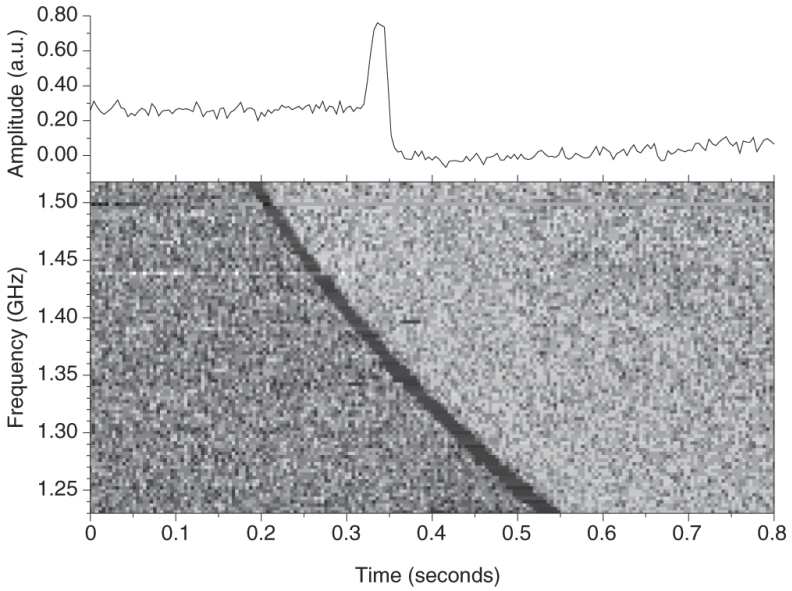


Figure 22.5: Different arrival time of a fast radio burst at different frequencies as a result of dispersion in the ISM: the first fast radio burst, FRB 010724. From D. R. Lorimer [12] (figure by E. Keane); © Springer Nature, reproduced for non-commercial educational use and not covered by the license of these notes.

Example

Consider the ionosphere of the Earth, which has $n_e \sim 10^6\ \mathrm{cm}^{-3}$. Its plasma frequency is $f_p \approx 9\,\mathrm{MHz}$. AM radio, with $f \sim 1\,\mathrm{MHz}$, is reflected by the ionosphere. This is why AM stations can be received beyond the horizon, especially at night. FM radio and TV, with $f \sim 100\,\mathrm{MHz}$, pass through the ionosphere. For the same reason, radio astronomy at frequencies below $\sim 10\,\mathrm{MHz}$ is only possible from space.

22.3 In-class worksheet

Fast radio bursts are thought to be millisecond-long broad-band radio flares, produced during particularly violent magnetar flares, and possibly other astrophysical events. As the radio waves propagate from the source to Earth through the interstellar medium, the signal is dispersed, as shown in Figure 22.6:

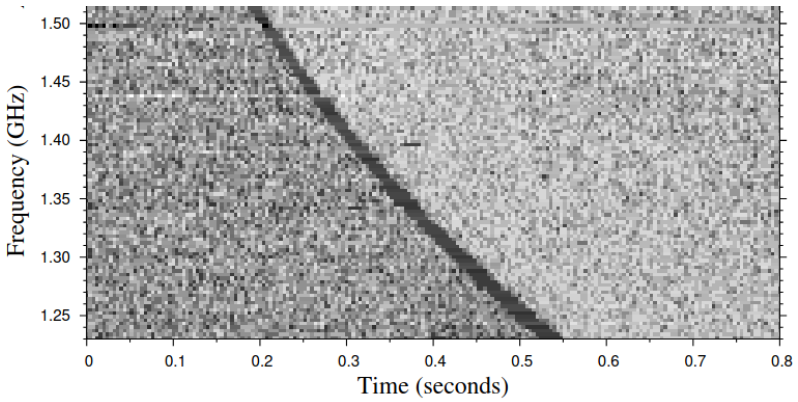


Figure 22.6: Time-frequency signal for FRB 010724. Adapted from D. R. Lorimer [12] (figure by E. Keane); © Springer Nature, reproduced for non-commercial educational use and not covered by the license of these notes.

The time of arrival of the peak of the burst at *angular* frequency ω, $t_p(\omega)$ is given by

$$t_p = \int_0^d \frac{\mathrm{d}s}{v_g(\omega)}, \tag{22.55}$$

where d is the distance to the source and $v_g(\omega)$ is the group velocity at frequency ω. The approximate arrival times for FRB 010724 are tabulated below:

t_p[s]	0.21	0.27	0.31	0.36	0.44
f[GHz]	1.50	1.44	1.39	1.35	1.30

1. Assume that the interstellar medium is a cold plasma, with plasma frequency

$$\omega_p = \sqrt{\frac{ne^2}{\epsilon_0 m_e}}, \tag{22.56}$$

where n is the number density of free electrons, and that $\omega \gg \omega_p$. Show that

$$t_p(\omega) \approx \frac{d}{c} + \frac{1}{2c\omega^2}\int_0^d \omega_p^2 \mathrm{d}s. \tag{22.57}$$

2. Show that

$$\frac{\mathrm{d}t_p}{\mathrm{d}\omega} = -\frac{e^2}{\epsilon_0 c m_e \omega^3}\mathrm{DM}, \tag{22.58}$$

where DM is the dispersion measure:

$$\mathrm{DM} = \int_0^d n\mathrm{d}s. \tag{22.59}$$

3. Compute the DM for FRB 010724. In the direction of this source, the DM due to free electrons within the Milky Way is estimated to be $\mathrm{DM} = 25\,\mathrm{cm}^{-3}$ pc. Is this source extragalactic? [Remember that $\omega = 2\pi f$. It is convenient to plot t_p against $1/f^2$.]

22.4 Exercises

Exercise 22.1 (Plasma oscillations and the Debye length). Consider a slab of plasma with uniform electron density n_e. The ions are at rest and have uniform charge density en_e. At $t = 0$, all electrons in the slab are displaced by a small distance x in the direction perpendicular to the slab.

1. Show that the displacement creates an electric field $E = n_e e x/\epsilon_0$ inside the slab.
2. Show that the electrons oscillate around their equilibrium positions with angular frequency $\omega_p = \sqrt{n_e e^2/(\epsilon_0 m_e)}$.
3. Show that $\lambda_D = v_{\mathrm{th}}/\omega_p$, where $v_{\mathrm{th}} = \sqrt{k_B T/m_e}$. Interpret this result.
4. Compute ω_p, $f_p = \omega_p/2\pi$, and λ_D for the ionosphere of the Earth ($n_e \sim 10^6\,\mathrm{cm}^{-3}$, $T \sim 10^3$ K) and for the warm interstellar medium ($n_e \sim 1\,\mathrm{cm}^{-3}$, $T \sim 10^4$ K).

22.5 Further reading

- C. Clarke and B. Carswell [1], Chapter 13 (plasmas).

23. The MHD Equations

23.1 Lorentz Force

These notes present a concise derivation of the equations of magnetohydrodynamics (MHD) and a discussion of the underlying assumptions. We consider a quasi-neutral gas of ions and electrons immersed in a magnetic field. Let $\mathbf{w}_e$ and $\mathbf{w}_i$ be the drift velocities of electrons and ions, respectively. The Lorentz force experienced by each electron is

$$\mathbf{F}_e = -e\mathbf{w}_e \times \mathbf{B}. \tag{23.1}$$

The force experienced by the ions is

$$\mathbf{F}_i = Ze\mathbf{w}_i \times \mathbf{B}. \tag{23.2}$$

So the total force per unit volume is

$$\mathbf{f} = n_e\mathbf{F}_e + n_i\mathbf{F}_i = e(n_i Z\mathbf{w}_i - n_e\mathbf{w}_e) \times \mathbf{B} =: \mathbf{j} \times \mathbf{B}. \tag{23.3}$$

Here $\mathbf{j}$ is the electric current density, measured in $\mathrm{A\,m^{-2}}$. Note that we have not included the electric force. The electric force per unit volume is $q\mathbf{E}$, where q is the charge density, and it is negligible because the plasma is quasi-neutral, $q \approx 0$ (Chapter 22).

The equations of MHD are just the "usual" hydrodynamics equations with the addition of the Lorentz force on the RHS:

$$\begin{aligned} \frac{\mathrm{d}\rho}{\mathrm{d}t} &= -\rho\nabla\cdot\mathbf{v}, \\ \rho\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} &= \boxed{\mathbf{j}\times\mathbf{B}} - \nabla p, \\ \rho\frac{\mathrm{d}\varepsilon}{\mathrm{d}t} &= -p\nabla\cdot\mathbf{v}. \end{aligned} \tag{23.4}$$

These equations look deceptively simple. The real complication is that $\mathbf{j}$ and $\mathbf{B}$ are not prescribed, but are self-consistently produced by the fluid. We have also implicitly made the following assumption:

Assumption 23.1 (Negligible drift momentum). *The drift motion of the charged particles in the plasma that gives rise to the current is sufficiently small that we can neglect its contribution to the macroscopic momentum $\rho\mathbf{v}$. In other words, in a frame in which $\mathbf{v} = 0$, we have $m_i n_i \mathbf{w}_i + m_e n_e \mathbf{w}_e \approx 0$.*

This assumption is satisfied in most situations encountered in terrestrial and astrophysical applications, because the currents are sufficiently weak and the charge densities sufficiently large that we can treat the plasma as being a single fluid. This can be violated in very tenuous plasmas, for example in the magnetosphere of dying pulsars, in which the field and spin have decayed to the point that pair production is no longer efficient. In this case, one needs to go beyond magnetohydrodynamics and use the multi-fluid magnetohydrodynamics equations, or employ a fully kinetic treatment. Another situation in which this assumption can fail is if the degree of ionization of the plasma is small, so there is drift between the charged component of the gas and the neutrals, *ambipolar diffusion*. As we will see in Chapter 24, ambipolar diffusion plays an important role, for example, in star formation.

23.2 Ohm's Law

We need two more equations, in addition to the equation of state, to close Equation (23.4). A first equation is the equation prescribing how currents $\mathbf{j}$ arise in the plasma. The simplest model for the current is to assume that it is proportional to the electric field.

Assumption 23.2 (Ohm's law). *The electric current in the plasma* $\mathbf{j}$ *is proportional to the electric field* $\mathbf{E}'$ *in the frame of the plasma (in which* $\mathbf{v} = 0$*):* $\mathbf{j} = \sigma \mathbf{E}'$.

The quantity σ is called the *electrical conductivity*, it is the inverse of the *electrical resistivity* $1/\sigma$, which you might be familiar with from electromagnetism. In SI units[10]

$$[\mathbf{E}] = \mathrm{V\,m^{-1}}, \qquad [\mathbf{j}] = \mathrm{A\,m^{-2}}, \tag{23.5}$$

so the units of electrical conductivity are

$$[\sigma] = \mathrm{A\,V^{-1}\,m^{-1}} = \mathrm{S\,m^{-1}}. \tag{23.6}$$

Here $\mathrm{S} = \mathrm{A\,V^{-1}}$ is the siemens, the inverse of the ohm. Again, note that in SI units the current is an independent quantity, measured in ampere, so the Maxwell and MHD equations contain the constants ϵ_0 and μ_0. Be very careful not to mix Gaussian and SI units, because the equations are different in the two cases (Chapter 22).

One subtlety with the new assumption is what we mean for the "electric field in the frame of the plasma". The hydrodynamics equations are invariant with respect to Galilean transformations, that is, if $\mathbf{u}$ is a velocity in the Eulerian frame, then in the fluid frame we have

$$\mathbf{u}' = \mathbf{u} - \mathbf{v}. \tag{23.7}$$

On the other hand, Maxwell equations are invariant with respect to Lorentz transformations. As such, Maxwell equations and the equations of non-relativistic hydrodynamics are fundamentally incompatible. To be able to put them together in a consistent way, we need to assume that $\mathbf{E}$ and $\mathbf{B}$ transform according to Lorentz transformations in the nonrelativistic limit.

Assumption 23.3 (Nonrelativistic limit). *We assume* $v \ll c$, $|\mathbf{E}| \ll c|\mathbf{B}|$, *and* $\left|\epsilon_0 \frac{\partial \mathbf{E}}{\partial t}\right| \ll |\mathbf{j}|$. *The second two assumptions are needed to ensure that charges are not relativistic and the plasma quasi-neutral.*

Under these assumptions, it is possible to show that

$$\mathbf{B}' = \mathbf{B}, \qquad \mathbf{E}' = \mathbf{E} + \mathbf{v} \times \mathbf{B}, \qquad \mathbf{j}' = \mathbf{j}. \tag{23.8}$$

Accordingly, Ohm's law becomes

$$\mathbf{j}' = \sigma \mathbf{E}' \Rightarrow \mathbf{j} = \sigma(\mathbf{E} + \mathbf{v} \times \mathbf{B}). \tag{23.9}$$

In Ampère's law,

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{j} + \frac{1}{c^2} \frac{\partial \mathbf{E}}{\partial t}, \tag{23.10}$$

where we have used $\mu_0 \epsilon_0 = 1/c^2$, we can drop the displacement current (Assumption 23.3). This makes sense, because $E \sim vB$, so if the fields change over a length L and a time $T \sim L/v$, the displacement current is of order $vB/(c^2 T) \sim (v/c)^2 B/L$, while $\nabla \times \mathbf{B}$ is of order B/L. So we obtain

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{j}. \tag{23.11}$$

Then the Lorentz force becomes

$$\boxed{\mathbf{f} = \mathbf{j} \times \mathbf{B} = \frac{1}{\mu_0} (\nabla \times \mathbf{B}) \times \mathbf{B}.} \tag{23.12}$$

With this, we have completely eliminated the current from the MHD equations:

[10] In Gaussian units the conductivity has units of $\mathrm{s^{-1}}$. To convert it to SI units, multiply by $4\pi\epsilon_0 \approx 1.11 \times 10^{-10}\,\mathrm{F\,m^{-1}}$, so $1\,\mathrm{s^{-1}}$ corresponds to $1.11 \times 10^{-10}\,\mathrm{S\,m^{-1}}$. The magnetic diffusivity, which in Gaussian units is $c^2/(4\pi\sigma)$, has the same value in both systems.

$$\boxed{\rho \frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = \frac{1}{\mu_0}(\nabla \times \mathbf{B}) \times \mathbf{B} - \nabla p.} \tag{23.13}$$

We still need another equation for $\mathbf{B}$ to completely close the MHD equations.

23.3 Faraday's Law

The missing equation is given by Faraday's law, also known as the induction equation:

$$\frac{\partial \mathbf{B}}{\partial t} = -\nabla \times \mathbf{E}. \tag{23.14}$$

Here, we use again Ohm's law to eliminate $\mathbf{E}$:

$$\frac{1}{\sigma}\mathbf{j} = \mathbf{E} + \mathbf{v} \times \mathbf{B} \Rightarrow \mathbf{E} = \frac{1}{\sigma}\mathbf{j} - \mathbf{v} \times \mathbf{B}. \tag{23.15}$$

We use again Ampère's law to remove $\mathbf{j} = \frac{1}{\mu_0}\nabla \times \mathbf{B}$:

$$\mathbf{E} = \frac{1}{\mu_0 \sigma}\nabla \times \mathbf{B} - \mathbf{v} \times \mathbf{B} =: \eta \nabla \times \mathbf{B} - \mathbf{v} \times \mathbf{B}. \tag{23.16}$$

Inserting this back into the induction equation yields:

$$\begin{aligned} \frac{\partial \mathbf{B}}{\partial t} &= -\nabla \times \mathbf{E} \\ &= \nabla \times (\mathbf{v} \times \mathbf{B}) - \nabla \times (\eta \nabla \times \mathbf{B}). \end{aligned} \tag{23.17}$$

If η **is constant**, then

$$\nabla \times (\nabla \times \mathbf{B}) = \nabla(\nabla \cdot \mathbf{B}) - \nabla^2 \mathbf{B}, \tag{23.18}$$

so we find

$$\boxed{\frac{\partial \mathbf{B}}{\partial t} = \nabla \times (\mathbf{v} \times \mathbf{B}) + \eta \nabla^2 \mathbf{B}.} \tag{23.19}$$

The coefficient $\eta = 1/(\mu_0 \sigma)$ has units of $\mathrm{m}^2\,\mathrm{s}^{-1}$ and is called the *magnetic diffusivity*. Note that some texts use η for the resistivity $1/\sigma$ instead. The magnetic diffusivity is extremely small in most astrophysical systems.

Example

Consider the warm interstellar medium, with $T \approx 10^4\,\mathrm{K}$. Its magnetic diffusivity is $\eta \approx 500\,\mathrm{m}^2\,\mathrm{s}^{-1}$ (Equation (24.27)). Consider a region of size $L \sim 1\,\mathrm{pc}$ in which the gas moves with velocity $v \sim 10\,\mathrm{km\,s}^{-1}$. The ratio between the time it takes for the magnetic field to diffuse over the region, L^2/η, and the time it takes for the gas to cross it, L/v, is

$$\mathrm{R}_M = \frac{Lv}{\eta} \sim 6 \times 10^{17}. \tag{23.20}$$

This is the magnetic Reynolds number, which we will discuss in Chapter 24. For comparison, copper has a conductivity of about $6 \times 10^7\,\mathrm{S\,m}^{-1}$, so its magnetic diffusivity is $1/(\mu_0\sigma) \approx 1.3 \times 10^{-2}\,\mathrm{m}^2\,\mathrm{s}^{-1}$. A copper bar of length 1 m moving with velocity $1\,\mathrm{m\,s}^{-1}$ has $\mathrm{R}_M \sim 100$. In other words, resistivity is negligible in astrophysics, but not in the laboratory. Even liquid-metal experiments designed to study the generation of magnetic fields struggle to reach $\mathrm{R}_M \sim 10^3$.

This is the last equation. The full set of MHD equations is

$$\begin{aligned}
\frac{\mathrm{d}\rho}{\mathrm{d}t} &= -\rho\nabla\cdot\mathbf{v},\\
\rho\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} &= \frac{1}{\mu_0}(\nabla\times\mathbf{B})\times\mathbf{B} - \nabla p - \rho\nabla\Psi,\\
\rho\frac{\mathrm{d}\varepsilon}{\mathrm{d}t} &= -p\nabla\cdot\mathbf{v} + \frac{j^2}{\sigma},\\
\frac{\partial\mathbf{B}}{\partial t} &= \nabla\times(\mathbf{v}\times\mathbf{B}) + \eta\nabla^2\mathbf{B},\\
\nabla\cdot\mathbf{B} &= 0,
\end{aligned} \tag{23.21}$$

where $\mathbf{j} = \nabla\times\mathbf{B}/\mu_0$. The last term in the energy equation, $j^2/\sigma = \mu_0\eta j^2$, describes the heating of the plasma by the electric currents, also known as Joule (or Ohmic) heating. It vanishes in ideal MHD. The last equation, $\nabla\cdot\mathbf{B} = 0$, is a constraint on the initial data. If it is satisfied at $t = 0$, the induction equation guarantees that it is satisfied at all times, because the divergence of a curl vanishes (see the in-class worksheet).

The first three equations can be converted to the Eulerian frame with the usual substitution

$$\frac{\mathrm{d}}{\mathrm{d}t} \to \frac{\partial}{\partial t} + (\mathbf{v}\cdot\nabla). \tag{23.22}$$

If $\eta = 0$, these are known as the ideal-MHD equations. If $\eta > 0$, these are called the resistive-MHD equations.

23.4 Conservation Laws

It is possible to rearrange the terms of the momentum equation to obtain an equation that shows the conservation of the linear momentum:

$$\boxed{\begin{aligned}&\frac{\partial(\rho\mathbf{v})}{\partial t}\\ &+\nabla\cdot\left[\rho\mathbf{v}\otimes\mathbf{v} + \left(p + \frac{B^2}{2\mu_0}\right)\mathbf{I}\right.\\ &\qquad\left. - \frac{\mathbf{B}\otimes\mathbf{B}}{\mu_0}\right] = \mathbf{f},\end{aligned}} \tag{23.23}$$

where $\mathbf{f}$ is some external force, e.g., gravity. To derive this equation, we use the vector identity

$$(\nabla\times\mathbf{B})\times\mathbf{B} = (\mathbf{B}\cdot\nabla)\mathbf{B} - \nabla\left(\frac{B^2}{2}\right) \tag{23.24}$$

and the fact that, since $\nabla\cdot\mathbf{B} = 0$,

$$\nabla\cdot(\mathbf{B}\otimes\mathbf{B}) = (\mathbf{B}\cdot\nabla)\mathbf{B} + \mathbf{B}(\nabla\cdot\mathbf{B}) = (\mathbf{B}\cdot\nabla)\mathbf{B}. \tag{23.25}$$

So the Lorentz force can be written as the divergence of a tensor, $\frac{1}{\mu_0}(\nabla\times\mathbf{B})\times\mathbf{B} = -\nabla\cdot\left[\frac{B^2}{2\mu_0}\mathbf{I} - \frac{\mathbf{B}\otimes\mathbf{B}}{\mu_0}\right]$, and the rest of the derivation is the same as in Chapter 6. Here, we limit ourselves to discuss a couple of the consequences of this equation. The first is that the magnetic field does not contribute to the momentum $\rho\mathbf{v}$, consistently with Assumption 23.1. This is not the case in relativistic MHD, in which the inertia of the electromagnetic field is also taken into account. Second, we see that the Lorentz force splits into two components: one that resembles an additional pressure term $B^2/2\mu_0$, plus a term proportional to $\mathbf{B}\otimes\mathbf{B}$. We will discuss these terms in more detail in Chapter 25.

The energy conservation equation now takes into account both the gas energy and the energy in the electromagnetic field and can be obtained by combining the gas-dynamics energy equation with the Poynting theorem. For simplicity, we consider ideal MHD ($\eta = 0$). The equation describing the energy conservation for the gas (Equation (6.22)) is

$$\begin{aligned}\frac{\partial}{\partial t}\left[\rho\left(\varepsilon+\frac{1}{2}v^2\right)\right]+\nabla\cdot\left[\left(\rho\varepsilon+\frac{1}{2}\rho v^2+p\right)\mathbf{v}\right]\\ =\mathbf{f}\cdot\mathbf{v}+(\mathbf{j}\times\mathbf{B})\cdot\mathbf{v}.\end{aligned}\tag{23.26}$$

This is just the usual energy equation, with the inclusion of a new term representing the work done by the Lorentz force. The latter can be rewritten using Equation (23.16) as

$$(\mathbf{j}\times\mathbf{B})\cdot\mathbf{v}=-(\mathbf{v}\times\mathbf{B})\cdot\mathbf{j}=\mathbf{E}\cdot\mathbf{j}.\tag{23.27}$$

You might recognize this as being the energy sink term in the Poynting theorem (Equation (22.26)), which reads

$$\frac{\partial u_{\mathrm{EM}}}{\partial t}+\nabla\cdot\mathbf{S}=-\mathbf{j}\cdot\mathbf{E},\tag{23.28}$$

where u_{EM} is the electromagnetic field energy, which in our case is:

$$u_{\mathrm{EM}}=\frac{\epsilon_0 E^2}{2}+\frac{B^2}{2\mu_0}\approx\frac{B^2}{2\mu_0}=u_{\mathrm{B}},\tag{23.29}$$

consistently with our assumption that the dynamics is nonrelativistic. The Poynting vector is, using again Equation (23.16):

$$\mathbf{S}=\frac{1}{\mu_0}(\mathbf{E}\times\mathbf{B})=\frac{\mathbf{B}\times(\mathbf{v}\times\mathbf{B})}{\mu_0}.\tag{23.30}$$

Combining Equation (23.26) and Equation (23.28), we obtain

$$\begin{aligned}&\frac{\partial}{\partial t}\left[\rho\left(\varepsilon+\frac{1}{2}v^2\right)+u_{\mathrm{B}}\right]\\ &+\nabla\cdot\left[\left(\rho\varepsilon+\frac{1}{2}\rho v^2+p\right)\mathbf{v}+\mathbf{S}\right]=\mathbf{f}\cdot\mathbf{v}.\end{aligned}\tag{23.31}$$

In full form:

$$\boxed{\begin{aligned}&\partial_t\left[\rho\left(\varepsilon+\frac{1}{2}v^2\right)+\frac{B^2}{2\mu_0}\right]\\ &+\nabla\cdot\left[\left(\rho\varepsilon+\frac{1}{2}\rho v^2+p\right)\mathbf{v}\right.\\ &\qquad\left.+\frac{\mathbf{B}\times(\mathbf{v}\times\mathbf{B})}{\mu_0}\right]=\mathbf{f}\cdot\mathbf{v}.\end{aligned}}\tag{23.32}$$

This equation highlights the fact that magnetic and fluid energies are not separately conserved, but their sum is. In other words, it is possible to convert gas kinetic energy into magnetic energy (*dynamo effect*) or magnetic energy into kinetic and internal energy (*magnetic reconnection*, Chapter 24). In resistive MHD, Equation (23.32) still holds. In this case, Equation (23.27) has an additional term, $-j^2/\sigma$, and the Poynting flux has an additional term proportional to η. However, these are exactly compensated by the Joule heating in the energy equation of the gas. In other words, the magnetic energy dissipated by the electric currents is transferred to the internal energy of the gas.

23.5 In-class worksheet

1. Write the Maxwell equations in SI units.
2. Write Ampère's law in the nonrelativistic limit. Use it to eliminate the current in the Lorentz force per unit volume experienced by a magnetized fluid:

$$\mathbf{f}=\mathbf{j}\times\mathbf{B}.\tag{23.33}$$

3. Write Ohm's law for an electrically conducting, quasi-neutral fluid. Use it to eliminate the electric field in Faraday's law.
4. Show that, if $\nabla \cdot \mathbf{B} = 0$ at $t = 0$, then the induction equation guarantees that $\nabla \cdot \mathbf{B} = 0$ at all times.

23.6 Exercises

Exercise 23.1 (Angular momentum of the solar wind).

1. Starting from the equations of ideal MHD, show that the angular momentum of the plasma,

$$\mathbf{J} = \int_V \mathbf{r} \times \rho \mathbf{v} \mathrm{d}V, \tag{23.34}$$

subject to an external force per unit volume $\mathbf{f}$, satisfies

$$\frac{\mathrm{d}\mathbf{J}}{\mathrm{d}t} = \int_V \mathbf{r} \times \mathbf{f} \mathrm{d}V - \oint_S [(\mathbf{v} \cdot \mathrm{d}\mathbf{S})(\mathbf{r} \times \rho \mathbf{v}) + (p + B^2/2\mu_0)\mathbf{r} \times \mathrm{d}\mathbf{S} - (\mathbf{r} \times \mathbf{B})(\mathbf{B} \cdot \mathrm{d}\mathbf{S})/\mu_0]. \tag{23.35}$$

[Hint. Follow the steps of Exercise 6.1.]

2. Assume that the magnetic field is stationary in a frame rotating with the angular velocity of the Sun. Show that if the solar wind is infinitely conducting, then (in cylindrical coordinates R, ϕ, z)

$$B_\phi v_R = (v_\phi - \Omega R) B_R, \tag{23.36}$$

where Ω is the angular velocity of the Sun.

3. Let $\dot{J}$ be the z-component of the torque across a cylinder of radius R and height $2R$. Assume that $\dot{J}$ does not depend on R, and neglect the flux of angular momentum through the top and bottom of the cylinder. Show that $\dot{J}$ can be written as

$$-\dot{J} = 2\pi \int_{-R}^{R} \left(R\rho v_R v_\phi - R \frac{B_\phi B_R}{\mu_0} \right) R \mathrm{d}z. \tag{23.37}$$

Assume that the expression in parentheses is independent of z, and that $4\pi\rho R^2 v_R$, the mass flux of the solar wind through the cylinder, is a constant. Show that

$$-\frac{\dot{J}}{4\pi\rho R^2 v_R} = R v_\phi \left(1 - \frac{B_R^2}{\mu_0 \rho v_R^2} \right) + \frac{B_R^2}{\mu_0 \rho v_R^2} \Omega R^2. \tag{23.38}$$

4. There is a radius, R_A, called the Alfvén radius where the radial Alfvén speed $v_A = B_R/\sqrt{\mu_0 \rho}$ is equal to v_R. Evaluate the angular momentum loss per unit mass at this radius and show that it is equal to ΩR_A^2.
5. If R_A is 10 $R_\odot$, approximately what fraction of its mass must the Sun lose to slow its rotation down by a factor of 2? Take the moment of inertia of the Sun to be $0.06 M_\odot R_\odot^2$. What fraction of its mass has the Sun lost in 5 billion years if the solar wind was steady and the number density at the Earth is $10\,\mathrm{cm}^{-3}$ and the velocity is $300\,\mathrm{km/s}$?

23.7 Further reading

- C. Clarke and B. Carswell [1], Chapter 13 (plasmas).

24. Flux Freezing

Flux freezing is an important phenomenon in highly conducting plasmas. In these notes, we derive flux freezing in ideal magnetohydrodynamics (MHD), however it is important to remark that flux freezing also applies beyond ideal MHD, at least in an approximate way. We also discuss the role of flux freezing in the context of some important astrophysical systems. The key equation we will consider is the induction equation (or Faraday's Law), which reads (when $\eta = \text{const}$, see Equation (23.19)):

$$\frac{\partial \mathbf{B}}{\partial t} = \nabla \times (\mathbf{v} \times \mathbf{B}) + \eta \nabla^2 \mathbf{B}. \tag{24.1}$$

24.1 The Ideal-MHD Limit

Let L and V be characteristic length and velocity scales for the plasma. We can introduce the dimensionless time and space coordinates and velocity

$$\hat{t} = t\frac{V}{L}, \qquad \hat{x} = \frac{x}{L}, \qquad \hat{\mathbf{v}} = \frac{\mathbf{v}}{V}. \tag{24.2}$$

Faraday's law can be written as

$$\frac{V}{L}\frac{\partial \mathbf{B}}{\partial \hat{t}} = \frac{V}{L}\hat{\nabla} \times (\hat{\mathbf{v}} \times \mathbf{B}) + \frac{\eta}{L^2}\hat{\nabla}^2 \mathbf{B}. \tag{24.3}$$

That is

$$\frac{\partial \mathbf{B}}{\partial \hat{t}} = \hat{\nabla} \times (\hat{\mathbf{v}} \times \mathbf{B}) + \left(\frac{\eta}{LV}\right)\hat{\nabla}^2 \mathbf{B} = \hat{\nabla} \times (\hat{\mathbf{v}} \times \mathbf{B}) + \frac{1}{\mathrm{R}_M}\hat{\nabla}^2 \mathbf{B}. \tag{24.4}$$

We have introduced the *magnetic Reynolds number*, by analogy with the Reynolds number of Equation (21.10),

$$\mathrm{R}_M = \frac{LV}{\eta}. \tag{24.5}$$

The magnetic Reynolds number measures the relative importance of advection and resistivity in the evolution of the magnetic field. It is analogous to the "usual" Reynolds number, which measures the ratio of inertial to viscous forces in a fluid. When $\mathrm{R}_M \gg 1$, we say that we are in the *ideal-MHD limit*. As we have seen in Chapter 23, R_M is enormous in most astrophysical systems, for example $\mathrm{R}_M \sim 6 \times 10^{17}$ in the warm interstellar medium.

In the ideal-MHD limit, the Faraday law simplifies to

$$\frac{\partial \mathbf{B}}{\partial t} = \nabla \times (\mathbf{v} \times \mathbf{B}) \Rightarrow \frac{\partial \mathbf{B}}{\partial t} + \nabla \times (\mathbf{B} \times \mathbf{v}) = 0. \tag{24.6}$$

Remarkably, this is identical to the vorticity equation for a barotropic, inviscid fluid:

$$\frac{\partial \boldsymbol{\omega}}{\partial t} + \nabla \times (\boldsymbol{\omega} \times \mathbf{v}) = 0. \tag{24.7}$$

The analogy between vorticity and magnetic field plays an important role in our understanding of MHD phenomena such as dynamos and is sometimes known as the Batchelor's analogy.

Consider now a surface $S(t)$ that moves with the fluid, and the magnetic flux across it

$$\Phi(t) = \int_{S(t)} \mathbf{B} \cdot \mathrm{d}\mathbf{A}. \tag{24.8}$$

The flux can change for two reasons: because $\mathbf{B}$ changes with time, and because the surface moves. In a time $\mathrm{d}t$, each element $\mathrm{d}\mathbf{l}$ of the boundary $C(t)$ of the surface sweeps an area $\mathbf{v}\mathrm{d}t \times \mathrm{d}\mathbf{l}$, so the boundary adds a flux $\mathbf{B} \cdot (\mathbf{v}\mathrm{d}t \times \mathrm{d}\mathbf{l}) = -(\mathbf{v} \times \mathbf{B}) \cdot \mathrm{d}\mathbf{l}\,\mathrm{d}t$. Using Stokes' theorem, we find

$$\boxed{\begin{aligned}\frac{\mathrm{d}\Phi}{\mathrm{d}t} &= \int_{S(t)} \frac{\partial \mathbf{B}}{\partial t} \cdot \mathrm{d}\mathbf{A} - \oint_{C(t)} (\mathbf{v} \times \mathbf{B}) \cdot \mathrm{d}\mathbf{l} \\ &= \int_{S(t)} \left[\frac{\partial \mathbf{B}}{\partial t} - \nabla \times (\mathbf{v} \times \mathbf{B})\right] \cdot \mathrm{d}\mathbf{A} = 0,\end{aligned}} \tag{24.9}$$

where we have used Equation (24.6) in the last step. In other words, the magnetic flux across any surface co-moving with the flow is constant in ideal MHD. The same argument applied to the vorticity gives Kelvin's circulation theorem: the circulation $\mathcal{C} = \oint_{C(t)} \mathbf{v} \cdot \mathrm{d}\mathbf{l} = \int_{S(t)} \boldsymbol{\omega} \cdot \mathrm{d}\mathbf{A}$ around a curve moving with a barotropic, inviscid fluid is conserved.

This fact is called **flux freezing** and is one of the most important properties of MHD. To understand this consider a magnetic flux tube. As the name suggests, a flux tube is an imaginary bundle of magnetic field lines dragged by the plasma. This is illustrated in Figure 24.1.

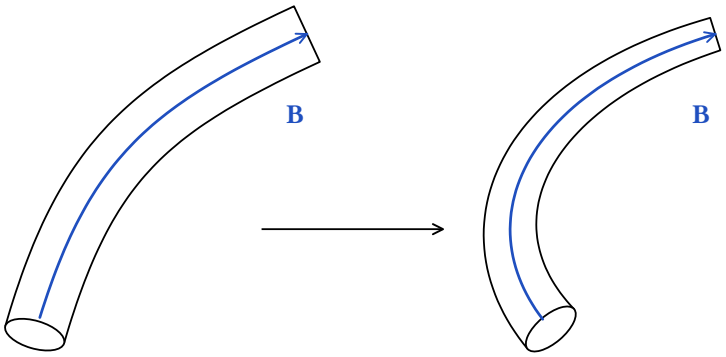


Figure 24.1: Effect of flux freezing.

Consider the surface bounding the flux tube on the sides. This surface is such that the magnetic flux across it is zero, since there is no component of the field perpendicular to it. If we let this surface move with the plasma velocity, the flux freezing condition requires the magnetic field to deform to stay always parallel to it, as illustrated in Figure 24.1. In other words, **the magnetic field is dragged by the plasma as if it was painted on the fluid elements.** As we will see shortly, flux freezing has other important consequences.

At the microscopic level, the reason why flux freezing occurs in highly conducting plasmas is that the charged particles are "stuck" to the magnetic field lines[11]. Another way to think about flux freezing is in terms of induced currents. In order for the magnetic field lines to drift with respect to the plasma with velocity $\mathbf{v}_d$, one would need to induce an electric field (in the plasma frame)

$$\mathbf{E}' = \mathbf{v}_d \times \mathbf{B} \tag{24.10}$$

which corresponds to a current $\mathbf{j}' = \sigma \mathbf{E}'$. However, $\sigma \to \infty$ (that is, $\eta \to 0$) in the ideal-MHD limit, meaning that an infinite current is required to cause the field lines to slip.

24.1.1 Magnetic Field Amplification

Flux freezing also implies that, if a flux tube is compressed by the flow, then its magnetic field intensity is amplified. To see this, consider the situation illustrated in Figure 24.2.

[11] More precisely, they are forced by the Lorentz force to move on circles around the field lines, with the Larmor radius (Equation (22.44)), while they drift along them.

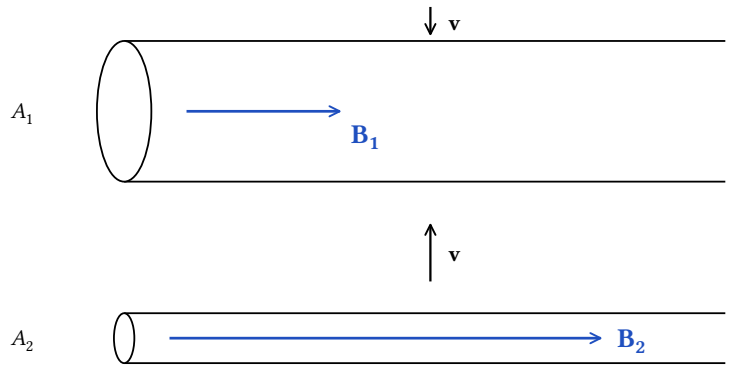


Figure 24.2: Field amplification from the compression of a magnetic flux tube: a wide tube of cross-section A_1 and field $\mathbf{B}_1$ is squeezed from the sides by a velocity $\mathbf{v}$ into a thinner tube of cross-section A_2 and field $\mathbf{B}_2$. Redrawn after R. M. Kulsrud [4].

When a flux tube is compressed and its area changes from A_1 to A_2, the B-field strength needs to change in such a way so that the magnetic flux remains unchanged:

$$A_1 B_1 = A_2 B_2. \tag{24.11}$$

So the magnetic field strength must be amplified:

$$B_2 = \frac{A_1}{A_2} B_1. \tag{24.12}$$

It is instructive to compare this with the rate at which the plasma density is changed during the same process. In this case, the quantity that is conserved is the mass of the plasma in the flux tube, so

$$\rho A \ell = \text{const} \Rightarrow \rho_1 A_1 \ell = \rho_2 A_2 \ell \Rightarrow \rho_2 = \frac{A_1}{A_2} \rho_1. \tag{24.13}$$

In this situation, we find that the ratio of magnetic field strength to density in the plasma stays constant:

$$\frac{B}{\rho} = \text{const}. \tag{24.14}$$

Things are very different if the magnetic flux tube is compressed in the direction parallel to the magnetic field, as illustrated in Figure 24.3.

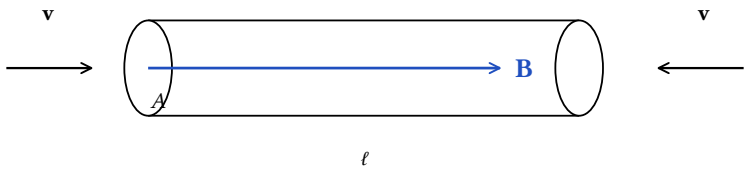


Figure 24.3: Compression along a flux tube.

In this case, the density of the plasma changes according to

$$\rho A \ell = \text{const} \Rightarrow \frac{\rho_2}{\rho_1} = \frac{\ell_1}{\ell_2}. \tag{24.15}$$

However, B remains constant, because the cross-section of the flux tube, A, is unchanged.

Example: Stellar collapse

A star with radius $r_1 \sim 10^9$ m collapses to form a neutron star with radius $r_2 \sim 10$ km, and all its magnetic flux ends up in the neutron star.

- By what factor does the magnetic field increase?
- By what factor does the density increase?
- If the initial field is $B_1 \sim 10^{-2}$ T (100 G), what will be the field of the neutron star?

Answer. The density evolves according to $\rho r^3 = \text{const}$, while the magnetic field evolves according to $Br^2 = \text{const}$ (assuming the field to be approximately radial), this means that

$$B_2 = \left(\frac{r_1}{r_2}\right)^2 B_1, \qquad \rho_2 = \left(\frac{r_1}{r_2}\right)^3 \rho_1, \qquad B \sim \rho^{2/3}. \tag{24.16}$$

For example, an initial field of 10^{-2} T (100 G) is amplified to 10^8 T (10^{12} G) as result of the collapse of the star. The strong amplification of the magnetic field during the collapse of a star is the reason why neutron stars have extremely strong magnetic fields, up to 10^{11} T (10^{15} G).

24.1.2 Resistive Evolution

If the resistivity $1/\sigma$ is finite, $\eta > 0$, then the magnetic field lines are not stuck to the plasma, but can drift. To understand how this process operates, we consider the evolution of a magnetic field under the sole influence of Ohmic diffusion:

$$\frac{\partial \mathbf{B}}{\partial t} = \eta \nabla^2 \mathbf{B}. \tag{24.17}$$

If the typical curvature radius of the field lines is L, then we can approximate the previous equation as

$$\frac{\partial B}{\partial t} \sim \frac{\eta}{L^2} B = \frac{1}{T_{\text{decay}}} B, \qquad T_{\text{decay}} = \frac{L^2}{\eta}. \tag{24.18}$$

For many astrophysical applications L is very large, so T_{decay} is very long. We will put actual numbers for η and L when we discuss the problem of star formation below. If the magnetic field has more than one characteristic scale, then the smaller scales will be dissipated more rapidly by the resistivity. This has the effect of "smoothing" the field lines. To quantify this, we can use the Fourier transform. Accordingly, we represent the magnetic field as a superposition of plane waves

$$\mathbf{B}(t, \mathbf{x}) = \int \mathrm{d}^3\mathbf{k} \hat{\mathbf{B}}_{\mathbf{k}}(t) \mathrm{e}^{\mathrm{i}\mathbf{k}\cdot\mathbf{x}}, \qquad \hat{\mathbf{B}}_{\mathbf{k}} = \frac{1}{(2\pi)^3} \int \mathbf{B}(t, \mathbf{x}) \mathrm{e}^{-\mathrm{i}\mathbf{k}\cdot\mathbf{x}} \mathrm{d}V. \tag{24.19}$$

Note that the $\mathbf{k} = 0$ mode of the B-field is proportional to the average value of the field in the volume V:

$$\hat{\mathbf{B}}_0 = \frac{1}{(2\pi)^3} \int \mathbf{B}(t, \mathbf{x}) \mathrm{d}V = \frac{V}{(2\pi)^3} \langle \mathbf{B} \rangle. \tag{24.20}$$

Inserting Equation (24.19) into Equation (24.17), we obtain:

$$\int \mathrm{d}^3\mathbf{k} \mathrm{e}^{\mathrm{i}\mathbf{k}\cdot\mathbf{x}} \left[\frac{\mathrm{d}\hat{\mathbf{B}}_{\mathbf{k}}}{\mathrm{d}t} + \eta k^2 \hat{\mathbf{B}}_{\mathbf{k}} \right] = 0 \tag{24.21}$$

The integral vanishes for all $\mathbf{x}$ only if the quantity in square brackets vanishes for every $\mathbf{k}$:

$$\frac{\mathrm{d}\hat{\mathbf{B}}_{\mathbf{k}}}{\mathrm{d}t} = -\eta k^2 \hat{\mathbf{B}}_{\mathbf{k}}. \tag{24.22}$$

So we have transformed Equation (24.17) into a simple system having an infinite number of decoupled equations, with solutions

$$\hat{\mathbf{B}}_{\mathbf{k}}(t) = \hat{\mathbf{B}}_{\mathbf{k}}(0)\exp(-\eta k^2 t). \tag{24.23}$$

We can see that different modes of the B-field decay exponentially and that large k modes are damped exponentially more quickly than smaller k modes. On the other hand, the $k = 0$ mode is not affected at all and stays constant for all times. This means that the action of the resistivity is not really to make the field decay, but rather to straighten out all field lines, as illustrated in Figure 24.4.

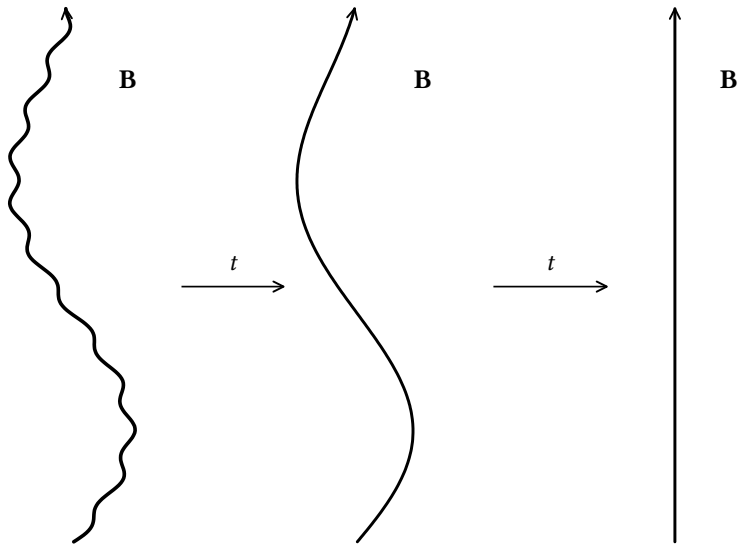


Figure 24.4: Evolution of the magnetic field under the effect of resistivity.

24.2 Star Formation

One situation in which flux freezing must be violated is star formation. Consider the collapse of a region of a molecular cloud to form a star. If flux freezing were to hold, using the same approach as for the stellar collapse example, we would find

$$\begin{aligned} \rho &: 10^{-24}\,\mathrm{g\,cm^{-3}} \quad \rightarrow 1\,\mathrm{g\,cm^{-3}} \\ B &: 10^{-10}\,\mathrm{T}\,(1\,\mu\mathrm{G}) \rightarrow 10^{6}\,\mathrm{T}\,(10^{10}\,\mathrm{G}) \end{aligned} \tag{24.24}$$

That is, the protostar would be embedded with a pulsar-like magnetic field. This is actually impossible. Indeed, the gravitational binding energy of the protostar would be

$$W \sim -\frac{3GM^2}{5R} \sim 2\times 10^{48} M_1^2\,\mathrm{erg}, \tag{24.25}$$

where M_1 is the protostar mass in solar masses. The magnetic energy would be

$$U_B \sim \frac{4}{3}\pi R^3 \cdot \frac{B^2}{2\mu_0} \sim 8\times 10^{51}\,\mathrm{erg} \gg |W|! \tag{24.26}$$

The protostar would be unbound, meaning that the magnetic stresses would tear it apart. In reality, the magnetic field halts the collapse of an ideal-MHD cloud well before these levels of magnetic field strength can be achieved. To see why, note that both the gravitational binding energy and the magnetic energy scale as $1/R$ during the collapse (see the in-class worksheet). So, if flux freezing holds, the ratio $U_B/|W|$ does not change during the collapse. With the numbers above, the cloud already has $U_B \gg |W|$ before it starts to collapse. In other words, if flux freezing holds, a cloud for which the magnetic energy exceeds

the binding energy can never collapse. To form a star, the cloud must get rid of most of its magnetic flux. Obviously, Nature has a way to form stars from magnetized clouds, so how does it defeat flux freezing?

It turns out that resistivity is not sufficient to do the trick. The magnetic diffusivity $\eta = 1/(\mu_0\sigma)$ of ionized hydrogen plasma depends weakly on density and can be approximated as [4]

$$\eta \approx \frac{420}{T_{\rm eV}^{3/2}}\,{\rm m}^2\,{\rm s}^{-1}. \tag{24.27}$$

For a star $T_{\rm eV} \approx 10^2$ and $L \sim R_\odot \approx 7 \times 10^8$ m, so the decay time is

$$T_{\rm decay} = \frac{L^2}{\eta} \approx 1.2 \times 10^{18}\,{\rm s} \approx 40\,{\rm Gyr}. \tag{24.28}$$

This is longer than the age of the Universe, so the magnetic fields of stars are also frozen in the plasma. For a molecular cloud core, with $T \approx 10\,{\rm K}$ and $L \approx 0.1\,{\rm pc}$, the decay time is even longer, $\sim 10^{16}$ yr. So resistivity is clearly not sufficient to release the magnetic flux and allow stars to form.

The mechanism currently thought to be responsible for the drift between magnetic fields and the plasma is *ambipolar diffusion*. The details go beyond the scope of this course, but the key observation is that molecular clouds are only weakly ionized. The neutral component of the gas is not affected by the Lorentz force, so it can flow perpendicular to field lines. The Lorentz force on the ions, on the other hand, is orders of magnitude larger than gravity, so the ions are stuck on the field lines. The mutual friction between ions and neutrals has the effect of regulating the rate of collapse and determines the time for the formation of protostars. This physics is illustrated in Figure 24.5.

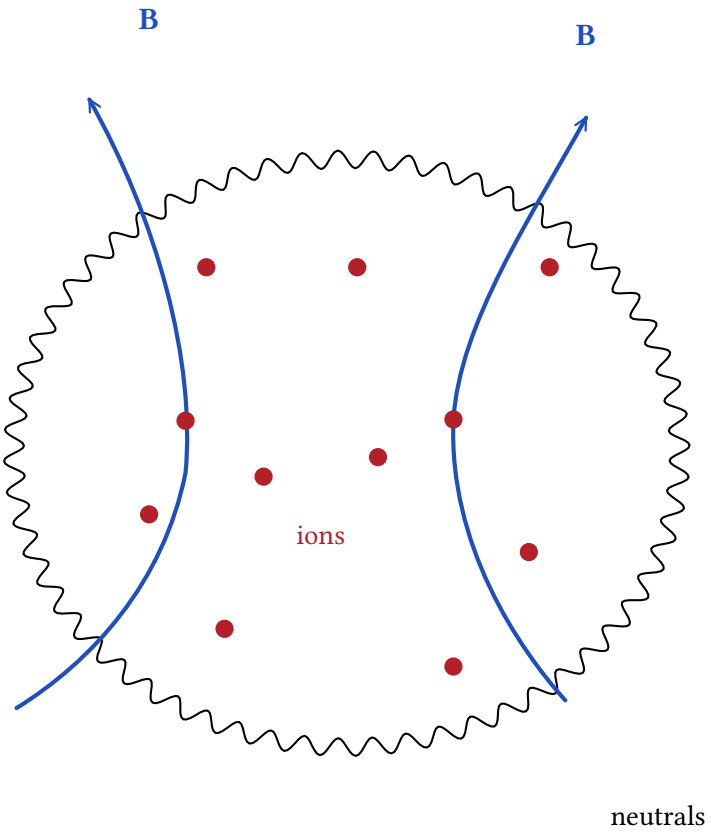


Figure 24.5: Ambipolar diffusion.

Magnetic fields do not just regulate the collapse of clouds into protostars. They also play a key role enabling it. Indeed, magnetic fields allow the protostar to lose angular momentum and prevent the gas from getting stuck at the centrifugal barrier (see Exercise 26.1). This happens because the collapsing cloud forces a large column of gas, parallel to the mean magnetic field, to corotate with it. This allows the

forming protostar to dissipate angular momentum. We explore this in a more quantitative way in Exercise 23.1, for the case of the solar wind.

24.3 Io and Jupiter

The unipolar-inductor model for the interaction between Jupiter and its moon Io is one of the early successes of plasma astrophysics. This model has also been adopted to explain a variety of other astrophysical systems, including the launching of electromagnetic waves by black holes moving through plasma. The main idea of this model is that the motion of a conductor, in this case Io, in an external magnetic field, in this case Jupiter's field, generates an electric current.

The original motivation for this model was the discovery in the 1960s of radio flares from Jupiter in phase with the rise and setting of Io. More precisely, Jupiter has a rotation period of about 10 hours, while Io has an orbital period of about 42 hours. Radio flares are preferentially detected when Io has a longitude of 90° or 240° with respect to Jupiter, as shown in Figure 24.6, taken from R. A. Duncan [13].

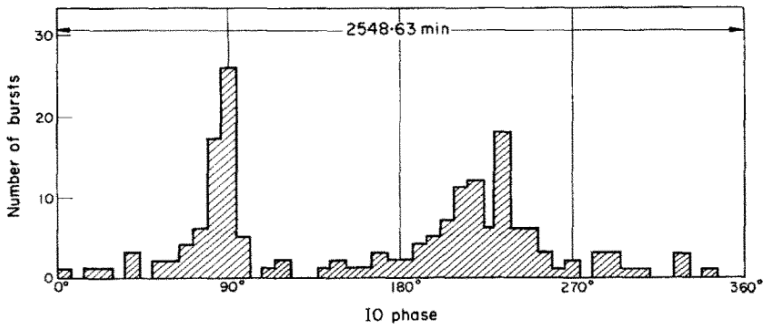


Figure 24.6: Number of radio bursts detected from Jupiter as a function of the phase of Io. From R. A. Duncan [13]; © Elsevier, reproduced for non-commercial educational use and not covered by the license of these notes.

In 1969, Goldreich and Lynden-Bell [14] proposed a model to explain:

1. the mechanism generating the radiation;
2. the reason for the phase difference between the rising and setting of Io and the radiation bursts.

We explain this model in a qualitative way and then build a semi-quantitative simplified model below.

Io is primarily composed of silicates and iron, so it has a good electrical conductivity. This means that there is no electric field inside Io, in Io's frame of reference, and the magnetic field lines of Jupiter are stuck to it (flux freezing). However, because in Jupiter's frame of reference Io completes a revolution every 13 hours, the field lines cannot be frozen everywhere, otherwise the magnetic energy in the magnetosphere would grow without bound[12]. This means that the field lines must slip somewhere. It turns out that this happens in Jupiter's atmosphere, where the gas resistivity is the highest, due to the presence of a neutral gas. The geometry of the magnetic flux tube connecting Jupiter and Io is illustrated in Figure 24.7.

[12]Imagine a single flux tube wound around Jupiter multiple times. The magnetic flux in the tube must be conserved, so the intensity of the field is constant. However, as the flux tube is wound around, more and more of the magnetosphere is filled by the coils of the flux tube, so the integrated magnetic energy ($B^2/2\mu_0$) grows linearly with time over the winding timescale.

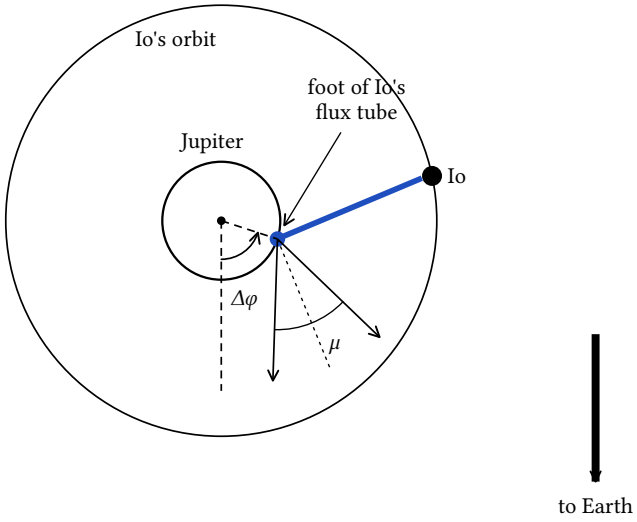


Figure 24.7: Magnetic flux tube connecting Io and Jupiter. Redrawn after P. Goldreich and D. Lynden-Bell [14].

The slippage of the field lines on Jupiter's surface drives a current along the flux tube connecting the planet to the moon. Electrons gyrate in the field lines producing cyclotron radiation, which is beamed primarily in the direction perpendicular to the field lines, explaining the offset between Io's rises and settings and the radio bursts. This is illustrated in more detail in Figure 24.8.

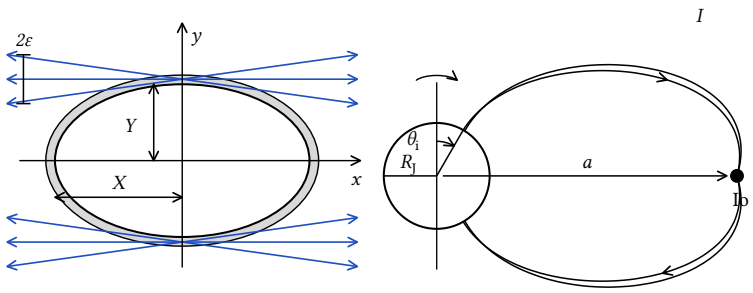


Figure 24.8: Left panel: footprint of the magnetic-flux tube connecting Io and Jupiter on Jupiter's atmosphere. Right panel: cross-section of the magnetic-flux tube connecting Jupiter and Io and direction in which the current I flows between the planet and the moon. Redrawn after P. Goldreich and D. Lynden-Bell [14].

A strong electric field is created in the direction perpendicular to the B-field, in the Y-direction shown in the figure. The potential difference across the flux tube is $\Delta\Phi \sim vBD$, where $v \approx 57\,\mathrm{km\,s^{-1}}$ is the velocity of Io with respect to Jupiter's corotating magnetic field, $B \approx 2\,\mu\mathrm{T}$ (0.02 G) is the magnetic field at Io, and $D \approx 3600\,\mathrm{km}$ is the diameter of Io. This gives $\Delta\Phi \sim 4 \times 10^5\,\mathrm{V}$[13]. However, the motion of electrons perpendicular to the field lines is suppressed by the Lorentz force, so a current is driven along the field lines, through the moon, and back to the planet, as shown in the right panel of Figure 24.8. The current intensity is $I \sim 10^6\,\mathrm{A}$, which is sufficient to produce cyclotron and synchrotron radiation.

Here, we consider a toy model in which we see how the electric field is created by the slippage of the field lines. We consider a conductor moving in an external B-field anchored in a wall with infinite conductivity, as shown in Figure 24.9.

[13] Goldreich and Lynden-Bell estimated $\Delta\Phi \approx 7 \times 10^5\,\mathrm{V}$, with the values of Jupiter's magnetic field known at the time.

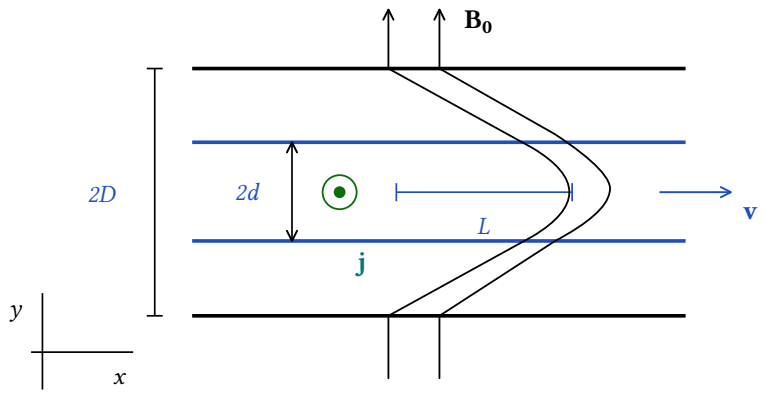


Figure 24.9: Toy model for the Io and Jupiter problem.

In our analogy, the conductor is Jupiter's atmosphere, while the wall represents Io. We imagine that initially the field is vertical and the conductor is at rest. At $t = 0$, we start moving the conductor with velocity $\mathbf{v}$ in the x-direction. The field is dragged up to a distance L and then starts to slip, realizing the configuration shown in Figure 24.9.

There is no external electric field, so the electric field in the frame of the conductor is

$$\mathbf{E}' = \mathbf{E} + \mathbf{v} \times \mathbf{B} = \mathbf{v} \times \mathbf{B} = v_x B_y \hat{\mathbf{z}}, \tag{24.29}$$

and the associated current is $\mathbf{j} = \sigma(\mathbf{v} \times \mathbf{B}) = \sigma v_x B_y \hat{\mathbf{z}}$. The current is responsible for the creation of the x-component of the B-field (with $\eta = 1/\mu_0\sigma$):

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{j} \Rightarrow -\partial_y B_x = \mu_0 j_z \Rightarrow B_x = -\mu_0 j_z y = -\frac{1}{\eta} v_x B_y y. \tag{24.30}$$

On the other hand, from the geometry of the system, we have:

$$B_x = \frac{B_y L}{D}\Big|_{y=d} \Rightarrow L = \frac{v_x D d}{\eta} = v_x T_{\text{decay}}, \tag{24.31}$$

where now T_{decay} is computed as in Equation (24.18), with L^2 replaced by Dd.

This shows the two key features of the system. The relative motion of the conductor and the walls generates an electric field (a consequence of Faraday's law). The electric field drives a current in the system. The current, in turn, causes the field lines to bend and slip (from Ampère's law).

i Going further: Pulsars

Pulsars were discovered by Jocelyn Bell Burnell and Antony Hewish in 1967. It was later realized that pulsars are rotating neutron stars endowed with strong magnetic fields. In 1969, Goldreich and Julian [15] developed the first model of a pulsar. Like the unipolar induction model, Goldreich and Julian's model applies to a variety of astrophysical systems and shows another key physical effect associated with flux freezing. Namely, the fact that rotating neutron stars fill their surroundings with a dense plasma, creating the so-called magnetosphere.

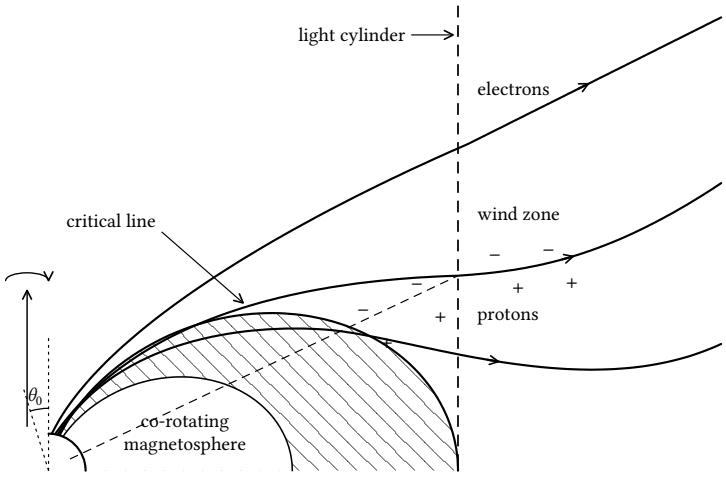


Figure 24.10: Schematic illustration of a pulsar magnetosphere. Redrawn after P. Goldreich and W. H. Julian [15].

In this model, shown in Figure 24.10, we neglect the misalignment between magnetic North pole and rotational axis, which would be needed to produce the pulsed emission. However, this model contains all required physical ingredients. We assume at first that the pulsar is a neutron star with a dipole magnetic field in vacuum. The magnetic field is then given by

$$\mathbf{B} = B_0 \left(\frac{R}{r}\right)^3 (2\cos\theta \hat{\mathbf{r}} + \sin\theta \hat{\boldsymbol{\theta}}). \tag{24.32}$$

The angular velocity of the star is given by

$$\boldsymbol{\Omega} = \Omega \hat{\mathbf{z}}, \qquad \hat{\mathbf{z}} = \cos\theta \hat{\mathbf{r}} - \sin\theta \hat{\boldsymbol{\theta}}. \tag{24.33}$$

In the previous equation $\hat{\mathbf{z}}$ is the unit vector in the z-direction. Since neutron stars are essentially perfect conductors, there should be no electric field in their frame $\mathbf{E}' = 0$. This means that in the inertial frame there is an electric field inside the star given by

$$\mathbf{E} = -\mathbf{v} \times \mathbf{B} = -(\boldsymbol{\Omega} \times \mathbf{r}) \times \mathbf{B} = B_0 \Omega r \left(\frac{R}{r}\right)^3 (\sin^2\theta \hat{\mathbf{r}} - 2\cos\theta \sin\theta \hat{\boldsymbol{\theta}}). \tag{24.34}$$

In the previous equation we have used the fact that

$$\boldsymbol{\Omega} \times \mathbf{r} = \Omega r \sin\theta \hat{\boldsymbol{\phi}} \tag{24.35}$$

and

$$\hat{\boldsymbol{\phi}} \times (2\cos\theta \hat{\mathbf{r}} + \sin\theta \hat{\boldsymbol{\theta}}) = 2\cos\theta \hat{\boldsymbol{\theta}} - \sin\theta \hat{\mathbf{r}}. \tag{24.36}$$

If there is vacuum in the exterior (and assuming axisymmetry), then the external electric potential is

$$\Phi(r,\theta) = \sum_{\ell=0}^{\infty} \frac{\phi_\ell}{r^{\ell+1}} P_\ell(\cos\theta), \qquad r > R \tag{24.37}$$

where ϕ_ℓ are fixed by the fact that the tangential component of the electric field should be continuous on the surface of the star[14]

[14] In other words, we are matching the interior solution with the exterior solution.

$$[\mathbf{E}_\parallel = 0 \Rightarrow E_\theta(r=R) = -\frac{1}{R}\partial_\theta \Phi(r=R). \tag{24.38}$$

Inside the star, $E_\theta \propto \sin\theta\cos\theta \propto \partial_\theta P_2(\cos\theta)$, so only the $\ell = 2$ term can match it (the $\ell = 0$ term is fixed by the total charge of the star, which we assume to be zero). So the external field is quadrupolar, meaning that

$$\Phi(r,\theta) = \frac{\phi_2}{r^3} P_2(\cos\theta), \tag{24.39}$$

then Equation (24.38) implies

$$\boxed{\Phi = -\frac{2B_0\Omega R^5}{3r^3} P_2(\cos\theta).} \tag{24.40}$$

So we find that there is an enormous electrostatic potential at the surface of the star:

$$|\Phi| \approx 4\times 10^{16} (B_8 R_4^2/P)\,\mathrm{V}, \tag{24.41}$$

where $B_8 = B_0/10^8$ T (10^{12} G), $R_4 = R/10^4$ m, and $P = 2\pi/\Omega$ is the period in seconds. For example, the ratio of electrostatic to gravitational force for a neutron star with $M = 1.4\,M_\odot$ is

$$\frac{|\mathbf{F}_{\rm EM}|}{|\mathbf{F}_{\rm G}|} \sim \frac{\Phi e}{\frac{GM}{R} m} = \begin{cases} 2\times 10^8, & \text{protons,} \\ 4\times 10^{11}, & \text{electrons} \end{cases} \gg 1. \tag{24.42}$$

The electromagnetic forces are strong enough to pull charges out of the surface of the neutron star and fill all the region around it with plasma. The field is so strong that electron-positron cascade production is triggered once the first particles are extracted. This is the so-called *magnetosphere*. The magnetosphere creates currents that to leading order short out the electric potential to infinity.

In the extreme conditions found in pulsar or black-hole magnetospheres, the electromagnetic energy density is so large compared to the plasma inertia that the plasma is forced to move in such a way as to cancel the Lorentz force, $q\mathbf{E} + \mathbf{j}\times\mathbf{B} = 0$. This is the so-called *force-free electrodynamics* approximation. Electrons, positrons, and ions can flow freely along the B-field lines, so they short out any electric field along $\mathbf{B}$. If, in addition, the plasma moves with velocity $\mathbf{v}$, the electric field is given by the ideal-MHD condition:

$$\mathbf{E}\cdot\mathbf{B} = 0, \qquad \mathbf{E} + \mathbf{v}\times\mathbf{B} = 0. \tag{24.43}$$

From the first condition, if the system is axisymmetric, we have that

$$\mathbf{E} = E_0\hat{\boldsymbol{\phi}}\times\mathbf{B} \Rightarrow \mathbf{E} = -\Omega r\sin\theta\hat{\boldsymbol{\phi}}\times\mathbf{B}. \tag{24.44}$$

From the last two equations, we find

$$0 = (\mathbf{v} - \Omega r\sin\theta\hat{\boldsymbol{\phi}})\times\mathbf{B} \Rightarrow \mathbf{v} = \Omega r\sin\theta\hat{\boldsymbol{\phi}} + v_d\hat{\mathbf{B}}. \tag{24.45}$$

So the plasma velocity has two components. The plasma is forced to co-rotate with the pulsar (first term in the first equation). However, when $\Omega r\sin\theta \geq c$, the drift velocity v_d of plasma along the field lines must be non-zero so that $|\mathbf{v}| < c$. The region at which $\Omega r\sin\theta = c$ is called the light cylinder. It is the boundary of the region in which magnetic field lines can be closed. Outside the light cylinder, the B-field lines bend and reach infinity so that plasma can flow away from/to the pulsar (depending on its charge) creating a current.

The plasma charge density is given by Gauss' law of electrodynamics:

$$q = \epsilon_0\nabla\cdot\mathbf{E}. \tag{24.46}$$

Note that

$$\frac{q}{\epsilon_0} = \nabla\cdot\mathbf{E} = \nabla\cdot[-\mathbf{v}\times\mathbf{B}] = -\mathbf{B}\cdot(\nabla\times\mathbf{v}) + \mathbf{v}\cdot(\nabla\times\mathbf{B}). \tag{24.47}$$

This can be computed using

$$\nabla \times \mathbf{v} = \nabla \times (\mathbf{\Omega} \times \mathbf{r}) = \mathbf{\Omega}(\nabla \cdot \mathbf{r}) - (\mathbf{\Omega} \cdot \nabla)\mathbf{r} = 2\mathbf{\Omega} \tag{24.48}$$

and

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{j} = \mu_0 q \mathbf{v} \tag{24.49}$$

So the charge density is

$$\frac{q}{\epsilon_0} = -2\mathbf{B} \cdot \mathbf{\Omega} + \frac{q}{\epsilon_0}\frac{v^2}{c^2} \Rightarrow q_{\mathrm{GJ}} = -2\epsilon_0 \mathbf{\Omega} \cdot \mathbf{B} \frac{1}{1 - v^2/c^2}. \tag{24.50}$$

Note that $\mathbf{\Omega} \cdot \mathbf{B}$ changes sign across a critical line in the magnetosphere. This means that there is charge separation in the magnetosphere.

The *minimum* electron number density is the so-called Goldreich–Julian density

$$n_{\mathrm{GJ}} = \frac{|q_{\mathrm{GJ}}|}{e} \approx 7 \times 10^{10} (B_8/P)\,\mathrm{cm}^{-3}. \tag{24.51}$$

This is ~ 10 orders of magnitude larger than the typical density of electrons in the ISM. More recent theoretical models show that the plasma number density can be several orders of magnitude larger than the Goldreich-Julian density. This is because the plasma is rich in electron-positron pairs and n_{GJ} only measures the *net* density of charges. For example, see Figure 24.11.

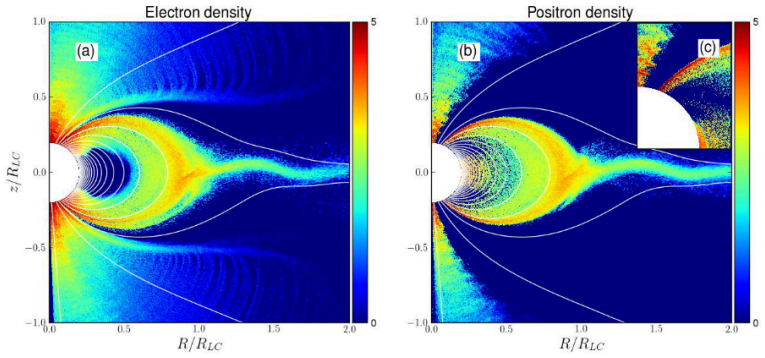


Figure 24.11: $\log_{10}(n/n_{\mathrm{GJ}})$ in a particle-in-cell simulation with approximate treatment of pair creation. Coherent plasma waves are thought to be responsible for the production of the radio pulses. From A. A. Philippov, B. Cerutti, A. Tchekhovskoy, and A. Spitkovsky [16]; © The American Astronomical Society, reproduced for non-commercial educational use and not covered by the license of these notes.

24.4 In-class worksheet

Spherical Collapse. A spherical distribution of plasma (a star or a molecular cloud) is endowed with a dipolar magnetic field, as shown in Figure 24.12.

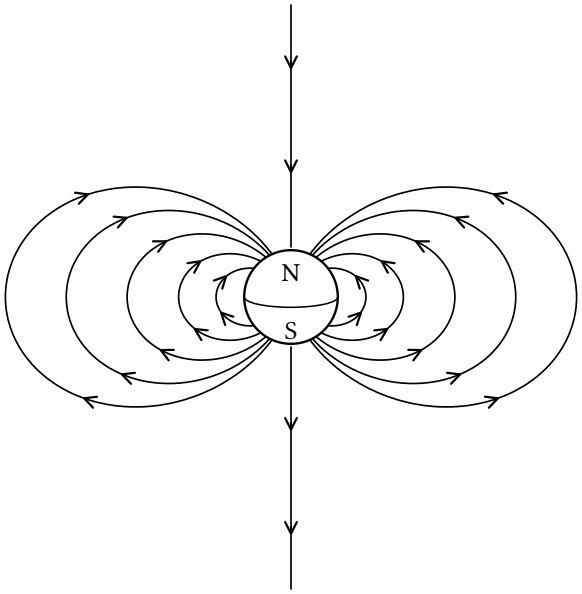


Figure 24.12: Dipolar magnetic field of a sphere.

The sphere contracts from an initial radius R_0 to a new radius $R_1 < R_0$.

1. How do the gravitational binding energy W and the magnetic energy U_B change? What does this imply for the collapse of a magnetized cloud?
2. In the process of forming a neutron star, an object with mass $M \sim 10^{31}\,\mathrm{kg}$ contracts from an initial radius $R_0 \sim 10^9\,\mathrm{m}$ to $R_1 \sim 10\,\mathrm{km}$. By what factor is the magnetic field amplified? If the typical magnetic field strength for a star is $B \sim 10^{-2}\,\mathrm{T}$ (100 G), what is the field strength expected for a neutron star?
3. Gas in a molecular cloud with density $\rho \sim 10^{-24}\,\mathrm{g\,cm^{-3}}$ and typical magnetic field $B \sim 0.1\,\mathrm{nT}$ ($1\,\mu\mathrm{G}$) contracts to form a star with $\rho \sim 1\,\mathrm{g\,cm^{-3}}$. What is the strength of the resulting field if flux freezing holds?

24.5 Exercises

Exercise 24.1 (Flux freezing in an Alfvén wave). In this exercise we check, for a simple wave, that the magnetic field lines move together with the plasma. Consider a plasma with uniform density ρ and pressure p, in the uniform magnetic field $B_0\hat{\mathbf{z}}$. We perturb it with a transverse wave travelling along the field, with velocity and magnetic field

$$\mathbf{v} = \epsilon v_A \sin(kz - \omega t)\hat{\mathbf{x}}, \qquad \mathbf{B} = B_0\hat{\mathbf{z}} - \epsilon B_0 \sin(kz - \omega t)\hat{\mathbf{x}}, \tag{24.52}$$

where $\epsilon \ll 1$ and $v_A = B_0/\sqrt{\mu_0\rho}$.

1. Show that Equation (24.52) solves the ideal-MHD equations to first order in ϵ, provided that $\omega = kv_A$. In other words, we neglect all terms of order ϵ^2 and higher.
2. The ideal-MHD equations neglect the displacement current. Use the ideal-MHD electric field to compute the displacement current of this wave, and compare it with the term $\mu_0\mathbf{j}$ in Ampère's law. Show that the displacement current is negligible when $v_A \ll c$.
3. Now consider a plasma with finite resistivity, that is, with magnetic diffusivity $\eta > 0$. Compare the resistive term in the induction equation with the term $\nabla \times (\mathbf{v} \times \mathbf{B})$, and show that the resistive term is negligible when $\omega T_{\mathrm{decay}} \gg 1$, where $T_{\mathrm{decay}} = 1/(k^2\eta)$ is the decay time of Equation (24.18) for $L = 1/k$.
4. To first order in ϵ, the field lines are the curves $x(z)$ that solve

$$\frac{\mathrm{d}x}{\mathrm{d}z} = \frac{B_x}{B_0}. \tag{24.53}$$

Find these curves. Then compute the displacement of the fluid elements,

$$x(z,t) = \int v_x \mathrm{d}t, \tag{24.54}$$

and show that it gives the same curves. This is what we expect if the field lines are frozen into the plasma.

Adapted from R. M. Kulsrud [4], Problem 3.1.

24.6 Further reading

- C. Clarke and B. Carswell [1], Chapter 13 (plasmas).
- R. M. Kulsrud [4]

25. Magnetic Tension and Hydromagnetic Equilibria

Up to now we have been primarily concerned with the consequences of the induction equation (Faraday's law). We have seen that, in the ideal-MHD limit, the magnetic field lines are frozen in the flow and explored some of the consequences of this fact. Now, we consider the feedback of the magnetic field on the plasma, that is the impact of the $\mathbf{j} \times \mathbf{B}$ term in the momentum equation.

25.1 Magnetic Tension Force

A first somewhat obvious, but important remark is that the magnetic force $\mathbf{j} \times \mathbf{B}$ is always perpendicular to the magnetic field, as is obvious from the cross product. In other words, there is no impediment to the plasma flow along field lines. On the other hand, the field might resist or drive plasma motion perpendicular to the field lines. The latter, in turn, causes the field lines to move, in accordance with the induction equation. It is important to keep this in mind, because we are now going to rewrite the Lorentz force in a form in which this constraint is not manifest. Using the vector identity

$$\nabla(\mathbf{A}\cdot\mathbf{B}) = \mathbf{A}\times(\nabla\times\mathbf{B}) + \mathbf{B}\times(\nabla\times\mathbf{A}) + (\mathbf{A}\cdot\nabla)\mathbf{B} + (\mathbf{B}\cdot\nabla)\mathbf{A} \tag{25.1}$$

(see also Equation (23.24)) and Ampère's law ($\mu_0\mathbf{j} = \nabla\times\mathbf{B}$), we can rewrite the Lorentz force as

$$\mathbf{j}\times\mathbf{B} = \frac{(\nabla\times\mathbf{B})\times\mathbf{B}}{\mu_0} = \frac{(\mathbf{B}\cdot\nabla)\mathbf{B}}{\mu_0} - \nabla\frac{B^2}{2\mu_0}. \tag{25.2}$$

The second term is often improperly referred to as the magnetic pressure, because it is reminiscent of the $-\nabla p$ term in the momentum equation, while the first term is improperly referred to as magnetic tension. However, remember that magnetic pressure is only effective perpendicular to the field lines, as we have already remarked. To better understand the nature of the Lorentz force in MHD, we factor

$$\mathbf{B} = B\hat{\mathbf{b}}, \tag{25.3}$$

where $\hat{\mathbf{b}}$ is a unit vector. Accordingly, we can write

$$\begin{aligned}(\mathbf{B}\cdot\nabla)\mathbf{B} &= B(\hat{\mathbf{b}}\cdot\nabla)(B\hat{\mathbf{b}})\\ &= B^2(\hat{\mathbf{b}}\cdot\nabla)\hat{\mathbf{b}} + B\hat{\mathbf{b}}(\hat{\mathbf{b}}\cdot\nabla)B\\ &= B^2(\hat{\mathbf{b}}\cdot\nabla)\hat{\mathbf{b}} + \frac{1}{2}\hat{\mathbf{b}}(\hat{\mathbf{b}}\cdot\nabla)B^2.\end{aligned} \tag{25.4}$$

The last term on the RHS is, apart from a factor 1/2, the projection of ∇B^2 parallel to $\mathbf{B}$:

$$[\hat{\mathbf{b}}(\hat{\mathbf{b}}\cdot\nabla)B^2]_i = \sum_{j=1}^{3}\hat{b}_i\hat{b}_j\partial_j(B^2) = [(\hat{\mathbf{b}}\otimes\hat{\mathbf{b}})\nabla(B^2)]_i =: \nabla_\parallel B^2. \tag{25.5}$$

Plugging this back into Equation (25.2) yields

$$\mathbf{j}\times\mathbf{B} = \frac{B^2}{\mu_0}(\hat{\mathbf{b}}\cdot\nabla)\hat{\mathbf{b}} - [\mathbf{I}-\hat{\mathbf{b}}\otimes\hat{\mathbf{b}}]\frac{\nabla B^2}{2\mu_0} = \frac{B^2}{\mu_0}(\hat{\mathbf{b}}\cdot\nabla)\hat{\mathbf{b}} - \nabla_\perp\frac{B^2}{2\mu_0}. \tag{25.6}$$

The last term is the component of $\nabla(B^2/2\mu_0)$ perpendicular to the field lines. This is the correct form of the magnetic pressure, while the first term on the RHS is the proper form of the magnetic tension.

We remark that $u_B = B^2/2\mu_0$ is the energy density of the magnetic field, so the magnetic pressure is

$$p_B = (\Gamma_B - 1)u_B, \tag{25.7}$$

which has the same form as the equation of state of an ideal gas (Equation (2.16)), with an effective adiabatic index $\Gamma_B = 2$. This makes sense, because, if we compress a flux tube perpendicular to the field, flux freezing (Chapter 24) gives $B \propto \rho$, so $p_B \propto \rho^2$. For ionized hydrogen $\Gamma = 5/3$, so magnetic energy is more effective at resisting compression than gas internal energy. Of course, magnetic pressure can only resist compression perpendicular to the field lines.

To understand the meaning of the first term on the RHS of Equation (25.6), note that

$$(\hat{\mathbf{b}} \cdot \nabla)\hat{\mathbf{b}} = \boldsymbol{\kappa}, \tag{25.8}$$

where $1/|\boldsymbol{\kappa}|$ is the curvature radius of the field lines and $\boldsymbol{\kappa}$ points towards the centre of curvature. In other words, $\boldsymbol{\kappa} = 0$ for straight field lines. For a circular field line, $1/|\boldsymbol{\kappa}|$ is the radius of the circle spanned by the line. This means that the magnetic field lines back-react on the plasma as if they were elastic strings under a tension B^2/μ_0 per unit area.

Summarizing, we find that there are two components of the Lorentz force. The first is the magnetic pressure, according to which the magnetic field lines resist being bundled together. The second is the magnetic tension, according to which the magnetic field lines resist bending. This situation is illustrated in Figure 25.1. We will see that magnetic tension is the restoring force for Alfvén waves (Chapter 26) and the driver of the magnetorotational instability (Chapter 27).

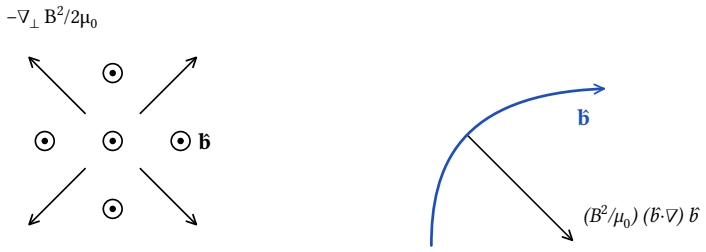


Figure 25.1: *Left panel:* effect of the magnetic pressure force. *Right panel:* effect of the magnetic tension force.

As we have already anticipated (see Equation (23.23)), the Lorentz force can be rewritten as the divergence of the magnetic stresses:

$$\frac{(\mathbf{B} \cdot \nabla)\mathbf{B}}{\mu_0} - \nabla\frac{B^2}{2\mu_0} = \nabla \cdot \left[\frac{\mathbf{B} \otimes \mathbf{B}}{\mu_0} - \frac{B^2}{2\mu_0}\mathbf{I}\right] =: \nabla \cdot \mathbf{T}_{\mathrm{EM}}. \tag{25.9}$$

This follows from Equation (23.25). Including both the fluid stresses $\mathbf{T}$ and the magnetic stresses, the momentum equation reads:

$$\rho\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = \nabla \cdot [\mathbf{T} + \mathbf{T}_{\mathrm{EM}}]. \tag{25.10}$$

For an ideal fluid $\mathbf{T} = -p\mathbf{I}$, however shear and bulk viscosity can also be included in $\mathbf{T}$ to obtain the viscous MHD equations. In conservation form, the momentum equation reads

$$\frac{\partial(\rho\mathbf{v})}{\partial t} + \nabla \cdot [\rho\mathbf{v} \otimes \mathbf{v} - \mathbf{T} - \mathbf{T}_{\mathrm{EM}}] = 0, \tag{25.11}$$

where $\rho\mathbf{v} \otimes \mathbf{v}$ is the flux of momentum carried by the flow[15].

Example: Solar prominence

Solar prominences are plasma arches which rise above the solar photosphere and are supported by the magnetic tension, as illustrated by Figure 25.2 and Figure 25.3. Prominences are made of cool

[15]This term is not a real force. It appears only because we are writing the equations in the Eulerian frame.

gas, $T \sim 10^4$ K, with densities $n \sim 10^{10}$–10^{11} cm^{-3}, about 100 times denser than the surrounding corona, which has $T \sim 10^6$ K and $n \sim 10^9$ cm^{-3}. The scale height of a gas at 10^4 K in the gravity of the Sun is only a few hundred km, while prominences extend for tens of thousands of km above the photosphere. So the gas pressure alone cannot support them against gravity.

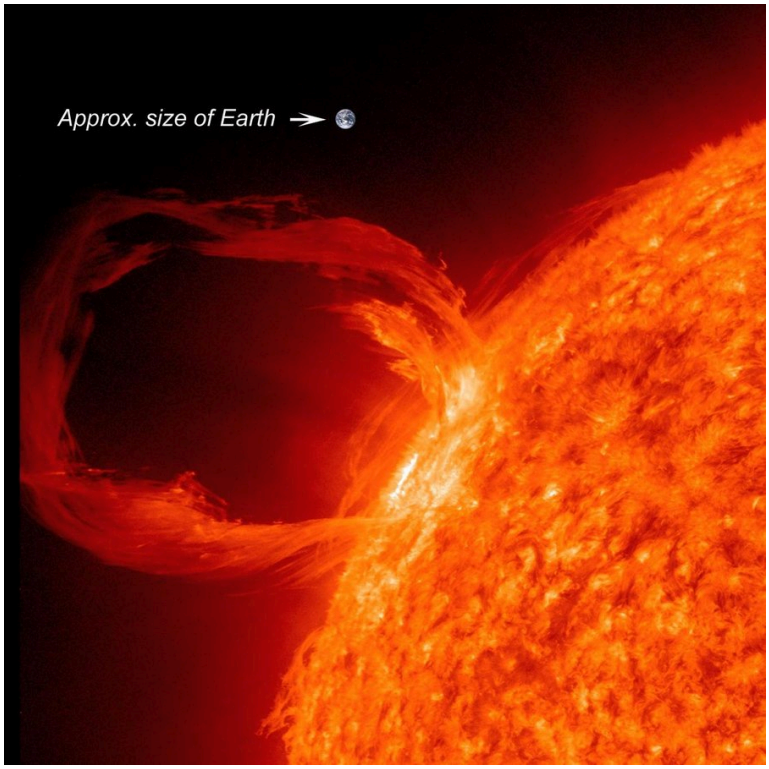


Figure 25.2: Solar prominence in UV. Credit: NASA (What is a solar prominence?); not covered by the license of these notes.

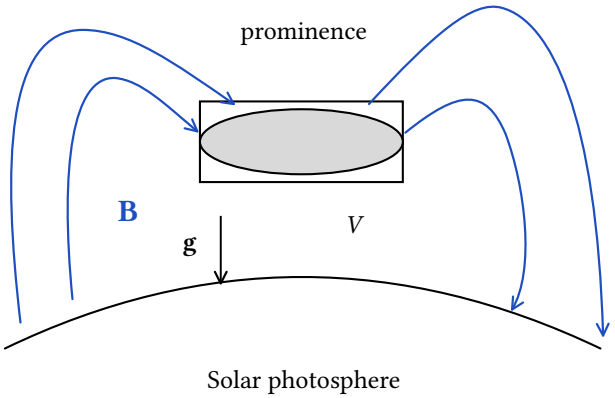


Figure 25.3: Force diagram for a Solar prominence.

To understand how prominences are supported against gravity, let us consider the balance of forces on the region in the volume V with boundary surface S:

$$
\begin{aligned}
0 &= \frac{\mathrm{d}}{\mathrm{d}t}\int_V \rho\mathbf{v}\mathrm{d}V \\
&= -\int_V \nabla\cdot[\rho\mathbf{v}\otimes\mathbf{v} - \mathbf{T} - \mathbf{T}_{\mathrm{EM}}]\mathrm{d}V \\
&\quad + \int_V \rho\mathbf{g}\mathrm{d}V \\
&= -\int_S \rho\mathbf{v}\mathbf{v}\cdot\mathrm{d}\mathbf{S} \\
&\quad - \int_S \left(p + \frac{B^2}{2\mu_0}\right)\mathrm{d}\mathbf{S} \\
&\quad + \int_S \frac{\mathbf{B}}{\mu_0}\mathbf{B}\cdot\mathrm{d}\mathbf{S} + \int_V \rho\mathbf{g}\mathrm{d}V.
\end{aligned}
\tag{25.12}
$$

For a prominence at rest, $\mathbf{v} = 0$ and the first term vanishes. Note that the field lines enter V going down on the left and leave it going up on the right. In other words, the prominence sits in a dip of the field lines, like a weight on a hammock. The main force balancing gravity is the magnetic tension, arising from the $\mathbf{B}\mathbf{B}\cdot\mathrm{d}\mathbf{S}$ term. On the left, $\mathbf{B}$ points into V, so $\mathbf{B}\cdot\mathrm{d}\mathbf{S} < 0$ and the force is opposite to $\mathbf{B}$. Since $\mathbf{B}$ points down, the force points up. On the right, $\mathbf{B}$ points out of V, so $\mathbf{B}\cdot\mathrm{d}\mathbf{S} > 0$ and the force is along $\mathbf{B}$, which also points up. The horizontal components cancel. The magnetic pressure, arising from the squeezing of the field lines below the prominence, also contributes, but is subdominant. We work out a simple model of this configuration in Exercise 25.1.

The attentive reader might have noticed that the term $\mathbf{B}\mathbf{B}\cdot\mathrm{d}\mathbf{S}$ gives a force along $\mathbf{B}$, with magnitude $B^2|\hat{\mathbf{b}}\cdot\mathrm{d}\mathbf{S}|/\mu_0$. How can this be possible? We have seen that the magnetic force is always perpendicular to $\mathbf{B}$. The point is that this term represents the net force from the magnetic tension on the whole volume V, not the force at a specific point. In other words, we should not confuse the force on individual fluid elements, which is always perpendicular to $\mathbf{B}$, with the net force on a control volume, which can be in any direction.

25.2 Hydromagnetic Equilibria

Consider the hydrostatic equilibrium of a plane-parallel, magnetized atmosphere in a gravitational potential $\Psi(z)$, as illustrated in Figure 25.4. This equilibrium configuration could be representative of the disk in a spiral galaxy, or of the photosphere of the Sun.

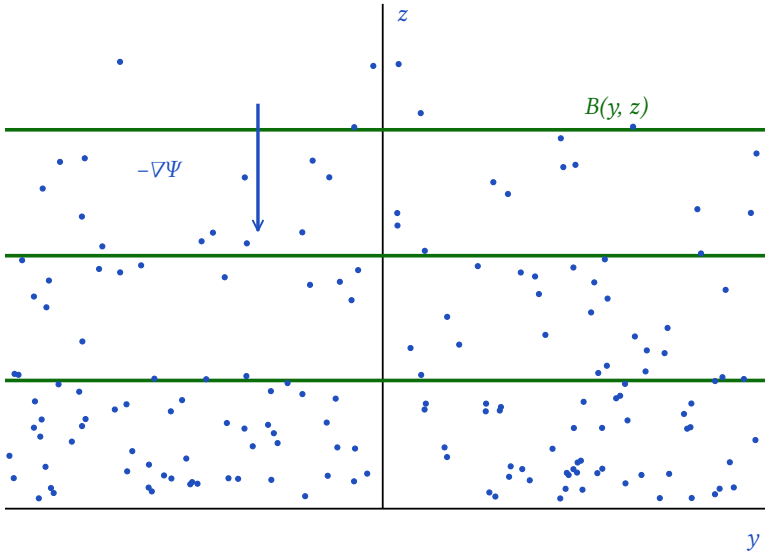


Figure 25.4: Force diagram for a plane-parallel magnetized atmosphere.

The force-balance equation for hydromagnetic equilibrium is

$$-\nabla p - \rho\nabla\Psi + \frac{1}{\mu_0}(\nabla\times\mathbf{B})\times\mathbf{B} = 0. \tag{25.13}$$

We will make two simplifying assumptions.

1. The gas is isothermal, so $p = c_{\rm iso}^2\rho$. This means that, in absence of magnetic fields,

$$\rho(z) = \rho_0\exp(-\Psi/c_{\rm iso}^2). \tag{25.14}$$

2. We assume that the magnetic field is planar and does not depend on x, meaning that

$$\mathbf{B} = B_y(y,z)\hat{\mathbf{y}} + B_z(y,z)\hat{\mathbf{z}}. \tag{25.15}$$

   This means that there exists a vector potential $\mathbf{A} = A(y,z)\hat{\mathbf{x}}$ such that $\nabla\times\mathbf{A} = \mathbf{B}$. In particular,

$$B_y = \partial_z A, \quad B_z = -\partial_y A \qquad\Leftrightarrow\qquad \mathbf{B} = \nabla A\times\hat{\mathbf{x}}. \tag{25.16}$$

   Note that $\mathbf{A}\perp\mathbf{B}$ and that the function $A(y,z)$ is constant along the field lines, because $\mathbf{B}\cdot\nabla A = 0$. So the isocontours of $A(y,z)$ are the field lines.

To compute the Lorentz force we use

$$\nabla\times\mathbf{B} = \nabla\times(\nabla\times\mathbf{A}) = -\nabla^2\mathbf{A} + \nabla(\nabla\cdot\mathbf{A}) = -\hat{\mathbf{x}}\nabla^2 A \tag{25.17}$$

and

$$(\nabla\times\mathbf{B})\times\mathbf{B} = -(\nabla^2 A)\hat{\mathbf{x}}\times(-\hat{\mathbf{x}}\times\nabla A) = -(\nabla^2 A)\nabla A. \tag{25.18}$$

For reasons that will be more clear later, we rewrite

$$\nabla p + \rho\nabla\Psi = \mathrm{e}^{-\Psi/c_{\rm iso}^2}\nabla q, \qquad q = p\,\mathrm{e}^{\Psi/c_{\rm iso}^2}. \tag{25.19}$$

The force equation finally reduces to

$$\boxed{-\frac{1}{\mu_0}(\nabla^2 A)\nabla A = \mathrm{e}^{-\Psi/c_{\rm iso}^2}\nabla q.} \tag{25.20}$$

The last equation shows that $\nabla q \parallel \nabla A$, which means that $q = q(A)$. Since $\nabla q = (\mathrm{d}q/\mathrm{d}A)\nabla A$, the force equation becomes

$$\nabla^2 A = -\mu_0 \mathrm{e}^{-\Psi/c_{\mathrm{iso}}^2} \frac{\mathrm{d}q}{\mathrm{d}A}. \tag{25.21}$$

This is a nonlinear Poisson equation for A, analogous to the Grad–Shafranov equation used to model the magnetic equilibria of tokamaks and of pulsar magnetospheres. One can then find equilibrium configurations by specifying $q = q(A)$, which encodes the amount of matter enclosed by the field lines A and $A + \mathrm{d}A$, and solving Equation (25.21) for A. We shall not follow this line of reasoning here, instead we will make two more simplifying assumptions, following E. N. Parker [17].

1. We assume that the magnetic field lines are horizontal, $\mathbf{B} = B(z)\hat{\mathbf{y}}$.
2. We assume that the ratio of magnetic to gas pressure is constant

$$\frac{B^2}{2\mu_0 p} = \alpha_0. \tag{25.22}$$

With these simplifications, the equation for equilibrium becomes

$$(1 + \alpha_0)\frac{\mathrm{d}p}{\mathrm{d}z} = -\frac{p}{c_{\mathrm{iso}}^2}\frac{\mathrm{d}\Psi}{\mathrm{d}z}. \tag{25.23}$$

Apart from the additional factor $1 + \alpha_0$, this is the usual equation for hydrostatic equilibrium of a plane-parallel isothermal atmosphere (Equation (10.2)). The solution is

$$\begin{aligned} p(z) &= p_0 \exp\left(-\frac{\Psi(z)}{(1+\alpha_0)c_{\mathrm{iso}}^2}\right), \\ B(z) &= B_0 \exp\left(-\frac{\Psi(z)}{2(1+\alpha_0)c_{\mathrm{iso}}^2}\right). \end{aligned} \tag{25.24}$$

For a uniform gravitational field, $\Psi = gz$, the scale height of the gas is $H = (1 + \alpha_0)c_{\mathrm{iso}}^2/g$. In other words, the magnetic field makes the atmosphere thicker, because part of the weight of the gas is supported by the magnetic pressure. The inverse of α_0,

$$\beta = \frac{2\mu_0 p}{B^2}, \tag{25.25}$$

is called the *plasma beta*. Plasmas with $\beta \gg 1$ are dominated by the gas pressure, while plasmas with $\beta \ll 1$ are dominated by the magnetic field.

Example: Magnetic support of the Galactic disk

Consider the gas in the disk of the Milky Way near the Sun. The magnetic field is $B \approx 0.5\,\mathrm{nT}$ ($5\,\mu\mathrm{G}$), so the magnetic pressure is $B^2/2\mu_0 \approx 1 \times 10^{-13}\,\mathrm{Pa}$. The thermal pressure of the gas is $p/k_B \approx 3000\,\mathrm{K\,cm^{-3}}$, that is $p \approx 4 \times 10^{-14}\,\mathrm{Pa}$. So $\alpha_0 \approx 2$, or $\beta \approx 0.4$. In other words, the magnetic field is as important as the gas pressure in supporting the disk against the gravity of the stars[16]. In the model of Equation (25.24), the field increases the scale height of the gas by a factor $1 + \alpha_0 \approx 3$.

Remarkably, the solution in Equation (25.24) turns out to be unstable for any $\alpha_0 > 0$ to long-wavelength perturbations, if the perturbations are also isothermal. This is the so-called *Parker instability* [17]. Note that, if the perturbations have adiabatic index Γ and are allowed to vary across the field lines, the atmosphere is unstable for $\Gamma < 1 + \alpha_0$ [18]. Without magnetic field, $\alpha_0 = 0$, this is the condition $N^2 < 0$ for the onset of convection in an isothermal atmosphere (Section 10.2). In other words, an isothermal atmosphere is marginally stable to isothermal perturbations, $N = 0$, and the magnetic field makes it unstable. The Parker instability is demonstrated in the MHD simulations shown in Figure 25.5.

[16] The turbulent motions of the gas and the cosmic rays also contribute pressures of the same order, so in reality the disk is supported by all of these components together.

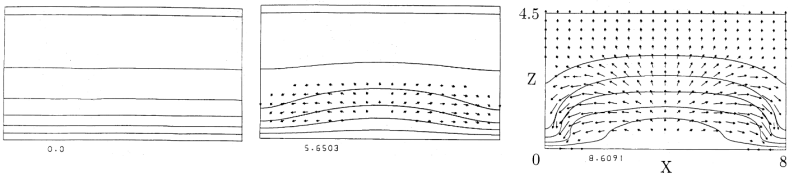


Figure 25.5: Development of the Parker instability in 3D MHD simulations, at three times, shown in a vertical slice. From R. Matsumoto and K. Shibata [19]; © Astronomical Society of Japan, reproduced for non-commercial educational use and not covered by the license of these notes.

Imagine starting from horizontal field lines, like in the left panel of the figure, and adding a small bend to them. The gas in the regions where the field lines are inclined can now slide down along them, because the magnetic force has no component along the field. In doing so, the gas loses gravitational potential energy and collects in the valleys, as shown by the arrows in the middle panel. The crests of the field lines are now lighter, so they are pushed further up by the magnetic pressure below them, the gas drains more efficiently, and the perturbation grows. This instability is responsible, for example, for the emergence of magnetic flux tubes from the solar interior, which gives rise to the type of prominence we have discussed previously.

In the case of the solar corona, the field lines at the foot of the prominences are anchored in the outer convection layers of the Sun. The motion of plasma in this region moves the field lines in random ways and builds enormous tension in the field. When field lines of opposite polarities find themselves sufficiently close they can annihilate in a so-called *reconnection event*. This might seem in contradiction with flux freezing, since the magnetic Reynolds number in the corona is enormous (Chapter 24). However, reconnection takes place in thin current sheets, in which the field changes direction over a very short distance. The magnetic Reynolds number computed with this length is small, so the field lines can diffuse through the plasma. Reconnection suddenly releases the magnetic tension and generates solar flares and coronal mass ejections. Solar eruptions can significantly impact satellites and space communication and can be hazardous for astronauts. The prediction of solar eruptions using ground- and space-based observations, in conjunction with MHD simulations, is a field of its own: *space weather modeling*.

25.3 In-class worksheet

Explosive Reconnection. Plasma is contained in a region bounded by two perfectly conducting plates. Two magnetic flux tubes are anchored on the plates and stretched by the application of an external force $\mathbf{f}$ as shown in panel (a) of Figure 25.6.

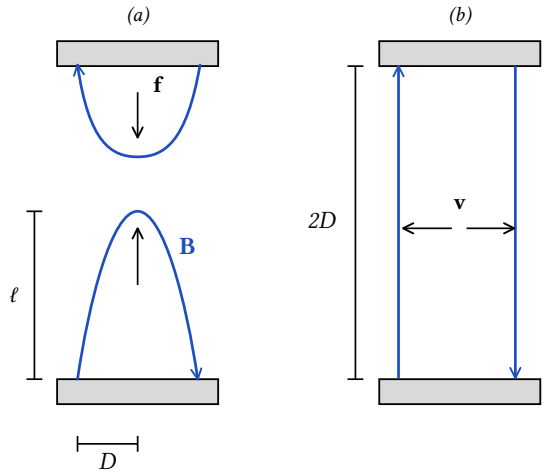


Figure 25.6: Explosive reconnection of two magnetic flux tubes anchored on conducting plates: (a) before and (b) after reconnection.

When $\ell \approx D$, the field lines reconnect and snap to a new configuration shown in panel (b). This process releases energy stored in the magnetic tension of the field to accelerate the plasma to a velocity $\mathbf{v}$, which reaches a maximum when the field lines are perfectly straight. Assume that the ideal-MHD approximation holds at all times, except when reconnection takes place.

1. Show that, apart from geometric factors of order unity, the force per unit volume necessary to hold the field lines bent to a height ℓ is
$$|\mathbf{f}| \approx \frac{B^2}{\mu_0}\frac{\ell}{D^2}. \tag{25.26}$$
2. Show that, apart from factors of order unity, the maximum velocity of the plasma after the reconnection event is
$$v \approx v_A, \tag{25.27}$$
where v_A is the Alfvén velocity
$$v_A = \frac{B}{\sqrt{\mu_0 \rho}}. \tag{25.28}$$
3. Magnetic fields in solar active regions have strengths $B \sim 0.03\,\mathrm{T}$ (300 G) and extend over regions of size $L \sim 3 \times 10^4$ km. Estimate the magnetic energy available. Compare it with the energy released by a large solar flare, $\sim 10^{32}$ erg $= 10^{25}$ J.

25.4 Exercises

Exercise 25.1 (A model for solar prominences). Consider the model of a solar prominence of R. Kippenhahn and A. Schlüter [20]. The prominence is a vertical sheet of isothermal gas, with isothermal sound speed c_{iso} (Equation (9.16)), in a uniform gravitational field $\mathbf{g} = -g\hat{\mathbf{z}}$. The magnetic field is
$$\mathbf{B} = B_x \hat{\mathbf{x}} + B_z(x)\hat{\mathbf{z}}, \tag{25.29}$$
where B_x is a constant, and all quantities depend only on x.

1. Show that $\nabla \cdot \mathbf{B} = 0$ and that the horizontal component of the force balance gives
$$p + \frac{B_z^2}{2\mu_0} = \text{const.} \tag{25.30}$$
2. Show that the vertical component of the force balance is
$$\rho g = \frac{B_x}{\mu_0}\frac{\mathrm{d}B_z}{\mathrm{d}x}. \tag{25.31}$$
Which component of the Lorentz force holds the gas up?
3. Assume that $p \to 0$ and $B_z \to \pm B_\infty$ for $x \to \pm\infty$. Show that
$$B_z = B_\infty \tanh(x/w), \qquad w = \frac{2B_x}{B_\infty}\frac{c_{\mathrm{iso}}^2}{g}, \tag{25.32}$$
and find $\rho(x)$. Sketch the field lines.
4. Show that the mass per unit area of the sheet is
$$\Sigma = \int_{-\infty}^{\infty} \rho \,\mathrm{d}x = \frac{2B_x B_\infty}{\mu_0 g}. \tag{25.33}$$
Evaluate w, the central number density $n(0)$ and Σ for $B_x = 1\,\mathrm{mT}$ (10 G), $B_\infty = 0.2\,\mathrm{mT}$ (2 G), $T = 7000\,\mathrm{K}$, mean molecular weight $\mu = 0.6$, and $g = 270\,\mathrm{m\,s^{-2}}$. Compare with observed prominences, which have widths of a few thousand km and densities $n \sim 10^{10}$–$10^{11}\,\mathrm{cm^{-3}}$.

25.5 Further reading

- C. Clarke and B. Carswell [1], Chapter 13 (plasmas).

26. MHD Waves

Waves play a very important role in the dynamics of magnetized plasmas. Particle-wave interaction with the interstellar medium regulates the propagation of cosmic rays in the Galaxy. In strong MHD turbulence (Chapter 21), the non-linear interaction between Alfvén waves acts as the main driver of the turbulence cascade. Torsional Alfvén waves extract angular momentum from proto-stellar cores allowing them to collapse and form stars (Exercise 26.1). These are just a few examples. In this chapter, we discuss the phenomenology of waves in compressible MHD.

26.1 Waves in a Homogeneous Plasma

We consider a homogeneous plasma described by the ideal-MHD equations with

$$\rho_0 = \text{const}, \qquad \mathbf{v}_0 = 0, \qquad \mathbf{B}_0 = B_0 \hat{\mathbf{z}}. \tag{26.1}$$

We proceed in the same way as when we derived the dispersion relation for sound waves in unmagnetized media (Chapter 9). However, as we will see shortly, the phenomenology of MHD waves is significantly more complex than that of sound waves. As before, we introduce a perturbation with a Lagrangian displacement $\boldsymbol{\xi}$, shown in Figure 26.1.

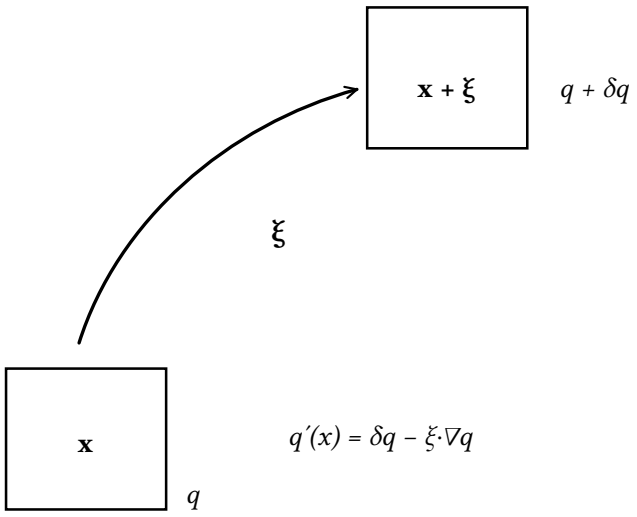


Figure 26.1: Lagrangian displacement of a fluid element from position $\mathbf{x}$ to position $\mathbf{x} + \boldsymbol{\xi}$.

The pressure and density perturbations in the Lagrangian frame are

$$\delta\rho = -\rho_0 \nabla \cdot \boldsymbol{\xi}, \qquad \delta p = \frac{\Gamma p_0}{\rho_0}\delta\rho = -\Gamma p_0 \nabla \cdot \boldsymbol{\xi}. \tag{26.2}$$

Since the background is uniform, the Lagrangian and Eulerian perturbations coincide. The magnetic field perturbations can be computed from the linearized induction equation:

$$\frac{\partial \delta\mathbf{B}}{\partial t} = \nabla \times \left(\frac{\partial \boldsymbol{\xi}}{\partial t} \times \mathbf{B}_0 \right) \Rightarrow \delta\mathbf{B} = \nabla \times (\boldsymbol{\xi} \times \mathbf{B}_0). \tag{26.3}$$

The linearized momentum equation becomes (note that the unperturbed current vanishes $\mathbf{j}_0 = 0$):

$$\rho\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = -\nabla p + \mathbf{j}\times\mathbf{B} \Rightarrow \rho_0\frac{\partial^2\boldsymbol{\xi}}{\partial t^2} = -\nabla\delta p + \delta\mathbf{j}\times\mathbf{B}_0. \tag{26.4}$$

The perturbed current is

$$\delta\mathbf{j} = \frac{1}{\mu_0}\nabla\times\delta\mathbf{B} = \frac{1}{\mu_0}\nabla\times[\nabla\times(\boldsymbol{\xi}\times\mathbf{B}_0)], \tag{26.5}$$

so putting everything together we get the oscillation equation

$$\boxed{\begin{aligned}\rho_0\frac{\partial^2\boldsymbol{\xi}}{\partial t^2} &= \Gamma p_0\nabla(\nabla\cdot\boldsymbol{\xi}) \\ &\quad + \frac{1}{\mu_0}\{\nabla\times[\nabla\times(\boldsymbol{\xi}\times\mathbf{B}_0)]\}\times\mathbf{B}_0.\end{aligned}} \tag{26.6}$$

Of course, if $\mathbf{B}_0 = 0$, this equation reduces to the equation for sound waves. All the new physics is in the last term on the RHS. To derive a dispersion relation, we take the usual ansatz

$$\boldsymbol{\xi}(t,\mathbf{x}) = \boldsymbol{\xi}_1 e^{\mathrm{i}(\mathbf{k}\cdot\mathbf{x}-\omega t)}, \tag{26.7}$$

which we plug into the oscillation equation. This yields

$$\begin{aligned}-\rho_0\omega^2\boldsymbol{\xi}_1 &= -\Gamma p_0\mathbf{k}(\mathbf{k}\cdot\boldsymbol{\xi}_1) \\ &\quad -\frac{1}{\mu_0}\{\mathbf{k}\times[\mathbf{k}\times(\boldsymbol{\xi}_1\times\mathbf{B}_0)]\}\times\mathbf{B}_0.\end{aligned} \tag{26.8}$$

As we have seen for sound waves (Chapter 9), the oscillation equation gives rise to an eigenvalue problem in the form

$$-\rho_0\omega^2\boldsymbol{\xi}_1 = \mathbf{F}(\mathbf{k})\boldsymbol{\xi}_1, \tag{26.9}$$

where $\mathbf{F}(\mathbf{k})$ is a matrix. This equation has solutions with $\boldsymbol{\xi}_1 \neq 0$ if and only if

$$\det[\mathbf{F}(\mathbf{k}) + \rho_0\omega^2\mathbf{I}] = 0. \tag{26.10}$$

The last equation defines the dispersion relation $\omega = \omega(\mathbf{k})$. The eigenvectors of $\mathbf{F}(\mathbf{k})$ define the polarization vector $\boldsymbol{\xi}_1$ of the wave.

26.2 Families of MHD Waves

In general, there are three solutions to the eigenvalue problem Equation (26.9), corresponding to three different modes. In the unmagnetized case, one of the modes is the sound wave, while the other two are the transverse modes with $\omega = 0$ we found in Chapter 9. In the magnetized case, the magnetic field provides a restoring force also for transverse displacements, and there are three types of waves: Alfvén waves, slow-, and fast-magnetosonic waves.

Without loss of generality, we can choose the axes so that $\mathbf{B}_0$ is along z and $\mathbf{k}$ is in the xz-plane, at an angle θ from $\mathbf{B}_0$. We will see that the y component of the displacement decouples from the other two. So the eigenvalue problem splits into a 1×1 problem (the Alfvén wave) and a 2×2 problem (the slow and fast waves). We consider these separately below.

26.2.1 Alfvén Waves

We first consider displacements along y, as depicted in Figure 26.2:

- $\mathbf{B}_0$ is along z
- $\mathbf{k}$ is in the xz-plane
- $\boldsymbol{\xi}$ is along y

Note that $\mathbf{k}\cdot\boldsymbol{\xi} = 0$, meaning that $\delta\rho = \delta p = 0$.

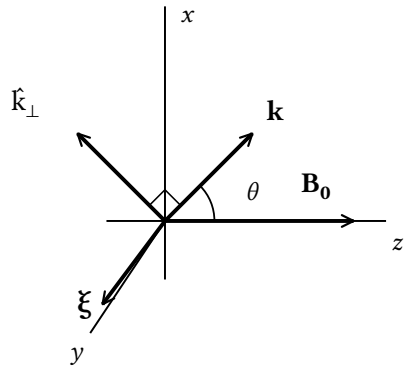


Figure 26.2: Geometry of the Alfvén wave: $\mathbf{B}_0$ along z, $\mathbf{k}$ in the xz-plane at an angle θ from $\mathbf{B}_0$, and displacement $\boldsymbol{\xi}$ along y.

To compute the Lorentz force we proceed as follows. In Fourier space, $\nabla \to \mathrm{i}\mathbf{k}$, so each derivative in Equation (26.3) and Equation (26.5) is replaced by a factor $\mathrm{i}\mathbf{k}$. First we note that

$$\boldsymbol{\xi} \times \mathbf{B}_0 = \xi_y B_0 \hat{\mathbf{x}}. \tag{26.11}$$

Next, we compute the magnetic field perturbation as

$$\delta\mathbf{B} = \mathrm{i}\mathbf{k} \times (\boldsymbol{\xi} \times \mathbf{B}_0) = \mathrm{i}k\xi_y B_0 \cos\theta \hat{\mathbf{y}}. \tag{26.12}$$

The current is given by

$$\begin{aligned} \mu_0 \delta\mathbf{j} &= \mathrm{i}\mathbf{k} \times \delta\mathbf{B} = -\mathbf{k} \times [\mathbf{k} \times (\boldsymbol{\xi} \times \mathbf{B}_0)] \\ &= k^2 \xi_y B_0 \cos\theta \hat{\mathbf{k}}_\perp. \end{aligned} \tag{26.13}$$

Finally, the Lorentz force is

$$\mu_0 \delta\mathbf{j} \times \mathbf{B}_0 = -k^2 B_0^2 \cos^2\theta \xi_y \hat{\mathbf{y}} = -(\mathbf{k} \cdot \mathbf{B}_0)^2 \boldsymbol{\xi}. \tag{26.14}$$

So $\boldsymbol{\xi}$ evolves as a mass attached to an elastic spring with spring constant $(\mathbf{k} \cdot \mathbf{B}_0)^2/\mu_0$. In formulas,

$$\begin{aligned} -\rho_0 \omega^2 \xi_y &= -\frac{1}{\mu_0} k^2 B_0^2 \cos^2\theta \xi_y \\ \Rightarrow \quad & \boxed{\omega = \pm \frac{B_0}{\sqrt{\mu_0 \rho_0}} k \cos\theta = \pm \mathbf{k} \cdot \mathbf{v}_A,} \end{aligned} \tag{26.15}$$

where $\mathbf{v}_A = \mathbf{B}_0/\sqrt{\mu_0\rho_0}$. We call $\omega_A = \mathbf{k} \cdot \mathbf{v}_A$ the Alfvén frequency. These waves are called Alfvén waves, after Alfvén (1942) (Physics Nobel Prize in 1970). Some comments are in order.

1. If $\theta = 0$, these waves propagate along the field lines with velocity (Alfvén velocity):

$$v_A = \frac{B_0}{\sqrt{\mu_0 \rho_0}}. \tag{26.16}$$

   If $\theta > 0$, the phase velocity is reduced to $v_A \cos\theta$. However, the group velocity is $\partial\omega/\partial\mathbf{k} = \pm v_A \hat{\mathbf{z}}$, for any θ. In other words, Alfvén waves always transport energy along the magnetic field lines, with velocity v_A.

2. The displacement $\boldsymbol{\xi}$ is perpendicular to the background field $\mathbf{B}_0$, so these waves are transverse. More precisely, Alfvén waves behave as transverse waves in an elastic string (the magnetic field). Indeed, the restoring force for these waves is magnetic tension (Chapter 25).

3. As already mentioned $\delta\rho = \delta p = 0$ for these waves. As such, Alfvén waves are incompressible and exist also in incompressible flows.

Although we have derived Alfvén waves as small perturbations, it turns out that Alfvén waves with $\delta\mathbf{B} = \mp\sqrt{\mu_0\rho}\,\delta\mathbf{v}$, propagating in the $\pm z$ direction, are an exact solution to the full non-linear equations, as long as the magnetic pressure $B^2/2\mu_0$ stays uniform. This means that MHD flows can support arbitrarily strong waves. We check the small-amplitude case in Exercise 24.1.

26.2.2 Magnetosonic Waves

We now consider displacements in the xz-plane, as depicted in Figure 26.3:

- $\mathbf{B}_0$ is along z
- $\mathbf{k}$ is in the xz-plane
- $\boldsymbol{\xi}$ is also in the xz-plane

Note that $\delta p, \delta\rho$ are no longer necessarily zero.

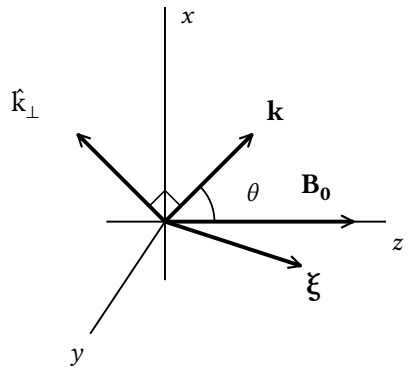


Figure 26.3: Geometry of the magnetosonic wave: $\mathbf{B}_0$ along z, $\mathbf{k}$ in the xz-plane at an angle θ from $\mathbf{B}_0$, and displacement $\boldsymbol{\xi}$ also in the xz-plane.

We can compute the current using the following vector identity

$$\mathbf{a}\times(\mathbf{b}\times\mathbf{c}) = \mathbf{b}(\mathbf{a}\cdot\mathbf{c}) - \mathbf{c}(\mathbf{a}\cdot\mathbf{b}) \tag{26.17}$$

and the fact that $\mathbf{k}\perp\boldsymbol{\xi}\times\mathbf{B}_0$, since the latter is in the y-direction, we can write

$$\mathbf{k}\times[\mathbf{k}\times(\boldsymbol{\xi}\times\mathbf{B}_0)] = \mathbf{k}[\mathbf{k}\cdot(\boldsymbol{\xi}\times\mathbf{B}_0)] - k^2(\boldsymbol{\xi}\times\mathbf{B}_0) = -k^2(\boldsymbol{\xi}\times\mathbf{B}_0). \tag{26.18}$$

Plugging this into the expression for the Lorentz force gives

$$\begin{aligned}\delta\mathbf{j}\times\mathbf{B}_0 &= -\frac{1}{\mu_0}\{\mathbf{k}\times[\mathbf{k}\times(\boldsymbol{\xi}\times\mathbf{B}_0)]\}\times\mathbf{B}_0\\ &= \frac{k^2}{\mu_0}(\boldsymbol{\xi}\times\mathbf{B}_0)\times\mathbf{B}_0 = -\frac{k^2B_0^2\xi_x}{\mu_0}\hat{\mathbf{x}}.\end{aligned} \tag{26.19}$$

The pressure force is

$$-\Gamma p_0\mathbf{k}(\mathbf{k}\cdot\boldsymbol{\xi}) = -k^2\Gamma p_0(\xi_x\sin\theta + \xi_z\cos\theta)(\sin\theta\hat{\mathbf{x}} + \cos\theta\hat{\mathbf{z}}). \tag{26.20}$$

Using $c_s^2 = \Gamma p_0/\rho_0$ and $v_A = B_0/\sqrt{\mu_0\rho_0}$, we find

$$\omega^2\begin{pmatrix}\xi_x\\ \xi_z\end{pmatrix} = k^2\begin{pmatrix}c_s^2\sin^2\theta + v_A^2 & c_s^2\sin\theta\cos\theta\\ c_s^2\sin\theta\cos\theta & c_s^2\cos^2\theta\end{pmatrix}\begin{pmatrix}\xi_x\\ \xi_z\end{pmatrix}. \tag{26.21}$$

The eigenvalues of this matrix are $\omega^2 = T/2 \pm \sqrt{T^2/4 - D}$, where $T = k^2(c_s^2 + v_A^2)$ and $D = k^4 c_s^2 v_A^2 \cos^2\theta$ are its trace and determinant. After some algebra, we find the dispersion relation

$$\boxed{\left(\frac{\omega}{k}\right)^2 = \frac{v_A^2 + c_s^2}{2} \pm \frac{1}{2}\sqrt{(v_A^2 - c_s^2)^2 + 4c_s^2 v_A^2 \sin^2\theta}.} \tag{26.22}$$

So there are two solutions and two corresponding polarization vectors $\boldsymbol{\xi}_s$ and $\boldsymbol{\xi}_f$. Note that $\boldsymbol{\xi}_s \perp \boldsymbol{\xi}_f$, since the matrix $\mathbf{F}(\mathbf{k})$ in Equation (26.21) is self-adjoint (see Equation (9.29)). Conventionally, $\boldsymbol{\xi}_s$ is taken to be the polarization corresponding to the minus sign in Equation (26.22), i.e., to the smaller phase velocity. This is the so-called slow magnetosonic wave. Conversely, $\boldsymbol{\xi}_f$ is called the fast magnetosonic wave. As we will see shortly, these have a mixed character of sound and magnetic waves.

26.3 Effect of Propagation Direction

We can get additional insight into the nature of the three modes by considering the properties of waves propagating along the magnetic field lines, or perpendicular to them. The general case will turn out to be intermediate between these two limits.

26.3.1 Parallel Propagating Waves ($\theta = 0$)

Let us consider the case of waves propagating along the background magnetic field lines, that is $\theta = 0$. In this case, the dispersion relation for Alfvén, slow, and fast modes are

$$\left(\frac{\omega}{k}\right)^2 = v_A^2, \qquad \left(\frac{\omega}{k}\right)^2 = \frac{1}{2}[v_A^2 + c_s^2 \pm |v_A^2 - c_s^2|]. \tag{26.23}$$

There are two different cases, depending on whether $v_A > c_s$ or $v_A < c_s$, shown in Figure 26.4.

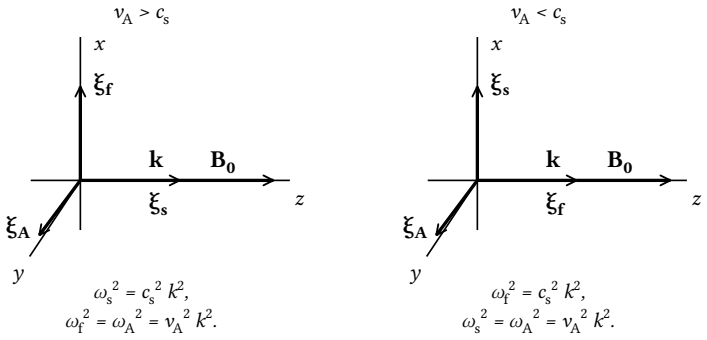


Figure 26.4: Polarization diagram for MHD waves propagating along the background magnetic field.

- If $v_A > c_s$, then
  - The slow mode polarization is along the magnetic field, the mode propagates at speed c_s and its restoring force is pressure.
  - Fast and Alfvén modes are orthogonal to the magnetic field and to each other and both have the character of Alfvén waves propagating with speed v_A. Magnetic tension is their restoring force.
- If $v_A < c_s$, then the situation is similar as before, but the slow and fast modes exchange character: the fast mode is now a sound wave, while the slow mode is an Alfvén wave.

26.3.2 Perpendicularly Propagating Waves ($\theta = 90°$)

Let us consider the case of waves propagating perpendicular to the field lines. In this case, Alfvén and slow modes cannot propagate (they have $\omega_{A,s} = 0$). The fast-magnetosonic wave satisfies the dispersion relation

$$\left(\frac{\omega}{k}\right)^2 = v_A^2 + c_s^2. \tag{26.24}$$

In this case $\boldsymbol{\xi}_f$ is parallel to $\mathbf{k}$. The restoring force for the mode is the sum of the magnetic and gas pressures, which explains why this mode is called the "magnetosonic" mode. In other words, the field lines are compressed, as in a sound wave, but they are not bent.

26.3.3 Friedrichs Diagrams

Friedrichs diagrams show the propagation velocity of the different MHD modes as a function of a generic propagation angle θ. The special cases of $\theta = 0, 90°$ discussed above correspond to the x- and y-axis in Figure 26.5. In general, slow and fast modes have a mixed character between sound waves and Alfvén waves.

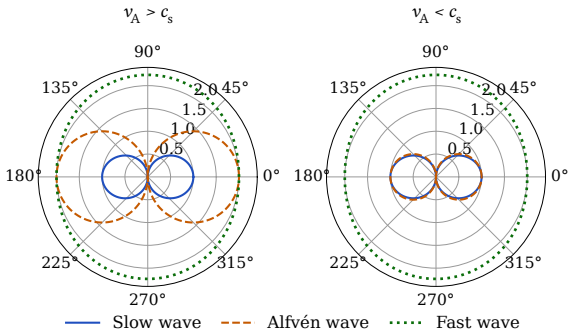


Figure 26.5: Friedrichs diagrams for ideal MHD, for $v_A = 2c_s$ (left) and $v_A = c_s/2$ (right). The phase speeds are in units of the smaller of v_A and c_s.

26.4 Weak and Strong Field Limits

Finally, it is also instructive to compare slow- and fast-magnetosonic waves in the limits of weak magnetic field ($c_s \gg v_A$) and strong magnetic field ($v_A \gg c_s$), as illustrated in Figure 26.6. Note that when v_A becomes a sizable fraction of the speed of light, we can no longer use Newtonian MHD, because the displacement current becomes important. In this limit, one needs to use the (significantly more complex) equations of relativistic magnetohydrodynamics.

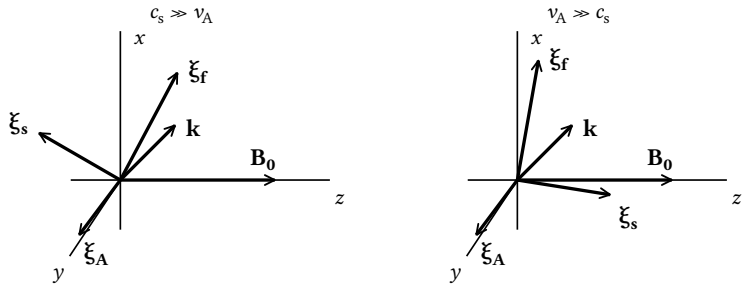


Figure 26.6: Slow and fast magnetosonic waves in the limits of weak (*left panel*) and strong (*right panel*) magnetic fields.

- *Left panel:* the fast mode is $\sim$ a sound wave, with $\omega \approx kc_s$, and the slow mode is nearly perpendicular to $\mathbf{k}$, with $\omega \approx kv_A \cos\theta$.
- *Right panel:* the fast mode is $\sim$ an Alfvén wave, with $\omega \approx kv_A$, and the slow mode is nearly parallel to $\mathbf{B}_0$, with $\omega \approx kc_s \cos\theta$. In other words, the slow mode is a sound wave guided along the magnetic field lines. This is the situation in the solar corona.

26.5 Shock Waves

As for unmagnetized gases, MHD waves can steepen into shock waves. Alfvén waves can steepen into *rotational discontinuities*, in which the magnetic field direction suddenly changes across an interface. Slow- and fast-magnetosonic waves can give rise to (slow and fast) shocks and rarefaction waves. The theory of MHD shocks is very complex and beyond the scope of this course. A good overview is given in Bazer and Ericson (1959). It is important to be aware that, even in the case of shocks propagating along magnetic field lines, MHD effects can be important. For example, a shock propagating along the magnetic field can create a component of the field parallel to the shock front (a *switch-on* shock), and the opposite process is also possible (a *switch-off* shock).

We treat in detail only one case that is relevant for the interstellar medium: a shock propagating in an isothermal flow with perpendicular magnetic field, as illustrated in Figure 26.7.

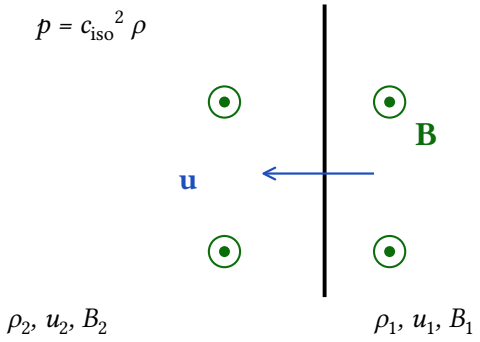


Figure 26.7: Isothermal shock propagating in a medium with perpendicular magnetic field.

The shock conditions for the shock shown in Figure 26.7 are:

$$[\![\rho u]\!] = 0, \quad [\![Bu]\!] = 0, \quad \left[\!\!\left[p + \rho u^2 + \frac{B^2}{2\mu_0} \right]\!\!\right] = 0, \tag{26.25}$$

where $[\![\,]\!]$ denotes the jump across the shock.

In the absence of a magnetic field, the shock conditions for an isothermal gas, $p = c_{\rm iso}^2 \rho$, are $\rho_1 u_1 = \rho_2 u_2$ and $c_{\rm iso}^2 \rho_1 + \rho_1 u_1^2 = c_{\rm iso}^2 \rho_2 + \rho_2 u_2^2$. Using the first condition to eliminate u_2 from the second, we find $c_{\rm iso}^2(\rho_2 - \rho_1) = \rho_1 u_1^2 (1 - \rho_1/\rho_2) = \rho_1 u_1^2 (\rho_2 - \rho_1)/\rho_2$. So the compression factor of an isothermal shock is

$$\frac{\rho_2}{\rho_1} = \left(\frac{u_1}{c_{\rm iso}} \right)^2, \tag{26.26}$$

which can become arbitrarily large for large upstream Mach numbers. We discuss isothermal shocks in more detail in Chapter 12. This is dramatically different when a magnetic field is present, as long as it is sufficiently strong:

$$u_1 \gg v_A = \frac{B_1}{\sqrt{\mu_0 \rho_1}} \gg c_{\rm iso}. \tag{26.27}$$

In this case, from $[\![\rho u]\!] = [\![Bu]\!] = 0$, we conclude that

$$\frac{B_1}{\rho_1} = \frac{B_2}{\rho_2}. \tag{26.28}$$

From the conservation of momentum, we have (using $u_1 \gg v_A \gg c_{\rm iso}$)

$$\begin{aligned}\frac{\rho_2}{\rho_1} &= \frac{c_{\rm iso}^2 + u_1^2 + \frac{1}{2}v_A^2}{c_{\rm iso}^2 + u_2^2 + \frac{B_2^2}{2\mu_0\rho_2}} \\ &\approx \frac{u_1^2}{c_{\rm iso}^2 + u_2^2 + \frac{B_2^2}{2\mu_0\rho_2^2}\rho_2} \\ &= \frac{u_1^2}{c_{\rm iso}^2 + u_2^2 + \frac{B_1^2}{2\mu_0\rho_1}\frac{\rho_2}{\rho_1}} \\ &\approx \frac{u_1^2}{\frac{v_A^2}{2}\frac{\rho_2}{\rho_1}}.\end{aligned} \tag{26.29}$$

Solving for ρ_2/ρ_1, we obtain

$$\boxed{\frac{\rho_2}{\rho_1} = \sqrt{2}\frac{u_1}{v_A}.} \tag{26.30}$$

This shows that the compression ratio now scales with the Alfvénic Mach number of the upstream gas, $\mathrm{Ma}_A = u_1/v_A$, instead of the square of its Mach number. This is another example showing that magnetic fields play an important role in the dynamics of the interstellar medium and in star formation.

Example: Shocks in molecular clouds

Consider a molecular cloud with temperature $T \approx 10\,\mathrm{K}$, number density $n \approx 100\,\mathrm{cm^{-3}}$ and magnetic field $B \approx 1\,\mathrm{nT}$ ($10\,\mu\mathrm{G}$). The gas has $\rho = \mu m_u n \approx 4\times10^{-22}\,\mathrm{g\,cm^{-3}} = 4\times10^{-19}\,\mathrm{kg\,m^{-3}}$ (with $\mu \approx 2.3$) and $c_{\rm iso} \approx 0.2\,\mathrm{km\,s^{-1}}$. The Alfvén velocity is

$$v_A = \frac{B}{\sqrt{\mu_0\rho}} \approx 1.4\,\mathrm{km\,s^{-1}}. \tag{26.31}$$

Note that $v_A \gg c_{\rm iso}$, so the magnetic pressure is much larger than the gas pressure. Suppose that the cloud is hit by a shock with $u_1 = 10\,\mathrm{km\,s^{-1}}$, for example driven by a nearby supernova remnant. Then $u_1 \gg v_A \gg c_{\rm iso}$, and Equation (26.30) applies. Without the magnetic field, the compression factor would be $(u_1/c_{\rm iso})^2 \approx 2500$. With the magnetic field, it is only $\sqrt{2}u_1/v_A \approx 10$. In other words, the magnetic field makes the shocked gas about 250 times less dense. So it is much harder to form dense cores in molecular clouds with shocks alone.

26.6 In-class worksheet

Fast Magnetosonic Waves. Consider a homogeneous plasma described by the ideal-MHD equations with

$$\rho_0 = \mathrm{const}, \qquad \mathbf{v}_0 = 0, \qquad \mathbf{B}_0 = B_0\hat{\mathbf{z}}. \tag{26.32}$$

1. Introduce a small perturbation in the form of a displacement $\boldsymbol{\xi}(t,\mathbf{x})$ of the fluid elements and show that the oscillation equation is

$$\begin{aligned}\rho_0\frac{\partial^2\boldsymbol{\xi}}{\partial t^2} &= \Gamma p_0\nabla(\nabla\cdot\boldsymbol{\xi}) \\ &+\frac{1}{\mu_0}\{\nabla\times[\nabla\times(\boldsymbol{\xi}\times\mathbf{B}_0)]\}\times\mathbf{B}_0.\end{aligned} \tag{26.33}$$

2. Introduce the ansatz $\boldsymbol{\xi}(t,\mathbf{x}) = \boldsymbol{\xi}_1 e^{\mathrm{i}(\mathbf{k}\cdot\mathbf{x}-\omega t)}$ and show that Equation (26.33) reduces to

$$
\begin{aligned}
-\rho_0\omega^2\boldsymbol{\xi}_1 &= -\Gamma p_0 \mathbf{k}(\mathbf{k}\cdot\boldsymbol{\xi}_1) \\
&\quad -\frac{1}{\mu_0}\{\mathbf{k}\times[\mathbf{k}\times(\boldsymbol{\xi}_1\times\mathbf{B}_0)]\}\times\mathbf{B}_0 .
\end{aligned}
\tag{26.34}
$$

[Hint. Use $\partial_t \to -\mathrm{i}\omega$ and $\nabla \to \mathrm{i}\mathbf{k}$.]

3. Consider waves with $\mathbf{k} \parallel \hat{\mathbf{x}}$ and $\boldsymbol{\xi} \parallel \mathbf{k}$. Find the dispersion relation $\omega = \omega(\mathbf{k})$ and compute group and phase velocities for these waves.

26.7 Exercises

Exercise 26.1 (Torsional Alfvén waves).

1. Consider the torsional Alfvén wave, in cylindrical coordinates (R, ϕ, z),

$$
\begin{aligned}
\mathbf{v} &= \epsilon v_A g(R) f(z - v_A t)\hat{\boldsymbol{\phi}}, \\
\mathbf{B} &= B_0\hat{\mathbf{z}} - \epsilon B_0 g(R) f(z - v_A t)\hat{\boldsymbol{\phi}},
\end{aligned}
\tag{26.35}
$$

where f and g are arbitrary functions and ϵ is a small parameter. Show that it is a solution of the ideal-MHD equations to first order in ϵ.

2. Consider a uniform plasma at rest, threaded by the magnetic field $B_0\hat{\mathbf{z}}$. A thin disk of radius a, made of a perfectly conducting material, lies in the xy-plane with its centre at the origin. At $t = 0$, the disk is set into rotation with a constant angular velocity Ω, with $\Omega a \ll v_A$. The plasma just above and below the disk must move with the same tangential velocity as the disk. Use the solution of part 1 to show that, at time t, the plasma with $R < a$ and $|z| < v_A t$ rotates with angular velocity Ω, while the rest of the plasma is still at rest.

3. Let Σ be the mass per unit area of the disk. Estimate how long it takes before the disk starts to slow down. [Hint. Compare the angular momentum of the disk with that of the plasma set in motion by the wave.]

Adapted from R. M. Kulsrud [4], Problem 3.4.

26.8 Further reading

- C. Clarke and B. Carswell [1], Chapter 13 (plasmas).
- J. Bazer and W. B. Ericson [22]

27. The Magnetorotational Instability

The magnetorotational instability (MRI) was first discovered by E. P. Velikhov [23] and S. Chandrasekhar [24], who studied the stability of a conducting fluid between two rotating cylinders, and then rediscovered in the context of accretion disks by S. A. Balbus and J. F. Hawley [25]. This is the instability thought to be responsible for the α-viscosity (Chapter 19) in many astrophysical systems: from X-ray binaries to AGN disks. It is one of the most important, among the many instabilities operating in plasmas, so we will cover it in detail in these notes.

It was long known that a way to transport angular momentum in the disk is through magnetic stresses, that is the $B_R B_\phi/\mu_0$ term in the momentum equation (Chapter 25). However, to explain the observed luminosities from accretion disks, the magnetic stresses must be similar in magnitude to the pressure in the disk p (the proportionality constant being the α-viscosity coefficient). Given a small component of the field in the radial direction, flux freezing (Chapter 24) can grow B_ϕ linearly in time. However, since B_R is small, this mechanism can transport angular momentum efficiently only if $B_\phi^2/2\mu_0 \gg p$. Such a strong toroidal field cannot be contained in the disk, because it is buoyant and rises out of it. The requirement for the magnetic stresses to do the job of transporting angular momentum is that $B_R B_\phi/\mu_0 \approx \alpha\rho c_s^2$, with $B_R \sim B_\phi$. Detailed simulations have shown that MRI is a mechanism that naturally produces the required field strengths over a short timescale. Note that a vertical field can also carry angular momentum away from the disk, through torsional Alfvén waves (Exercise 26.1).

27.1 Qualitative Picture

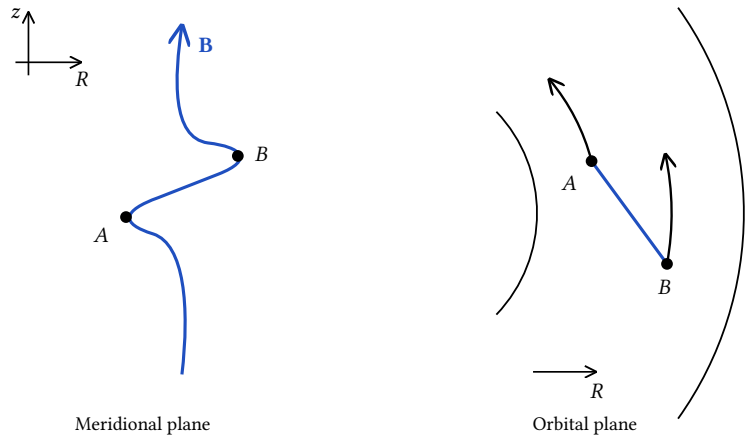


Figure 27.1: Qualitative description of the magnetorotational instability transporting angular momentum between fluid elements A and B. *Left panel:* meridional view. *Right panel:* orbital plane.

Consider an accretion disk with Keplerian angular velocity profile

$$\Omega(R) = \Omega_0 \left(\frac{R_0}{R}\right)^{3/2}, \tag{27.1}$$

initially threaded by a vertical magnetic field $B_0\hat{\mathbf{z}}$. Let A and B be two fluid elements connected by a magnetic field line (initially vertical) and displace their positions according to $\xi_r = \epsilon \cos kz$, as shown in Figure 27.1. Because inner orbits are faster, A moves ahead of B, as shown in the right panel of Figure 27.1. As a result, the field line that connects them is stretched in the azimuthal direction, and a toroidal component of the field, B_ϕ, is generated.

The magnetic tension (Chapter 25) then creates an attractive force between A and B, which has the effect of transporting angular momentum from A to B. This causes A to lose angular momentum and sink further inwards, while B gains angular momentum and starts moving outwards. In turn, because the angular velocity decreases with radius, A increases its angular velocity, while B slows down, further separating the two points. This in turn enhances the magnetic tension force. The cycle we have just described is unstable, so the initial perturbation grows exponentially in time, until the system becomes nonlinear.

The mechanism we have just described involves the magnetic tension force, so it is natural to expect a characteristic growth rate of

$$\lambda \approx k v_A, \qquad v_A = \frac{B_z}{\sqrt{\mu_0 \rho}}. \tag{27.2}$$

In our derivation, we have assumed the field to be sufficiently weak, so that the displaced fluid elements can acquire the angular velocity of their surroundings. This should be valid as long as

$$\omega_A = k v_A \lesssim \Omega, \tag{27.3}$$

where ω_A is the Alfvén frequency (Equation (26.15)). We will find that the exact condition for instability is $k v_A < \sqrt{3}\Omega$, so the maximum growth rate is achieved by taking k sufficiently large, so that $\lambda \approx k v_A \approx \Omega$. This is a very short timescale: the amplitude of the fastest mode grows by a factor of about 100 every orbit! Moreover, the maximum growth rate is independent of the field strength, as long as v_A is not so large that the required k to satisfy $k v_A < \sqrt{3}\Omega$ corresponds to perturbations with characteristic size larger than the vertical size of the disk. We will see that this requires $v_A \lesssim c_s$.

27.2 Detailed Calculation

The starting point is the momentum equation of ideal MHD (Equation (23.4)), with the Lorentz force written as the sum of magnetic pressure and tension (Equation (25.6)). We write it in vector form. We use cylindrical coordinates (R, ϕ, z).

$$\rho\left[\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v}\cdot\nabla)\mathbf{v}\right] = -\nabla\left(p + \frac{B^2}{2\mu_0}\right) + \frac{(\mathbf{B}\cdot\nabla)\mathbf{B}}{\mu_0} + \mathbf{g}\rho. \tag{27.4}$$

We consider a disk that is Keplerian and threaded by a vertical magnetic field:

$$\mathbf{v} = R\Omega\hat{\boldsymbol{\phi}} + \mathbf{u}, \qquad \mathbf{B} = B_0\hat{\mathbf{z}} + \mathbf{b}, \qquad p + \frac{B^2}{2\mu_0} = P_0 + P. \tag{27.5}$$

We will treat $\mathbf{u}$ and $\mathbf{b}$ as (small) Eulerian perturbations. We will only consider modes that are axisymmetric (no ϕ dependence) and incompressible

$$\nabla\cdot\mathbf{u} = \frac{1}{R}\frac{\partial(R u_R)}{\partial R} + \frac{\partial u_z}{\partial z} = 0. \tag{27.6}$$

In equilibrium, the centrifugal force balances gravity, $R\Omega^2 = GM/R^2$, and P_0 is constant. We neglect the self-gravity of the disk, so that $\mathbf{g}$ is not perturbed. To first order in $\mathbf{u}$, and using Kepler's law, we have

$$\begin{aligned}(\mathbf{v}\cdot\nabla)\mathbf{v} &\to (R\Omega)\hat{\boldsymbol{\phi}}\cdot\nabla\mathbf{u} + \mathbf{u}\cdot\nabla(R\Omega\hat{\boldsymbol{\phi}}) \\ &= -2\Omega u_\phi\hat{\mathbf{R}} + \left(R\frac{\mathrm{d}\Omega}{\mathrm{d}R} + 2\Omega\right)u_R\hat{\boldsymbol{\phi}} \\ &= \frac{1}{2}\Omega u_R\hat{\boldsymbol{\phi}} - 2\Omega u_\phi\hat{\mathbf{R}}.\end{aligned} \tag{27.7}$$

Similarly,

$$\begin{aligned}(\mathbf{B}\cdot\nabla)\mathbf{B} \to B_0\hat{\mathbf{z}}\cdot\nabla\mathbf{b} + \mathbf{b}\cdot\nabla(B_0\hat{\mathbf{z}}) = \\ B_0\frac{\partial b_R}{\partial z}\hat{\mathbf{R}} + B_0\frac{\partial b_\phi}{\partial z}\hat{\boldsymbol{\phi}} + B_0\frac{\partial b_z}{\partial z}\hat{\mathbf{z}}.\end{aligned} \tag{27.8}$$

Since the unit vectors do not depend on z, this is $B_0\,\partial\mathbf{b}/\partial z$. So the linearized momentum equation reads

$$\frac{\partial \mathbf{u}}{\partial t} + \frac{1}{2}\Omega u_R \hat{\boldsymbol{\phi}} - 2\Omega u_\phi \hat{\mathbf{R}} = -\frac{1}{\rho}\nabla P + \frac{B_0}{\mu_0 \rho}\frac{\partial \mathbf{b}}{\partial z}. \tag{27.9}$$

The induction equation, Equation (23.19) with $\eta = 0$, can be linearized in a similar way:

$$\begin{aligned}
&\nabla \times (\mathbf{v} \times \mathbf{B}) \\
&\quad \to \nabla \times \big[(R\Omega\hat{\boldsymbol{\phi}}) \times (b_R\hat{\mathbf{R}} + b_\phi\hat{\boldsymbol{\phi}} + b_z\hat{\mathbf{z}}) \\
&\qquad\quad + (u_R\hat{\mathbf{R}} + u_\phi\hat{\boldsymbol{\phi}} + u_z\hat{\mathbf{z}}) \times (B_0\hat{\mathbf{z}})\big] \\
&\quad = \nabla \times [-R\Omega b_R\hat{\mathbf{z}} + R\Omega b_z\hat{\mathbf{R}} \\
&\qquad\quad - u_R B_0\hat{\boldsymbol{\phi}} + u_\phi B_0\hat{\mathbf{R}}] \\
&\quad = \nabla \times [(R\Omega b_z + u_\phi B_0)\hat{\mathbf{R}} \\
&\qquad\quad - u_R B_0\hat{\boldsymbol{\phi}} - R\Omega b_R\hat{\mathbf{z}}] \\
&\quad = \left(\frac{\partial u_R}{\partial z}B_0\right)\hat{\mathbf{R}} + \Bigg(R\Omega\frac{\partial b_z}{\partial z} + \frac{\partial u_\phi}{\partial z}B_0 \\
&\qquad\quad - \frac{3}{2}\Omega b_R + \Omega\frac{\partial (R b_R)}{\partial R}\Bigg)\hat{\boldsymbol{\phi}} \\
&\qquad\quad - \frac{1}{R}\left[\frac{\partial (R u_R B_0)}{\partial R}\right]\hat{\mathbf{z}} \\
&\quad = \left(\frac{\partial u_R}{\partial z}B_0\right)\hat{\mathbf{R}} + \left(\frac{\partial u_\phi}{\partial z}B_0 - \frac{3}{2}\Omega b_R\right)\hat{\boldsymbol{\phi}} \\
&\qquad\quad - \frac{1}{R}\left[\frac{\partial (R u_R B_0)}{\partial R}\right]\hat{\mathbf{z}}.
\end{aligned} \tag{27.10}$$

In the last equation we have also used the condition that $\nabla \cdot \mathbf{b} = 0$.

We will consider a small patch of the disk, so that we can use the Fourier (WKB) ansatz:

$$\mathbf{u} = \mathbf{u}_0 \mathrm{e}^{\lambda t + \mathrm{i}(k_R R + k_z z)}, \qquad \mathbf{b} = \mathbf{b}_0 \mathrm{e}^{\lambda t + \mathrm{i}(k_R R + k_z z)} \tag{27.11}$$

where $k_R R, k_z R \gg 1$, so that we can neglect terms of order $1/R$. We also assume $k_z \gg k_R$, which simplifies the algebra and turns out to give the fastest-growing modes. Here $\lambda = -\mathrm{i}\omega$, so that modes with $\lambda^2 > 0$ grow exponentially, while modes with $\lambda^2 < 0$ oscillate. Substituting this into the linearized equations of motion yields

$$\begin{aligned}
\lambda u_R - 2\Omega u_\phi &= -\mathrm{i}k_R\frac{P}{\rho} + \frac{\mathrm{i}k_z B_0 b_R}{\mu_0\rho}, \\
\lambda u_\phi + \frac{\Omega}{2}u_R &= \frac{\mathrm{i}k_z B_0 b_\phi}{\mu_0\rho}, \\
\lambda u_z &= -\mathrm{i}k_z\frac{P}{\rho} + \frac{\mathrm{i}k_z B_0 b_z}{\mu_0\rho}.
\end{aligned} \tag{27.12}$$

We can eliminate P/ρ from the last equation and insert it into the first to obtain (in the limit $k_z \gg k_R$):

$$\lambda u_R - 2\Omega u_\phi = \mathrm{i}k_z\frac{B_0 b_R}{\mu_0\rho}. \tag{27.13}$$

The linearized induction equation yields

$$\begin{aligned}
\lambda b_R &= \mathrm{i}k_z B_0 u_R \\
\lambda b_\phi &= -\frac{3}{2}\Omega b_R + \mathrm{i}k_z B_0 u_\phi.
\end{aligned} \tag{27.14}$$

Eliminating b_R in the second equation yields

$$\lambda b_\phi = -\frac{3}{2}\frac{\mathrm{i}k_z B_0 \Omega}{\lambda} u_R + \mathrm{i}k_z B_0 u_\phi. \tag{27.15}$$

Eliminating b_ϕ and b_R in the momentum equations above and introducing the Alfvén frequency $\omega_A = k_z v_A$, with $\omega_A^2 = k_z^2 B_0^2/\mu_0\rho$, yields

$$\begin{aligned}(\lambda^2 + \omega_A^2)u_R &= 2\lambda\Omega u_\phi,\\ (\lambda^2 + \omega_A^2)u_\phi &= \left(-\frac{\lambda}{2}\Omega + \frac{3}{2}\frac{\omega_A^2\Omega}{\lambda}\right)u_R.\end{aligned} \tag{27.16}$$

Introducing

$$\mu^2 = \lambda^2 + \omega_A^2, \tag{27.17}$$

the dispersion relation becomes

$$f(\mu^2) = \mu^4 + \Omega^2\mu^2 - 4\Omega^2\omega_A^2 = 0. \tag{27.18}$$

The product of the two roots of $f(\mu^2)$ is negative, so there is one positive and one negative root. Since $\lambda^2 = \mu^2 - \omega_A^2$, the instability is present when the positive root is larger than ω_A^2. Since $f(0) < 0$, this occurs if

$$f(\omega_A^2) = \omega_A^4 - 3\Omega^2\omega_A^2 < 0 \Rightarrow \boxed{\frac{k_z B_0}{\sqrt{\mu_0\rho}} = \omega_A < \sqrt{3}\Omega.} \tag{27.19}$$

We find that if k_z is sufficiently small (long wavelengths), then the mode is unstable. However, if the background field is too large, then the condition in Equation (27.19) might only be satisfied for k_z that corresponds to modes larger than the disk thickness $H \approx c_s/\Omega$ (Exercise 19.1), so the disk is stable to the MRI. The unstable modes have wavelengths longer than $2\pi v_A/(\sqrt{3}\Omega)$, so they fit in the disk only if $v_A \lesssim c_s$. In other words, the MRI operates when the plasma β (Equation (25.25)) is larger than about one, that is, when the magnetic pressure is smaller than the gas pressure.

The growth rate of the instability is found by solving the dispersion relation to find

$$\lambda^2 = \frac{\Omega}{2}\left(-\Omega + \sqrt{\Omega^2 + 16\omega_A^2}\right) - \omega_A^2, \tag{27.20}$$

which is maximized when

$$\omega_A = \frac{\sqrt{15}}{4}\Omega \approx \Omega, \tag{27.21}$$

which corresponds to

$$k_{\max} = \frac{\sqrt{15}}{4}\frac{\Omega}{v_A} \approx \frac{\Omega}{v_A}. \tag{27.22}$$

The growth rate of the most unstable mode is

$$\boxed{\lambda_{\max} = \frac{3}{4}\Omega.} \tag{27.23}$$

Example: The MRI in a protoplanetary disk

Consider the disk of gas and dust around a young star like the Sun, at a distance $R = 1\,\mathrm{AU}$ from the star. In the minimum-mass solar nebula, a standard model for the early solar system, the gas at this distance has $T \approx 280\,\mathrm{K}$ and $\rho \approx 1.4 \times 10^{-9}\,\mathrm{g\,cm^{-3}} = 1.4 \times 10^{-6}\,\mathrm{kg\,m^{-3}}$. The gas is mostly molecular hydrogen, with $\mu \approx 2.3$, so $c_s \approx 1\,\mathrm{km\,s^{-1}}$. The angular velocity is $\Omega = (GM_\odot/R^3)^{1/2} \approx$

$2 \times 10^{-7}\,\mathrm{s}^{-1}$, corresponding to an orbital period of one year, and the thickness of the disk is $H \approx c_s/\Omega \approx 5 \times 10^9\,\mathrm{m} \approx 0.03\,\mathrm{AU}$.

Suppose that the disk is threaded by a vertical field $B_0 \approx 10^{-4}\,\mathrm{T}$ (1 G). The Alfvén speed is

$$v_A = \frac{B_0}{\sqrt{\mu_0 \rho}} \approx 75\,\mathrm{m\,s}^{-1}, \tag{27.24}$$

and the plasma $\beta \approx 360$. The fastest-growing mode has wavelength $2\pi/k_{\max} \approx 2.5 \times 10^9\,\mathrm{m} \approx H/2$, so it fits in the disk. Its growth time is $1/\lambda_{\max} = 4/(3\Omega) \approx 80$ days. In other words, even a field with magnetic pressure 360 times smaller than the gas pressure makes the disk unstable, and the perturbations grow by a factor of about 100 every year.

It is instructive to check the limit of our analysis when $B_0 \to 0$, which corresponds to $\omega_A \to 0$. If $\omega_A \equiv 0$, then Equation (27.16) reduces to

$$\lambda^2 u_R = -\Omega^2 u_R, \qquad \lambda^2 u_\phi = -\Omega^2 u_\phi. \tag{27.25}$$

In other words, the perturbations oscillate with the epicyclic frequency κ, which for a Keplerian disk equals Ω. More in general, an unmagnetized rotating fluid is stable to axisymmetric perturbations as long as its specific angular momentum $R^2\Omega$ increases outward. This is called the Rayleigh criterion, and it is satisfied by Keplerian disks.

In the limit of $0 < \omega_A \ll \Omega$ (long wavelength perturbations, or weak magnetic field), we can drop all the ω_A terms in Equation (27.16), with the exception of the term multiplied by Ω to obtain

$$\begin{aligned} \lambda^2 u_R &= 2\lambda\Omega u_\phi, \\ \lambda^2 u_\phi &= \left(-\frac{\lambda}{2}\Omega + \frac{3}{2}\frac{\omega_A^2 \Omega}{\lambda}\right) u_R. \end{aligned} \tag{27.26}$$

Eliminating u_ϕ, we obtain:

$$\lambda^2 u_R = \left(\frac{3\omega_A^2\Omega^2}{\lambda^2} - \Omega^2\right) u_R, \tag{27.27}$$

which has unstable solutions with $\lambda^2 > 0$ as long as $\omega_A > 0$. For $\omega_A \ll \Omega$, the growing solution is $\lambda \approx \sqrt{3}\omega_A$. This growth rate goes to zero for $B_0 \to 0$, at fixed k_z. However, the fastest-growing mode always grows at the rate $\frac{3}{4}\Omega$, for any $B_0 > 0$. In other words, an arbitrarily weak magnetic field makes the disk unstable, even though the disk is stable without the field. Note that the MRI only requires the angular velocity to decrease outward (see Section 27.3), while the Rayleigh criterion is about the angular momentum. This is why the MRI is so important for accretion disks.

27.3 In-class worksheet

There is a simple mechanical equivalent of the magnetorotational instability (MRI) in a disk with vertical magnetic field. This model is concerned with two particles of mass m orbiting a central object with angular velocity $\Omega = \Omega(R)$ (not necessarily Keplerian). The particles are connected by a spring, as shown in Figure 27.2. The spring pulls each mass towards the common center of mass with a force equal to $-mK$ times its displacement.

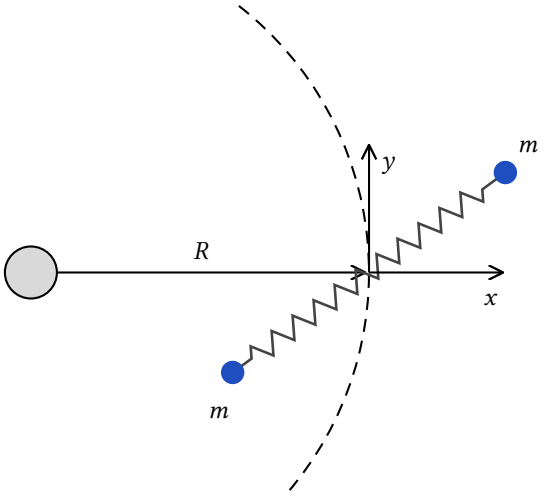


Figure 27.2: Mechanical analogue of the magnetorotational instability: two masses connected by a spring, orbiting a central object.

The coordinates x and y denote the displacement of one of the two masses from the common center of mass.

1. Assuming $|x|, |y| \ll R$, show that the displacement satisfies the differential equations

$$\begin{aligned} \ddot{x} - 2\Omega\dot{y} &= -\left(K + \frac{\mathrm{d}\Omega^2}{\mathrm{d}\ln R}\right) x, \\ \ddot{y} + 2\Omega\dot{x} &= -Ky. \end{aligned} \tag{27.28}$$

   Note that these are the same equations as those obtained for the MRI in a disk threaded by a vertical magnetic field, $k_R = 0$, and $K = (v_A k_z)^2 = \omega_A^2$.

2. Look for solutions in the form $x(t) = x_0 \mathrm{e}^{\lambda t}$, $y(t) = y_0 \mathrm{e}^{\lambda t}$. Find λ.

3. Under what conditions is the system unstable?

27.4 Exercises

Exercise 27.1 (Dead zones in protoplanetary disks). In this exercise we estimate how much ionization is needed for the MRI to operate in a protoplanetary disk.

1. Consider the MRI in a disk with finite resistivity, that is, with magnetic diffusivity $\eta > 0$. Argue that a mode with wavenumber k is damped by Ohmic diffusion on the time scale $T_{\mathrm{decay}} = 1/(k^2\eta)$ (see Equation (24.18)). Show that the fastest-growing mode, with $k \approx \Omega/v_A$, grows faster than it is damped only if

$$v_A^2 \gtrsim \eta\Omega. \tag{27.29}$$

2. In a weakly ionized gas, the magnetic diffusivity is approximately

$$\eta \approx 2.34 \times 10^{-2} \left(\frac{T}{\mathrm{K}}\right)^{1/2} x_e^{-1}\, \mathrm{m^2\,s^{-1}}, \tag{27.30}$$

   where $x_e = n_e/n$ is the ionization fraction. Find the minimum ionization fraction for the MRI to operate in the disk of the protoplanetary-disk Example ($T \approx 280\,\mathrm{K}$, $\Omega \approx 2 \times 10^{-7}\,\mathrm{s^{-1}}$, $v_A \approx 75\,\mathrm{m\,s^{-1}}$).

3. Close to the star, where $T \gtrsim 1000\,\mathrm{K}$, alkali metals such as sodium and potassium are thermally ionized, and x_e is large. Further out, the gas is ionized only by X-rays from the star and by cosmic rays. These ionize the surface layers of the disk, but they do not reach its dense midplane, unless the column density

of the disk is small. Where in the disk do you expect the MRI to operate? Sketch your answer in the Rz-plane.

27.5 Further reading

- C. Clarke and B. Carswell [1], Chapter 13, for the MHD equations used here.
- S. A. Balbus and J. F. Hawley [25], the paper that established the MRI as the origin of turbulence in accretion disks.
- S. A. Balbus and J. F. Hawley [26], a comprehensive review of the MRI and angular-momentum transport in disks.
- C. F. Gammie [27], on dead zones and layered accretion in protoplanetary disks.

Appendices

A. Vector Calculus

> **Note**
>
> This chapter is in preparation.

A.1 In-class worksheet

1. Compute the gradient of the function

$$f(x,y,z) = \frac{x}{\sqrt{x^2+y^2+z^2}}. \tag{A.1}$$

2. Consider the vector field

$$\mathbf{E} = \hat{\mathbf{r}} = \begin{pmatrix} \frac{x}{\sqrt{x^2+y^2+z^2}} \\ \frac{y}{\sqrt{x^2+y^2+z^2}} \\ \frac{z}{\sqrt{x^2+y^2+z^2}} \end{pmatrix}. \tag{A.2}$$

   Compute

   a. the gradient

$$\nabla\mathbf{E} = \begin{pmatrix} \partial_x E_x & \partial_x E_y & \partial_x E_z \\ \partial_y E_x & \partial_y E_y & \partial_y E_z \\ \partial_z E_x & \partial_z E_y & \partial_z E_z \end{pmatrix}; \tag{A.3}$$

   b. $\nabla \cdot \mathbf{E}$;

   c. $\nabla \times \mathbf{E}$.

3. Prove that

$$\nabla \cdot \nabla f = \nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}. \tag{A.4}$$

4. The divergence of a tensor field

$$\mathbf{T} = \begin{pmatrix} T_{xx} & T_{xy} & T_{xz} \\ T_{yx} & T_{yy} & T_{yz} \\ T_{zx} & T_{zy} & T_{zz} \end{pmatrix} \tag{A.5}$$

   is defined as

$$\nabla \cdot \mathbf{T} = \begin{pmatrix} \partial_x T_{xx} + \partial_y T_{xy} + \partial_z T_{xz} \\ \partial_x T_{yx} + \partial_y T_{yy} + \partial_z T_{yz} \\ \partial_x T_{zx} + \partial_y T_{zy} + \partial_z T_{zz} \end{pmatrix}. \tag{A.6}$$

   Prove that

$$\nabla \cdot (p\mathbf{I}) = \nabla \cdot \begin{pmatrix} p & & \\ & p & \\ & & p \end{pmatrix} = \nabla p. \tag{A.7}$$

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